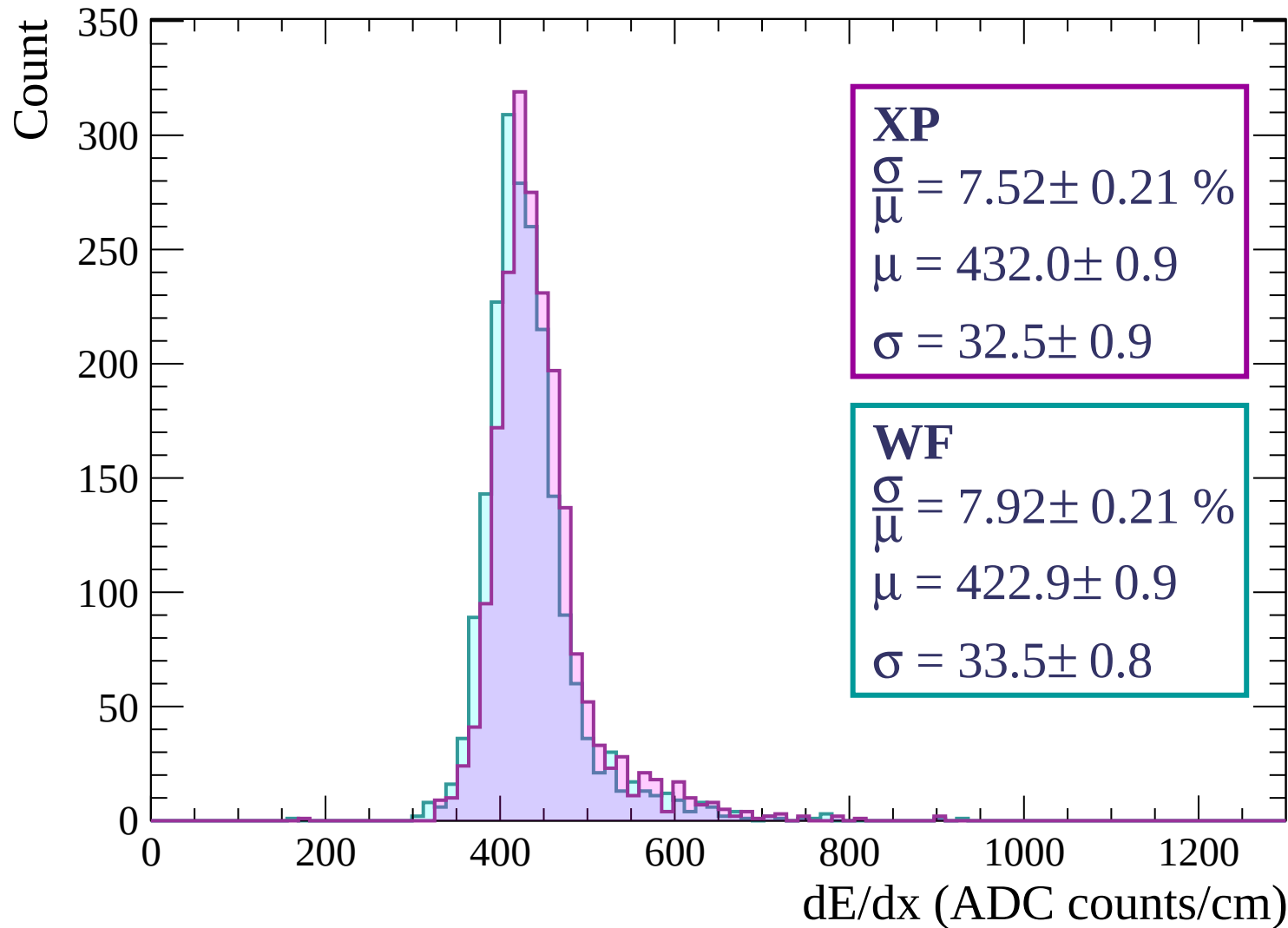
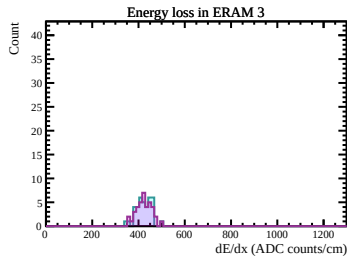
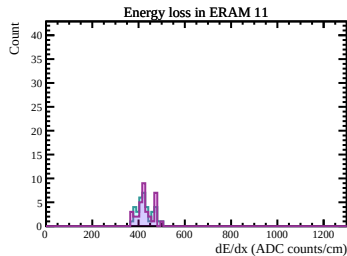
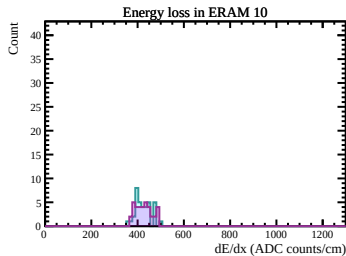
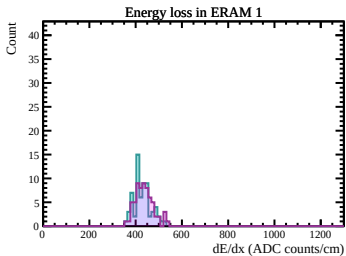
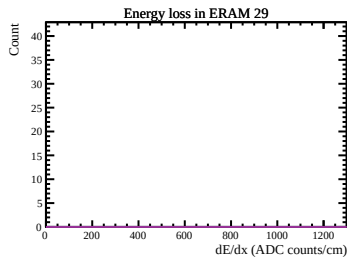
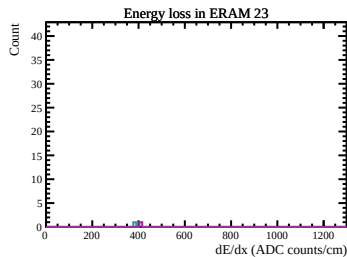
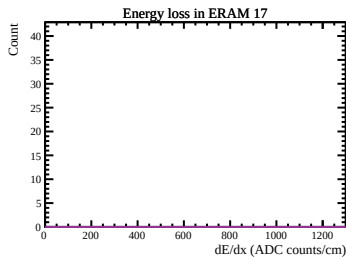
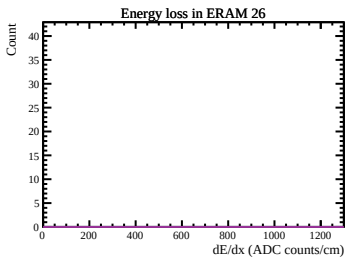
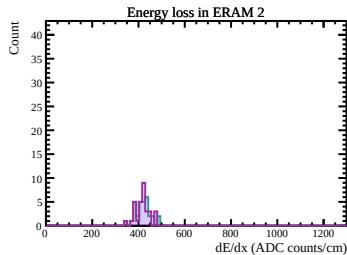
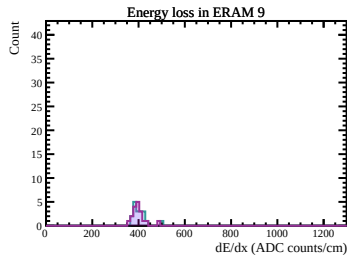
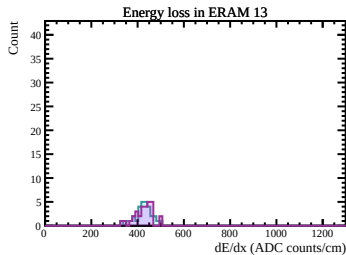
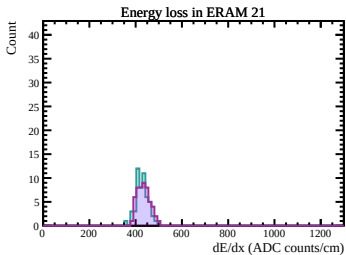
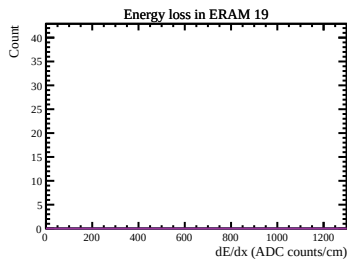
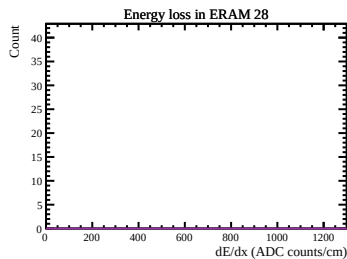
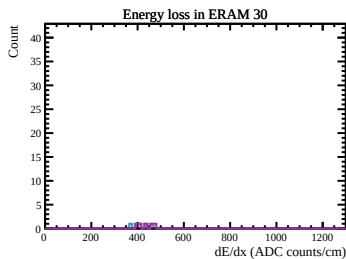
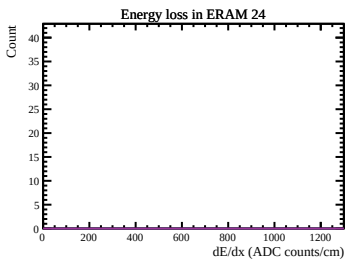
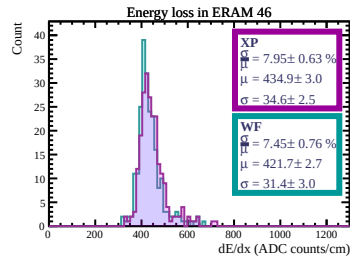
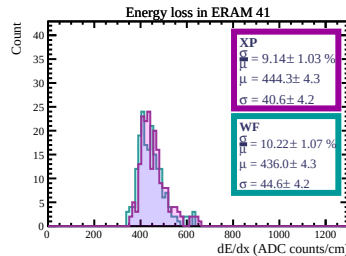
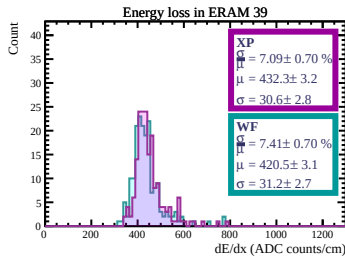
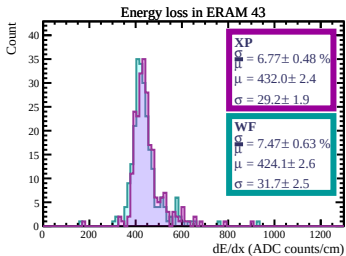
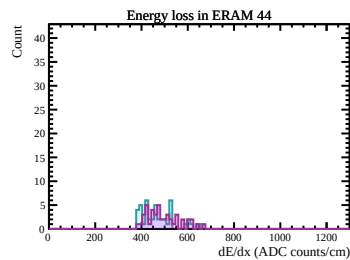
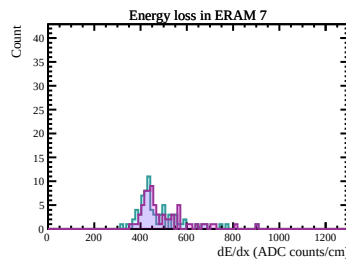
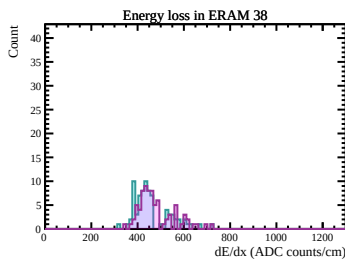
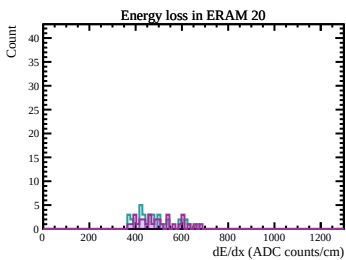
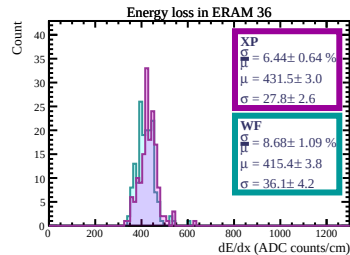
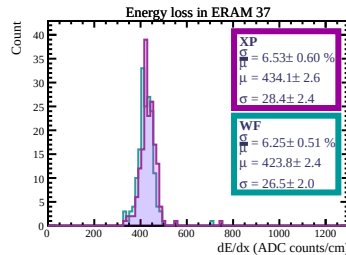
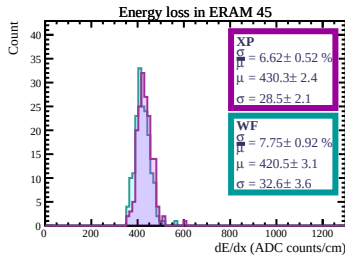
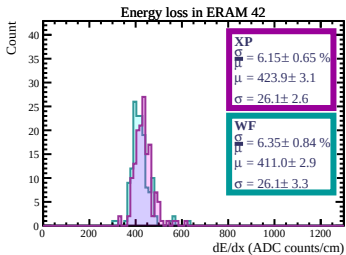
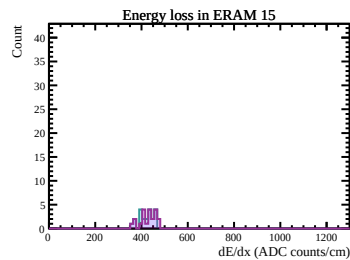
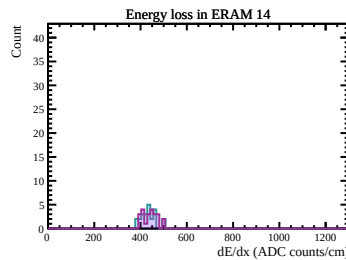
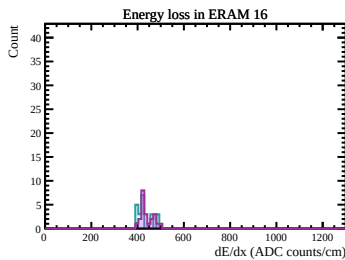
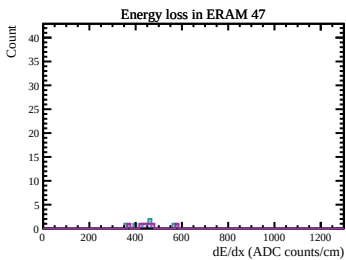
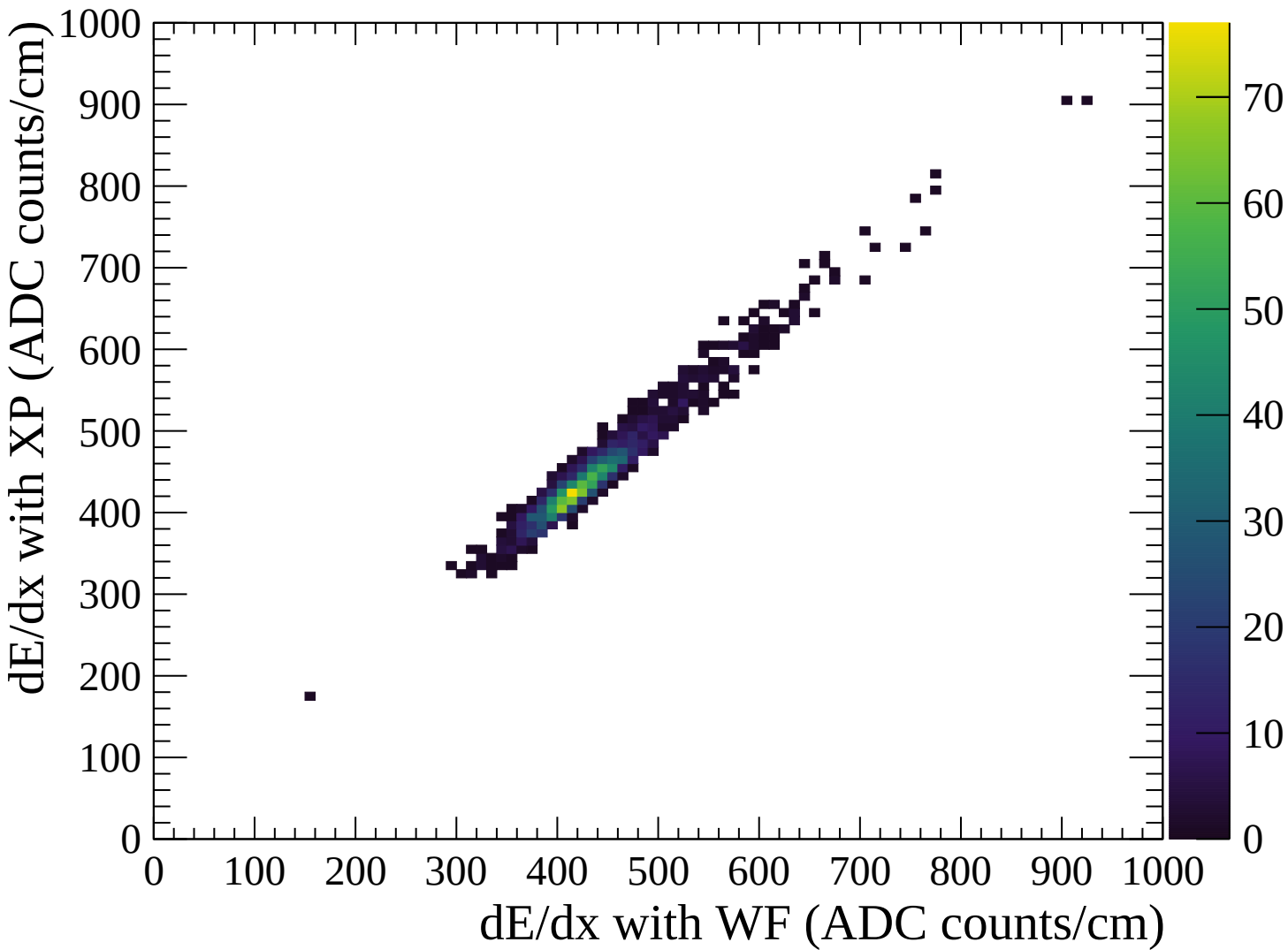
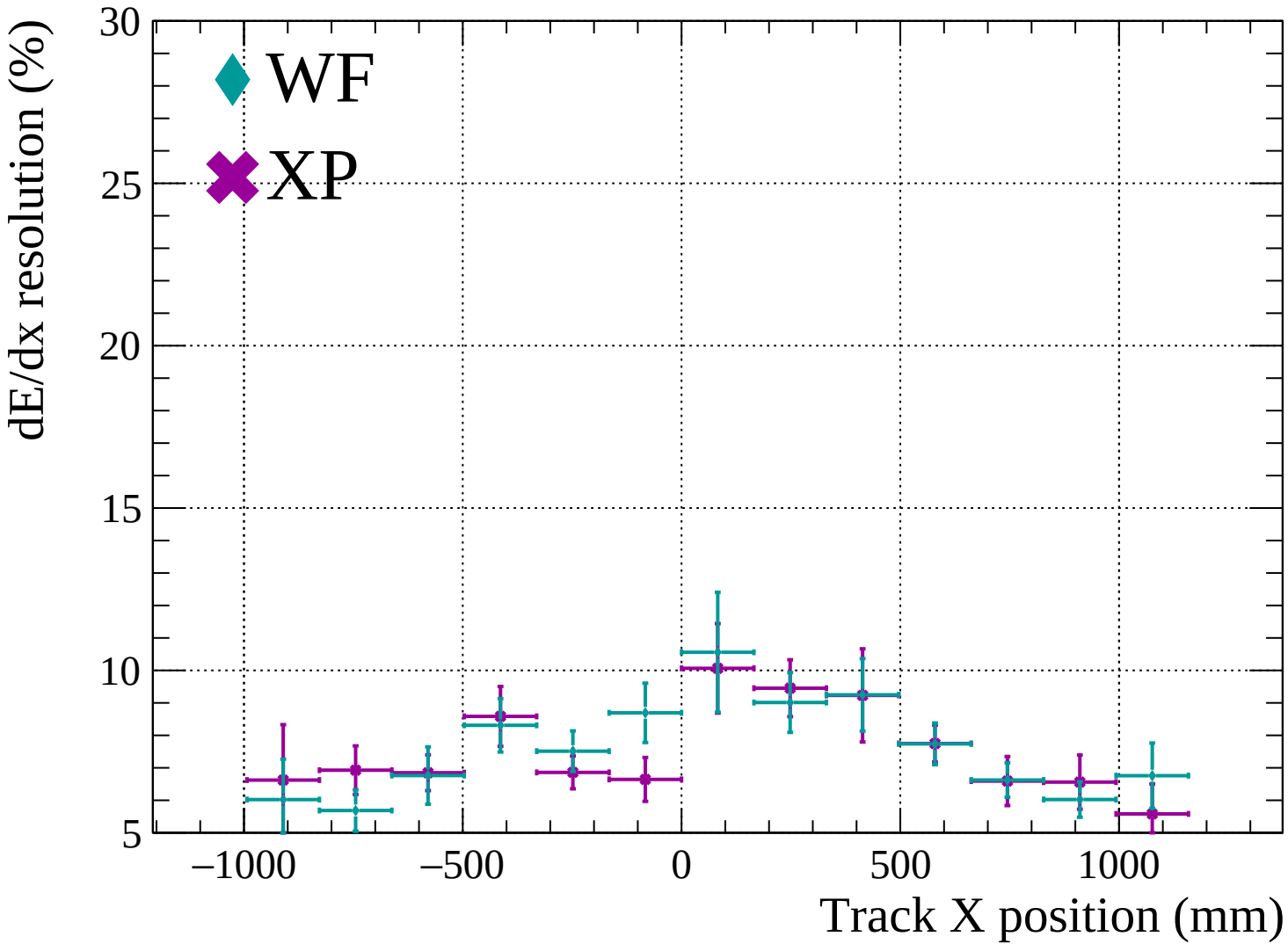


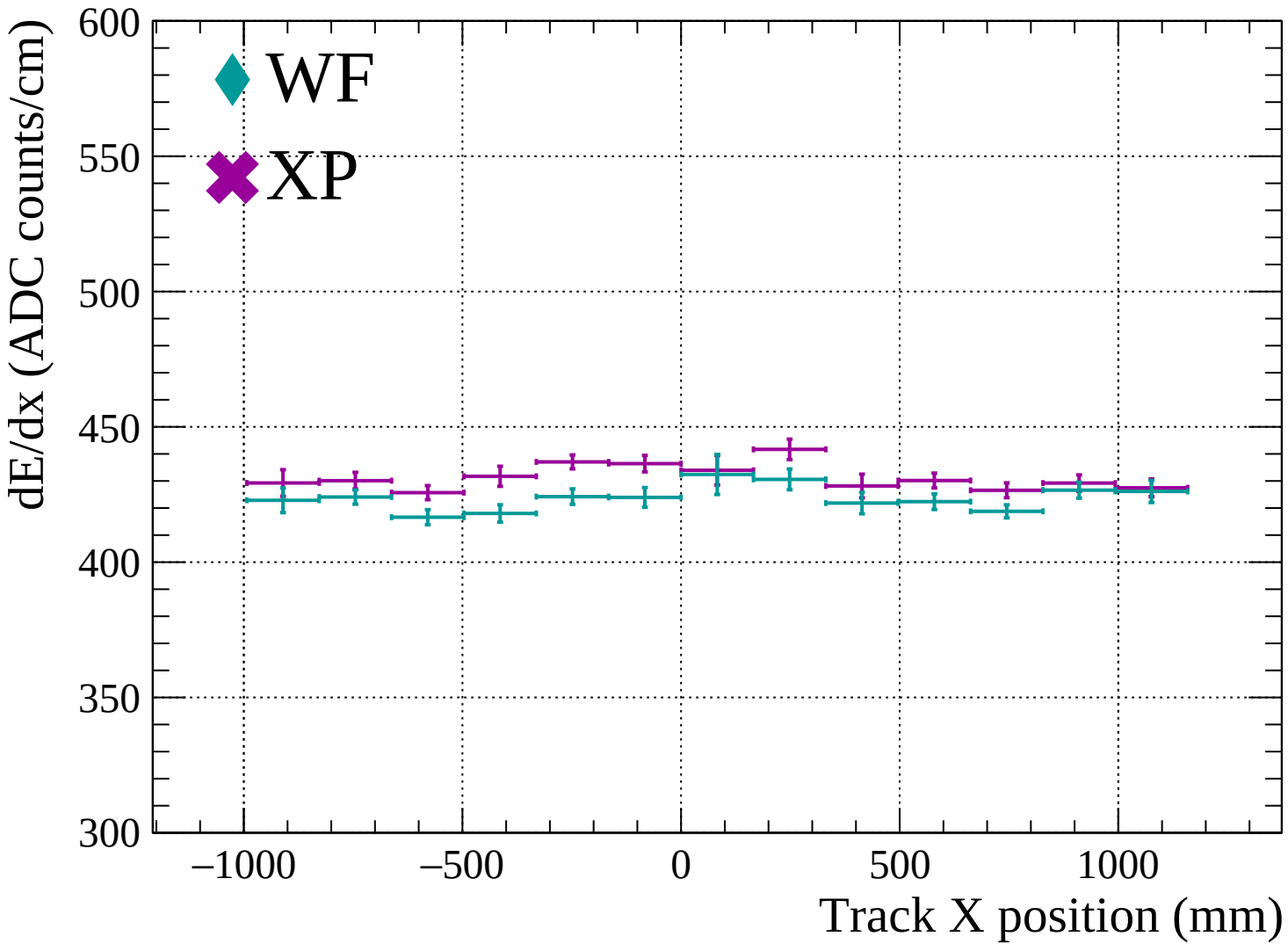


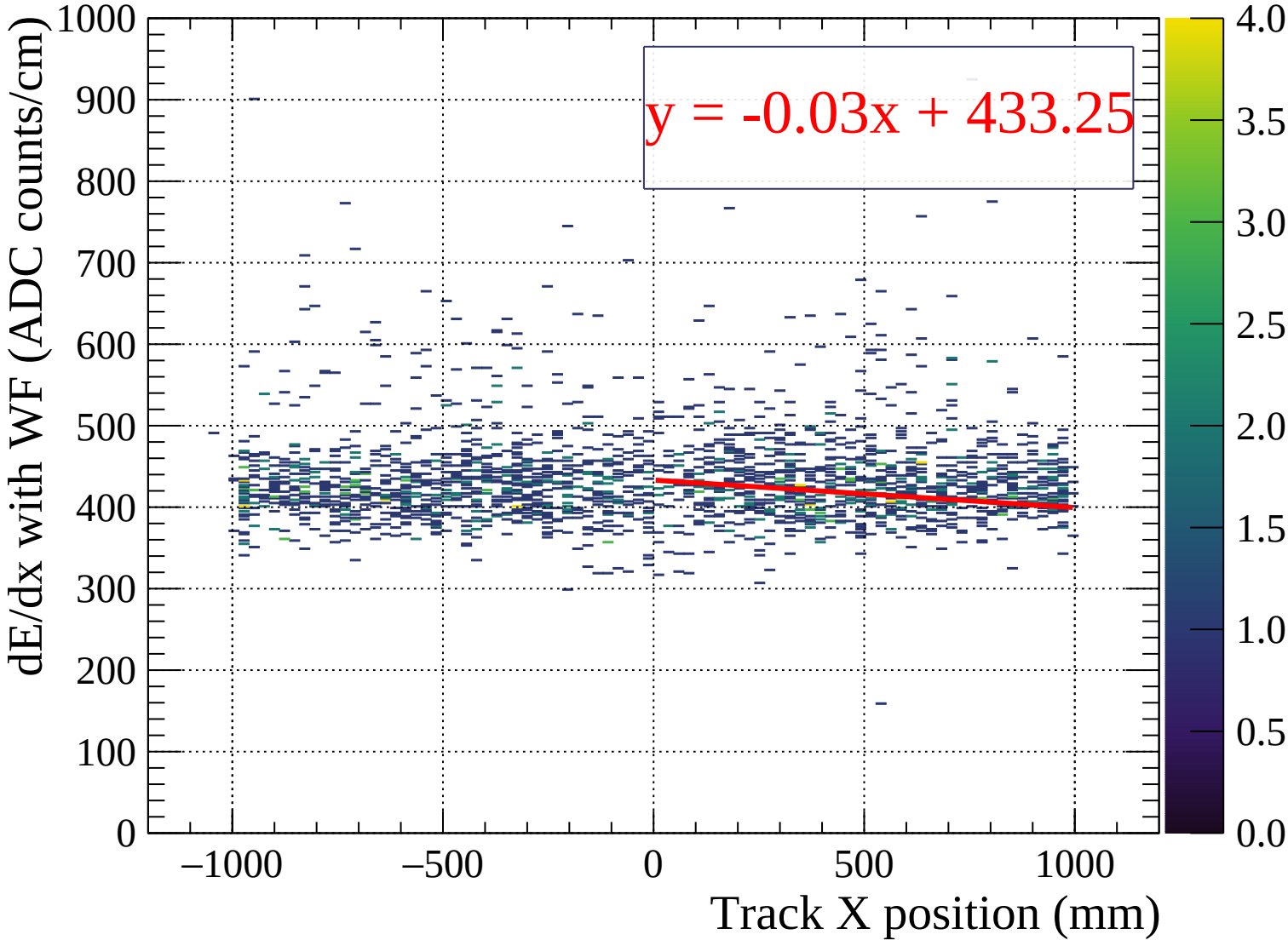


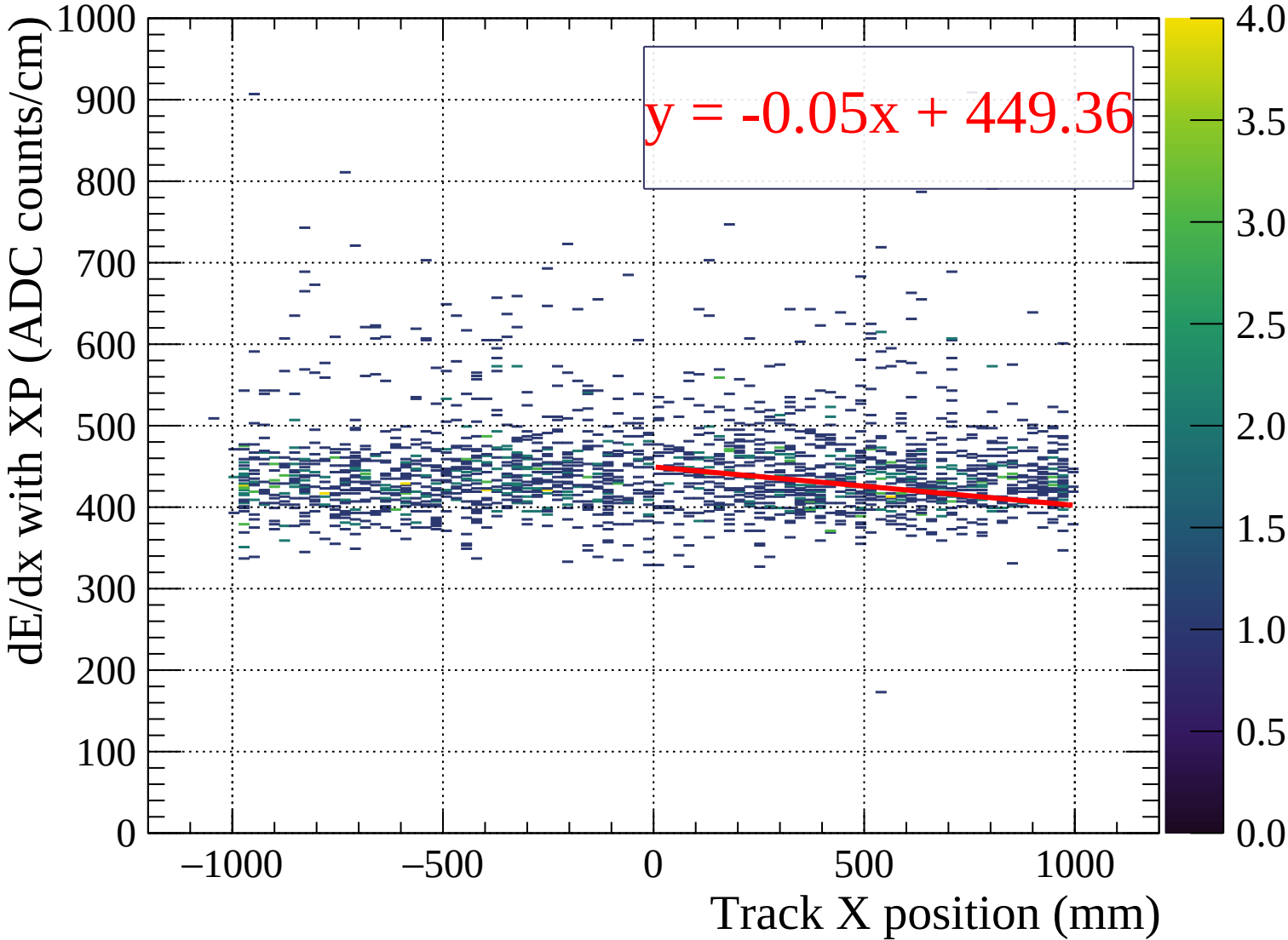


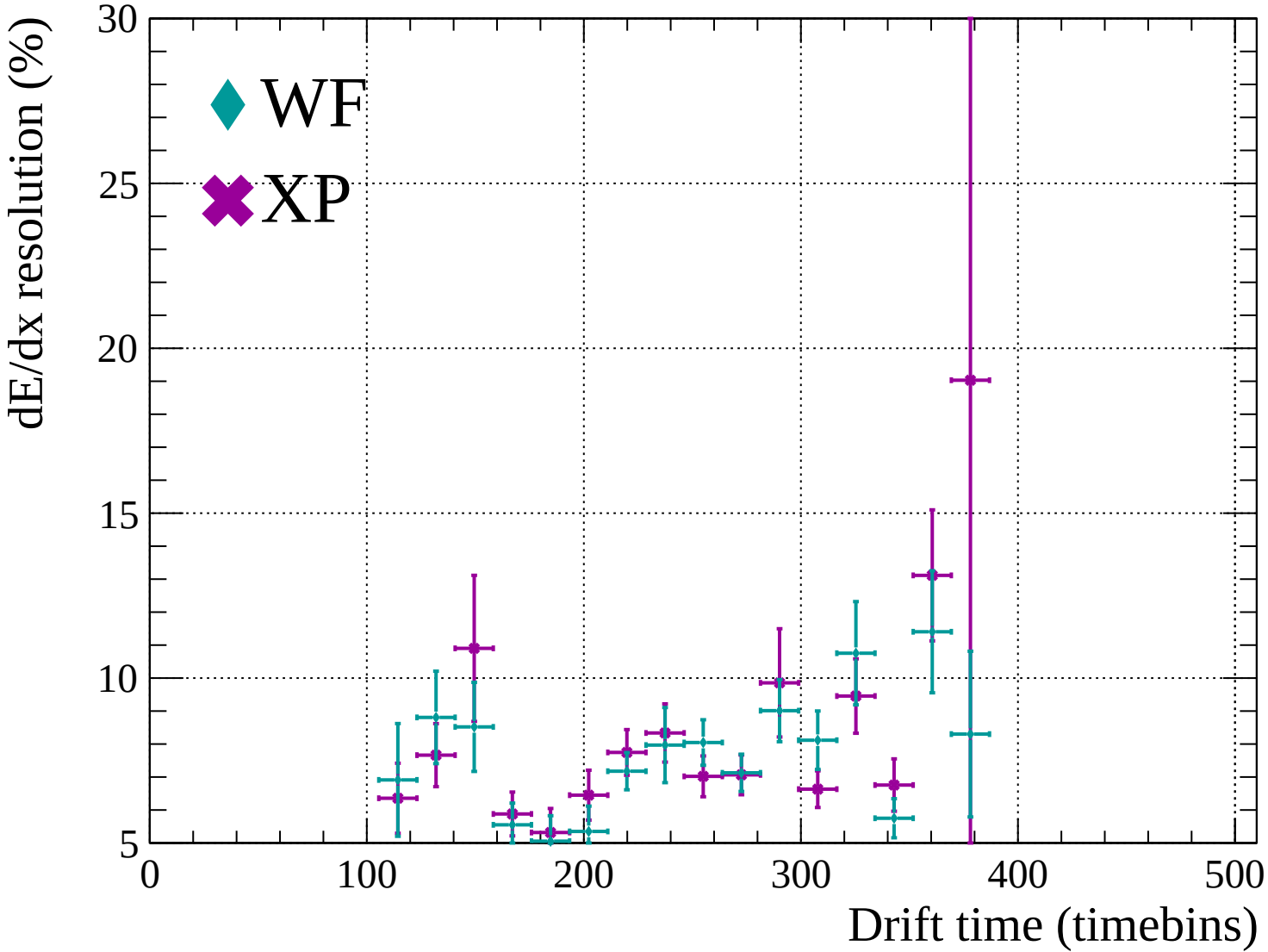


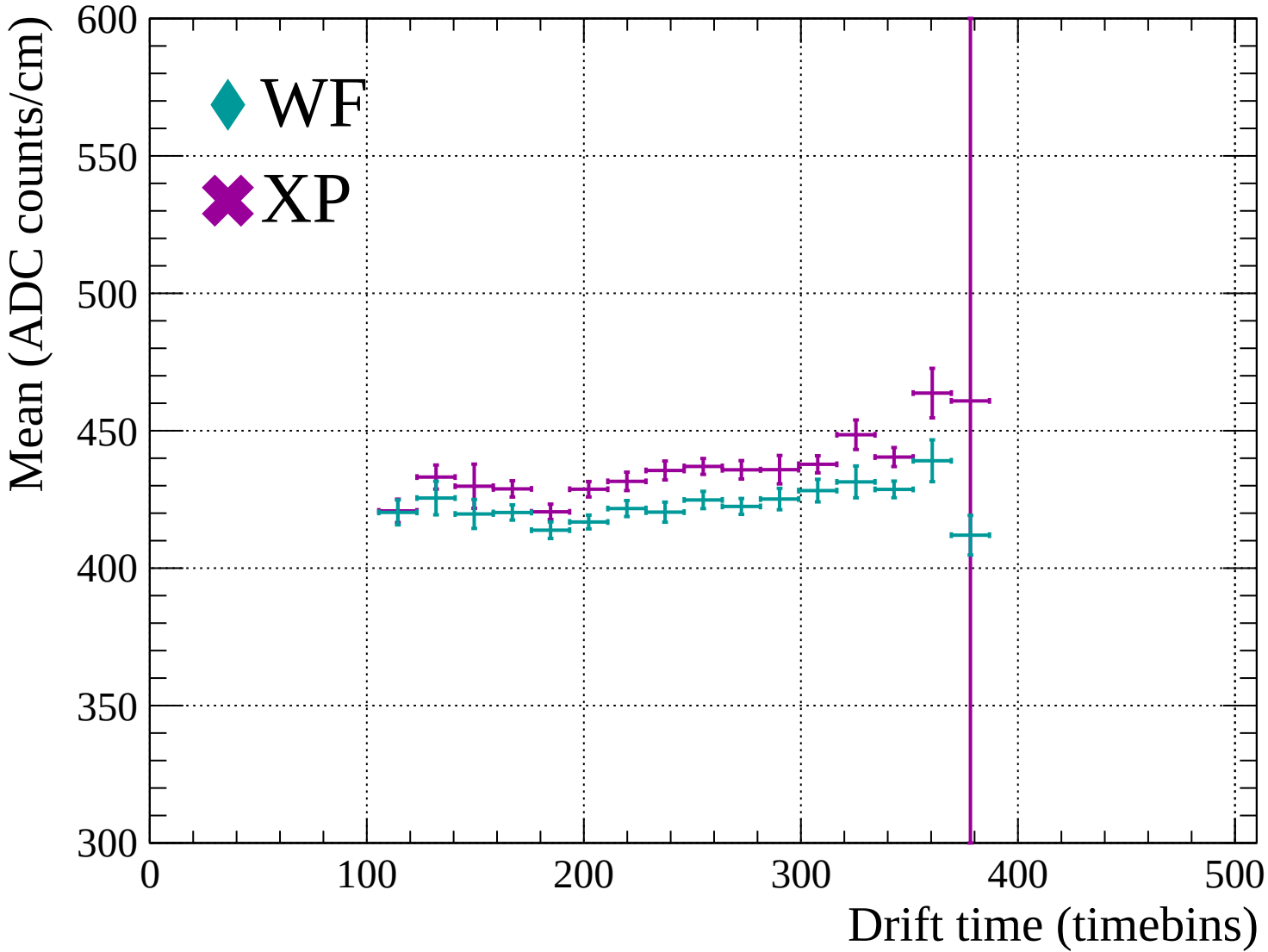


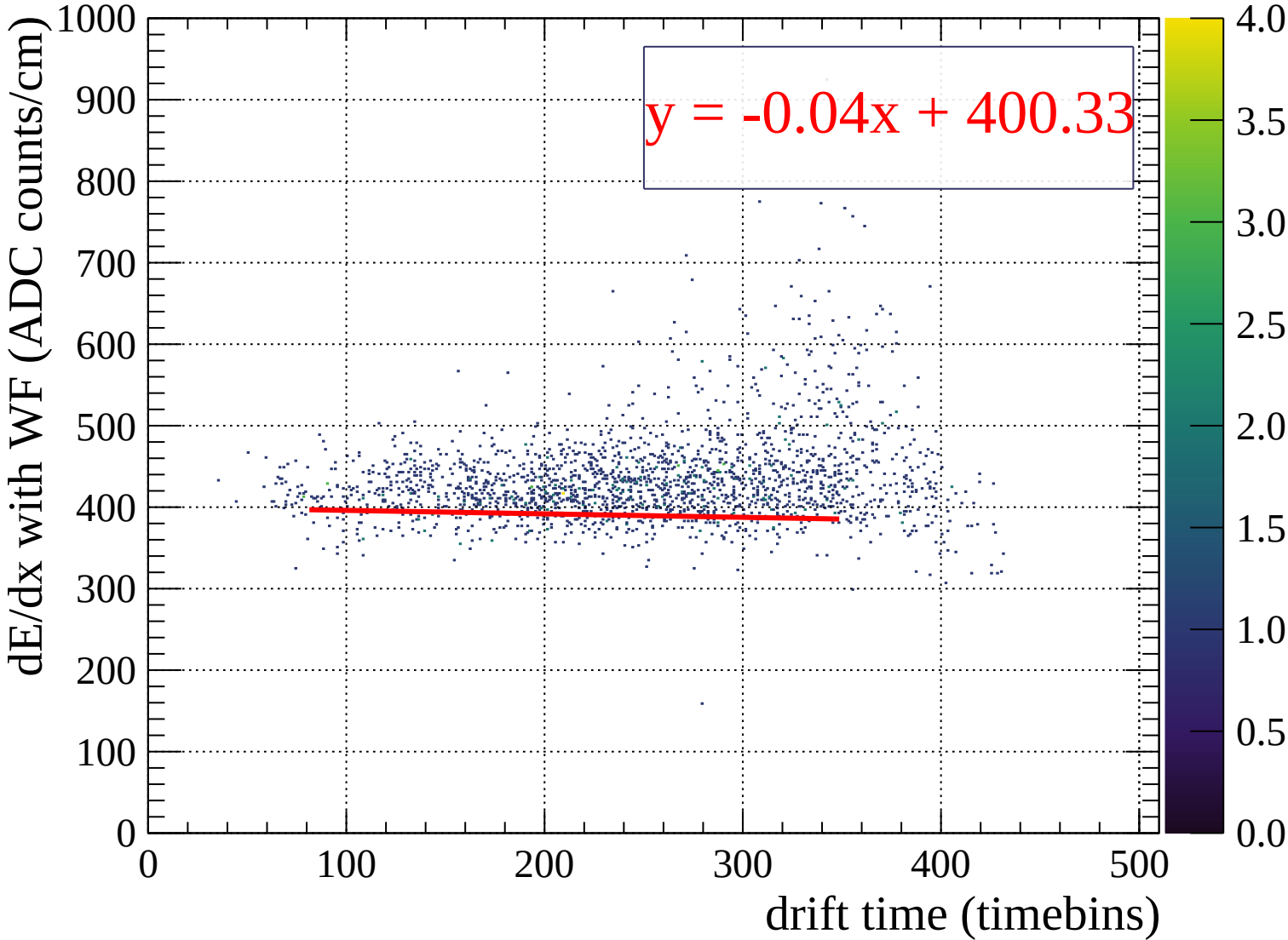


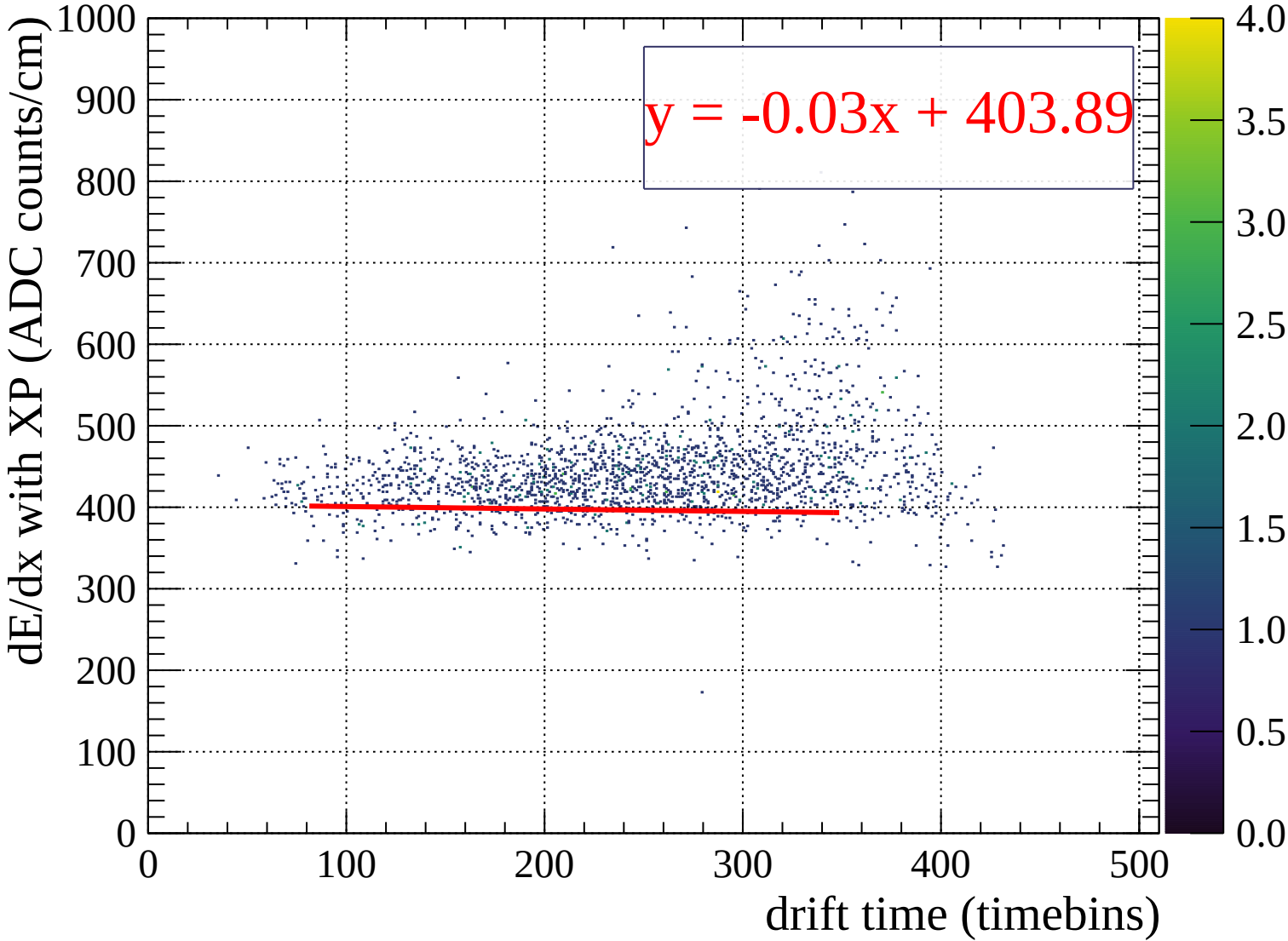


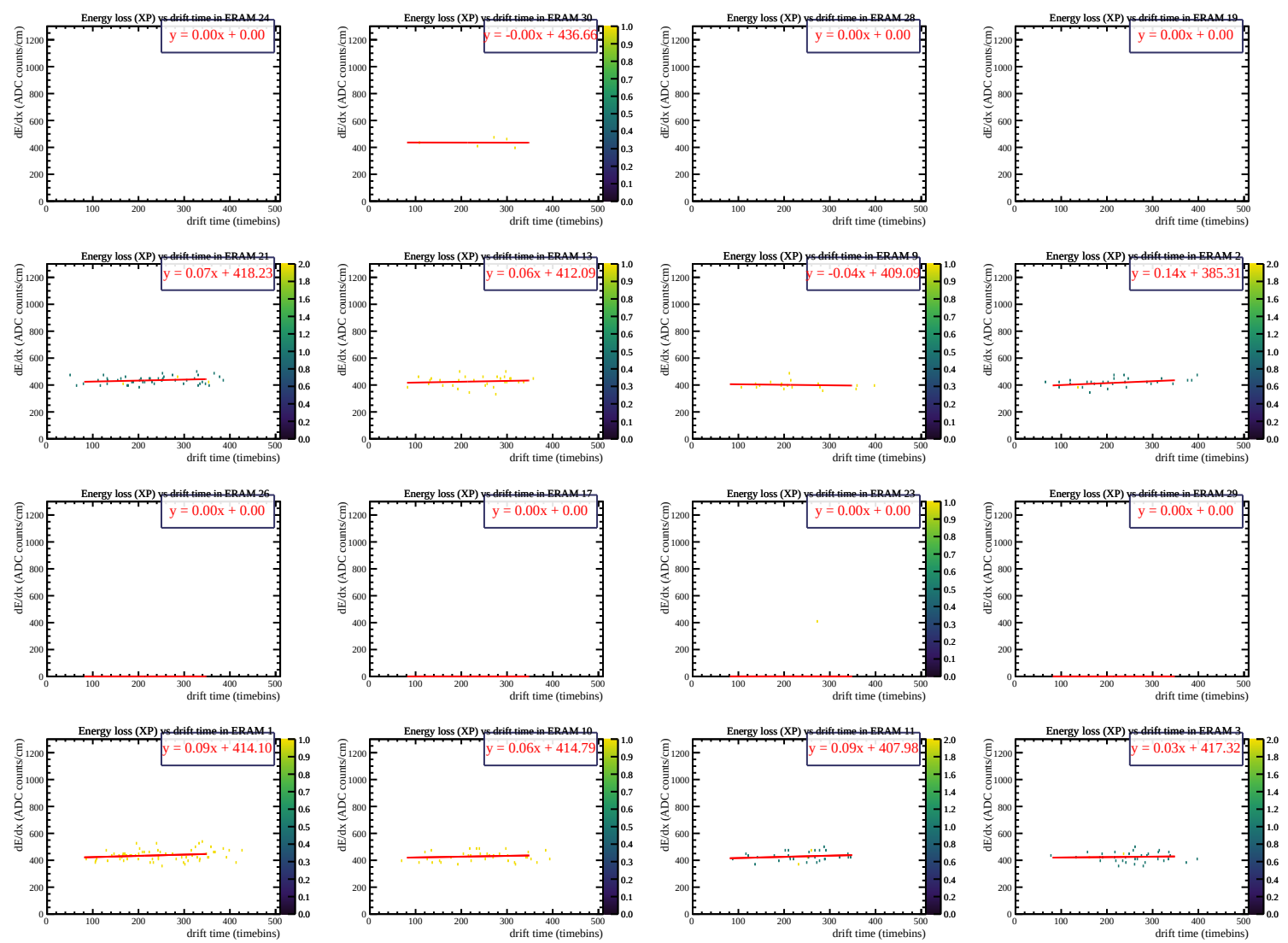


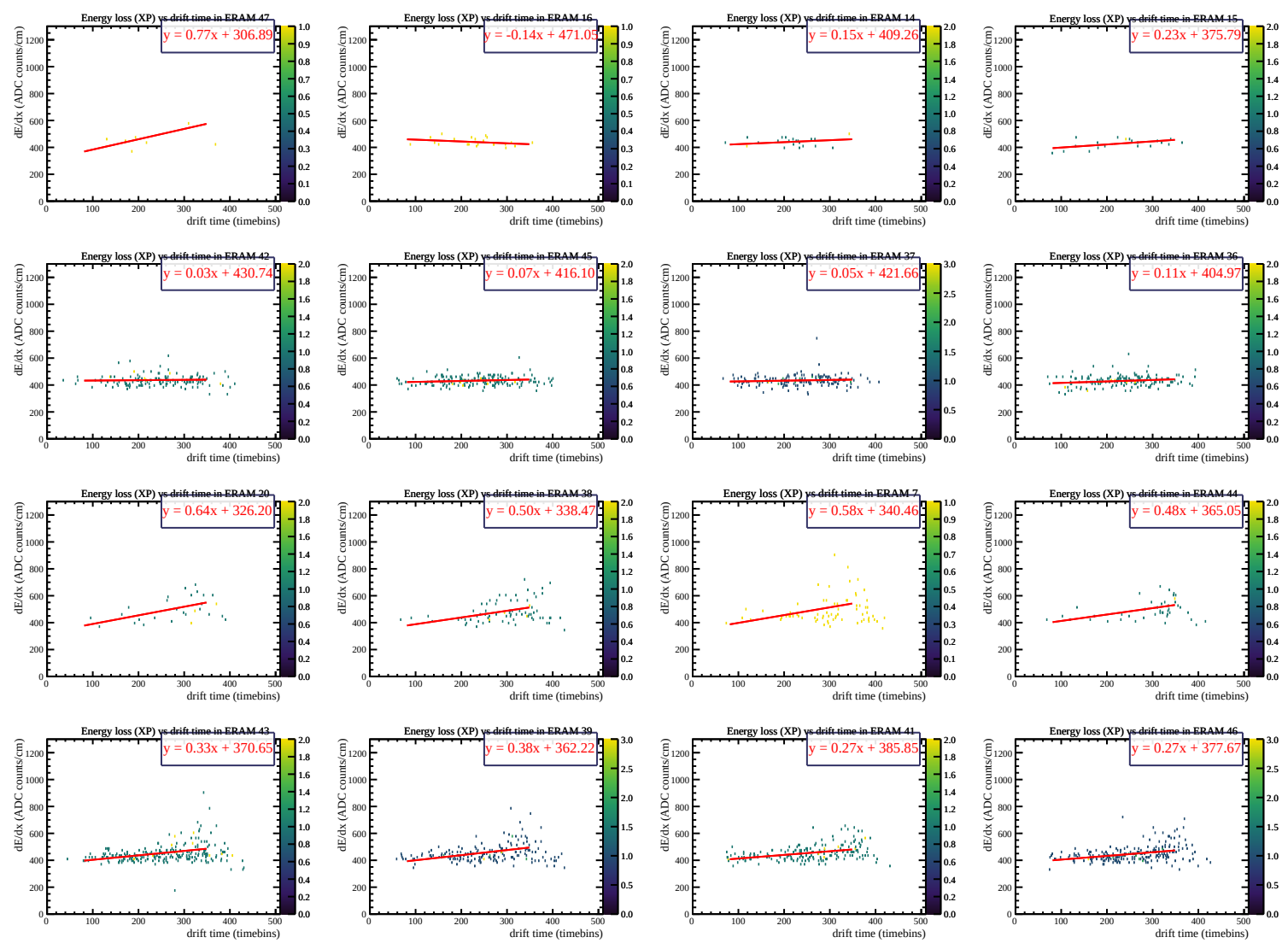


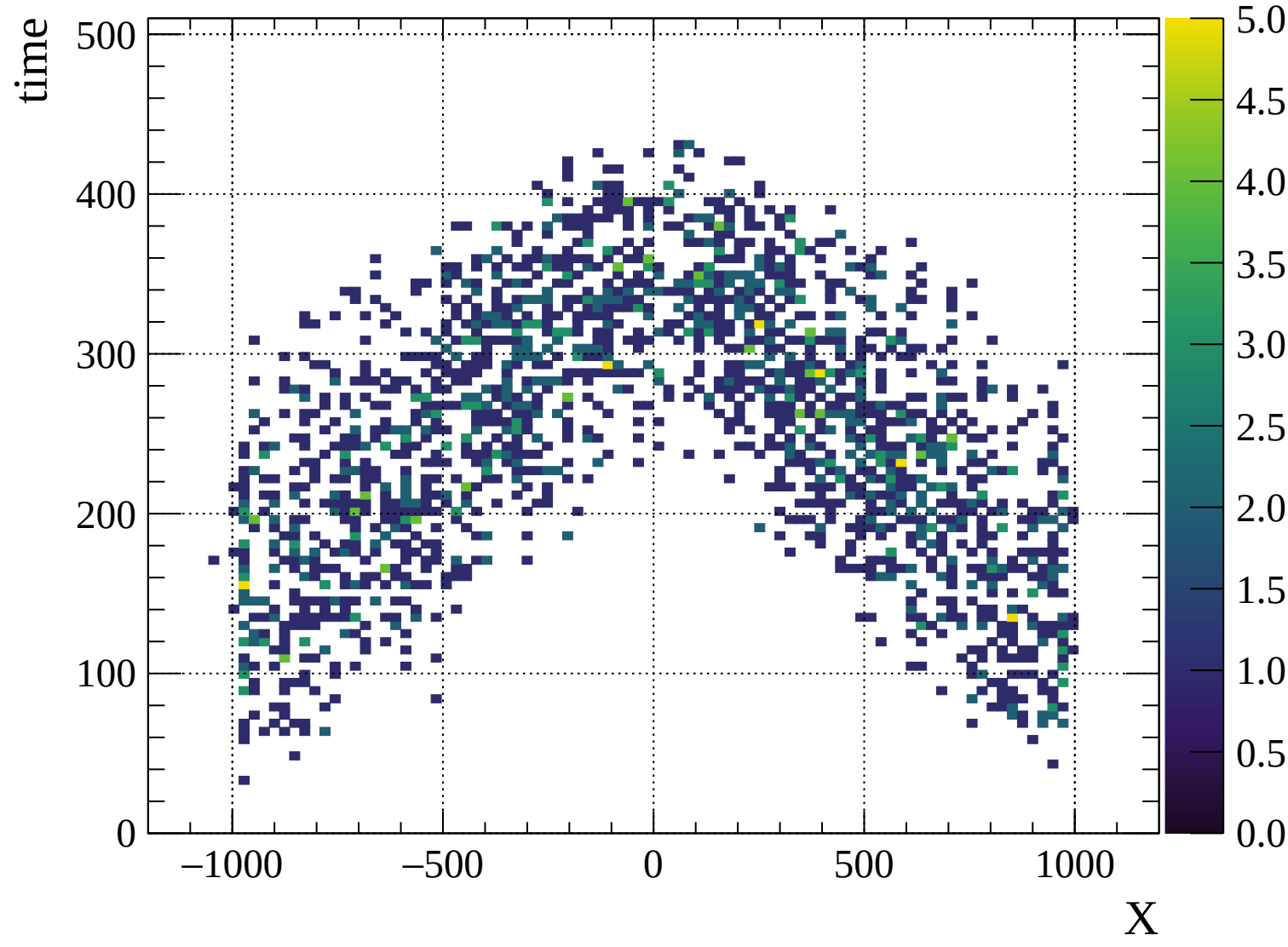


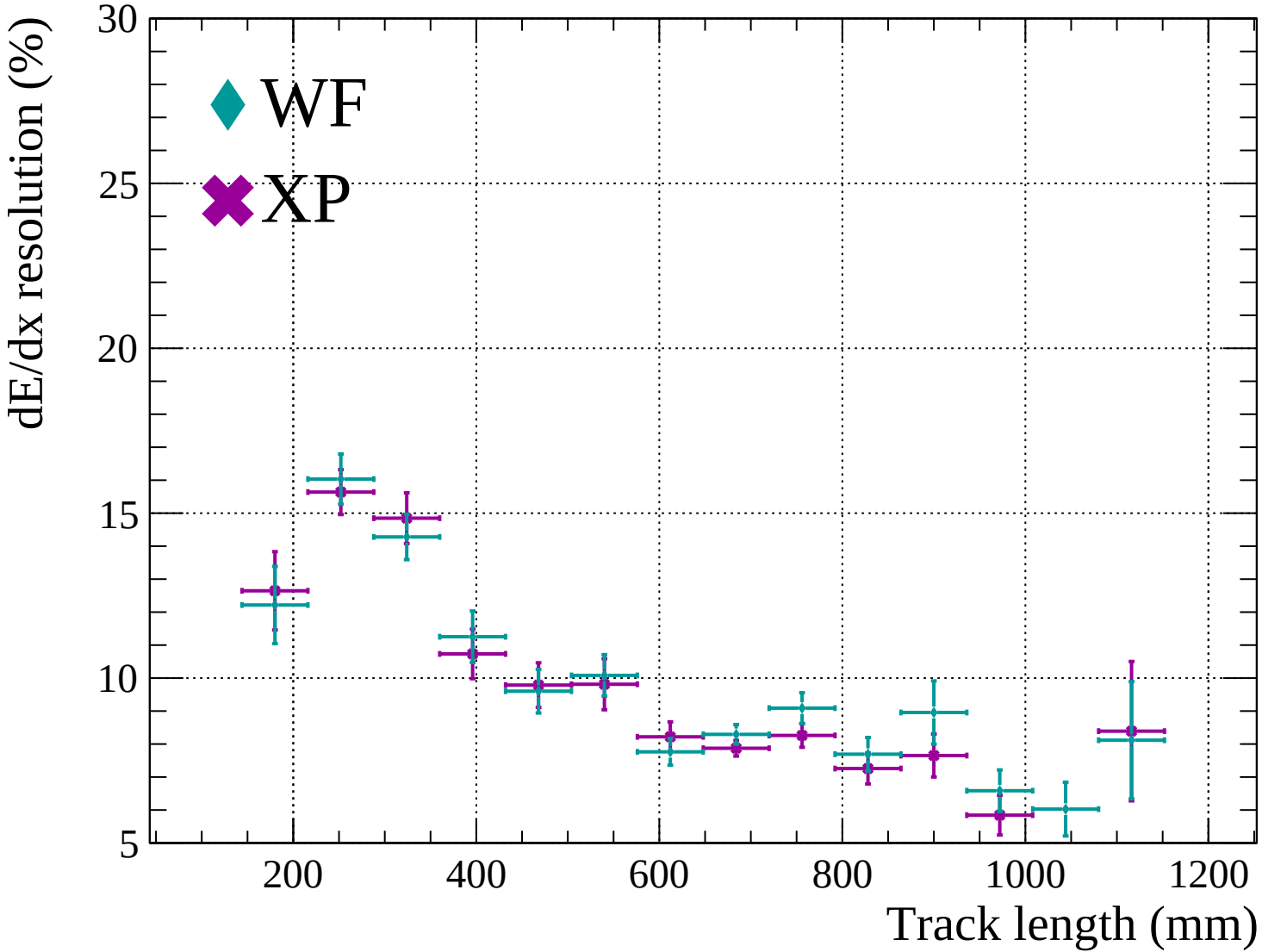


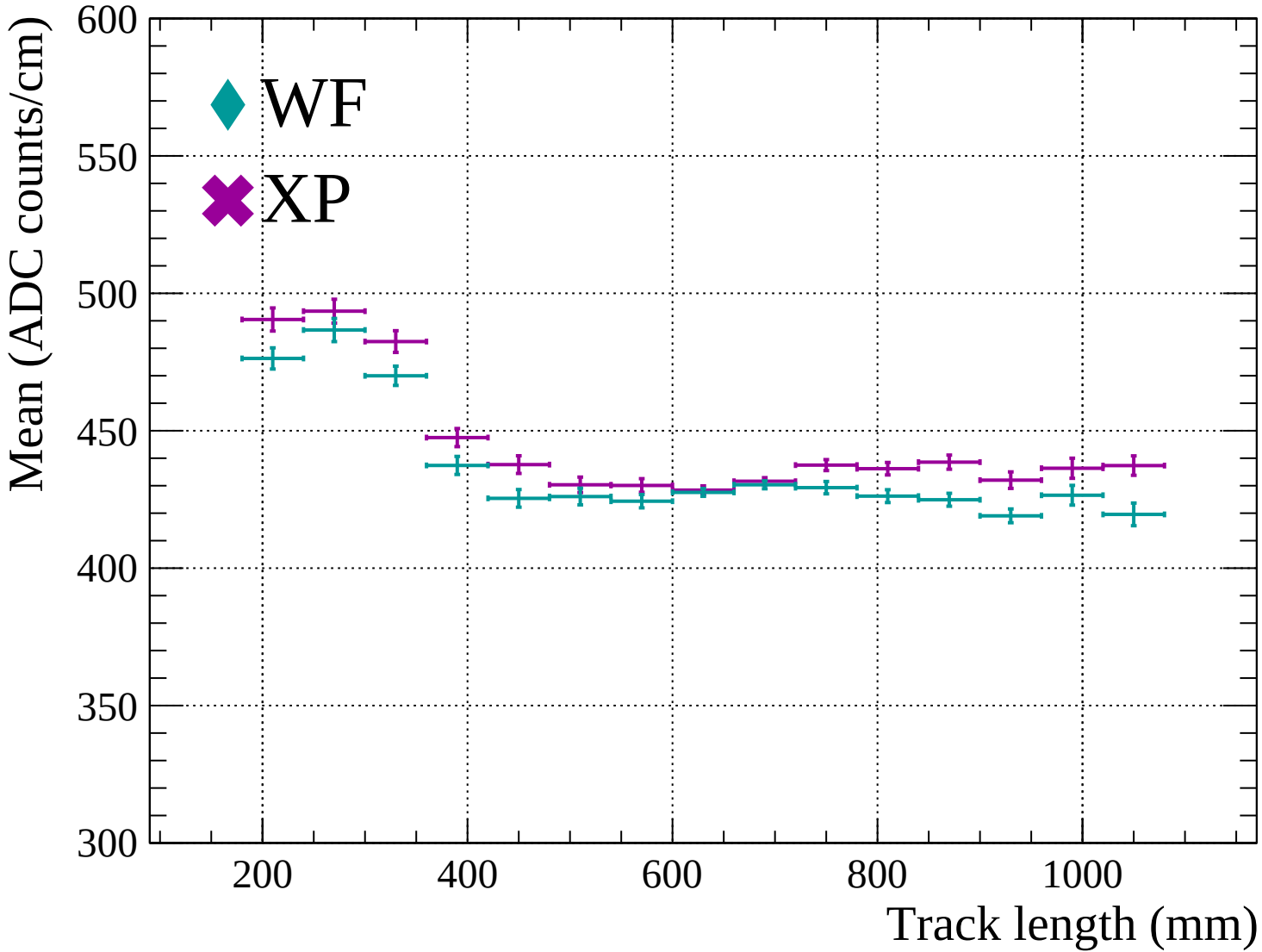


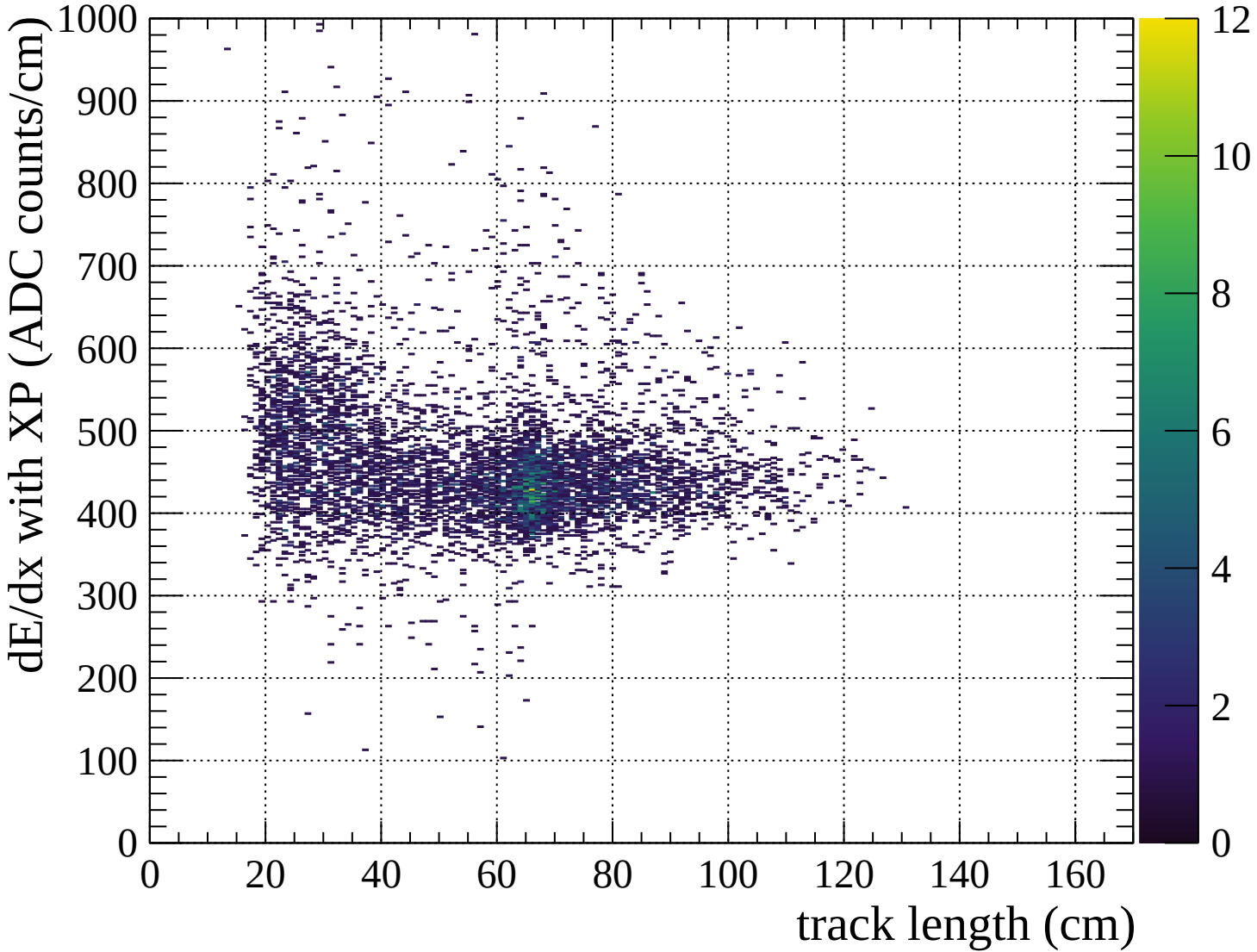


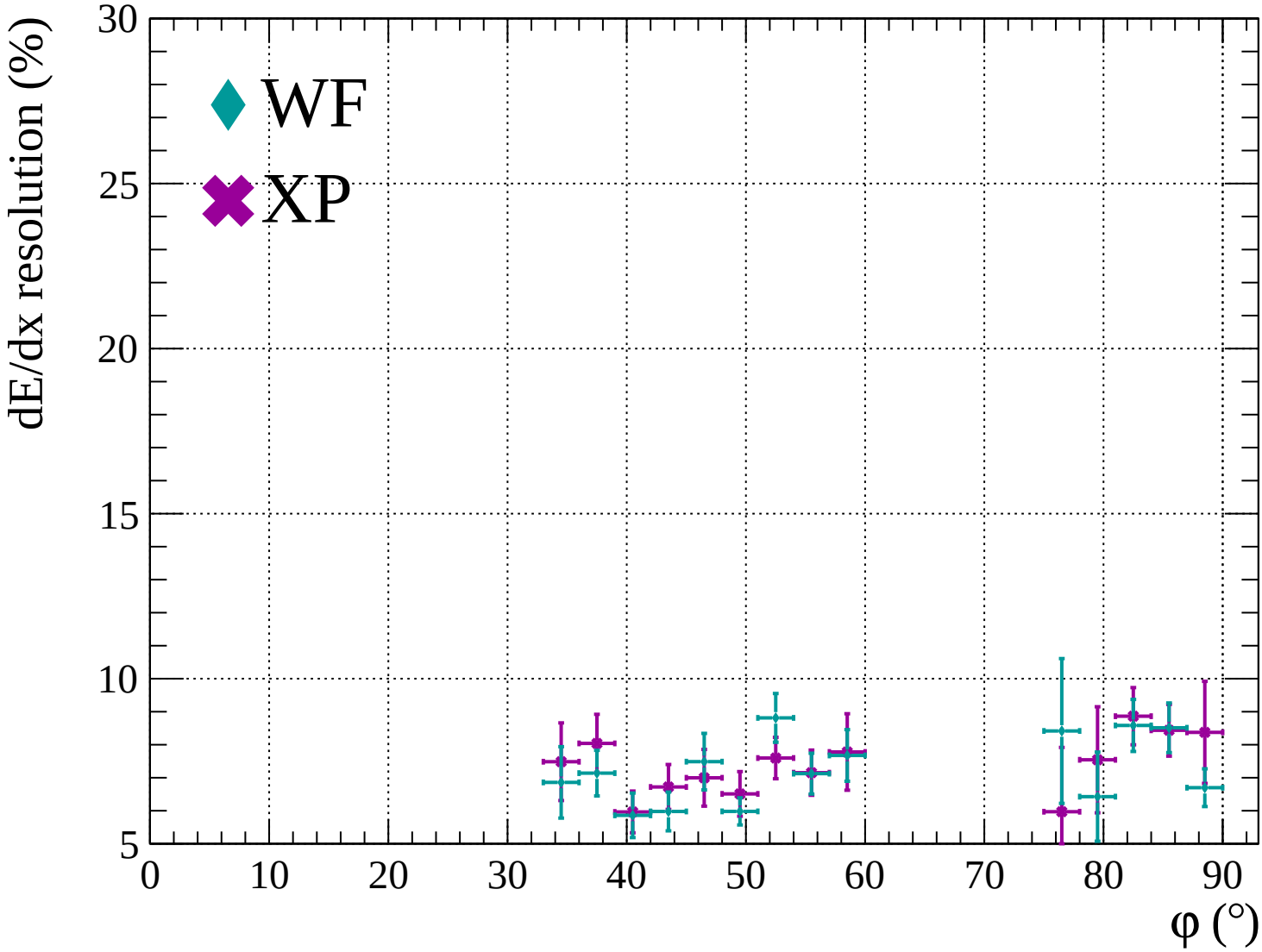


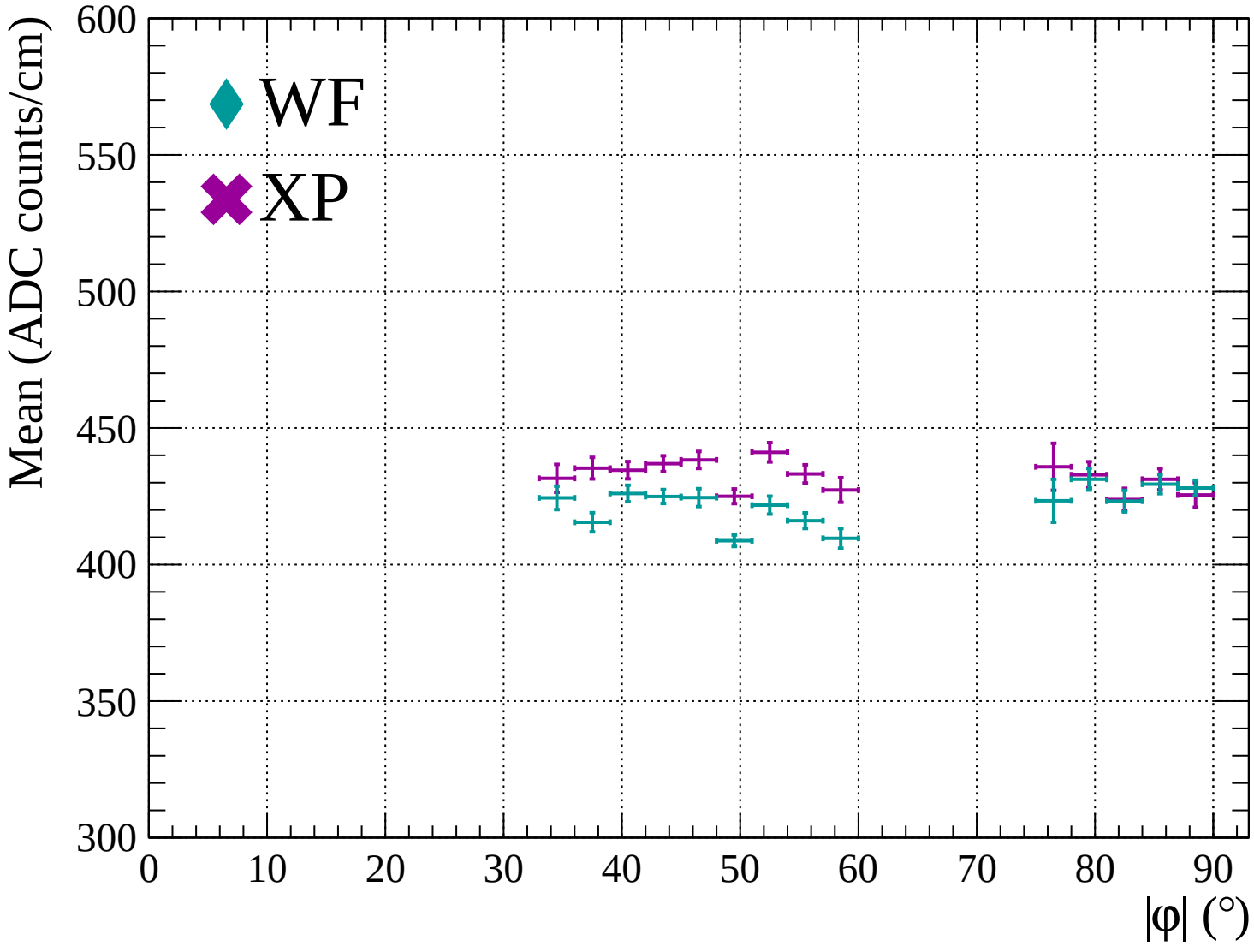


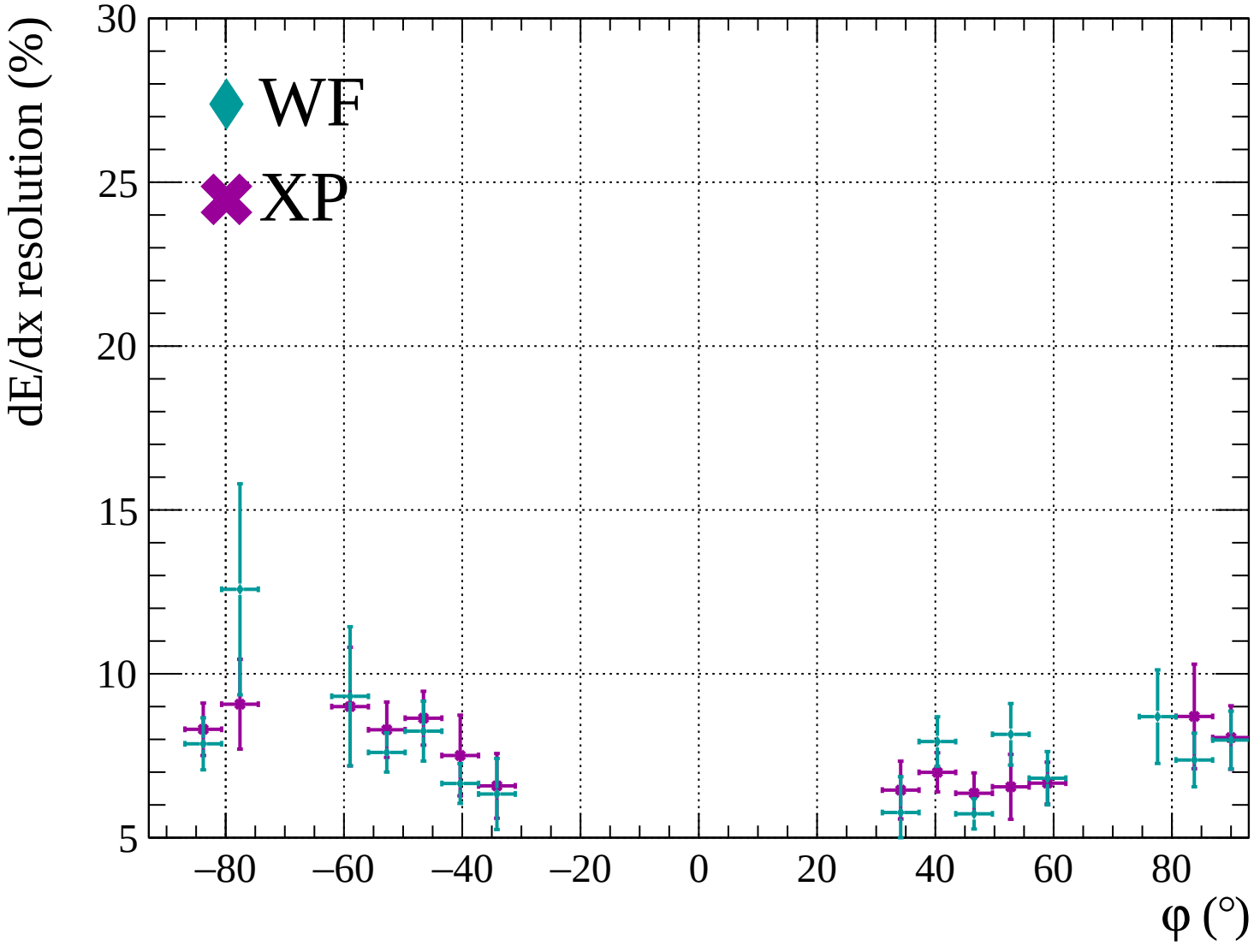


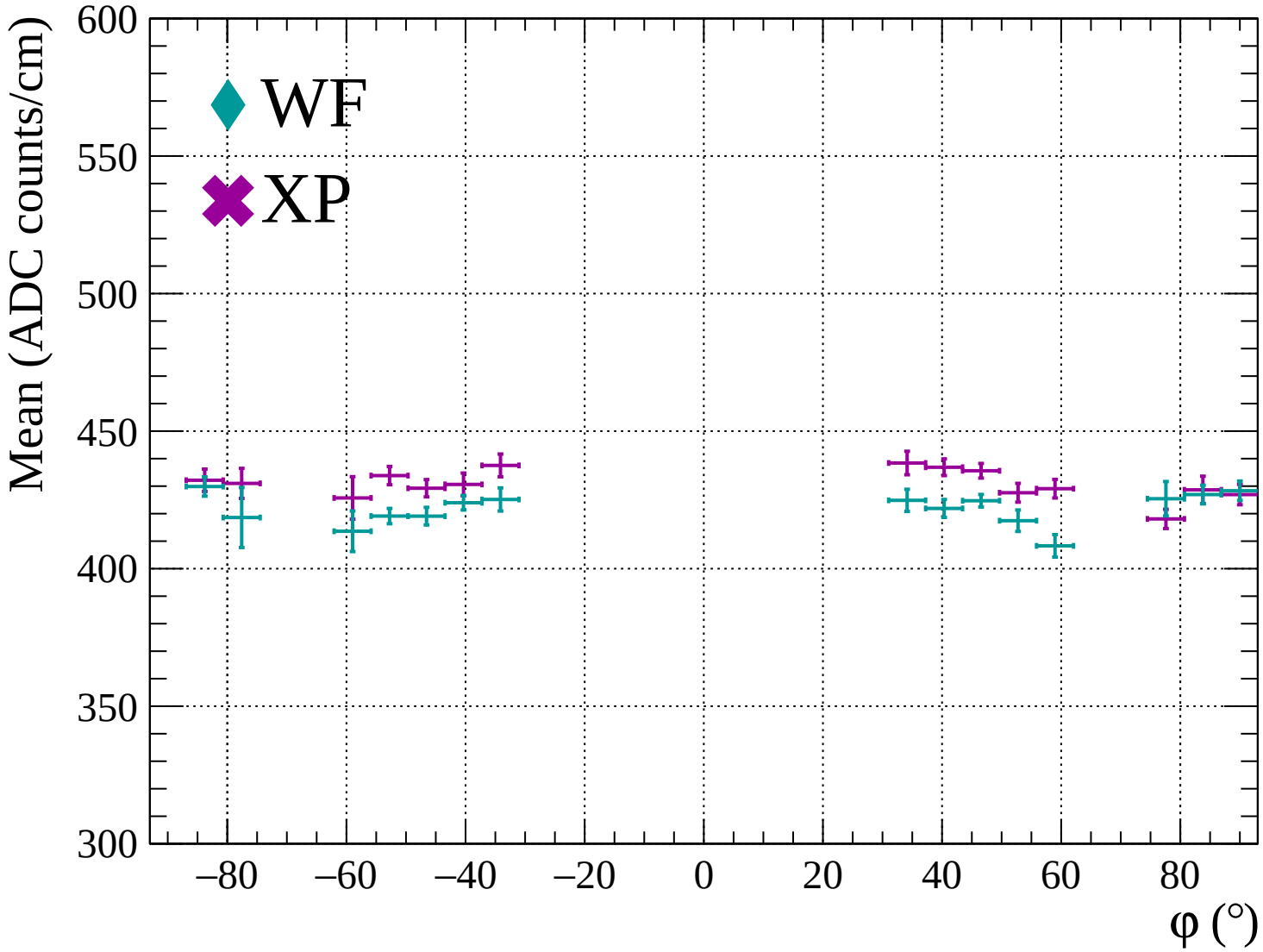


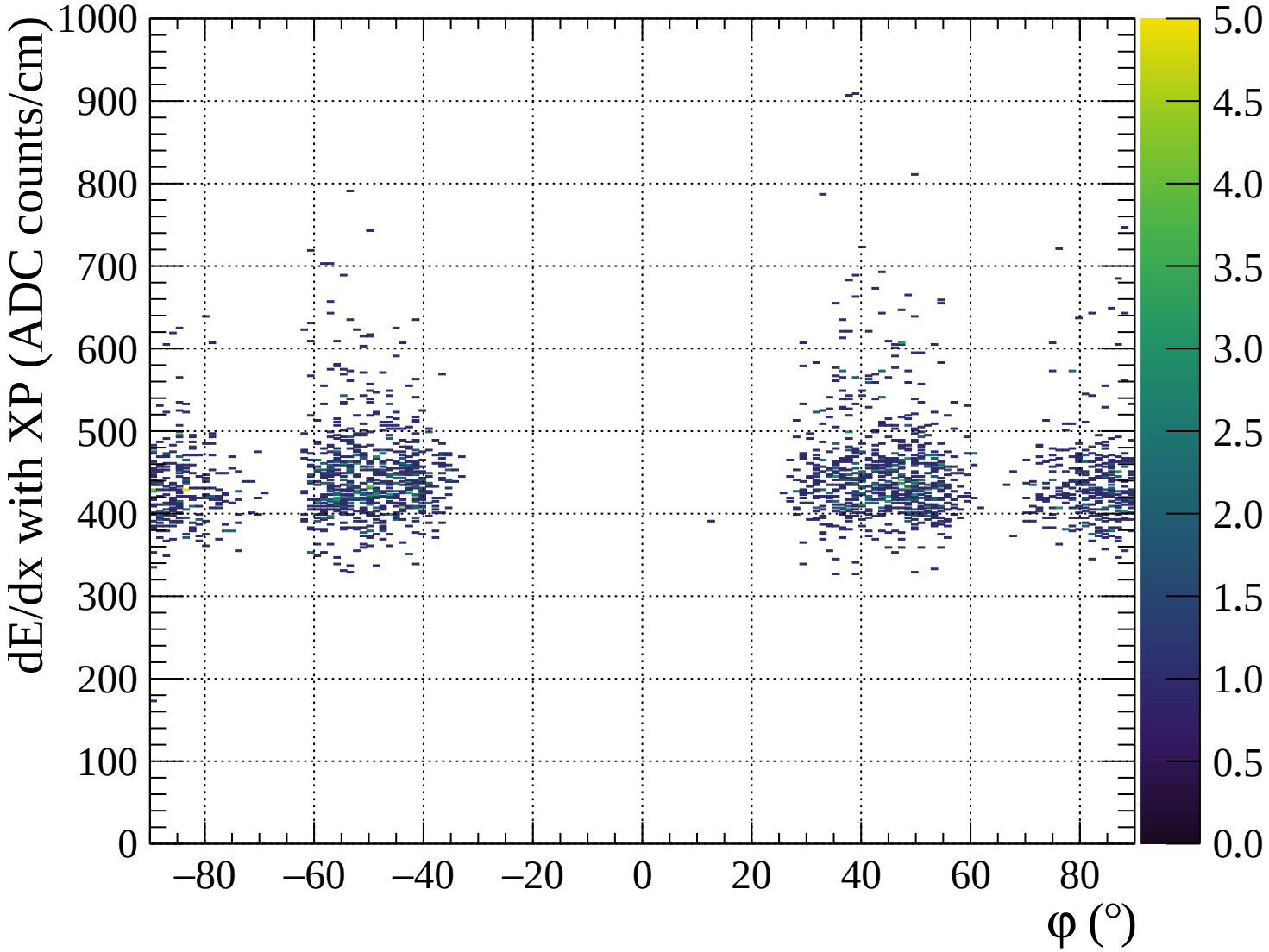


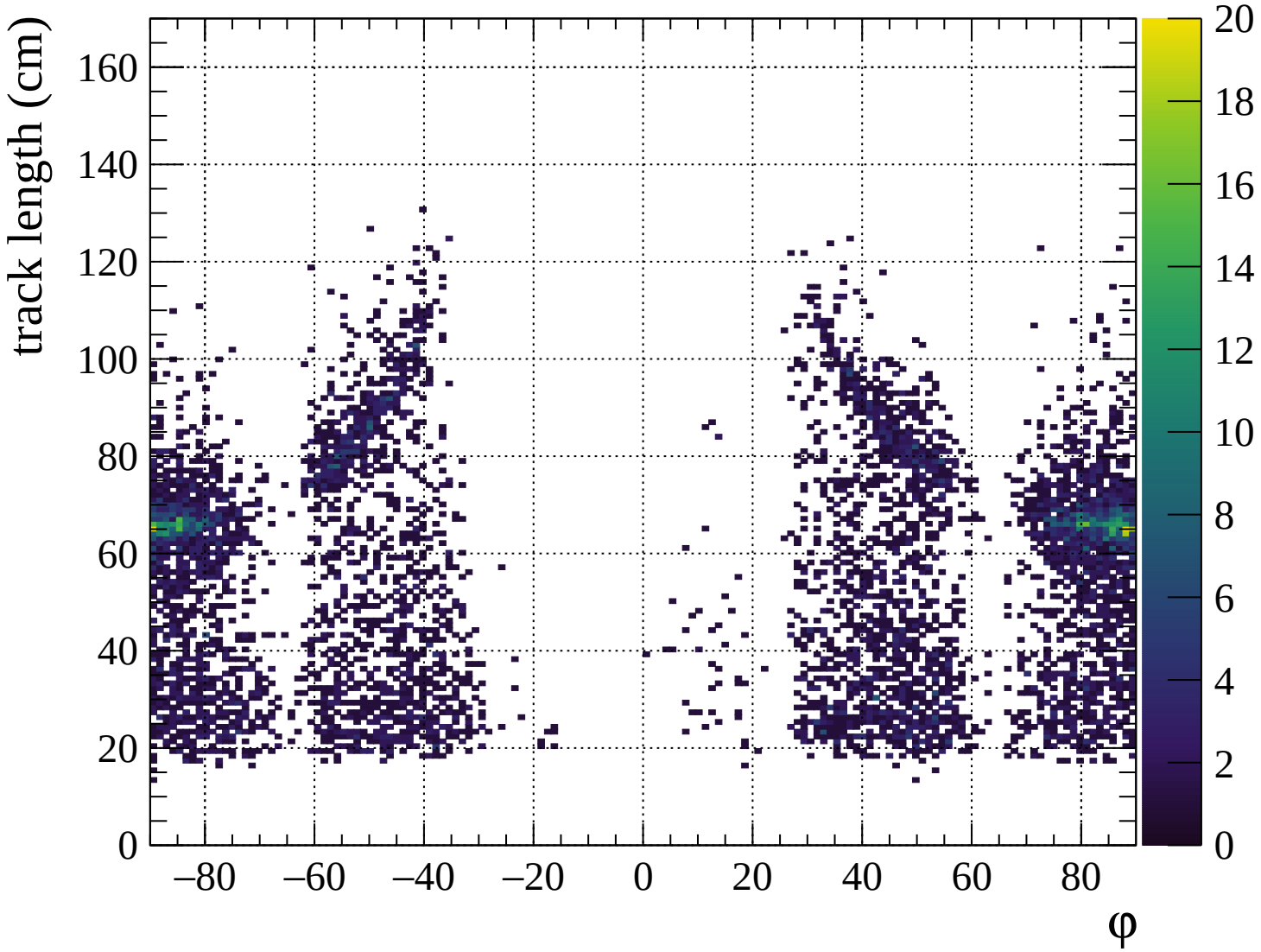


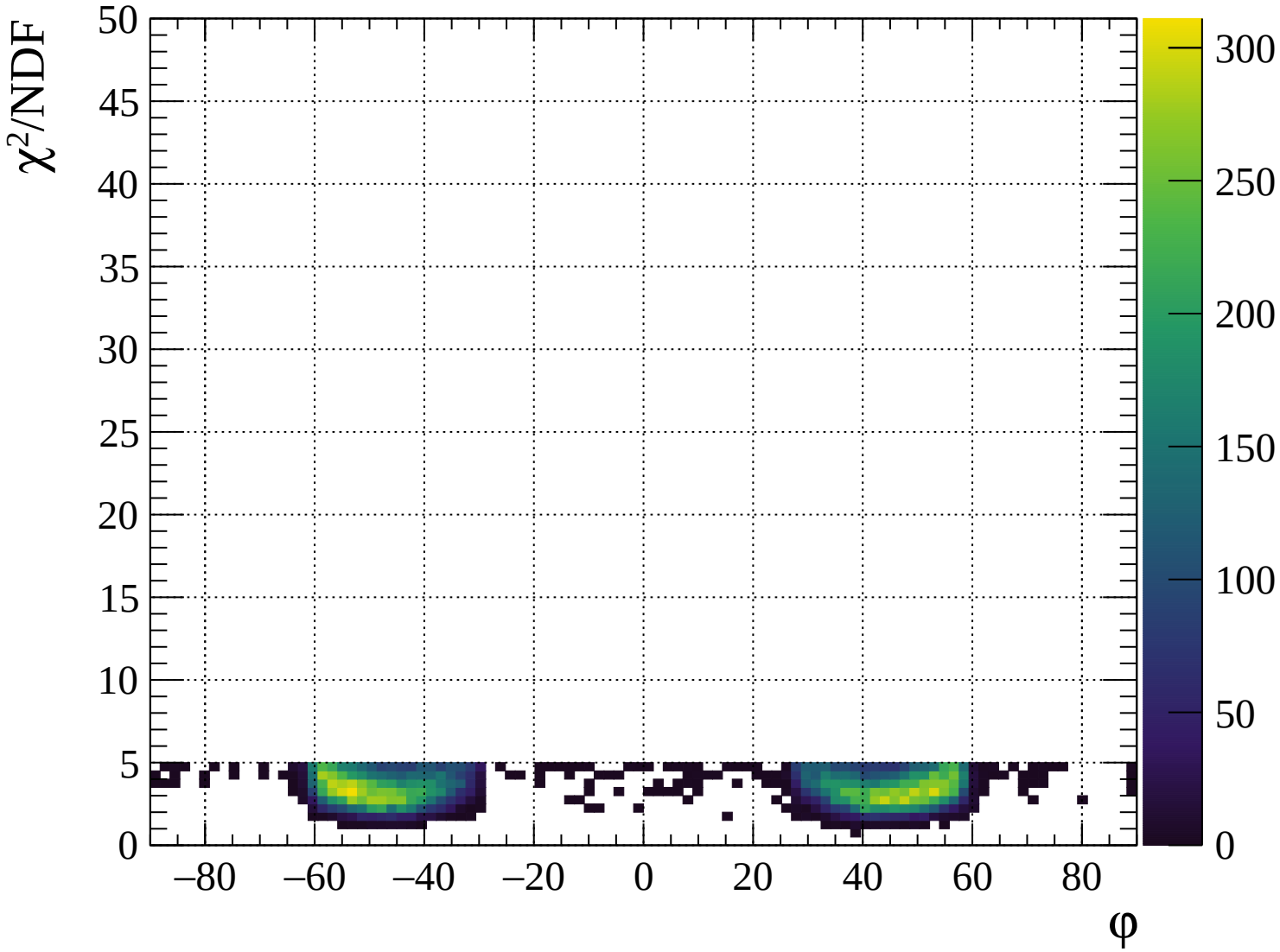


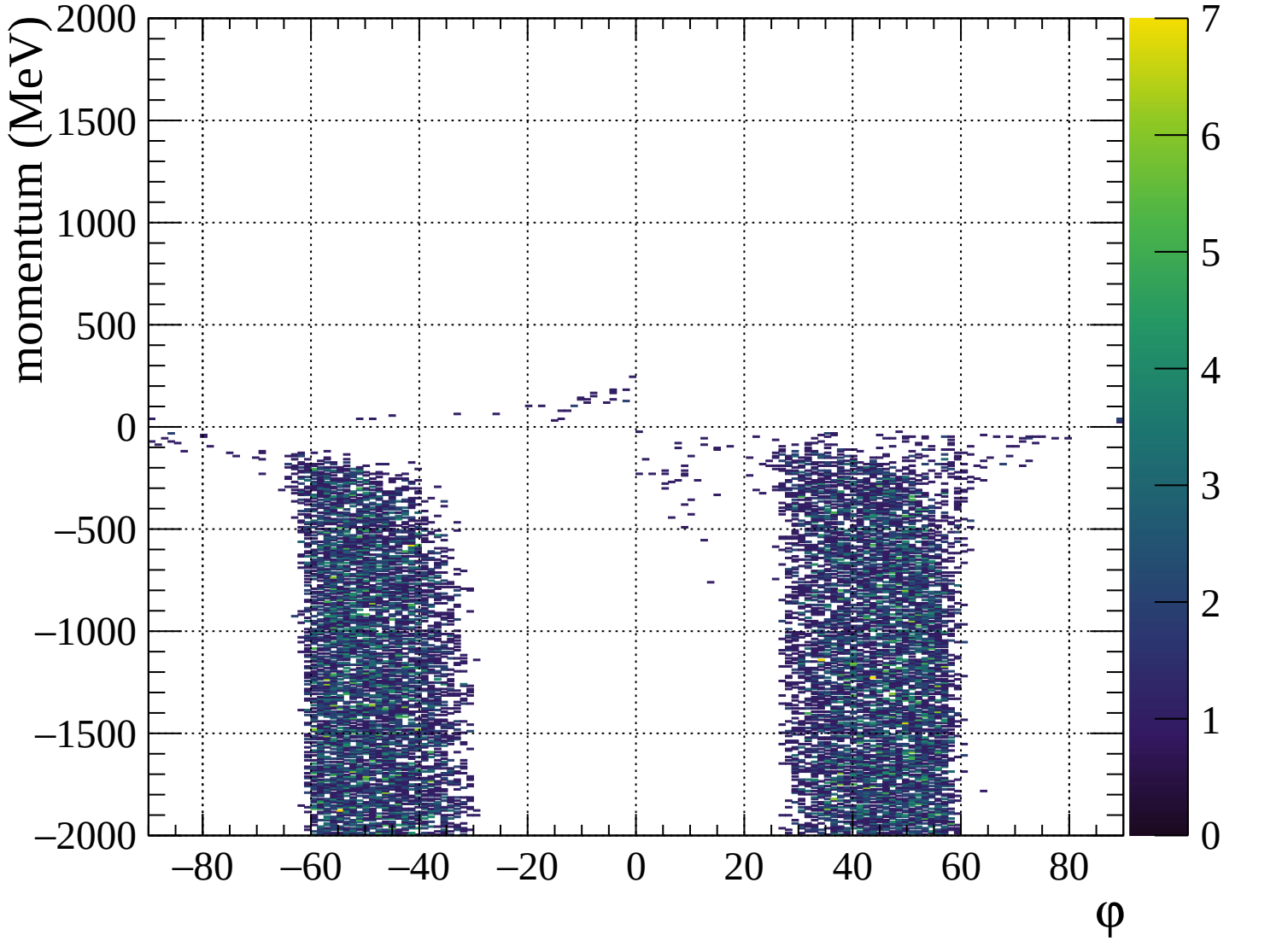


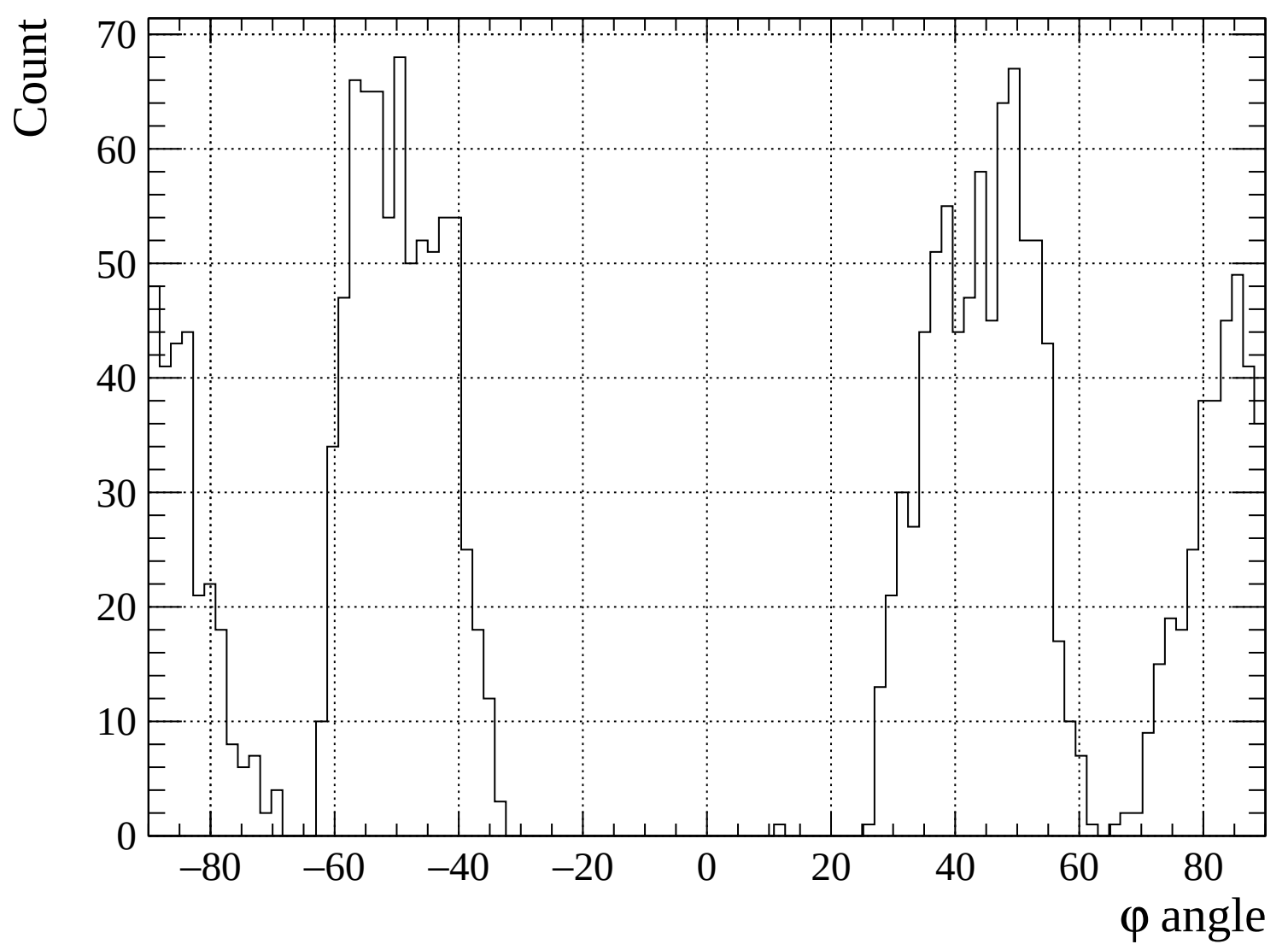


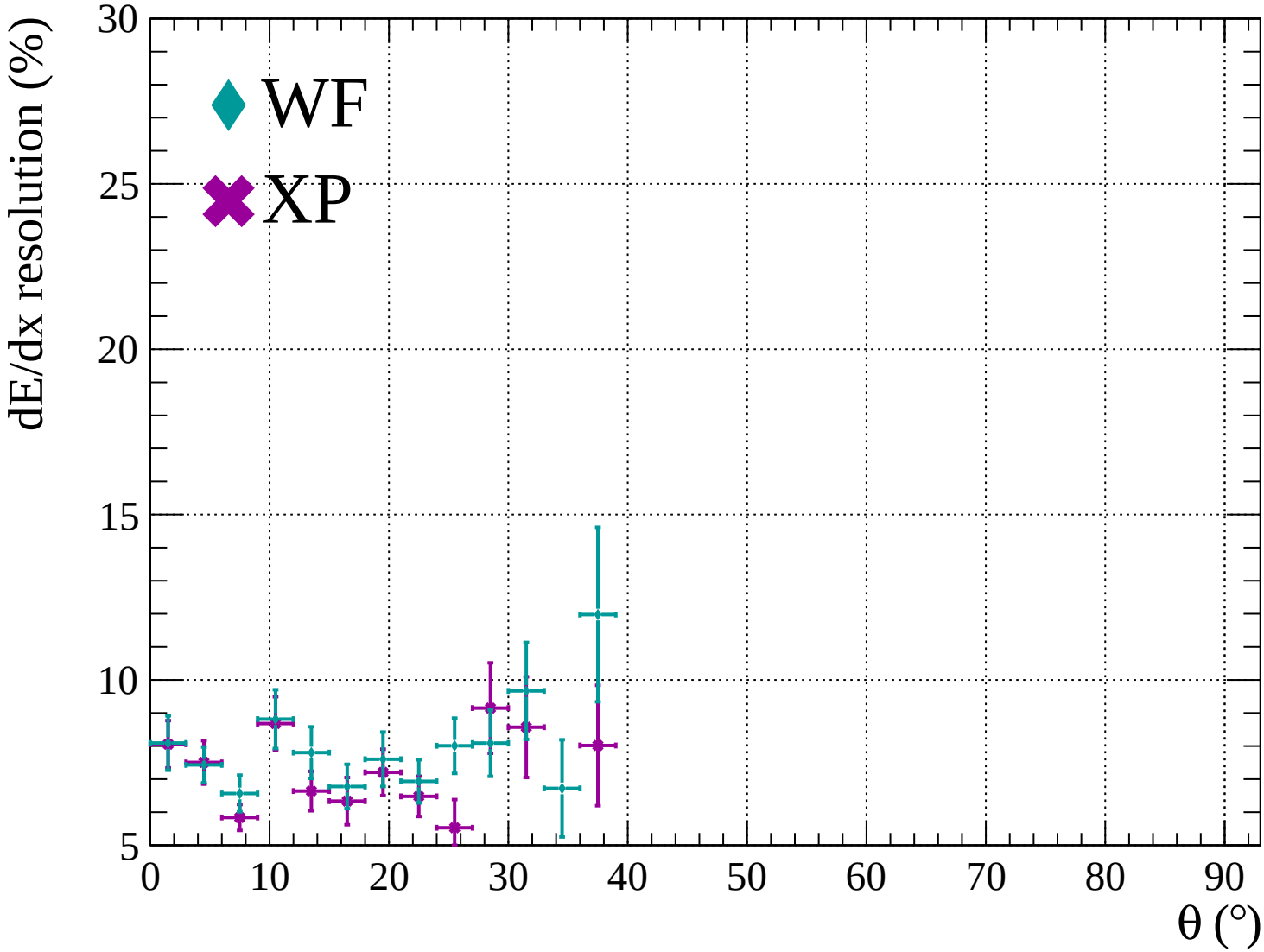


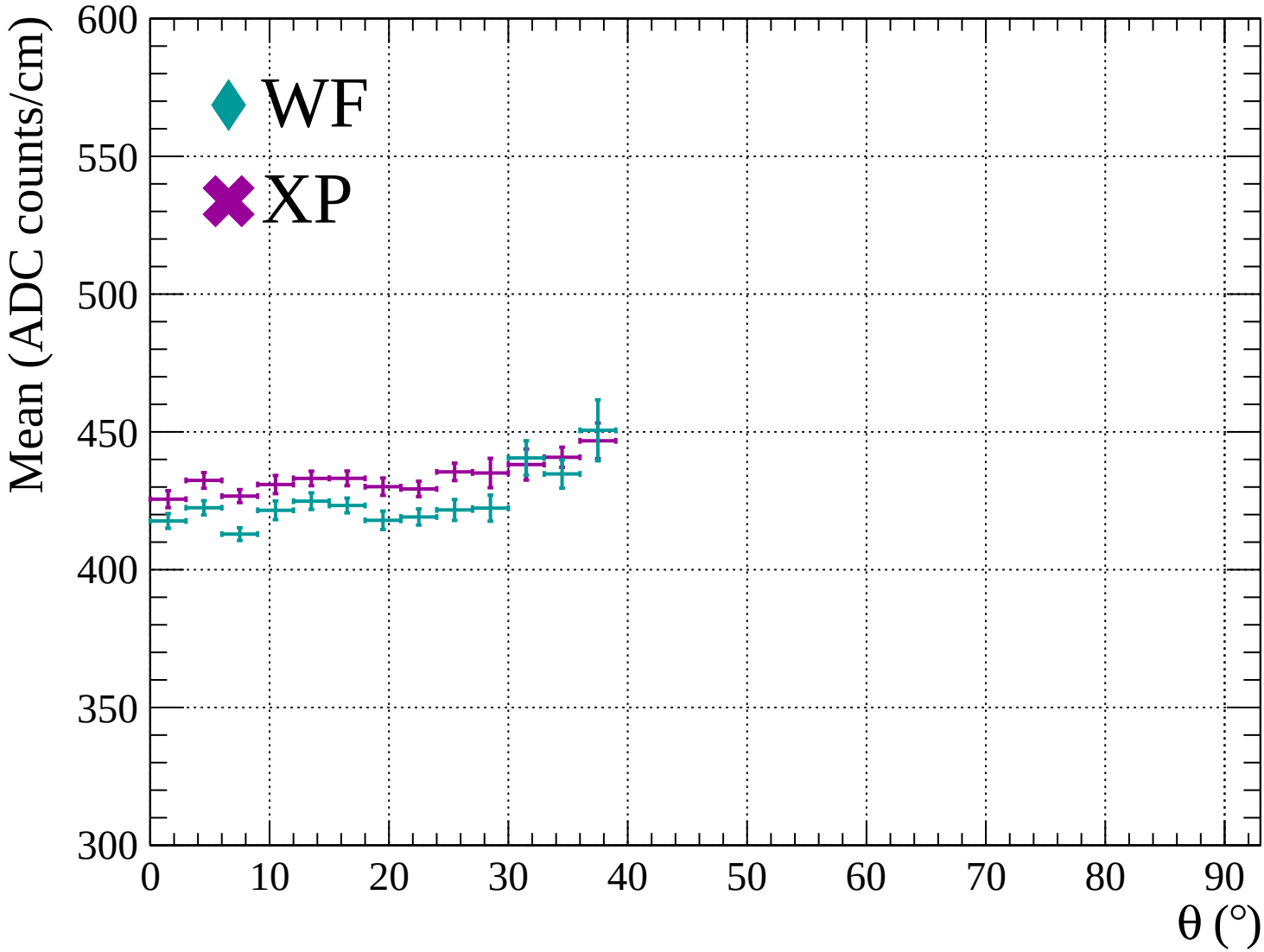


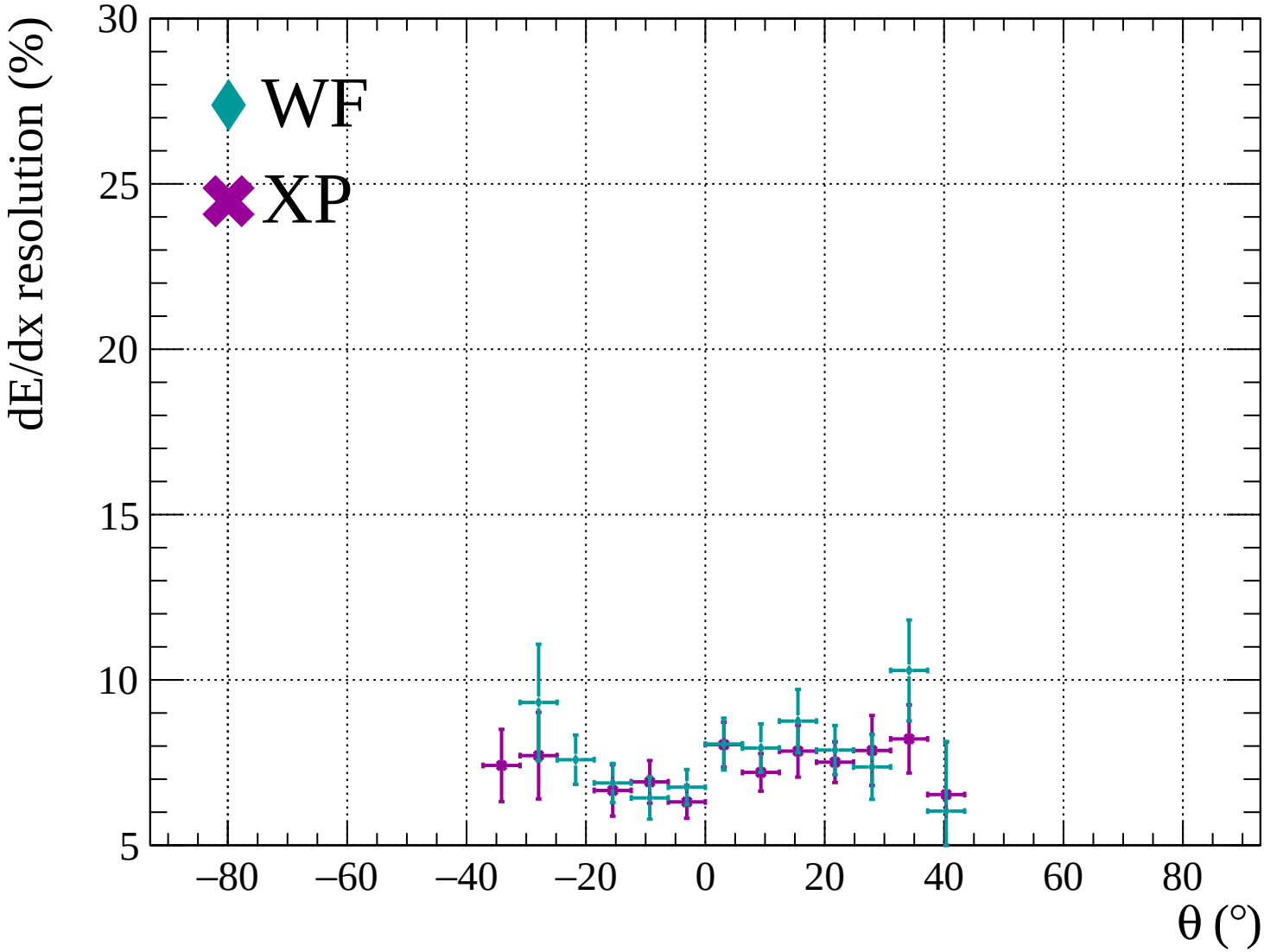


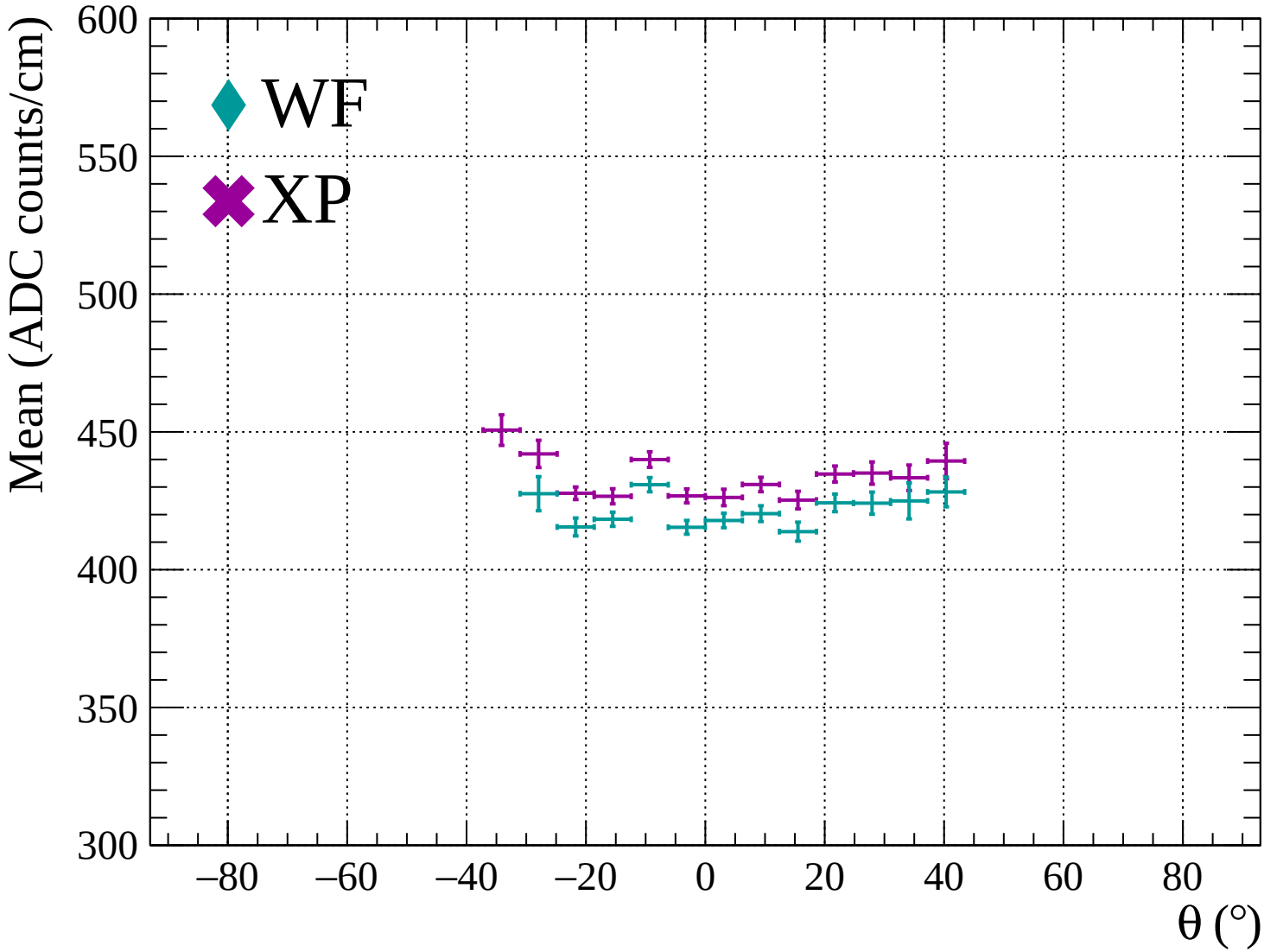


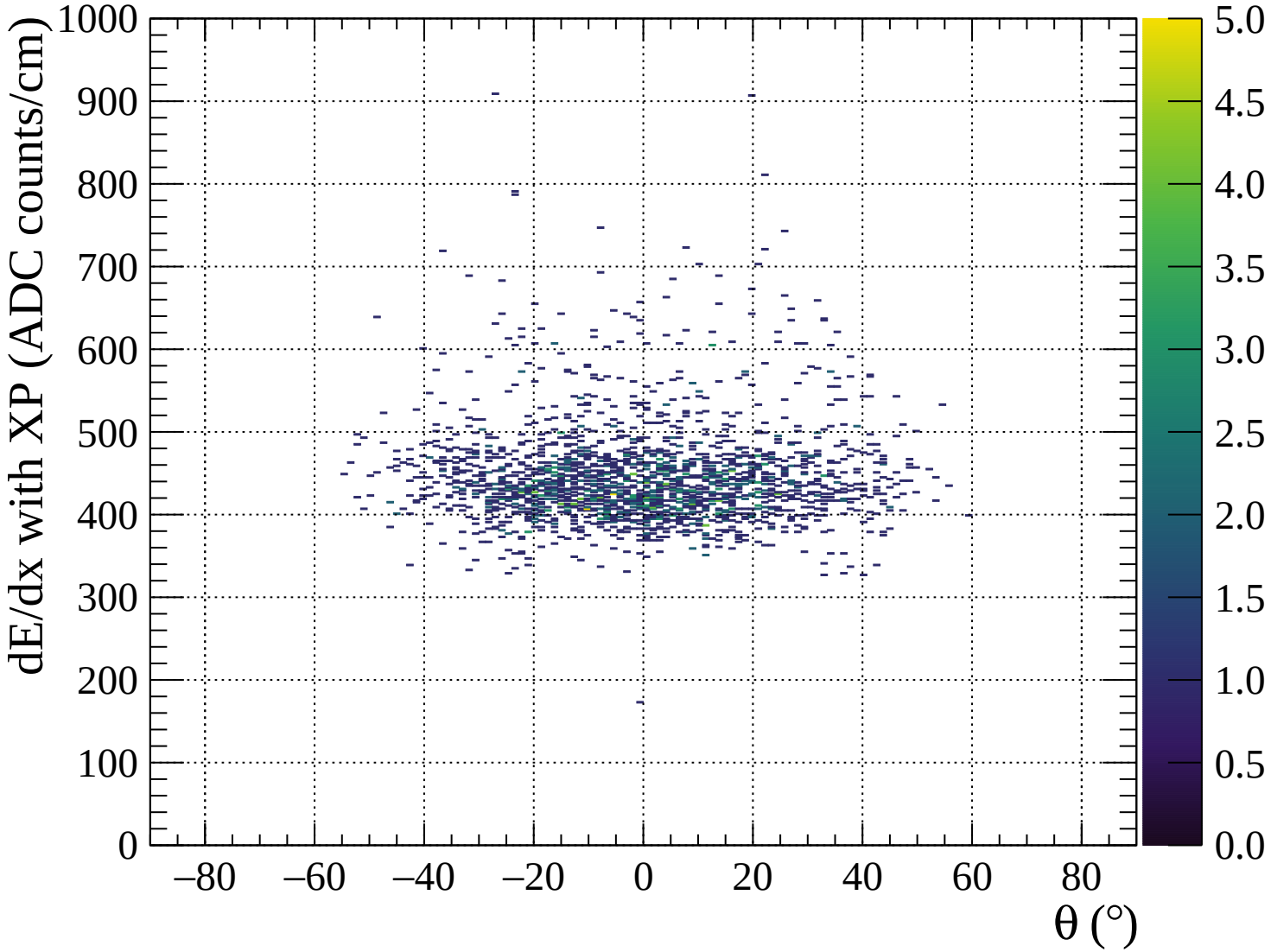


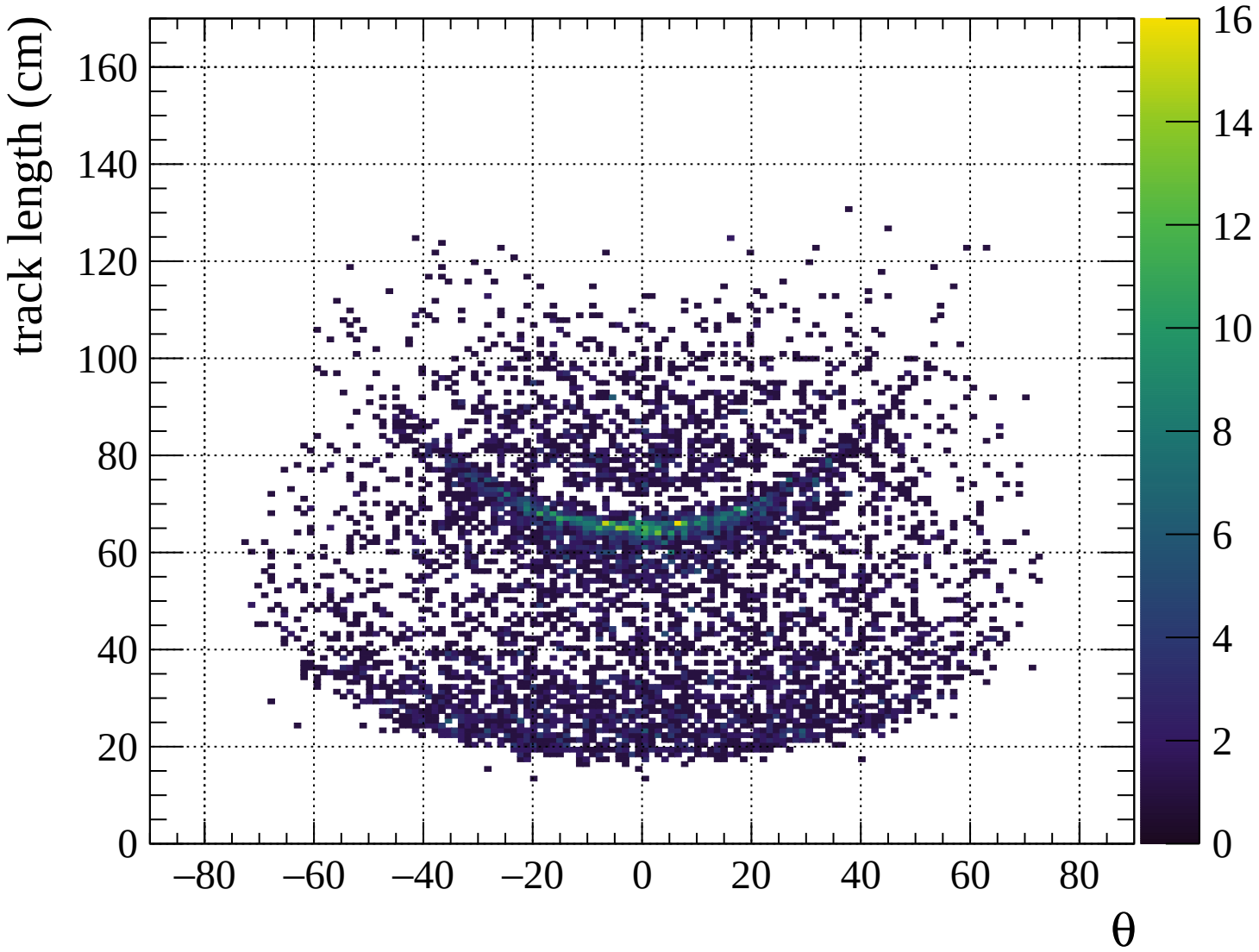


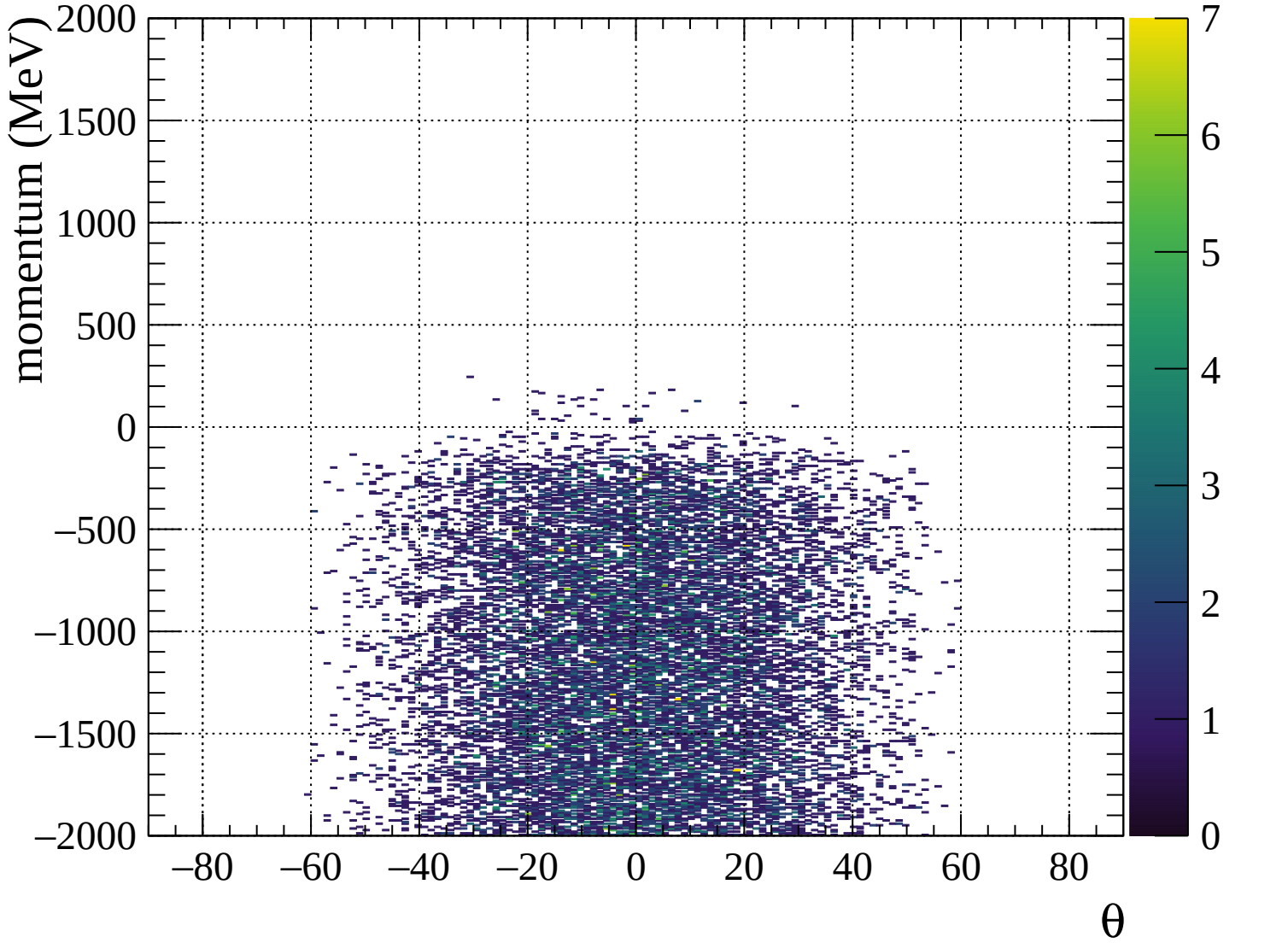


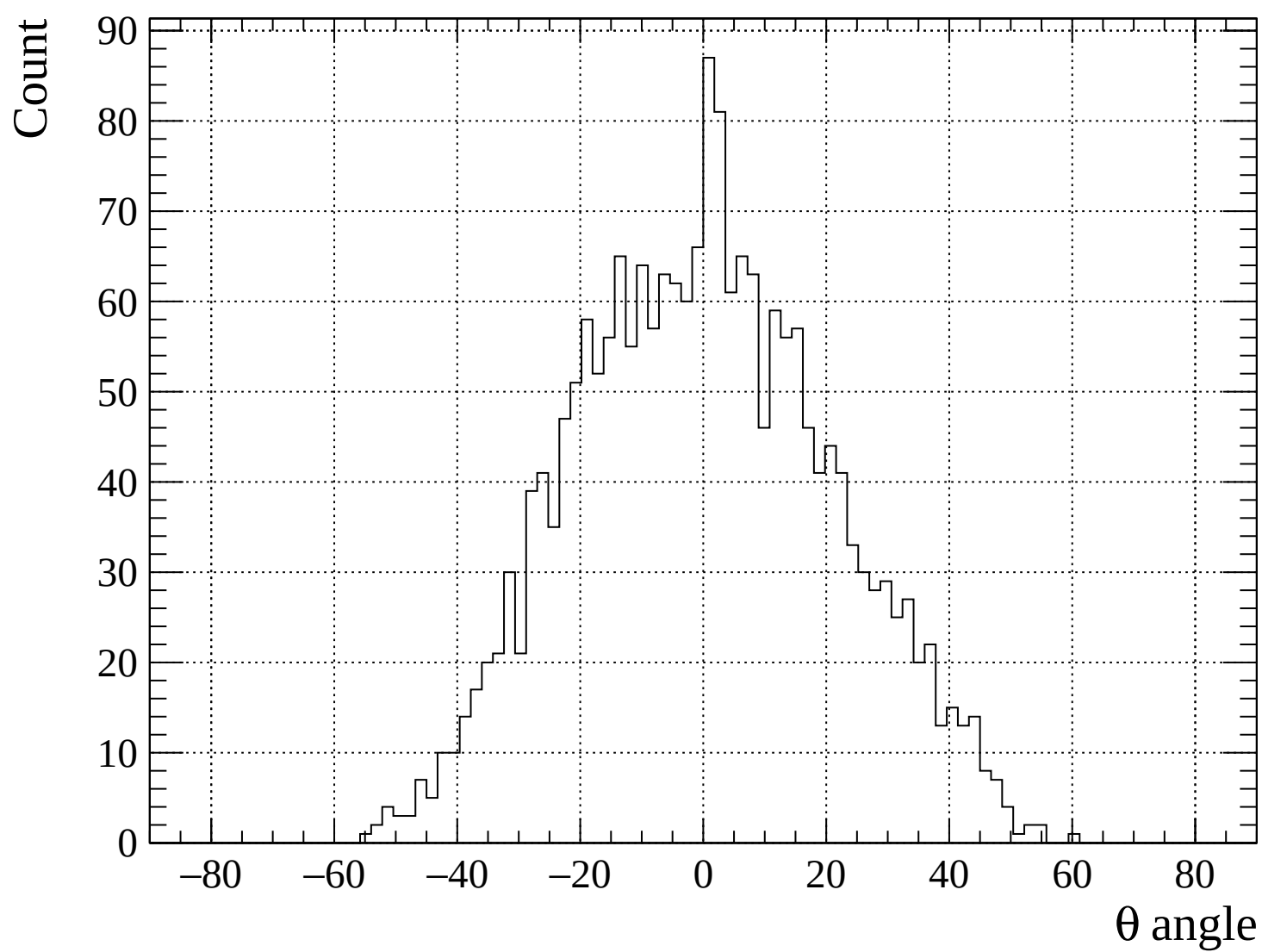


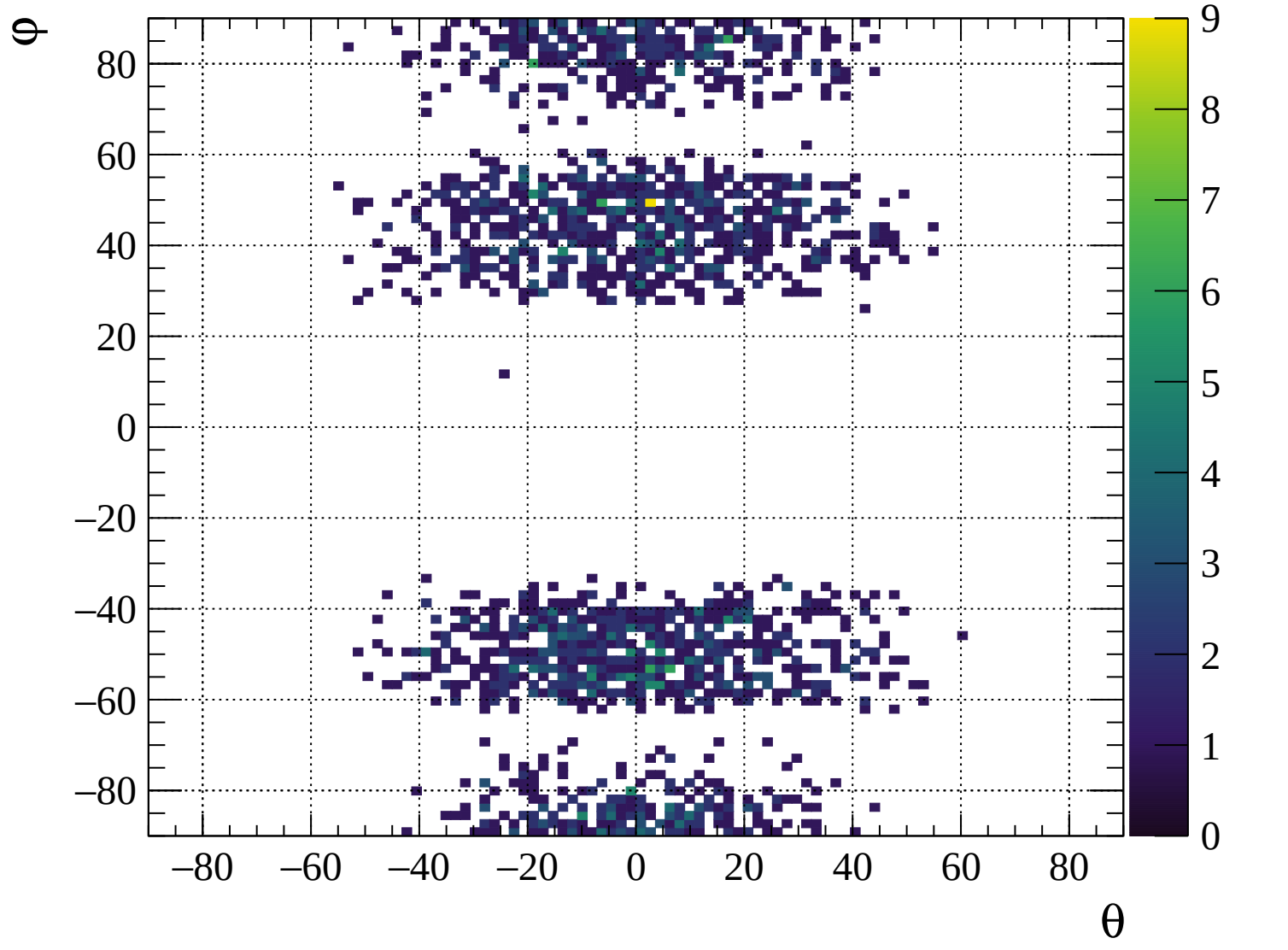


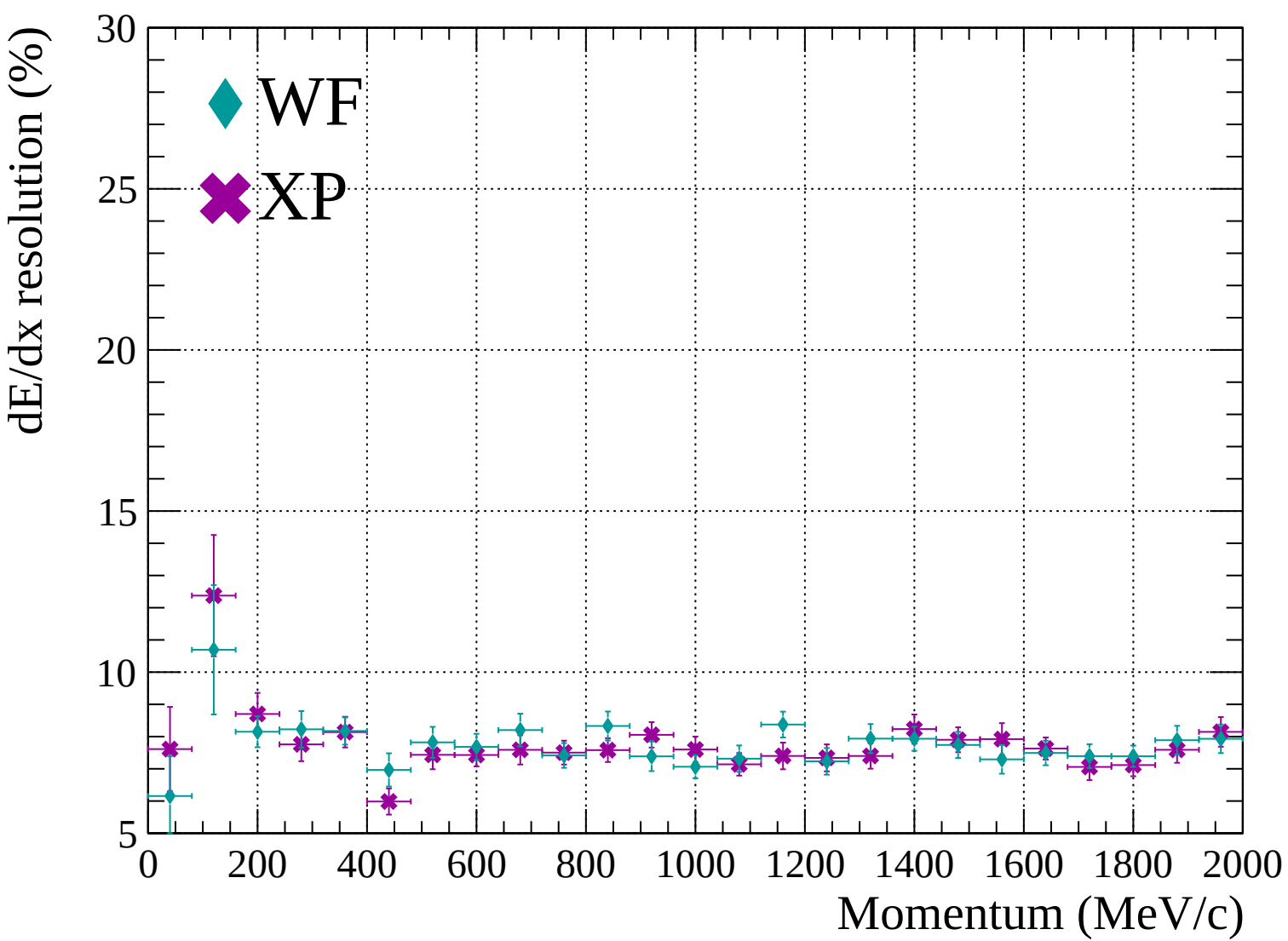


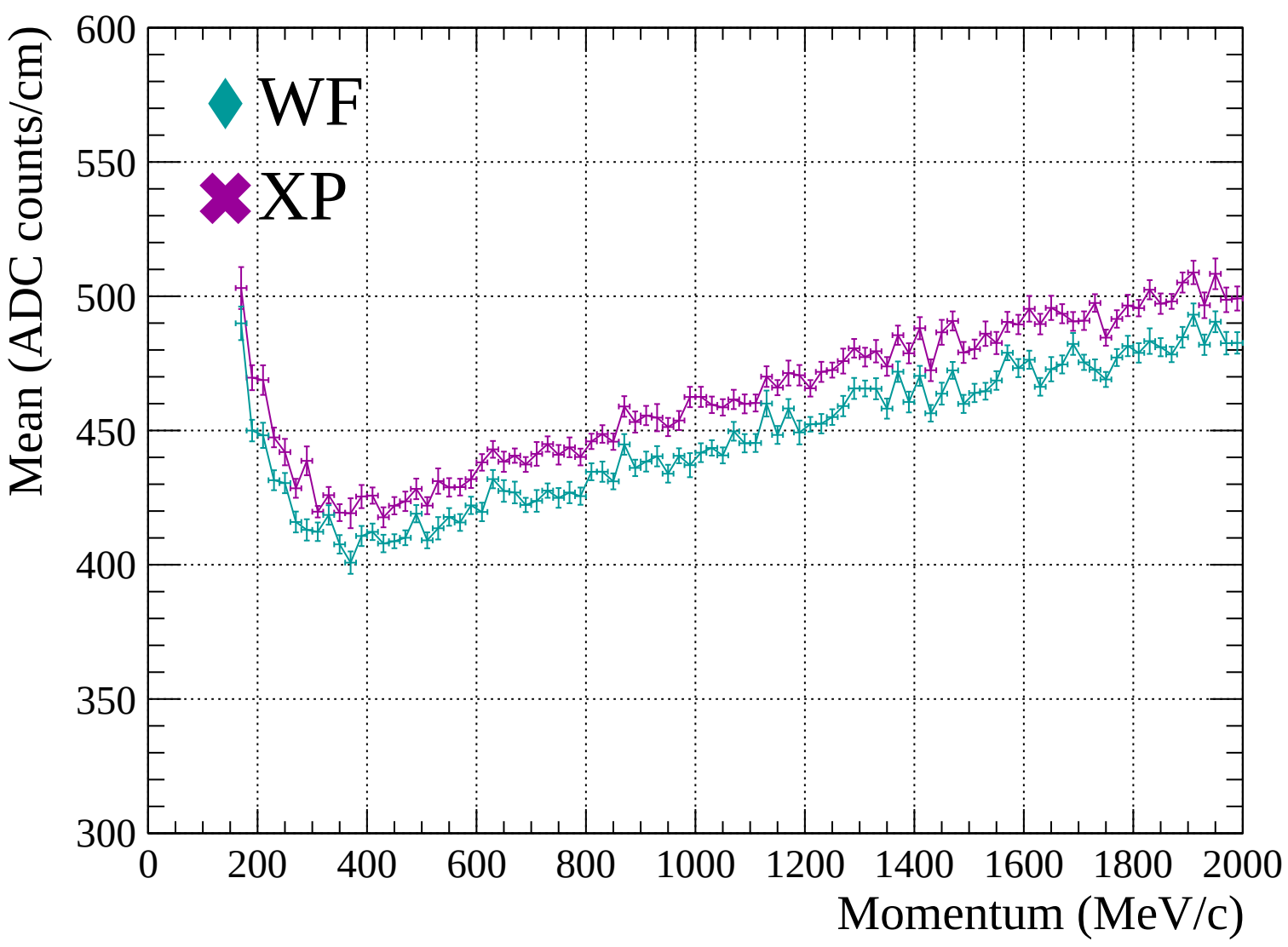


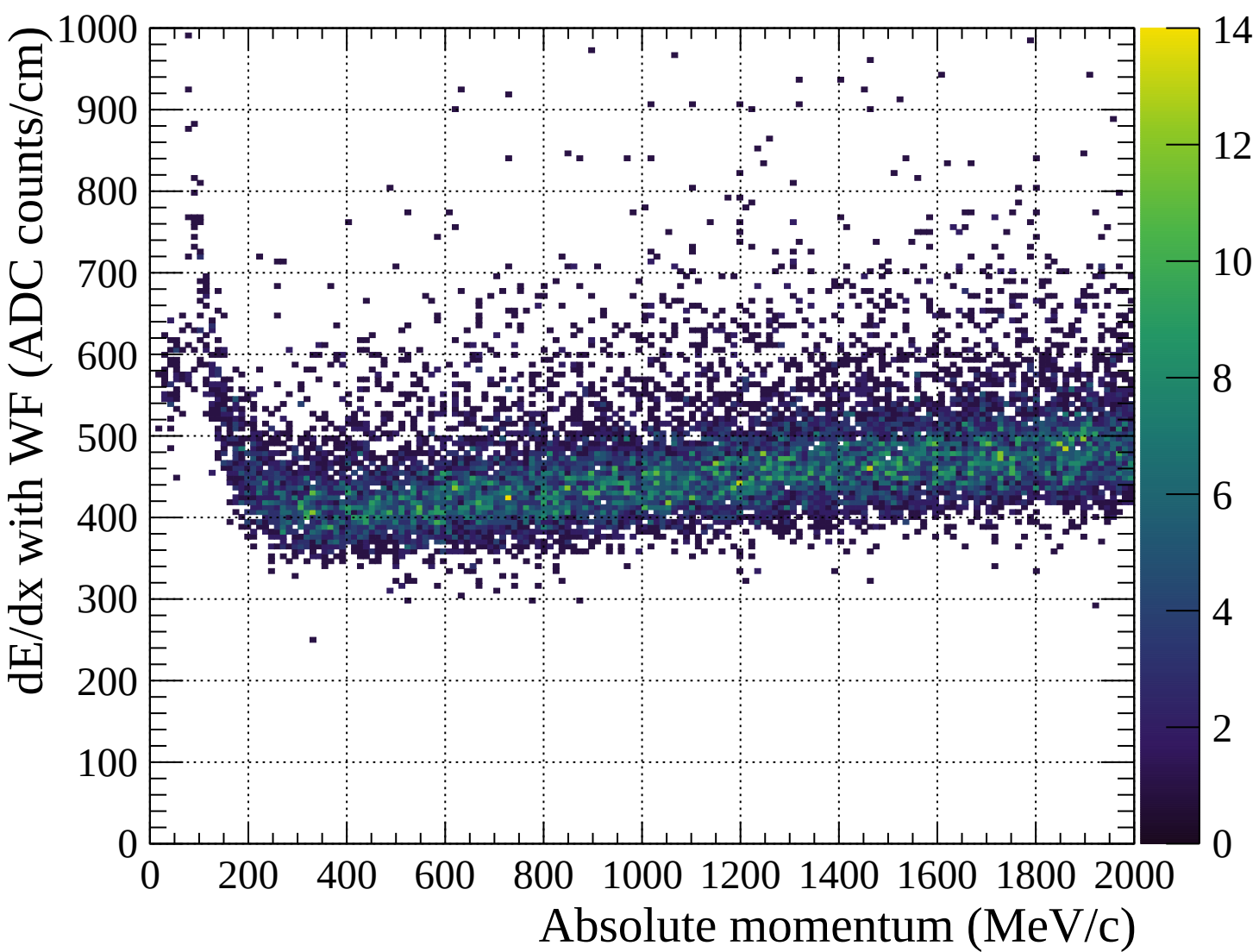


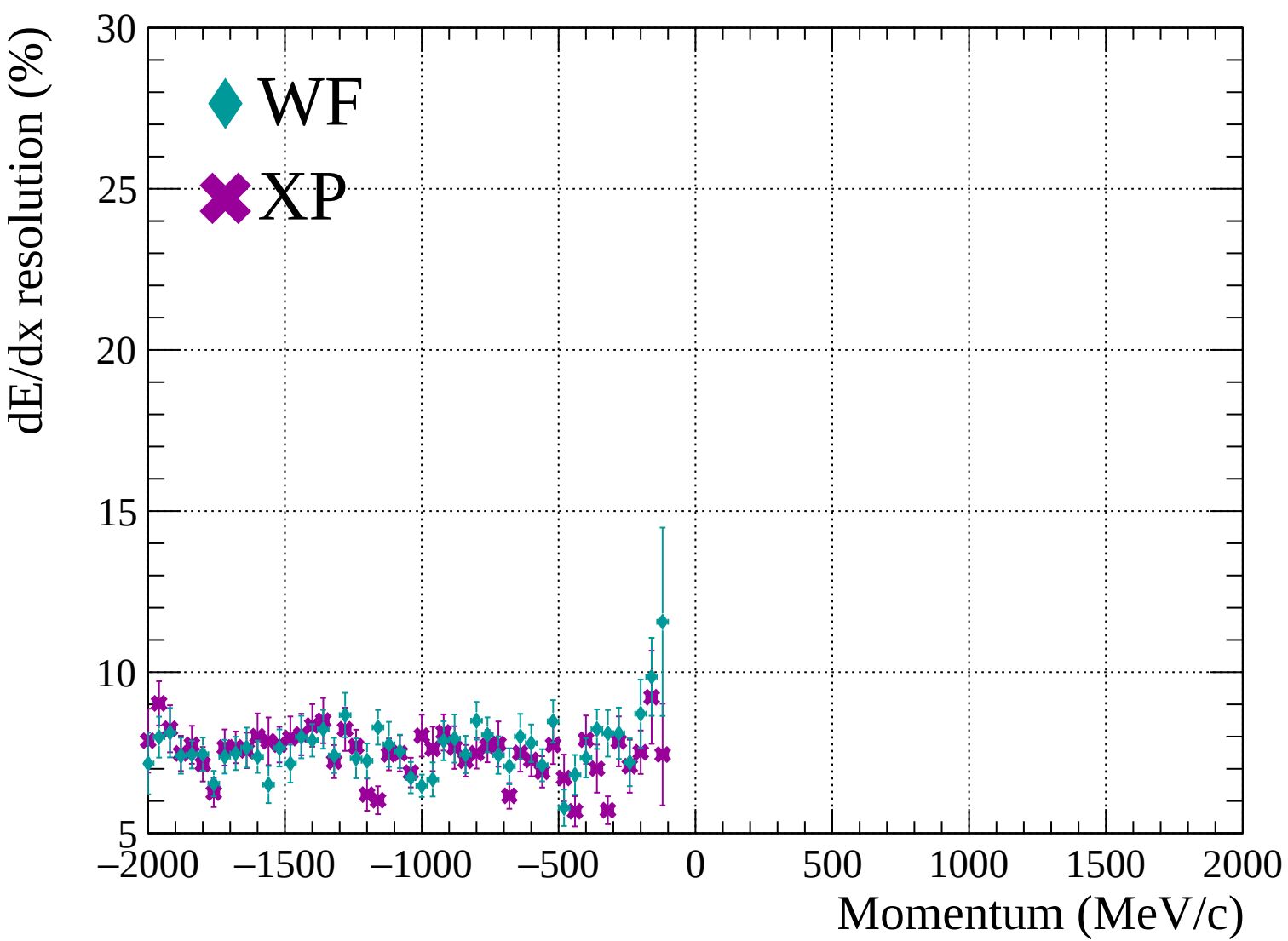


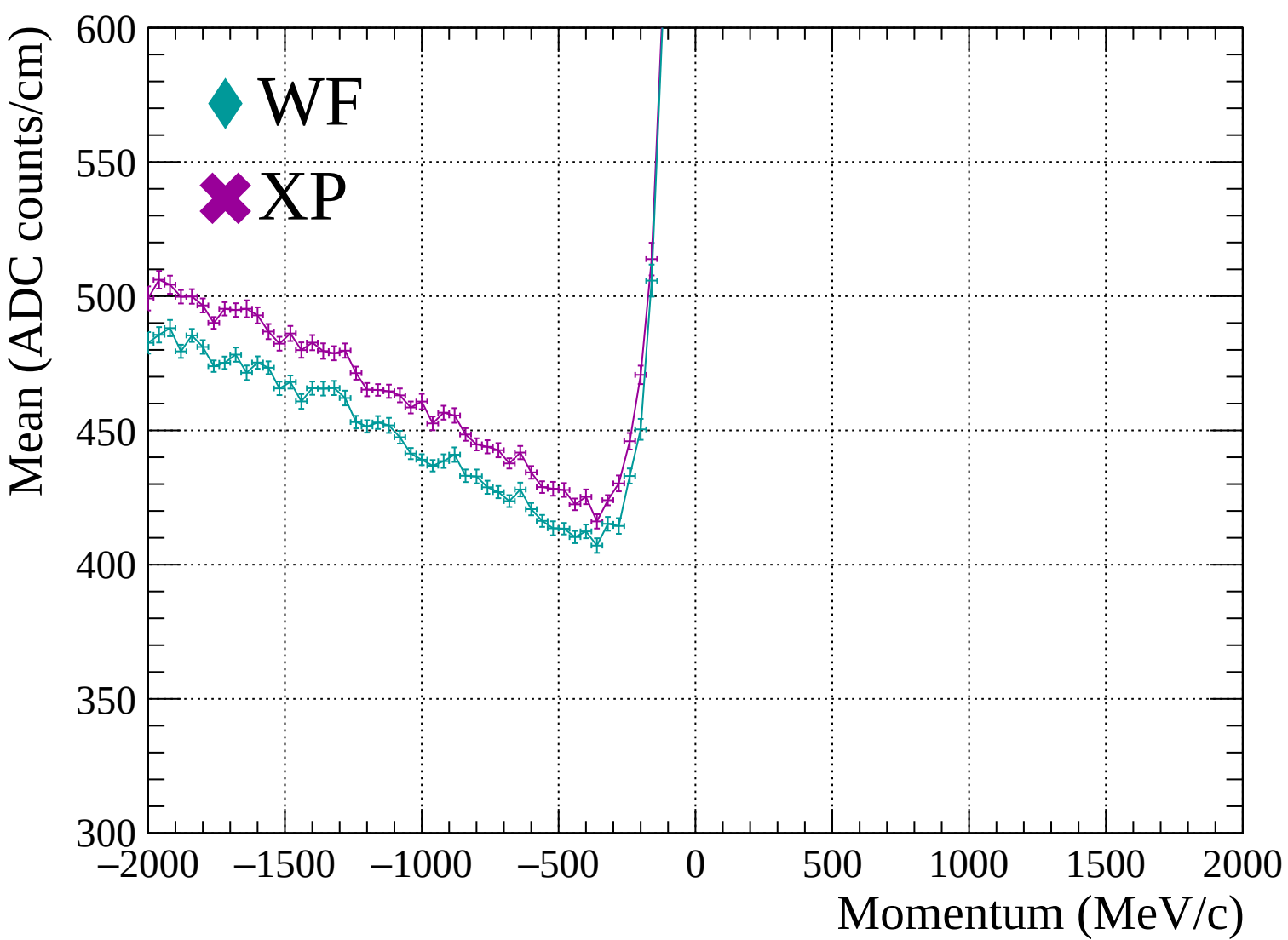


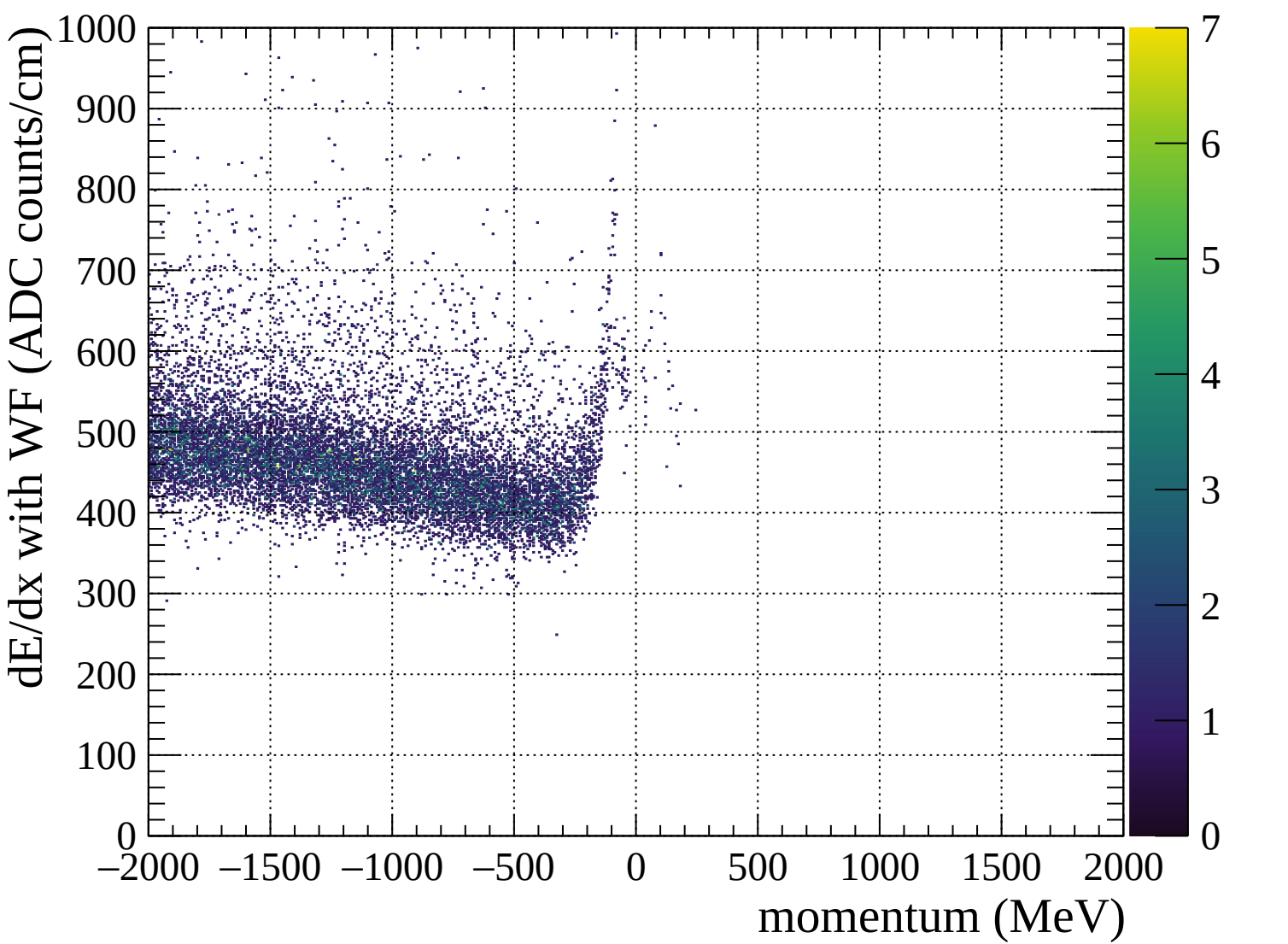


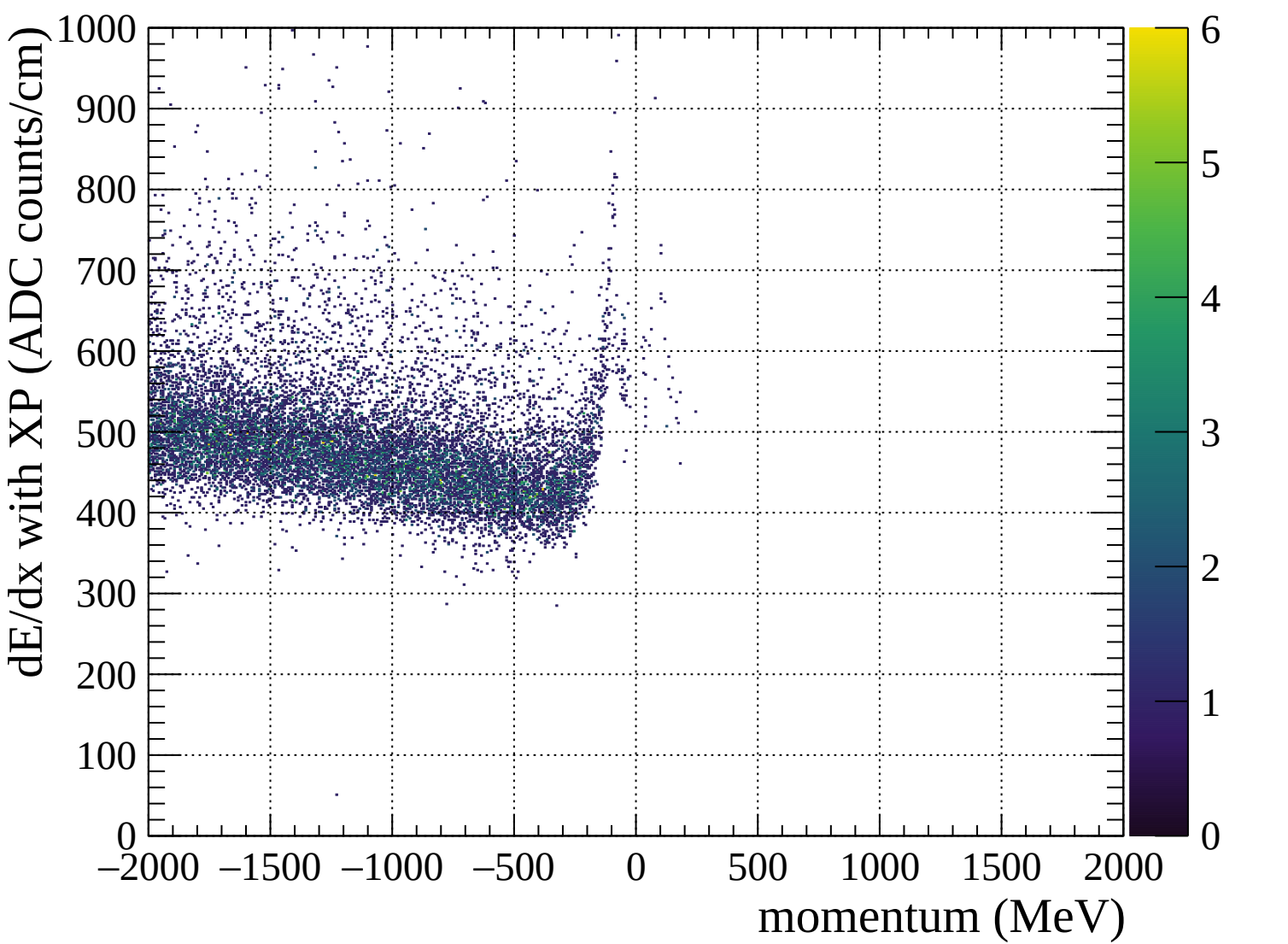


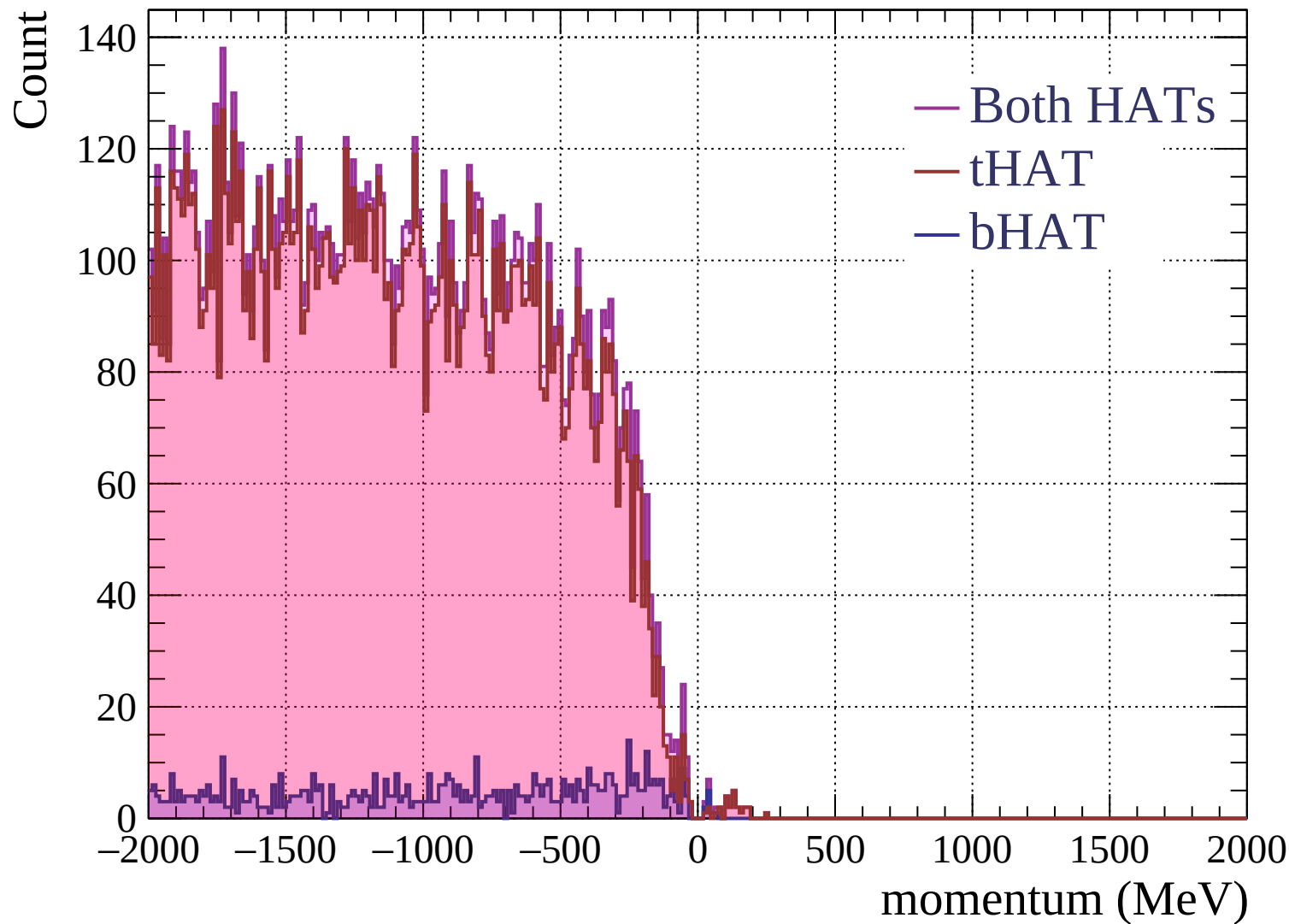


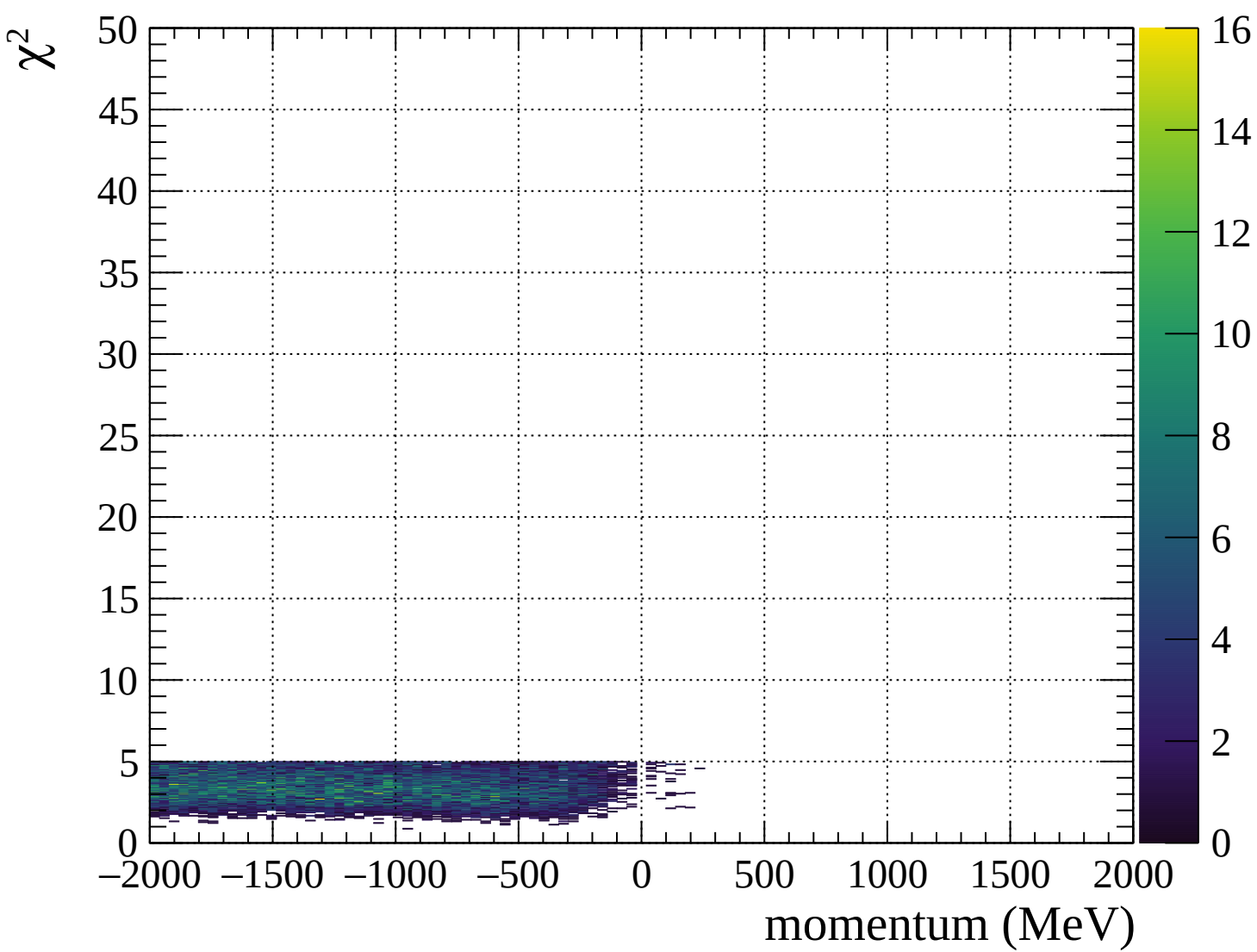




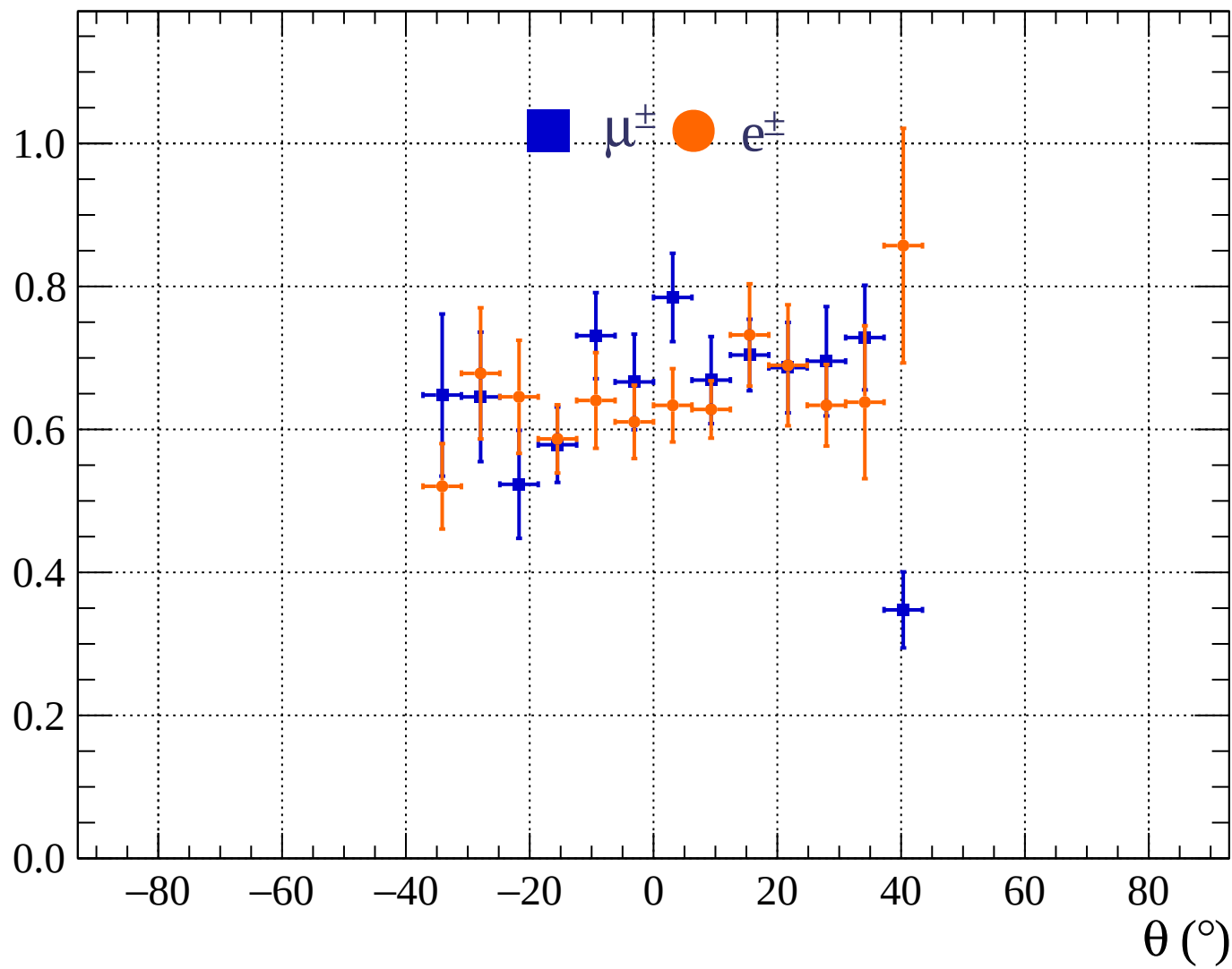


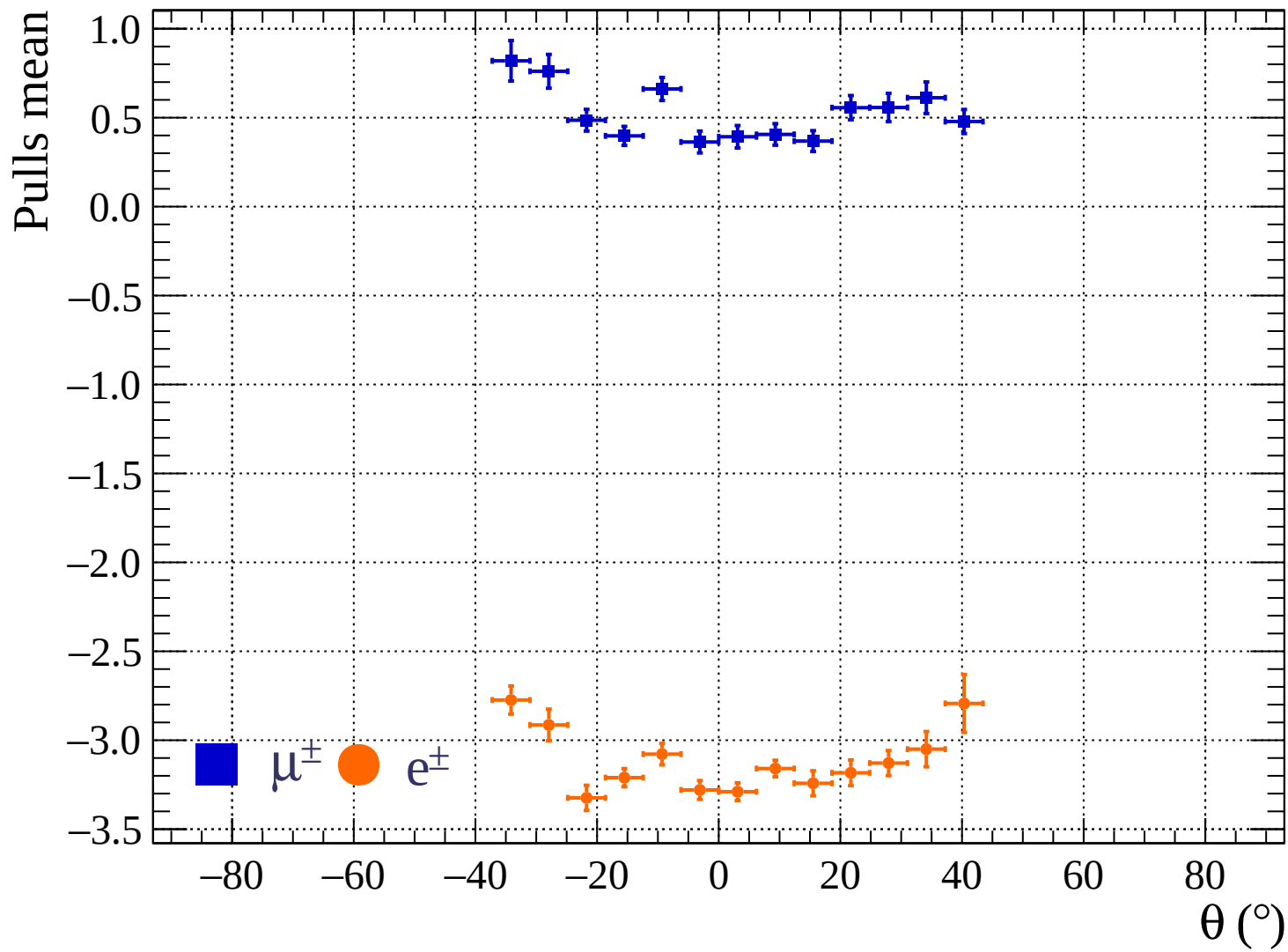




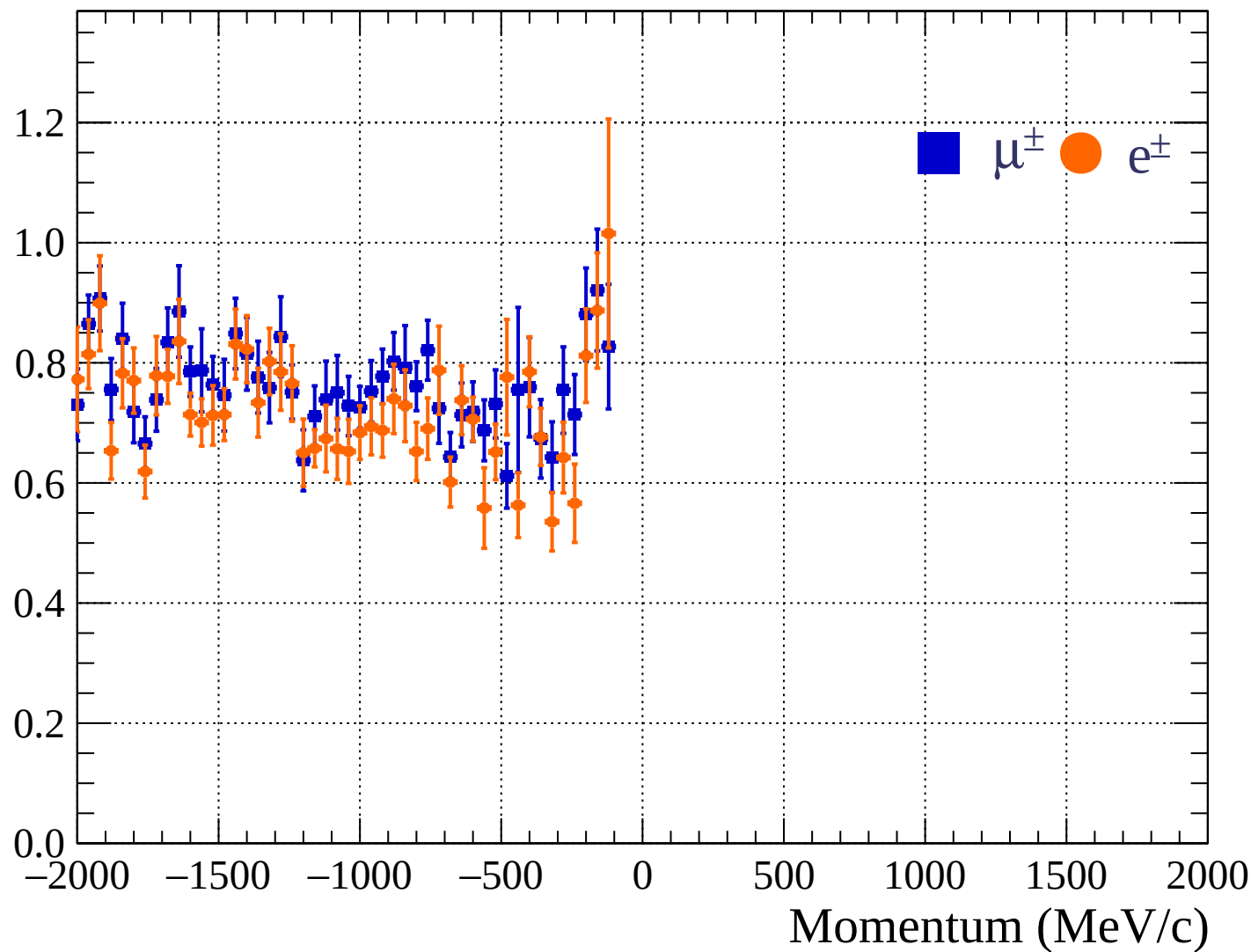


Pulls standard deviation

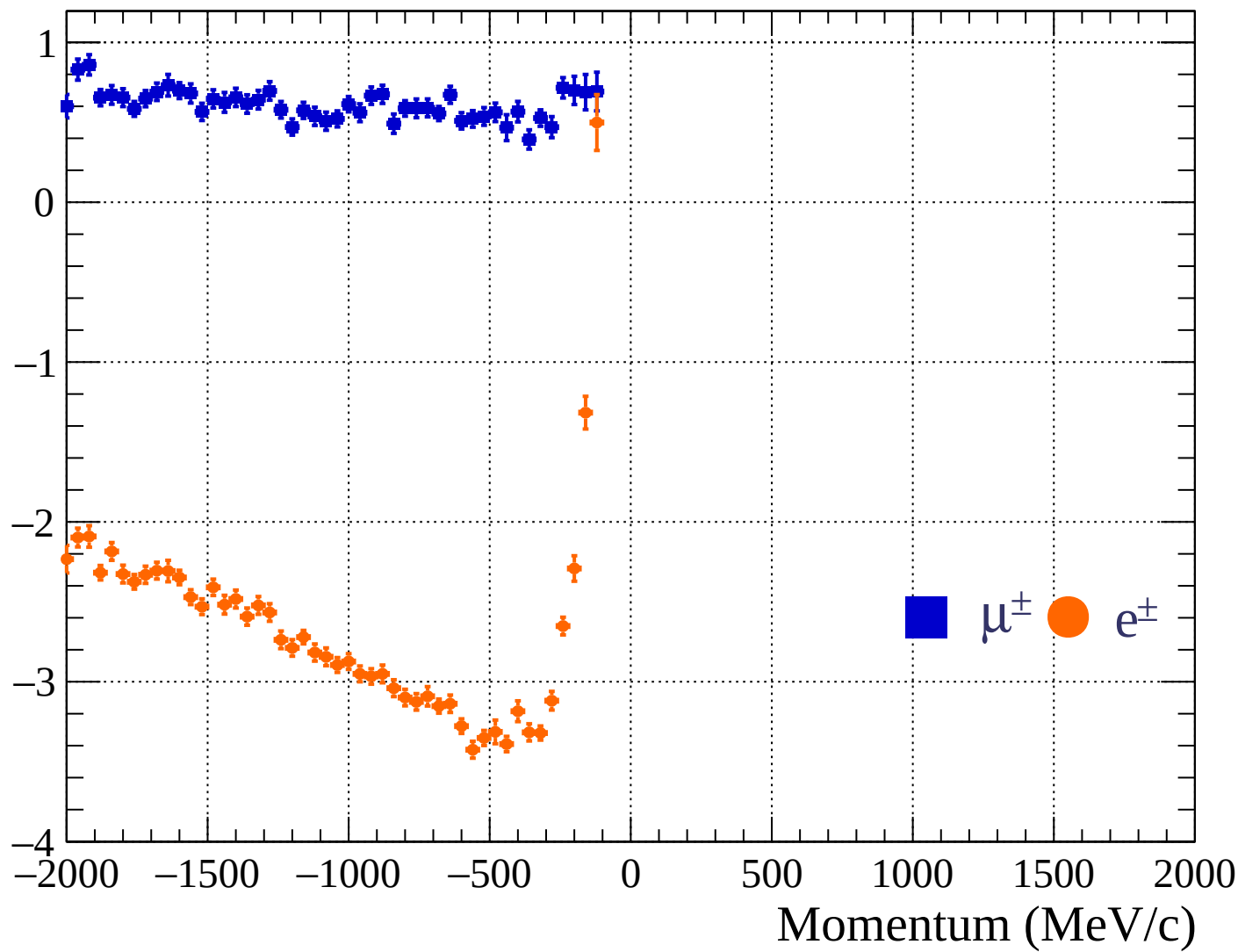


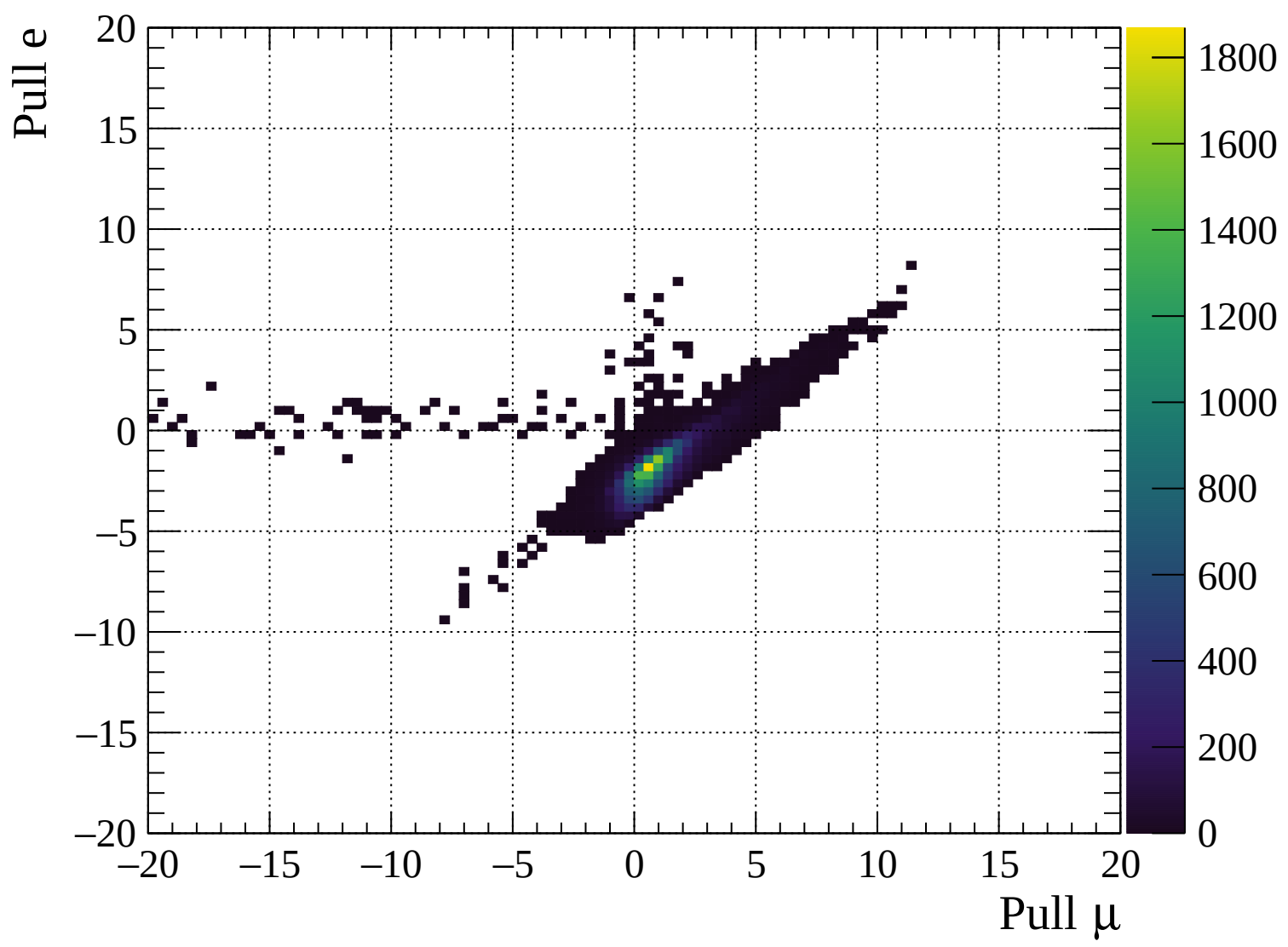


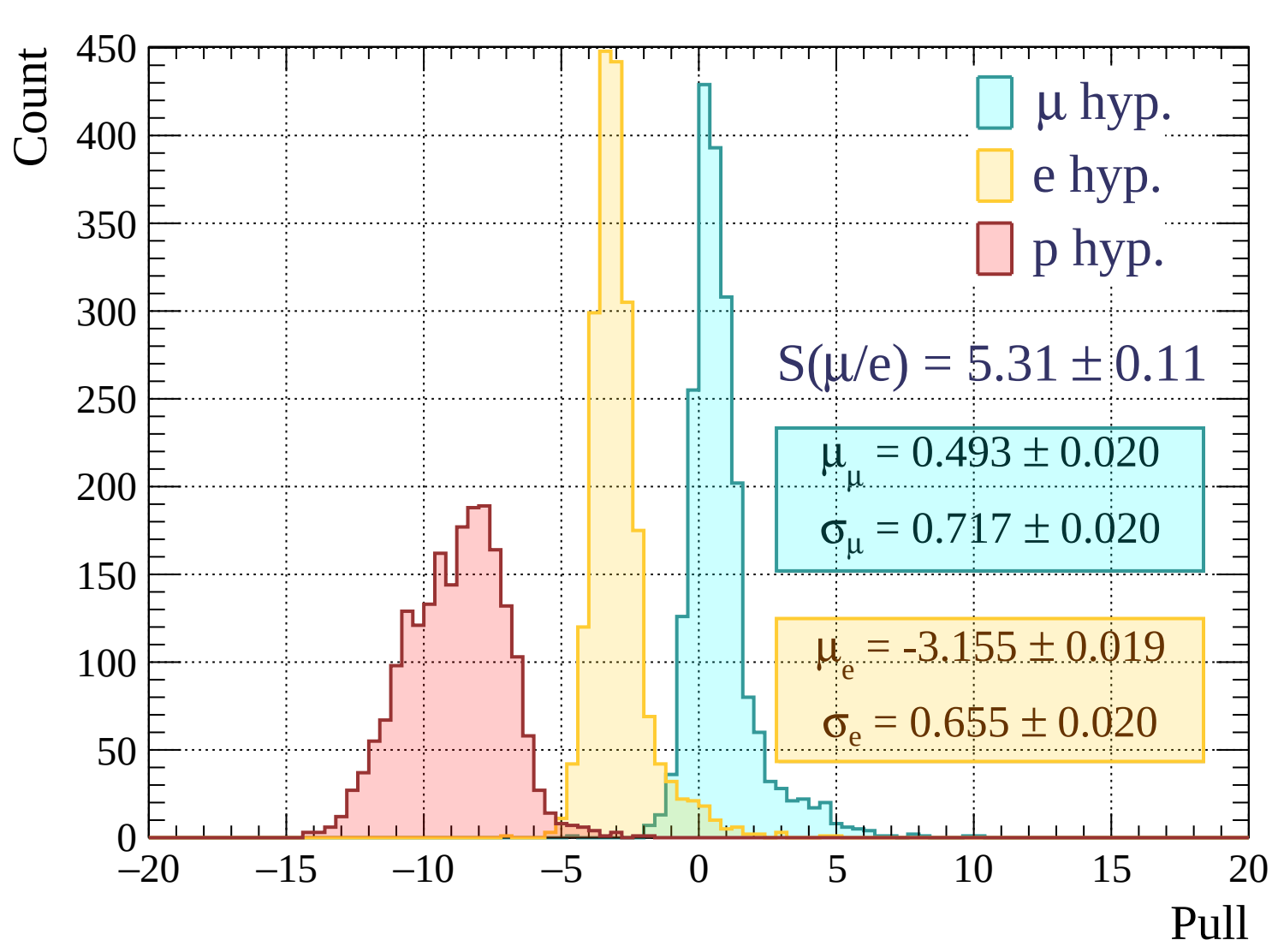
Pulls standard deviation

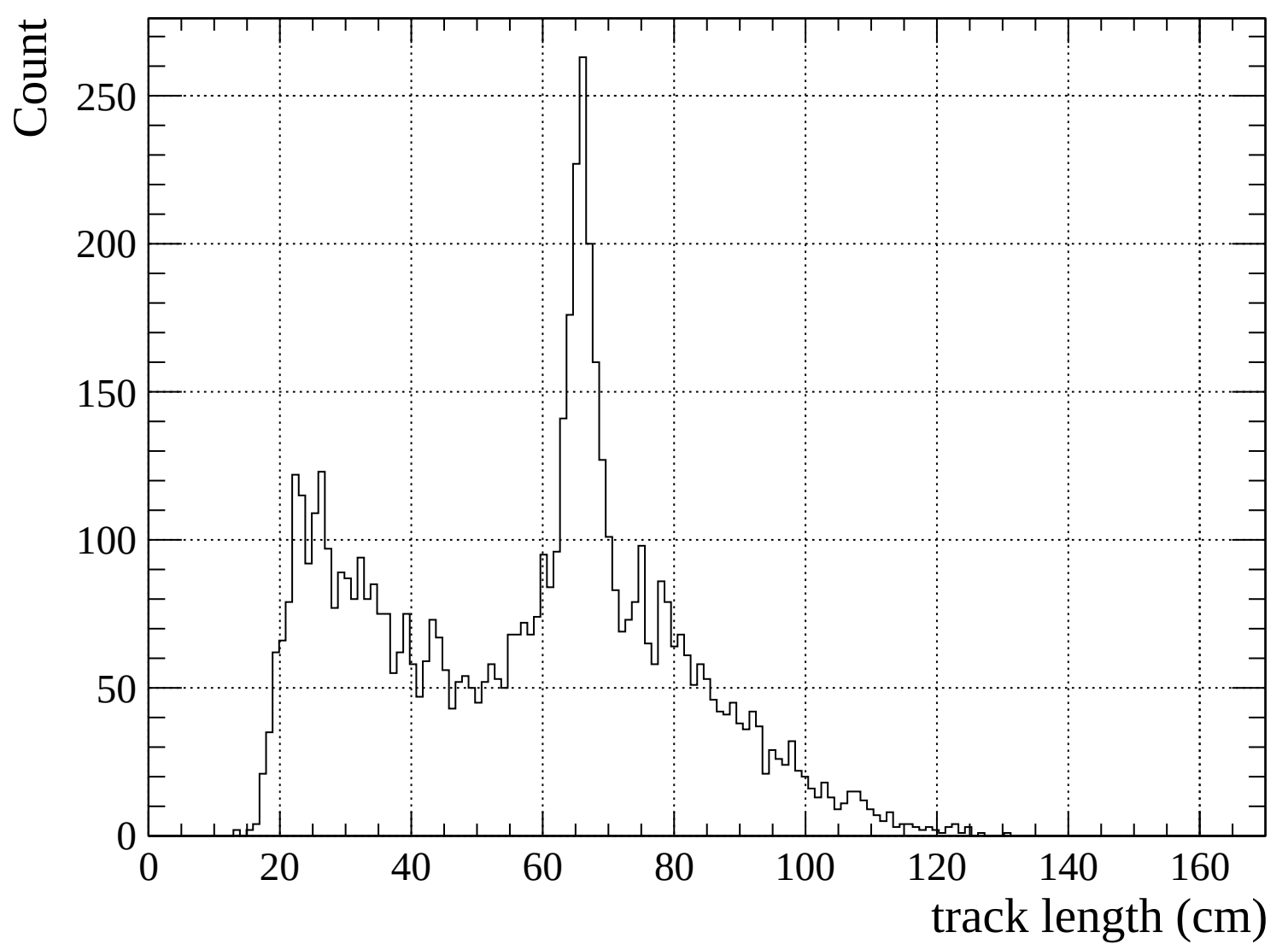


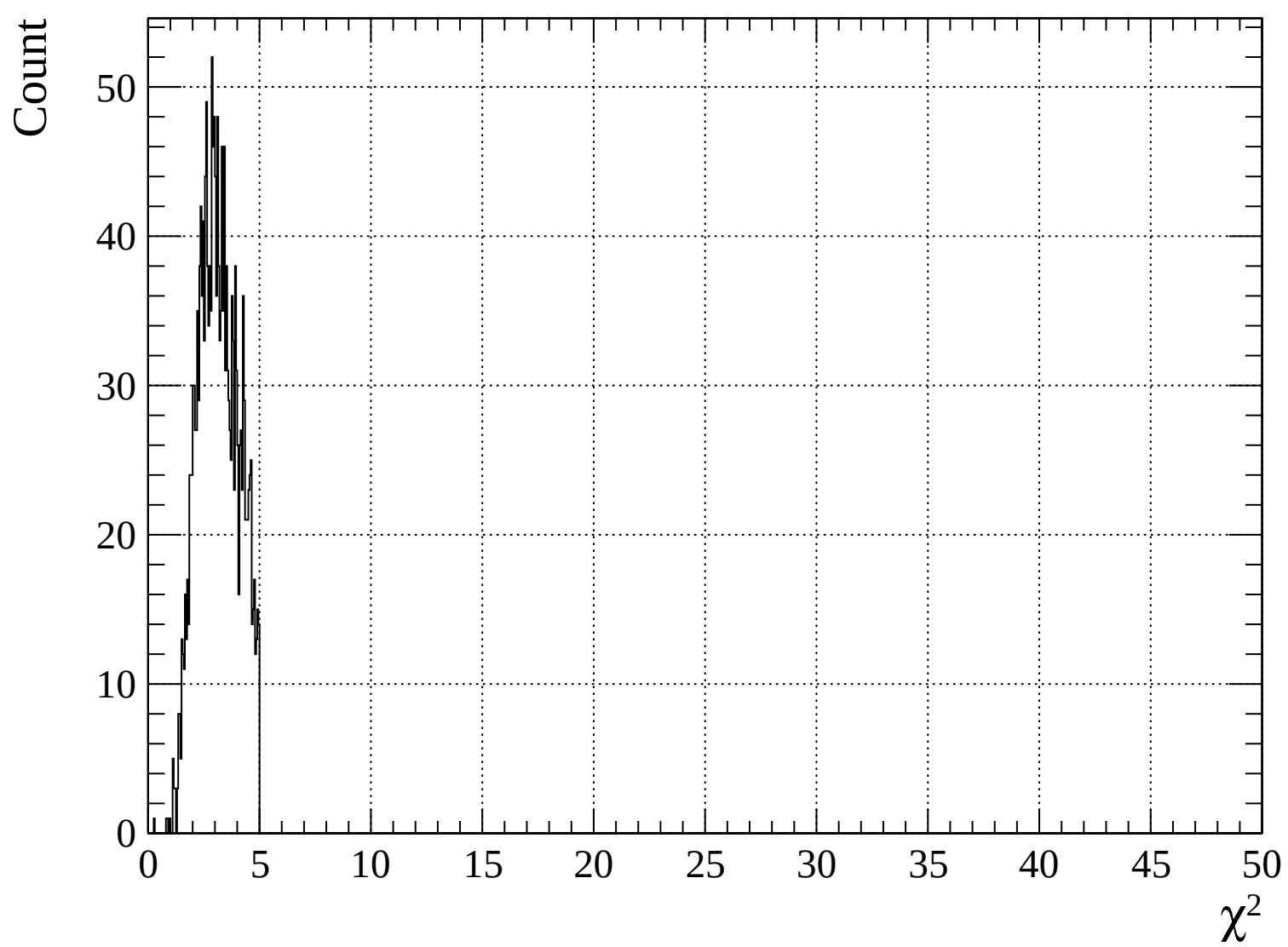
Pulls mean

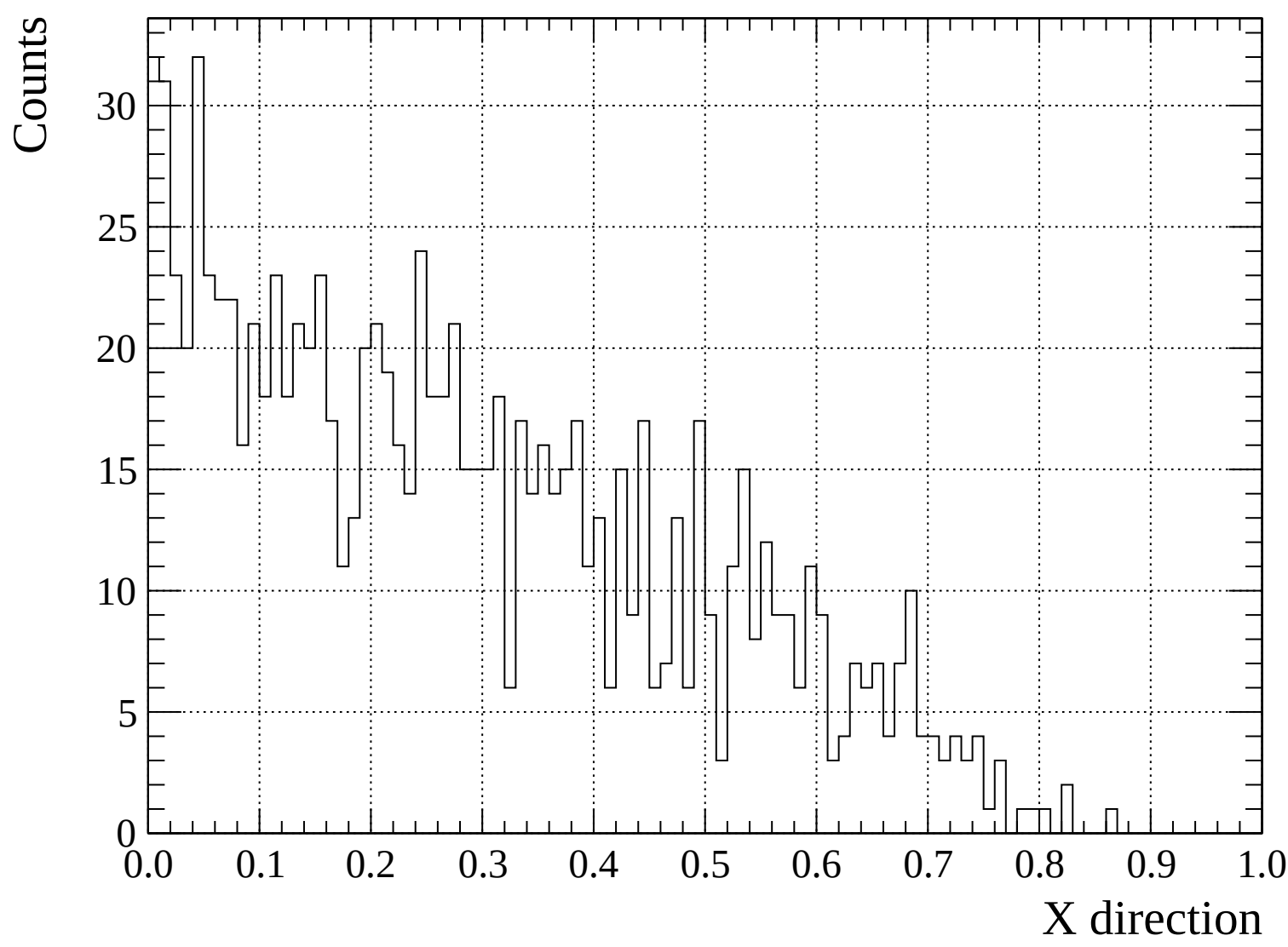


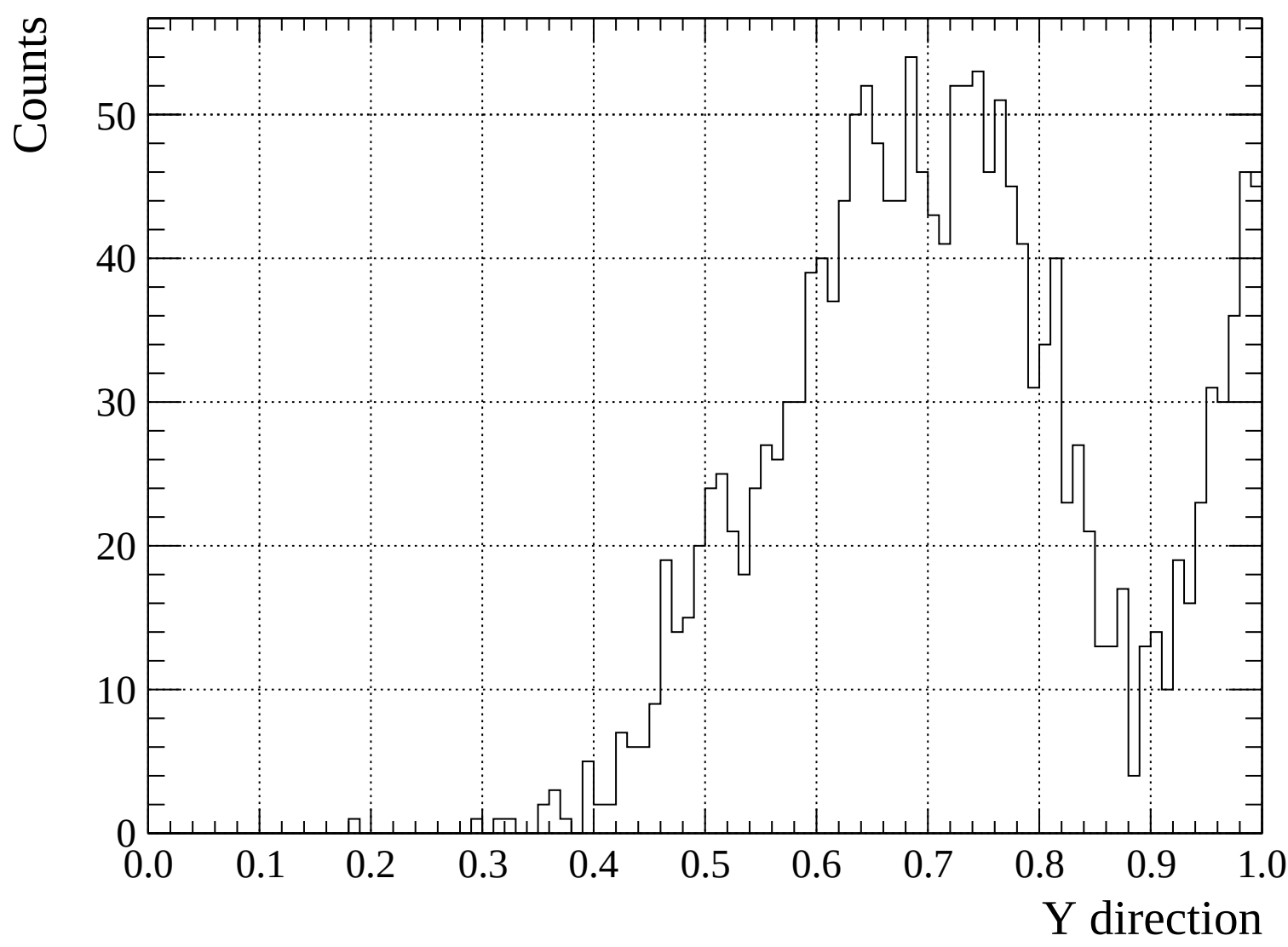


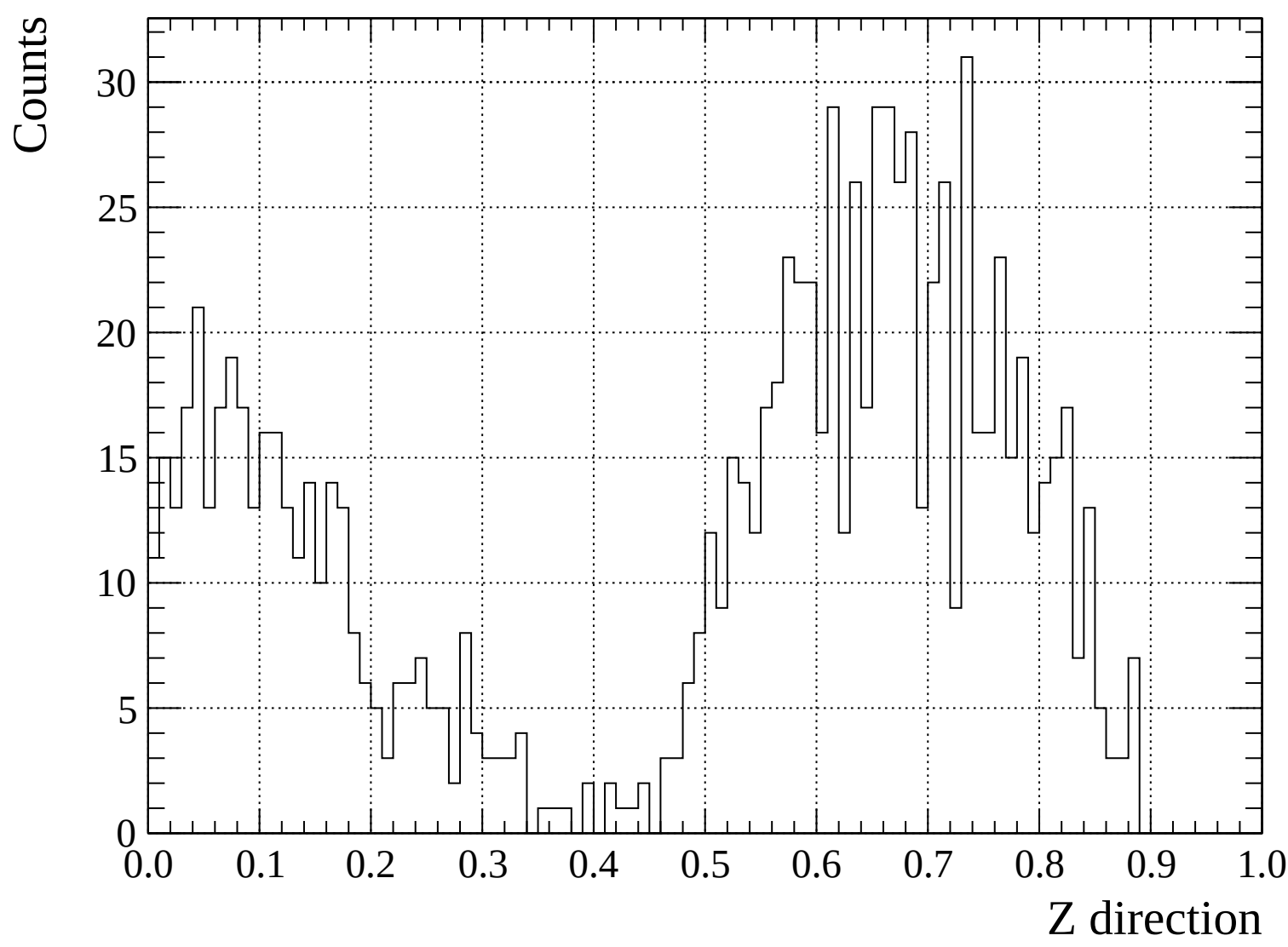


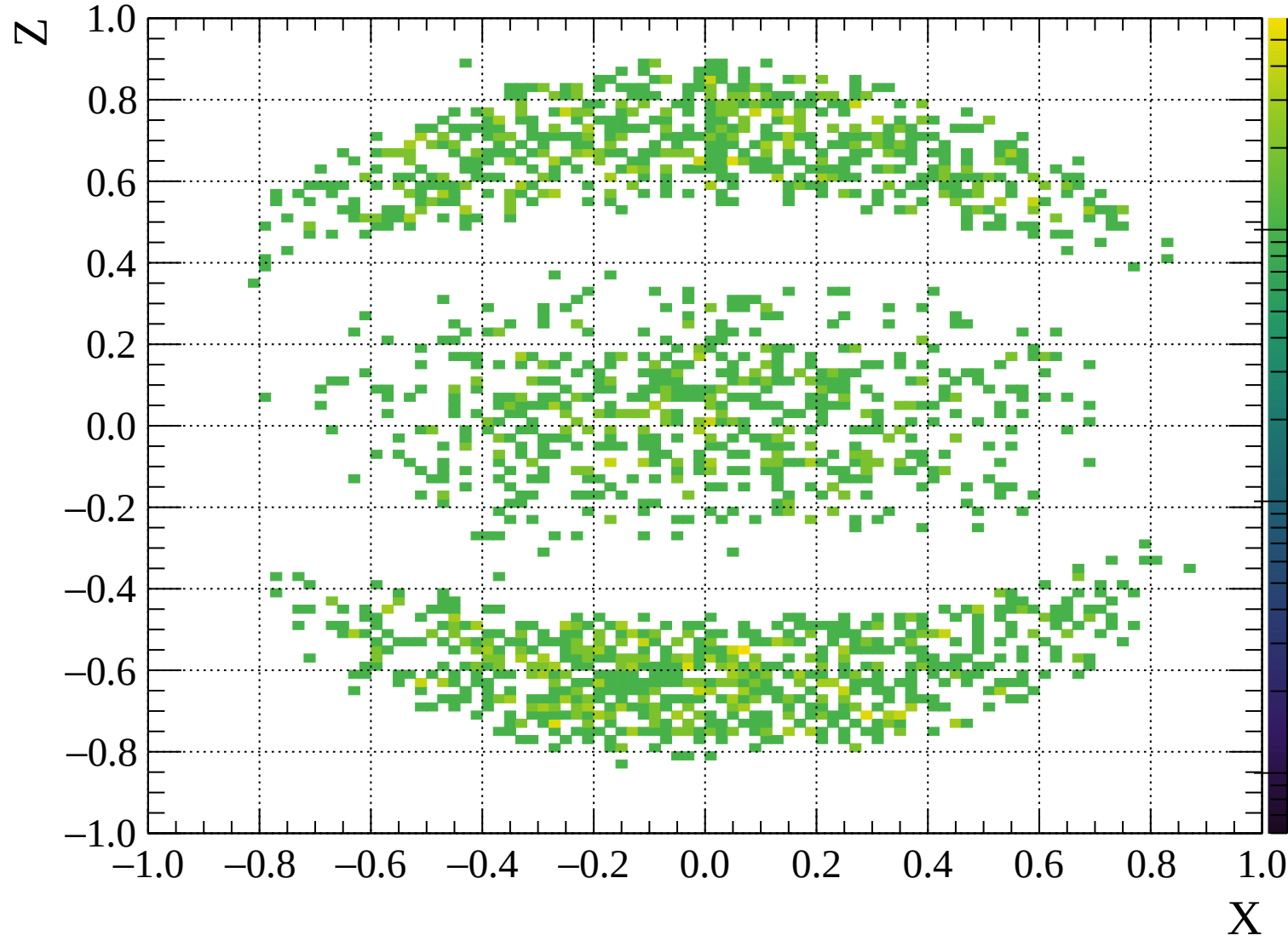


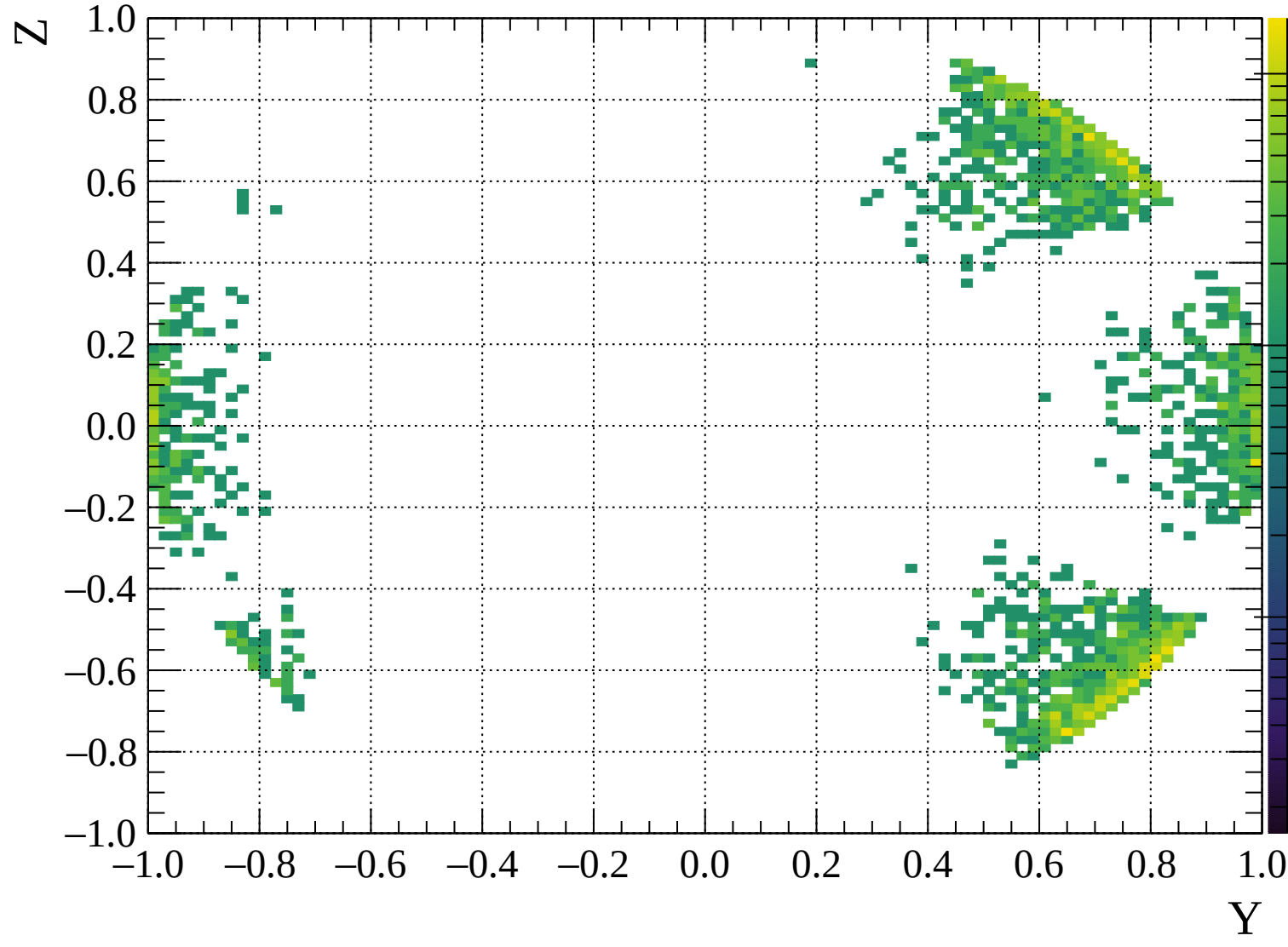


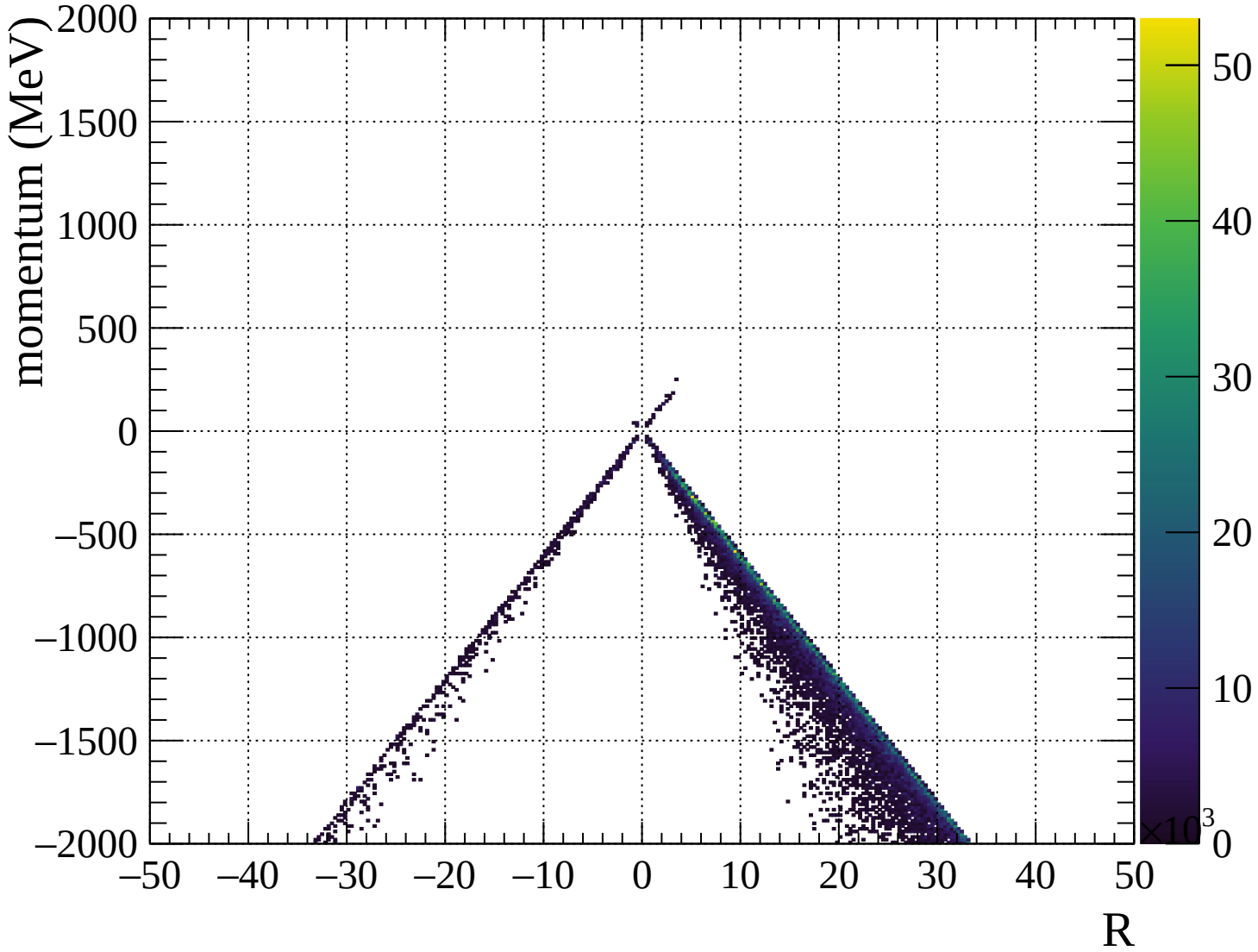


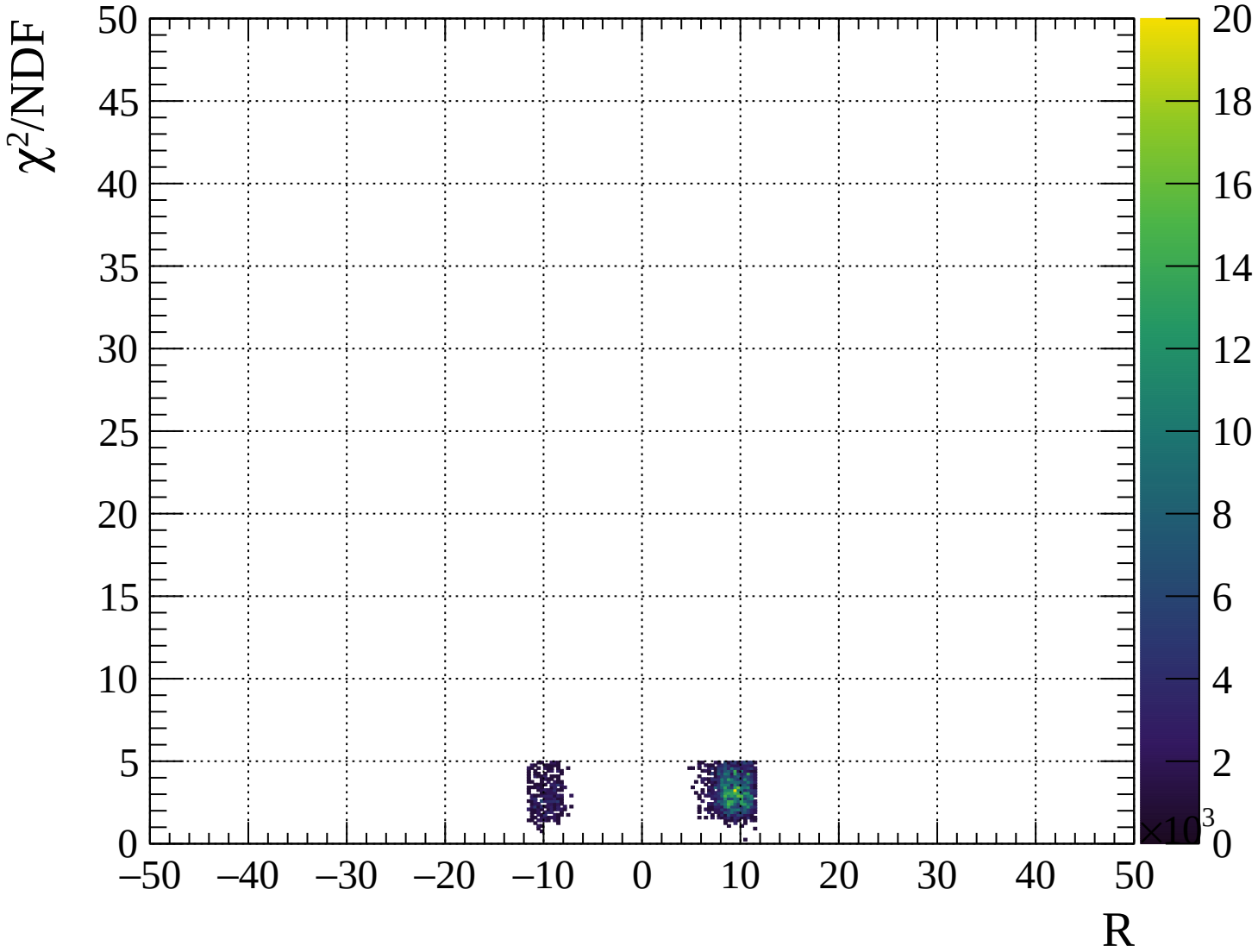






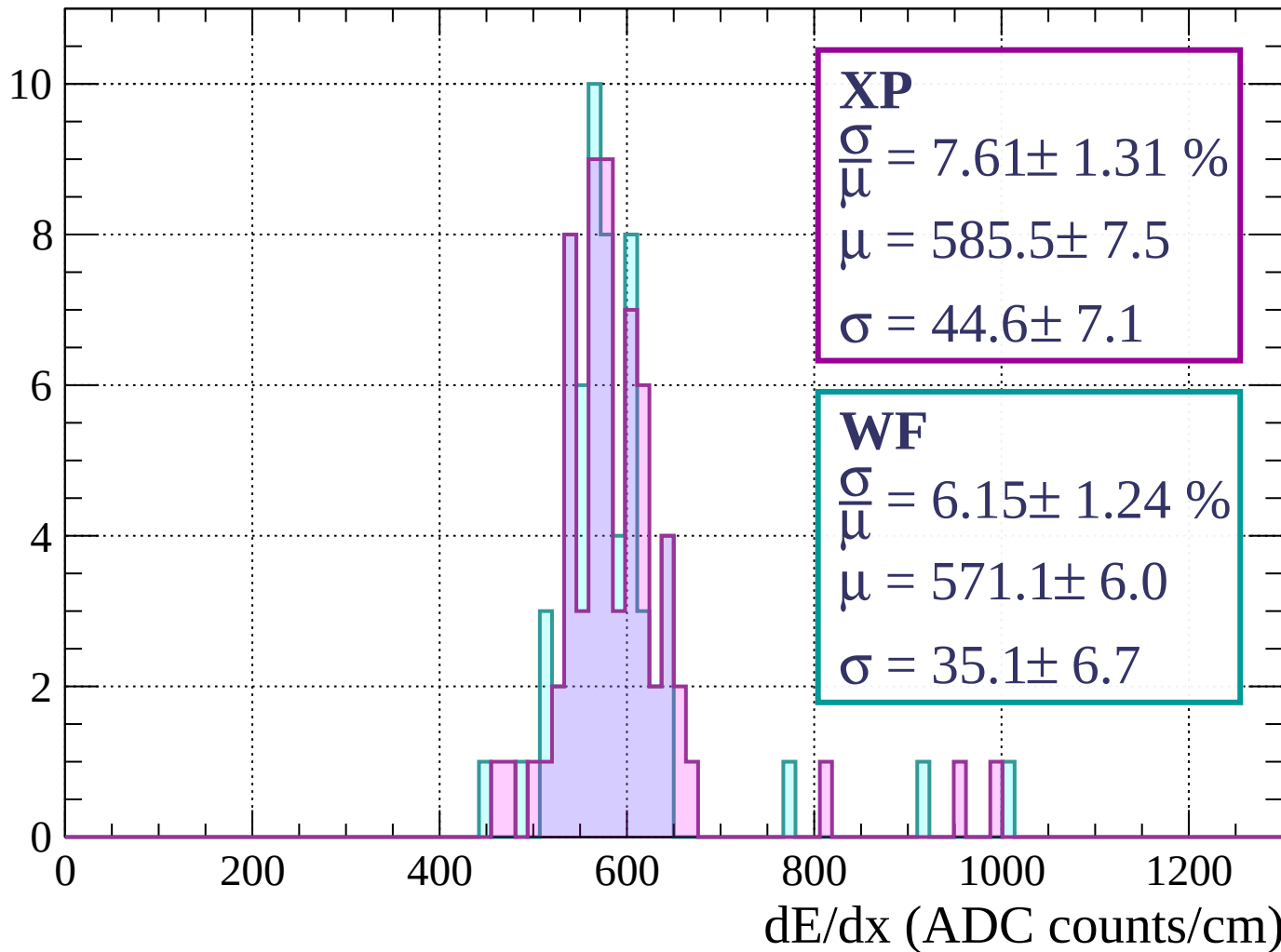






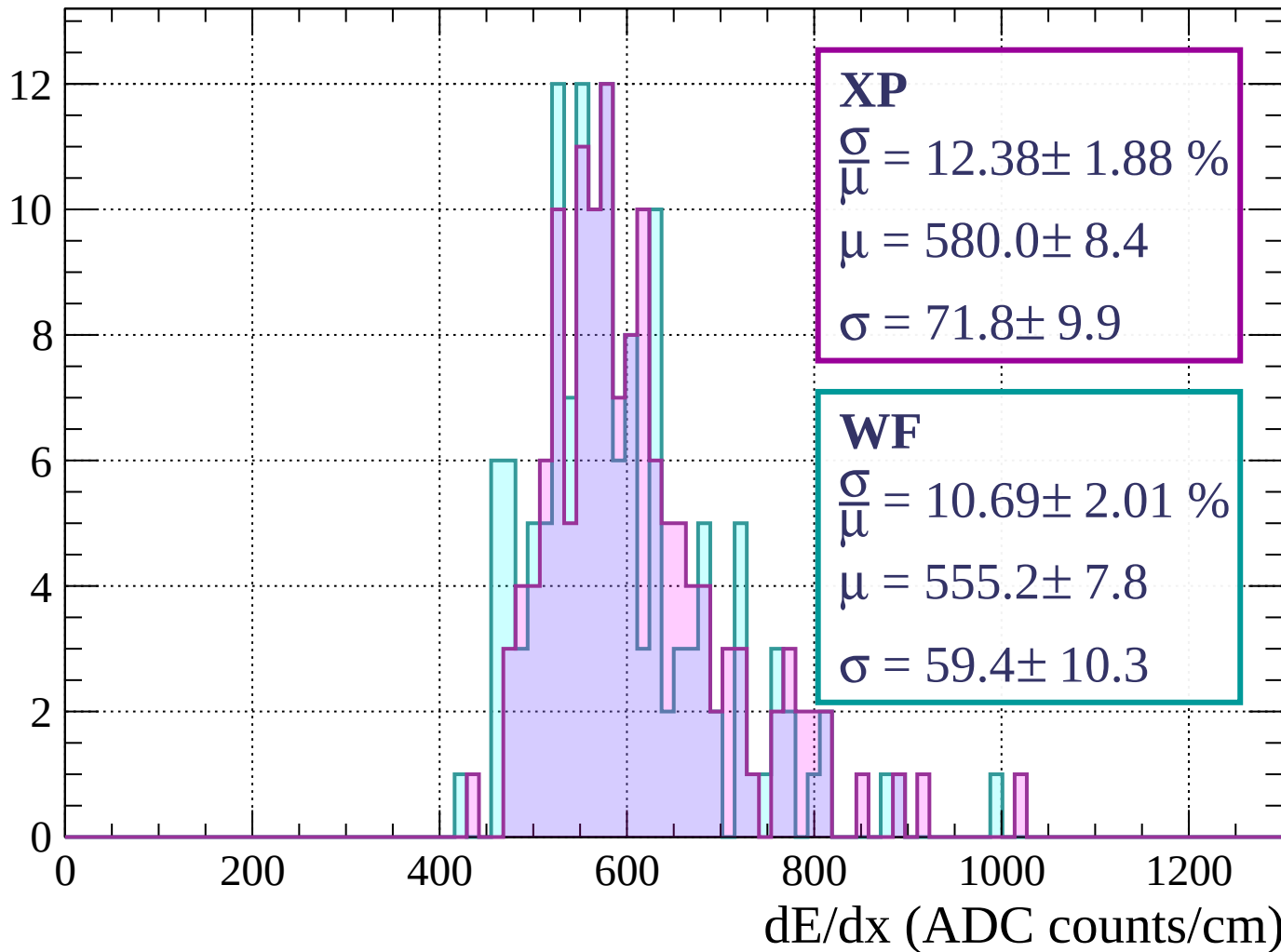
Energy loss | $0 < p < 80$

Count



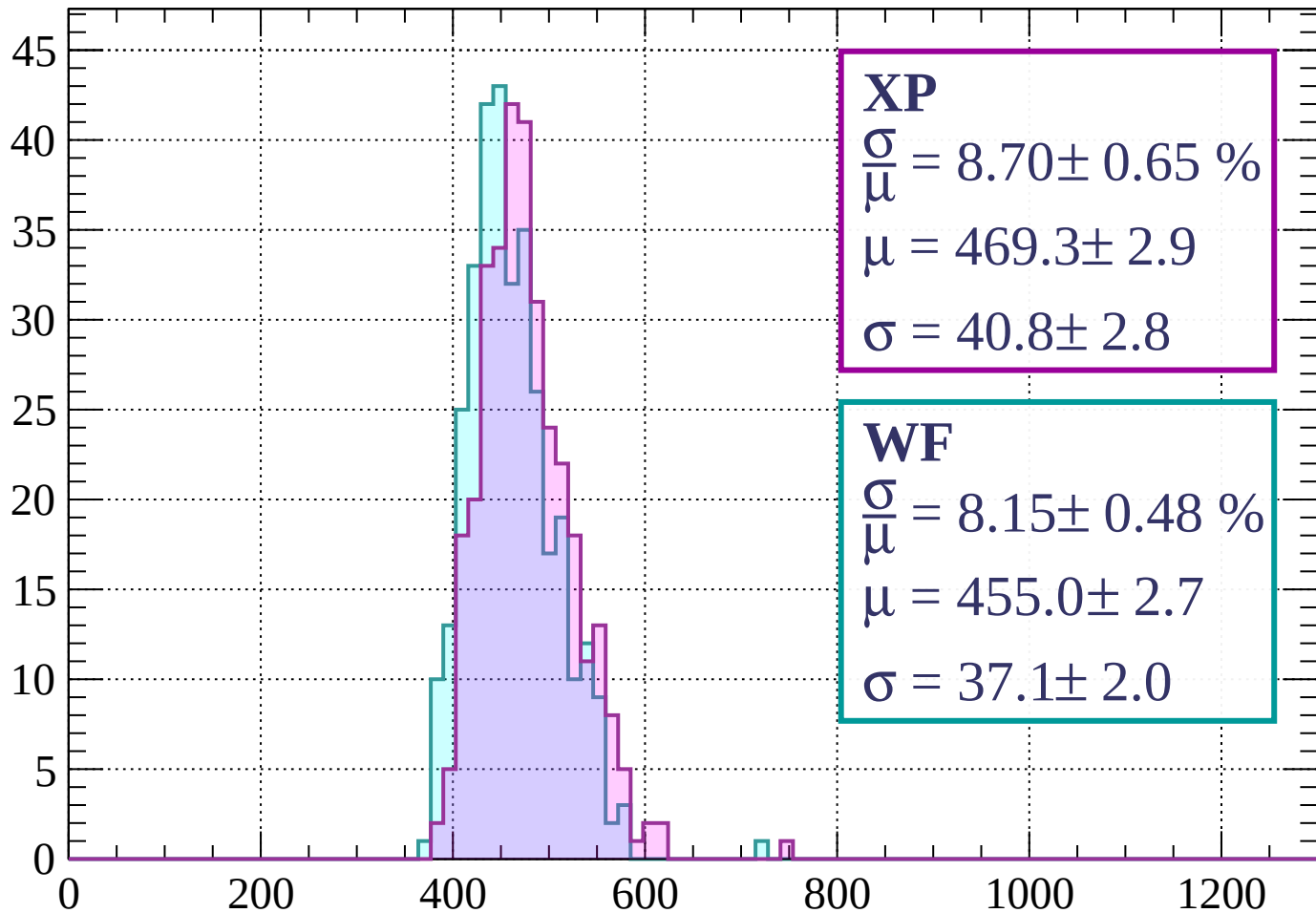
Energy loss | $80 < p < 160$

Count



Energy loss | $160 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 8.70 \pm 0.65 \%$$

$$\mu = 469.3 \pm 2.9$$

$$\sigma = 40.8 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 8.15 \pm 0.48 \%$$

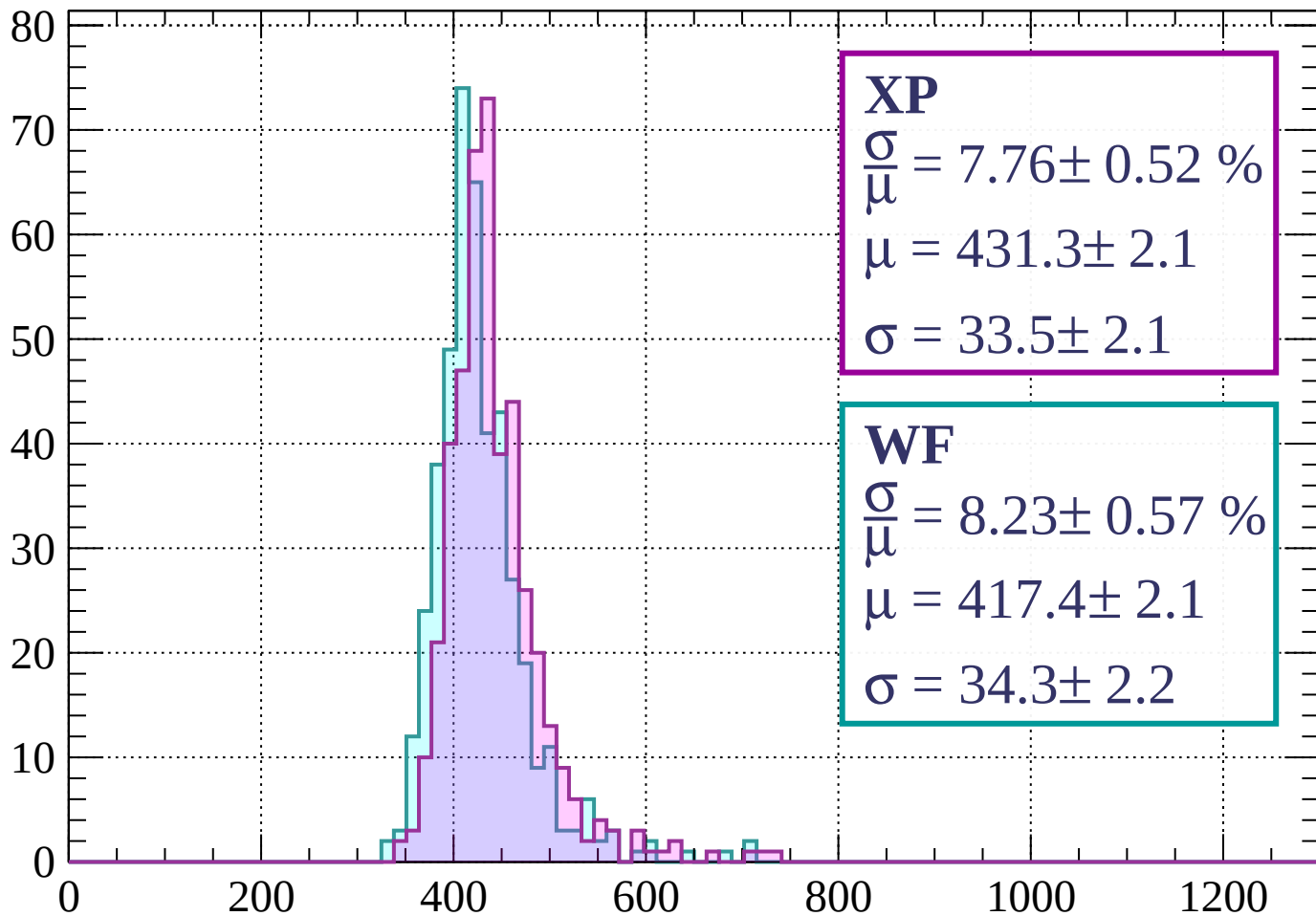
$$\mu = 455.0 \pm 2.7$$

$$\sigma = 37.1 \pm 2.0$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 320$

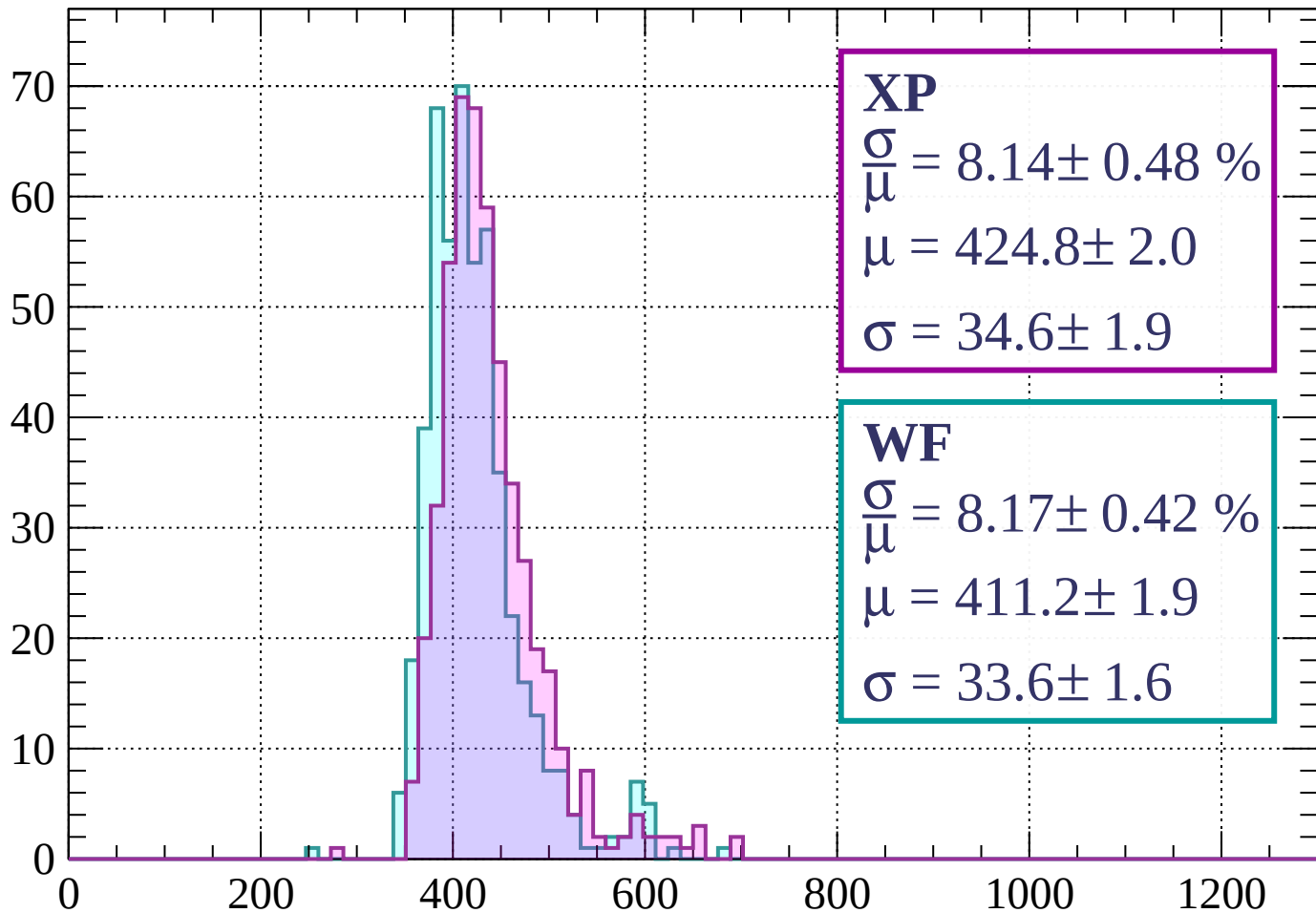
Count



dE/dx (ADC counts/cm)

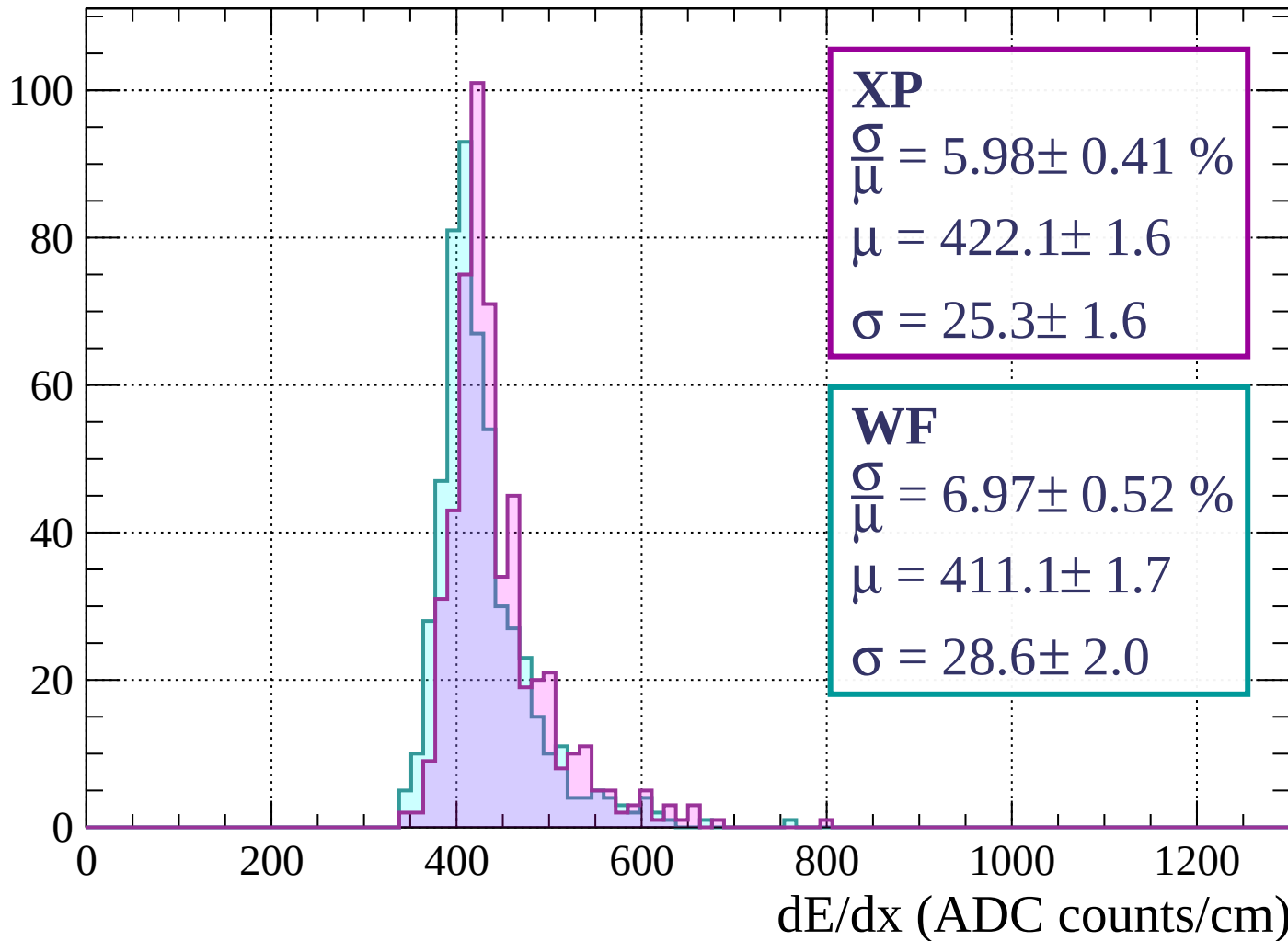
Energy loss | $320 < p < 400$

Count

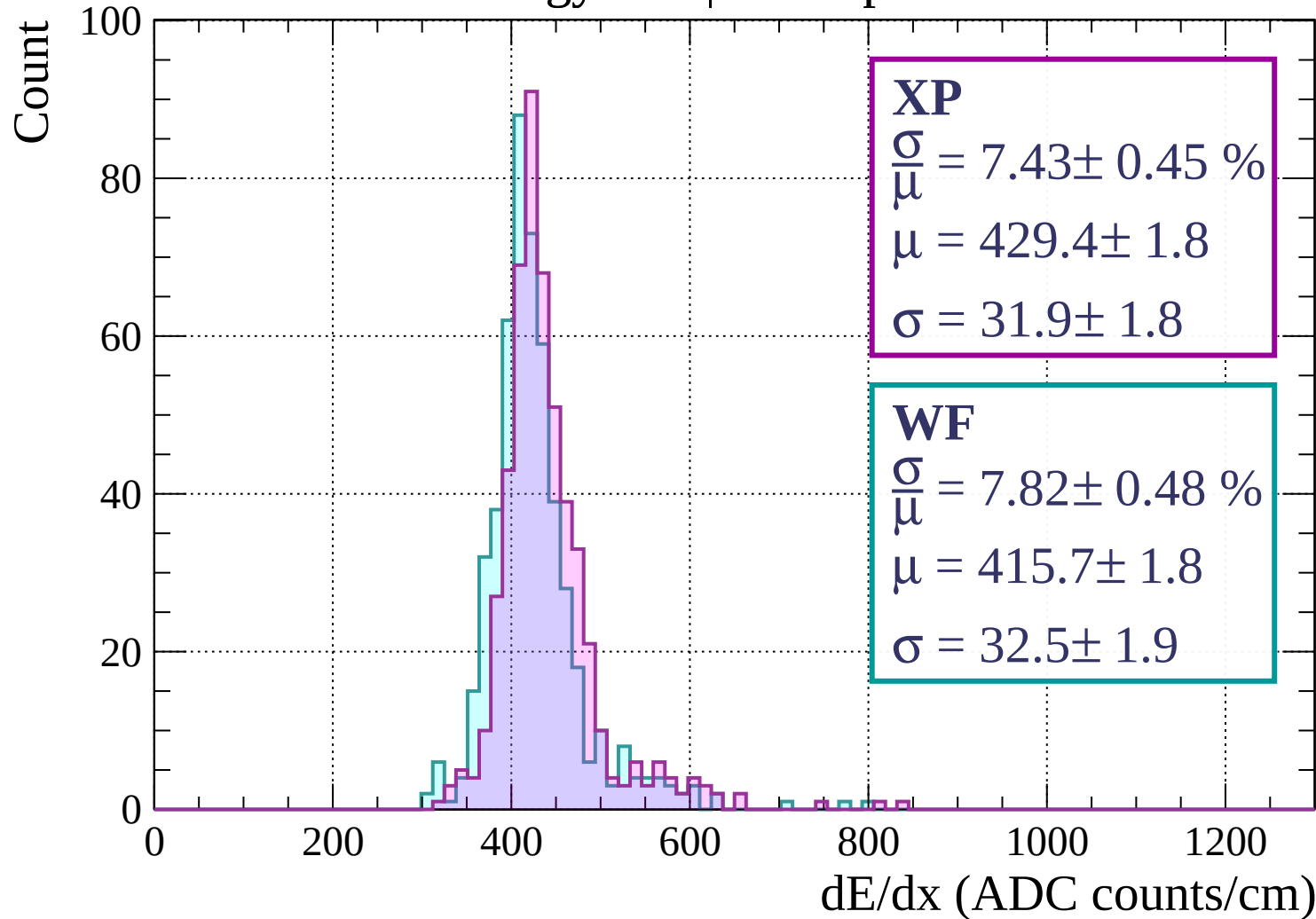


Energy loss | $400 < p < 480$

Count

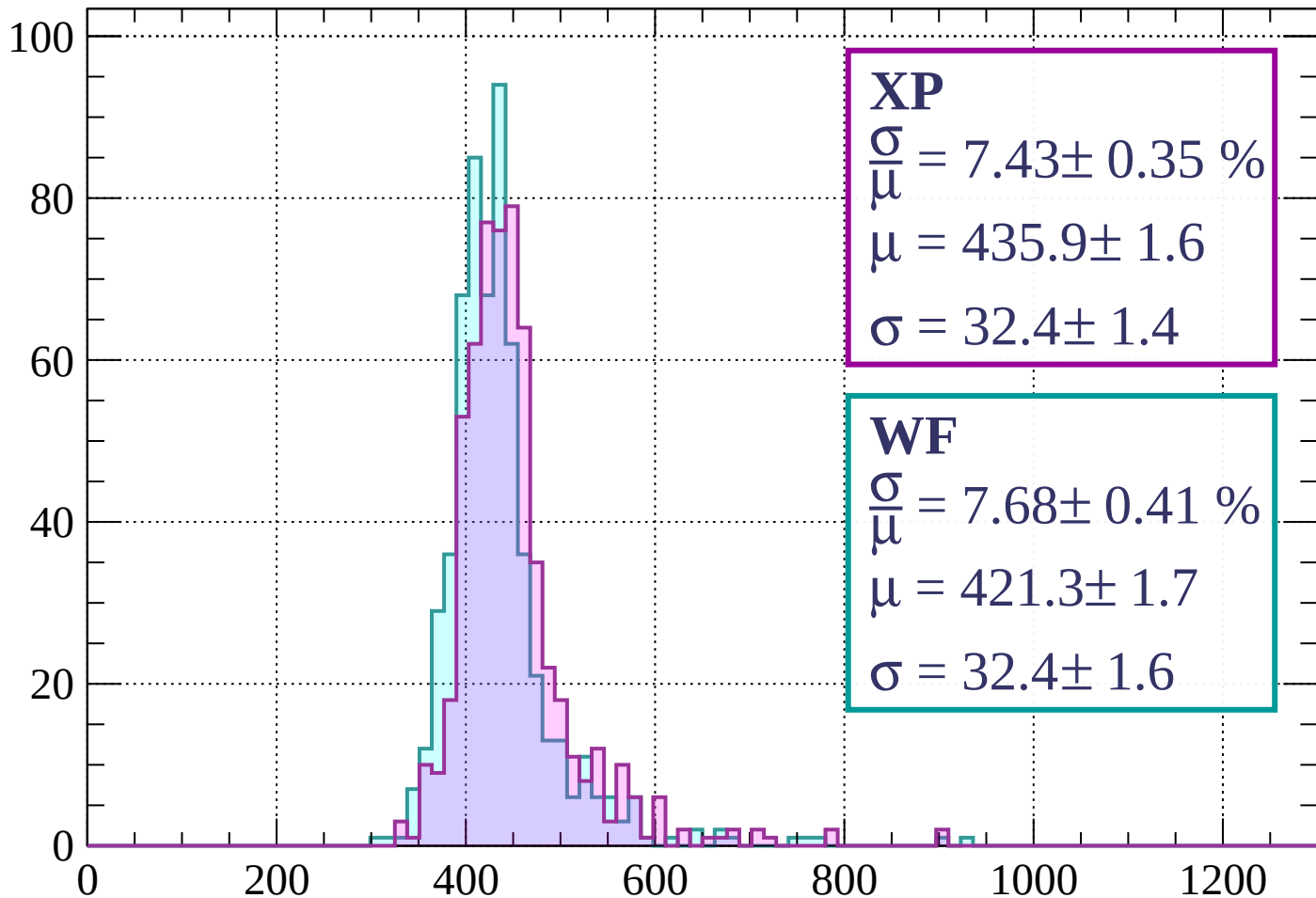


Energy loss | $480 < p < 560$



Energy loss | $560 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 7.43 \pm 0.35 \%$$

$$\mu = 435.9 \pm 1.6$$

$$\sigma = 32.4 \pm 1.4$$

WF

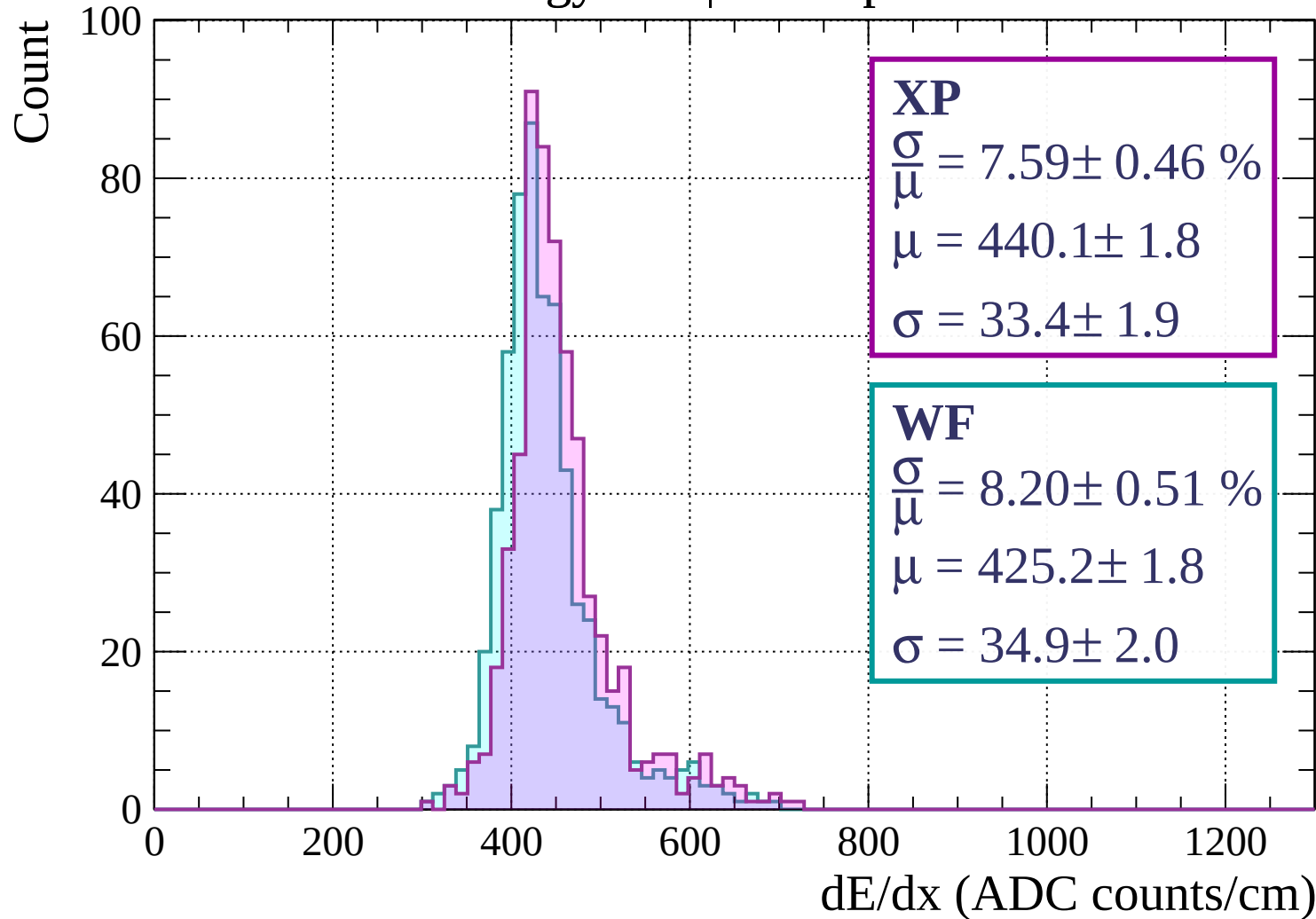
$$\frac{\sigma}{\mu} = 7.68 \pm 0.41 \%$$

$$\mu = 421.3 \pm 1.7$$

$$\sigma = 32.4 \pm 1.6$$

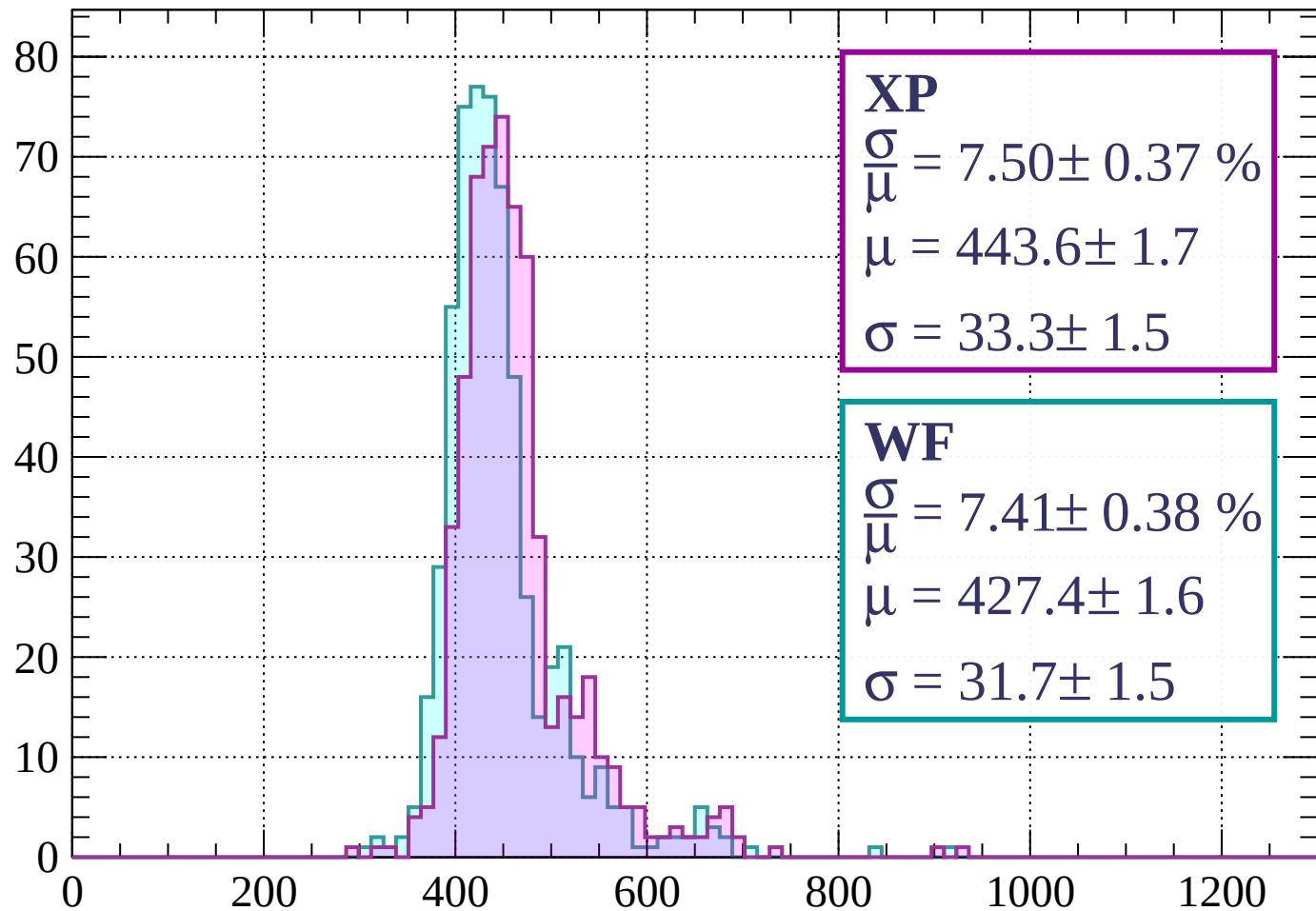
dE/dx (ADC counts/cm)

Energy loss | $640 < p < 720$



Energy loss | $720 < p < 800$

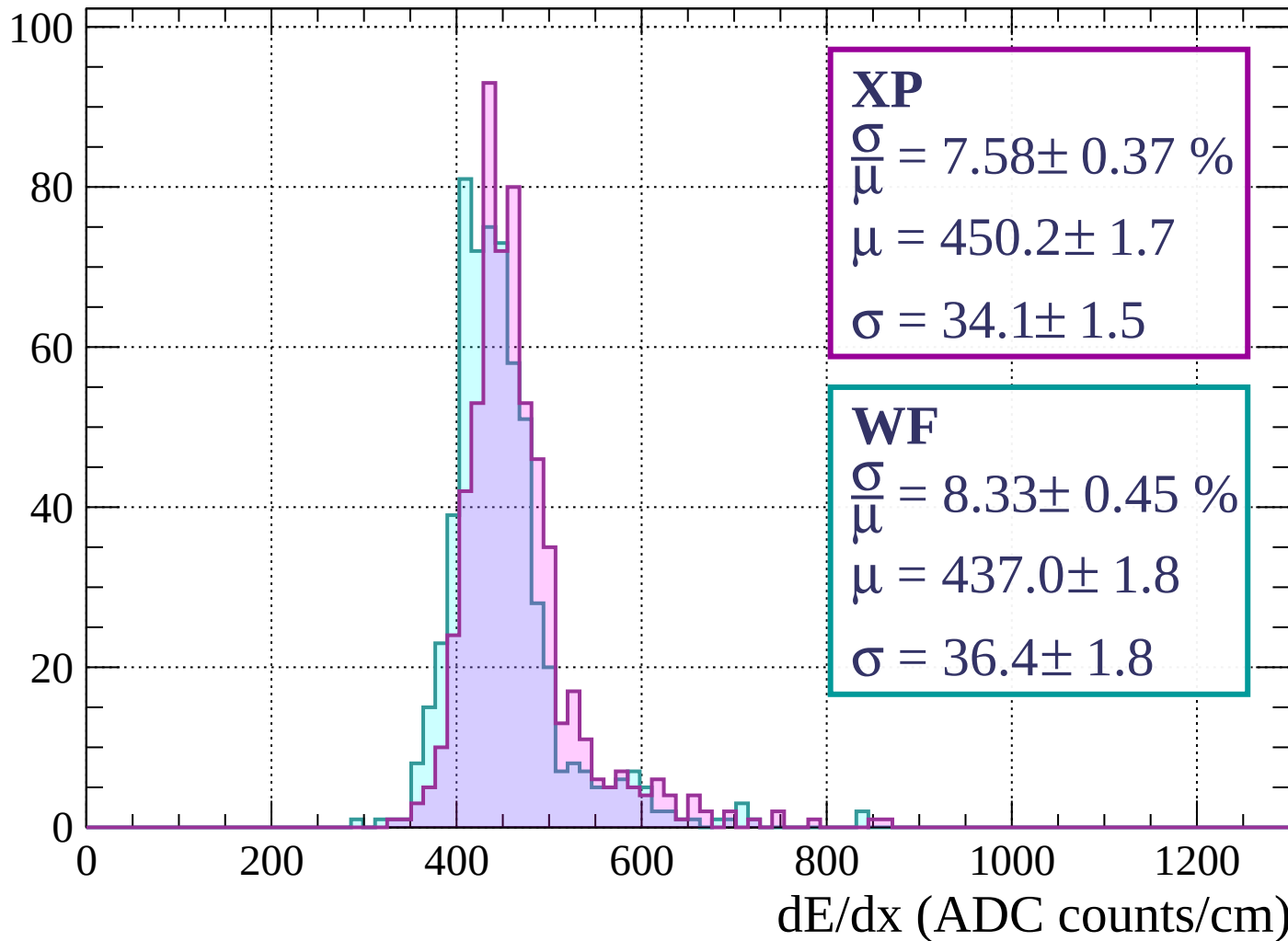
Count



dE/dx (ADC counts/cm)

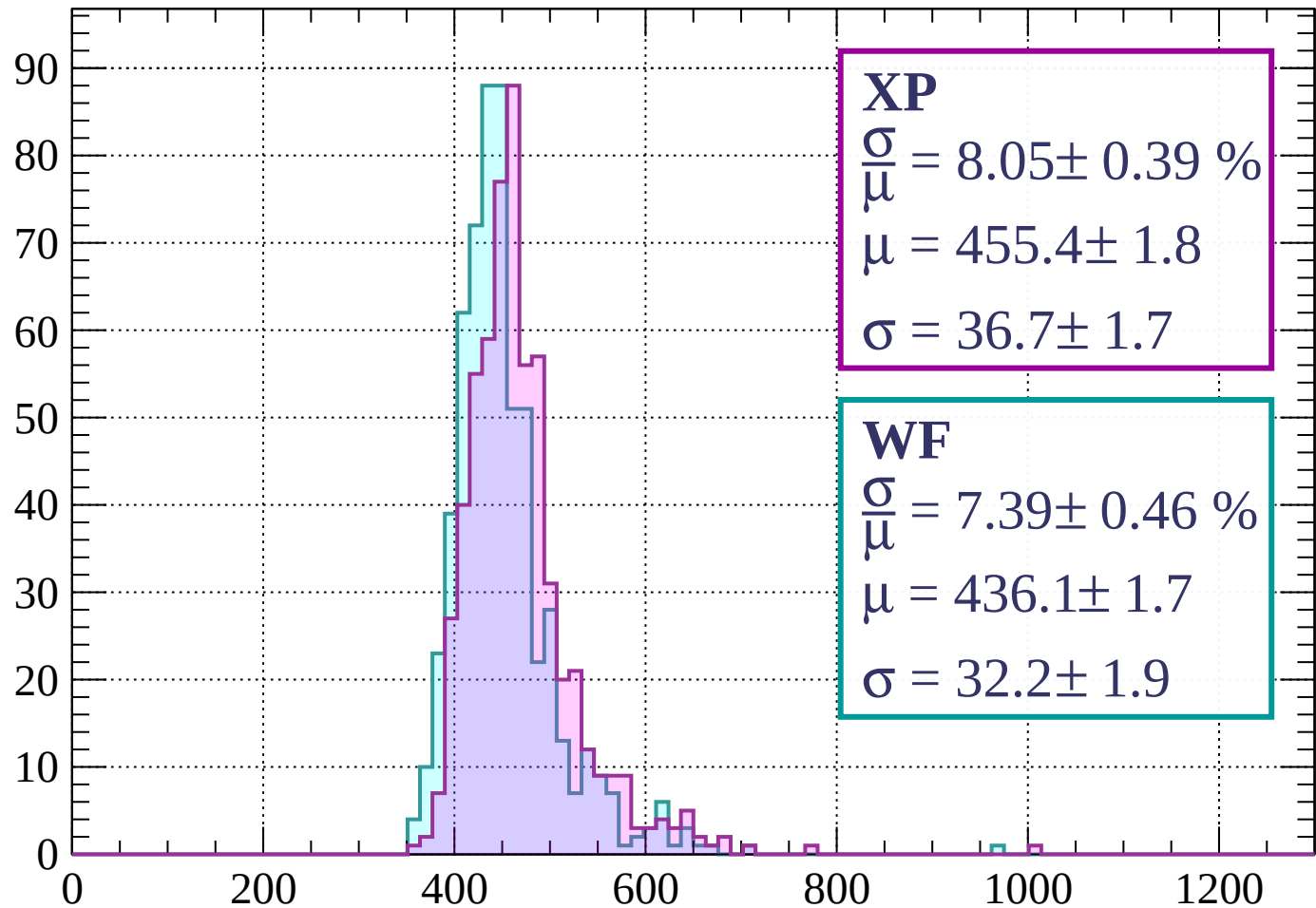
Energy loss | $800 < p < 880$

Count



Energy loss | $880 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 8.05 \pm 0.39 \%$$

$$\mu = 455.4 \pm 1.8$$

$$\sigma = 36.7 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 7.39 \pm 0.46 \%$$

$$\mu = 436.1 \pm 1.7$$

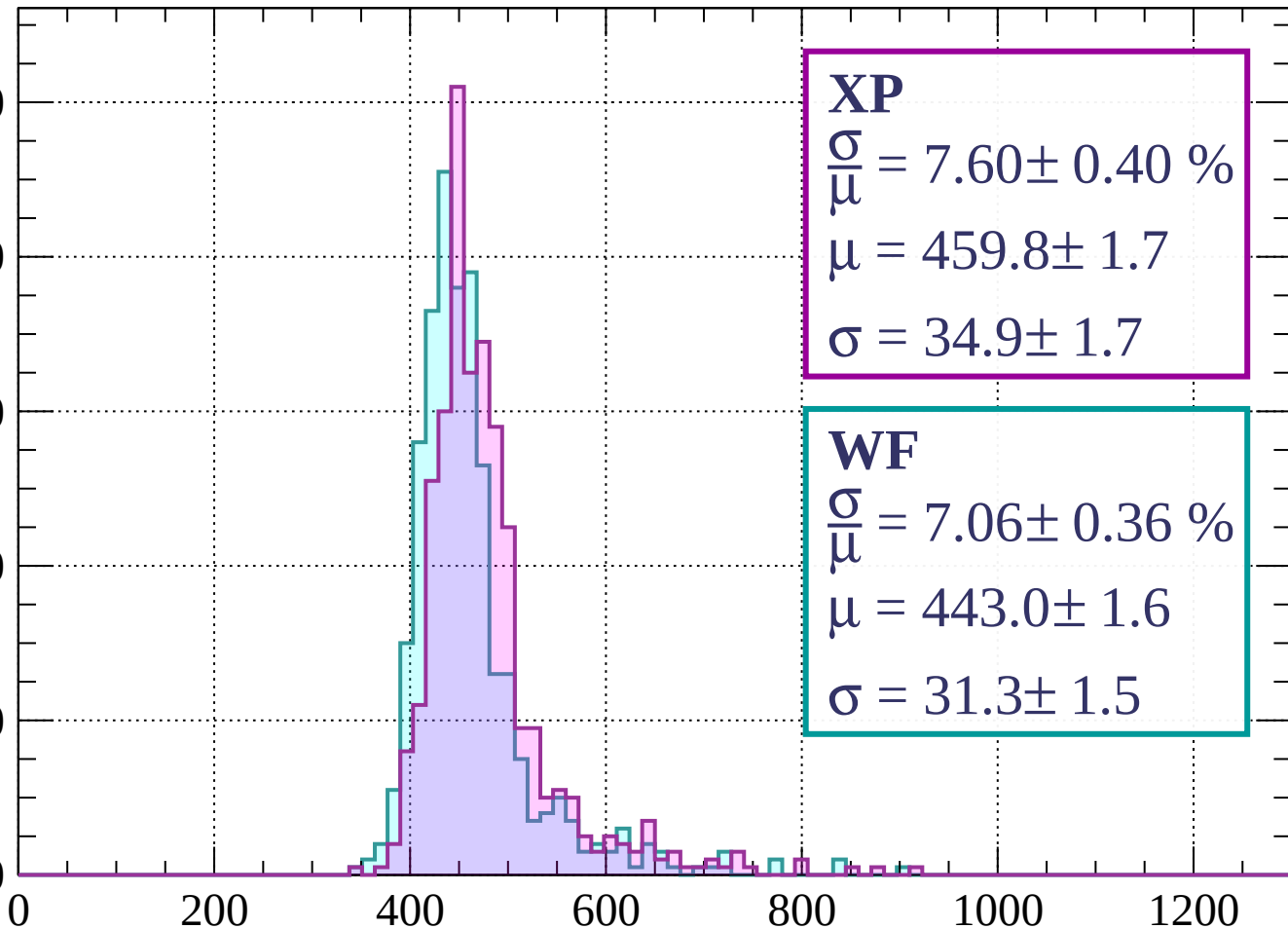
$$\sigma = 32.2 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1040$

Count

100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 7.60 \pm 0.40 \%$$

$$\mu = 459.8 \pm 1.7$$

$$\sigma = 34.9 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 7.06 \pm 0.36 \%$$

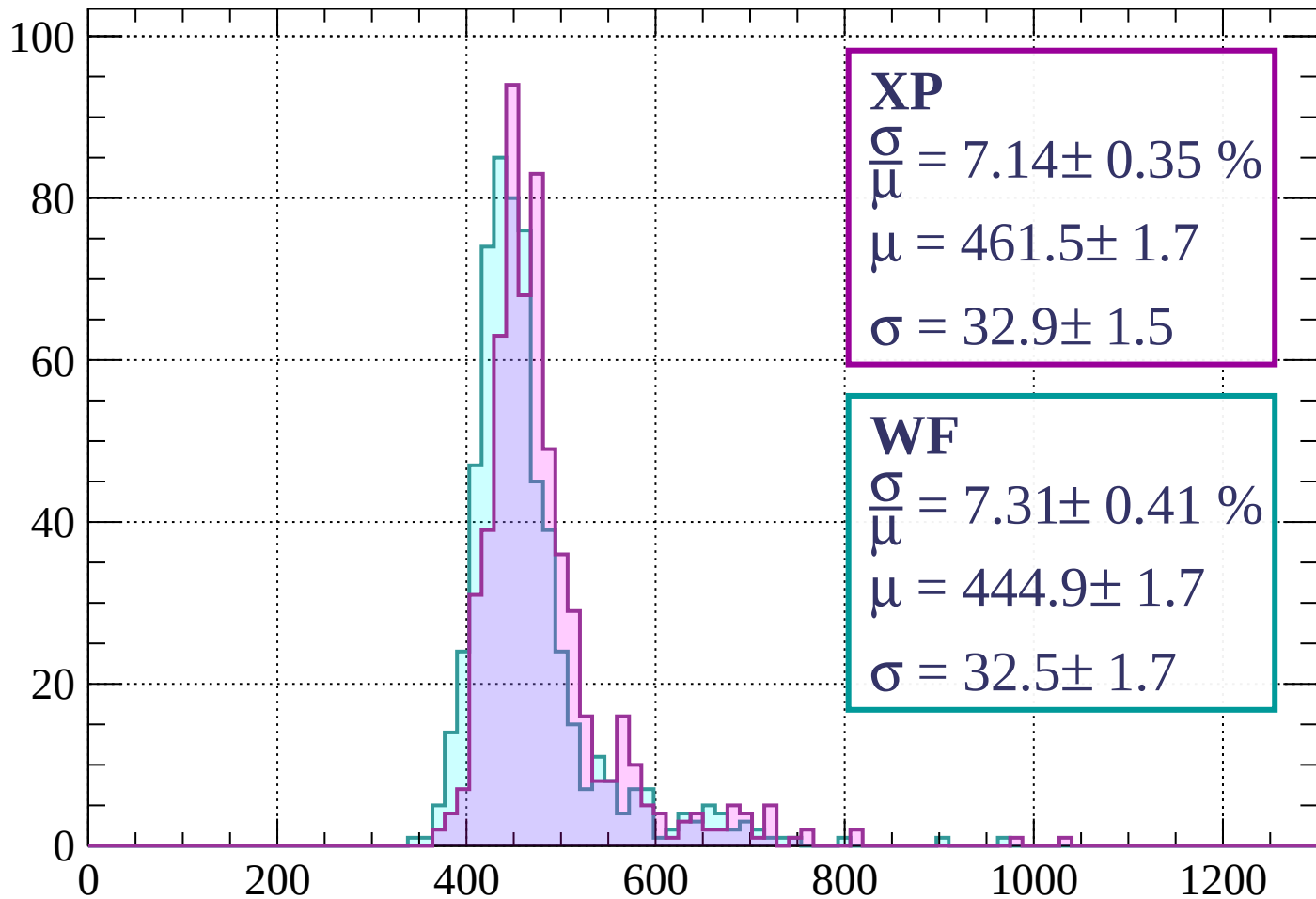
$$\mu = 443.0 \pm 1.6$$

$$\sigma = 31.3 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $1040 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 7.14 \pm 0.35 \%$$

$$\mu = 461.5 \pm 1.7$$

$$\sigma = 32.9 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 7.31 \pm 0.41 \%$$

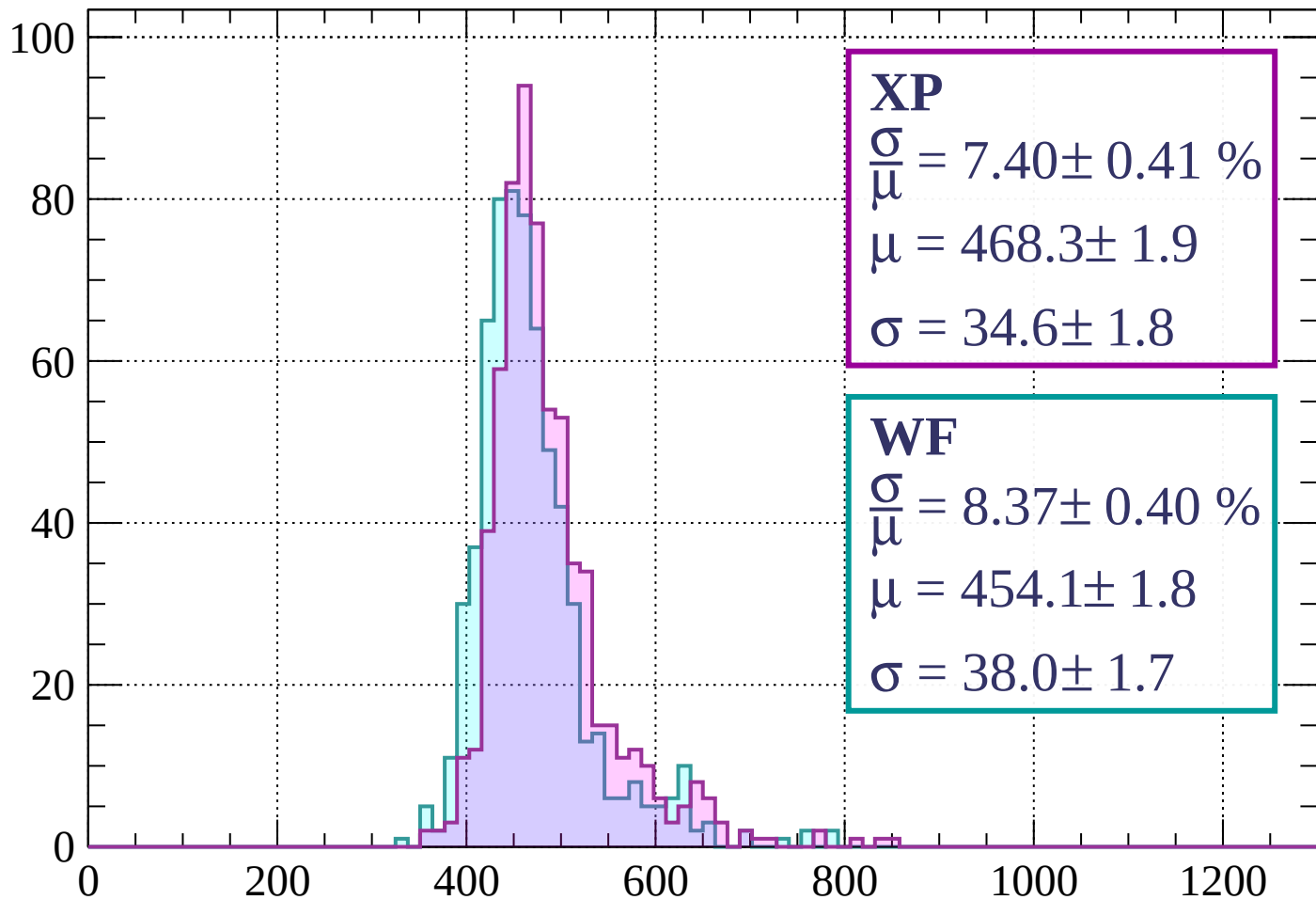
$$\mu = 444.9 \pm 1.7$$

$$\sigma = 32.5 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 7.40 \pm 0.41 \%$$

$$\mu = 468.3 \pm 1.9$$

$$\sigma = 34.6 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 8.37 \pm 0.40 \%$$

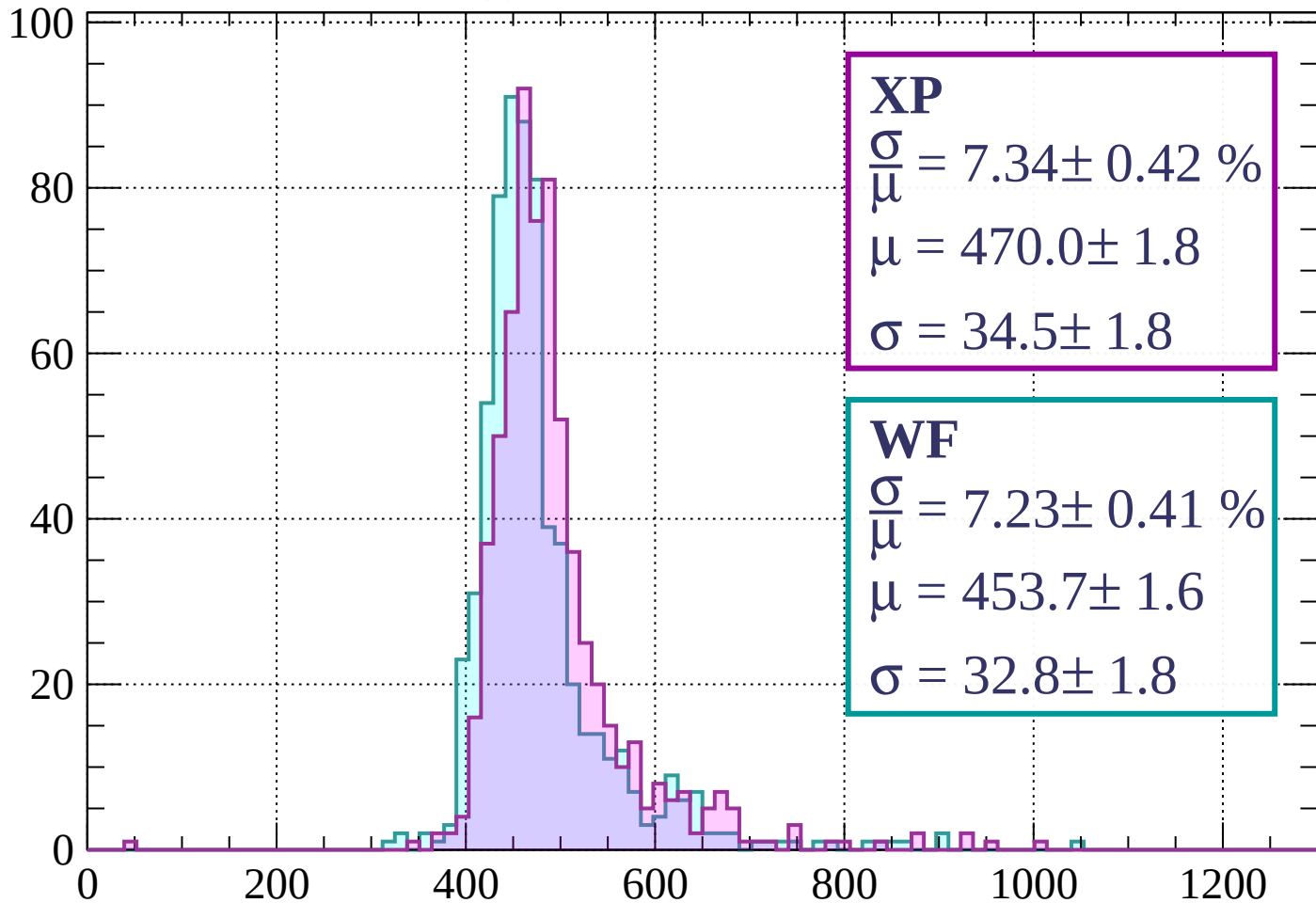
$$\mu = 454.1 \pm 1.8$$

$$\sigma = 38.0 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 7.34 \pm 0.42 \%$$

$$\mu = 470.0 \pm 1.8$$

$$\sigma = 34.5 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 7.23 \pm 0.41 \%$$

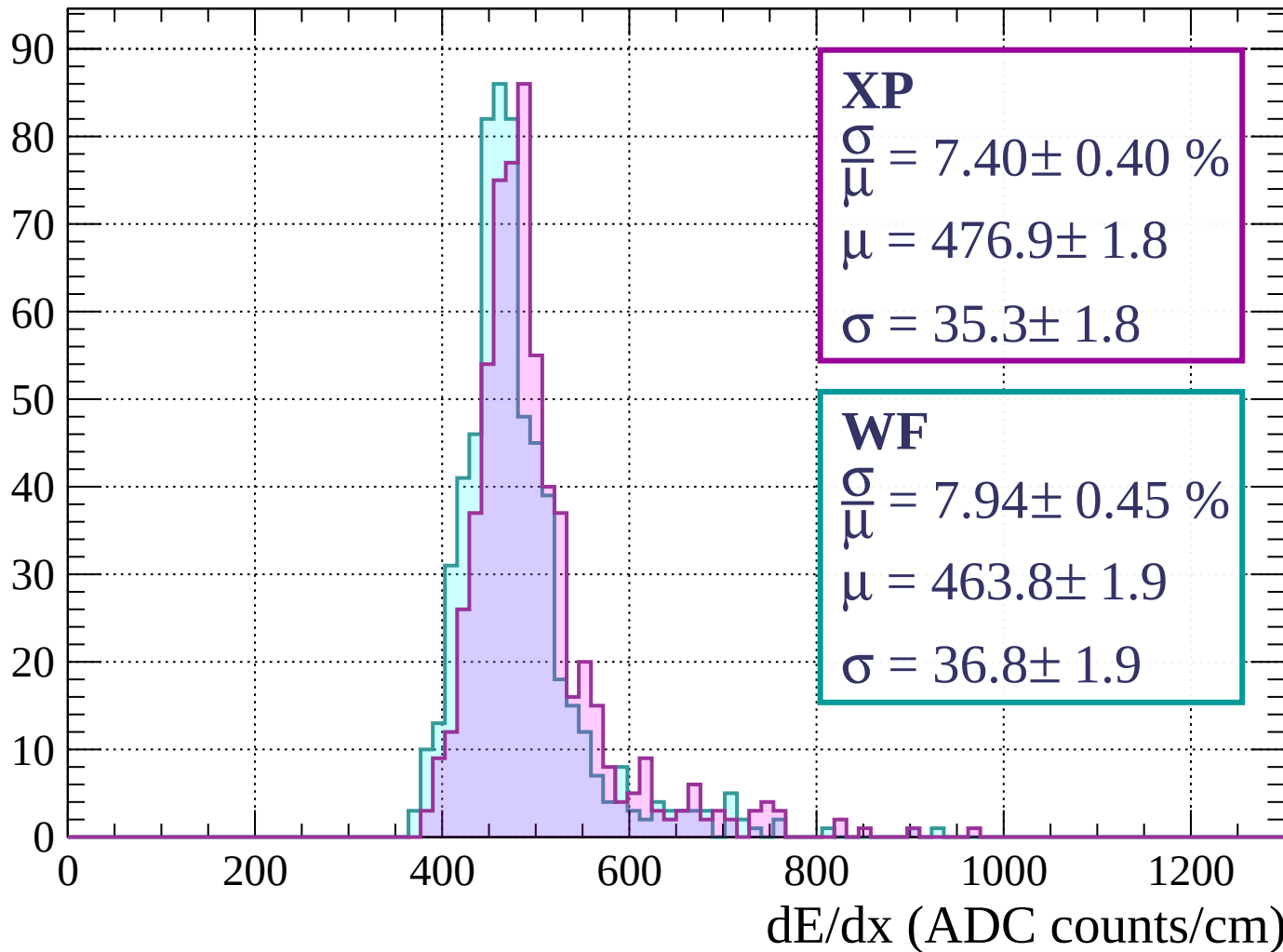
$$\mu = 453.7 \pm 1.6$$

$$\sigma = 32.8 \pm 1.8$$

dE/dx (ADC counts/cm)

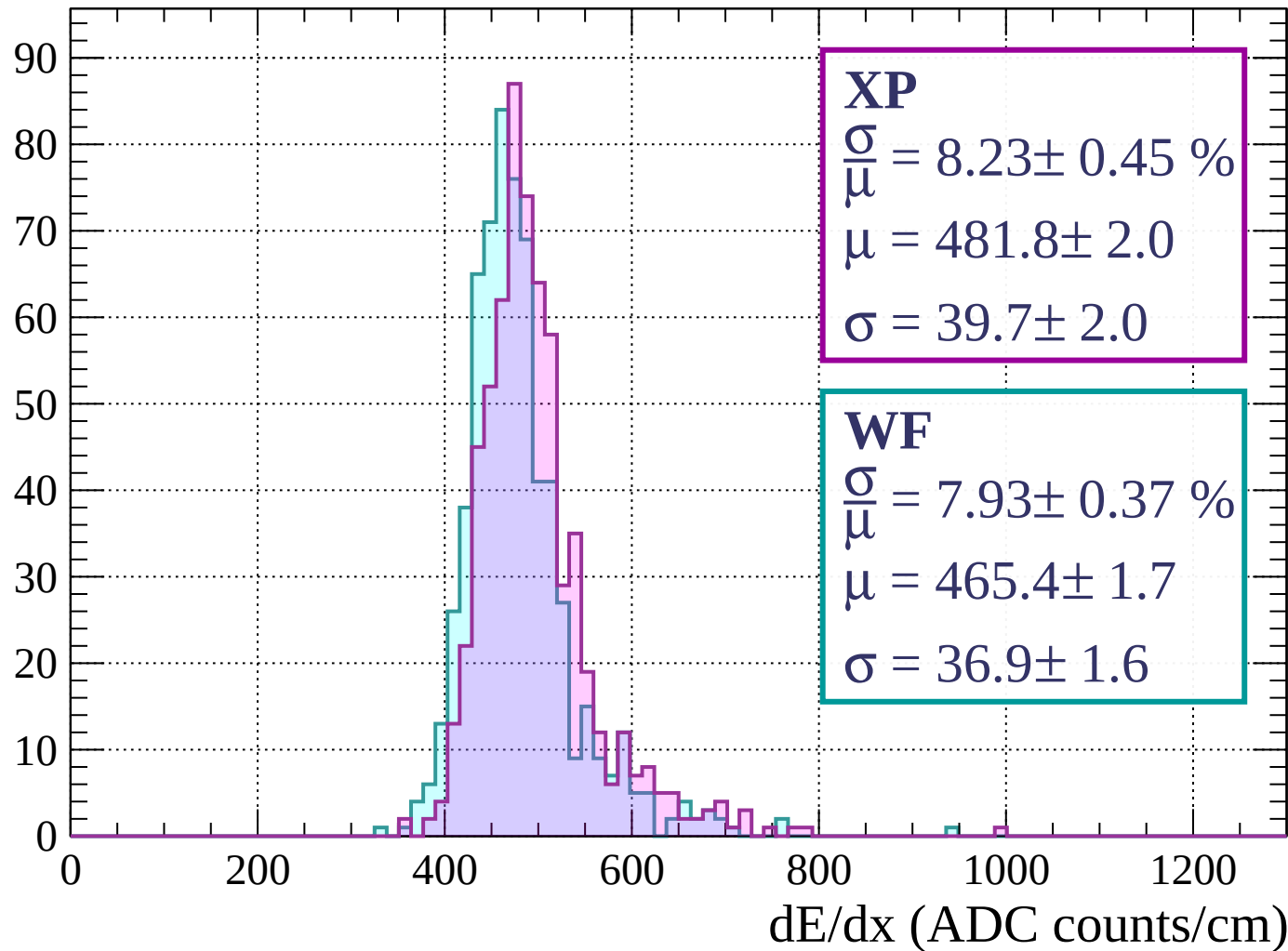
Energy loss | $1280 < p < 1360$

Count



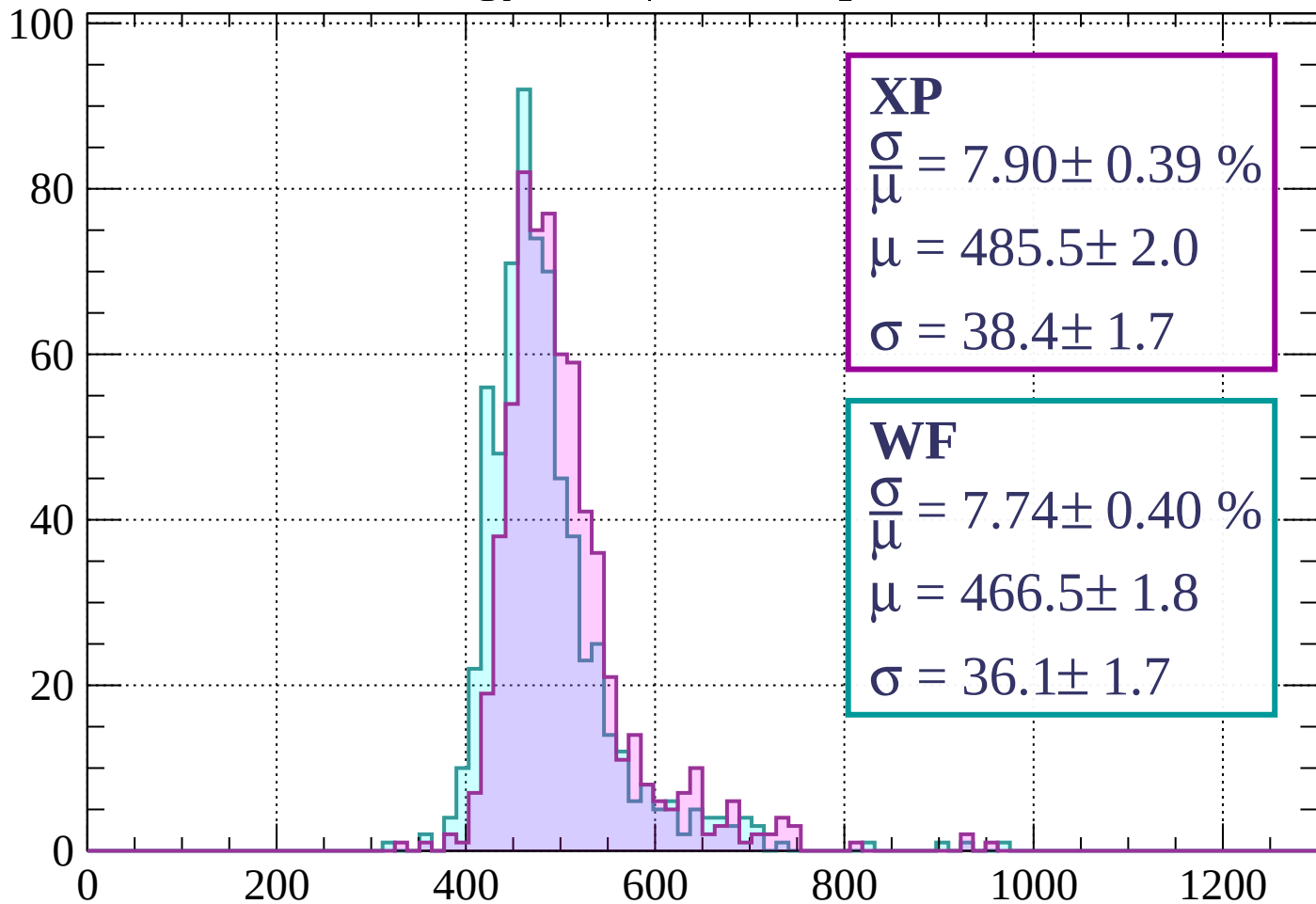
Energy loss | $1360 < p < 1440$

Count



Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 7.90 \pm 0.39 \%$$

$$\mu = 485.5 \pm 2.0$$

$$\sigma = 38.4 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 7.74 \pm 0.40 \%$$

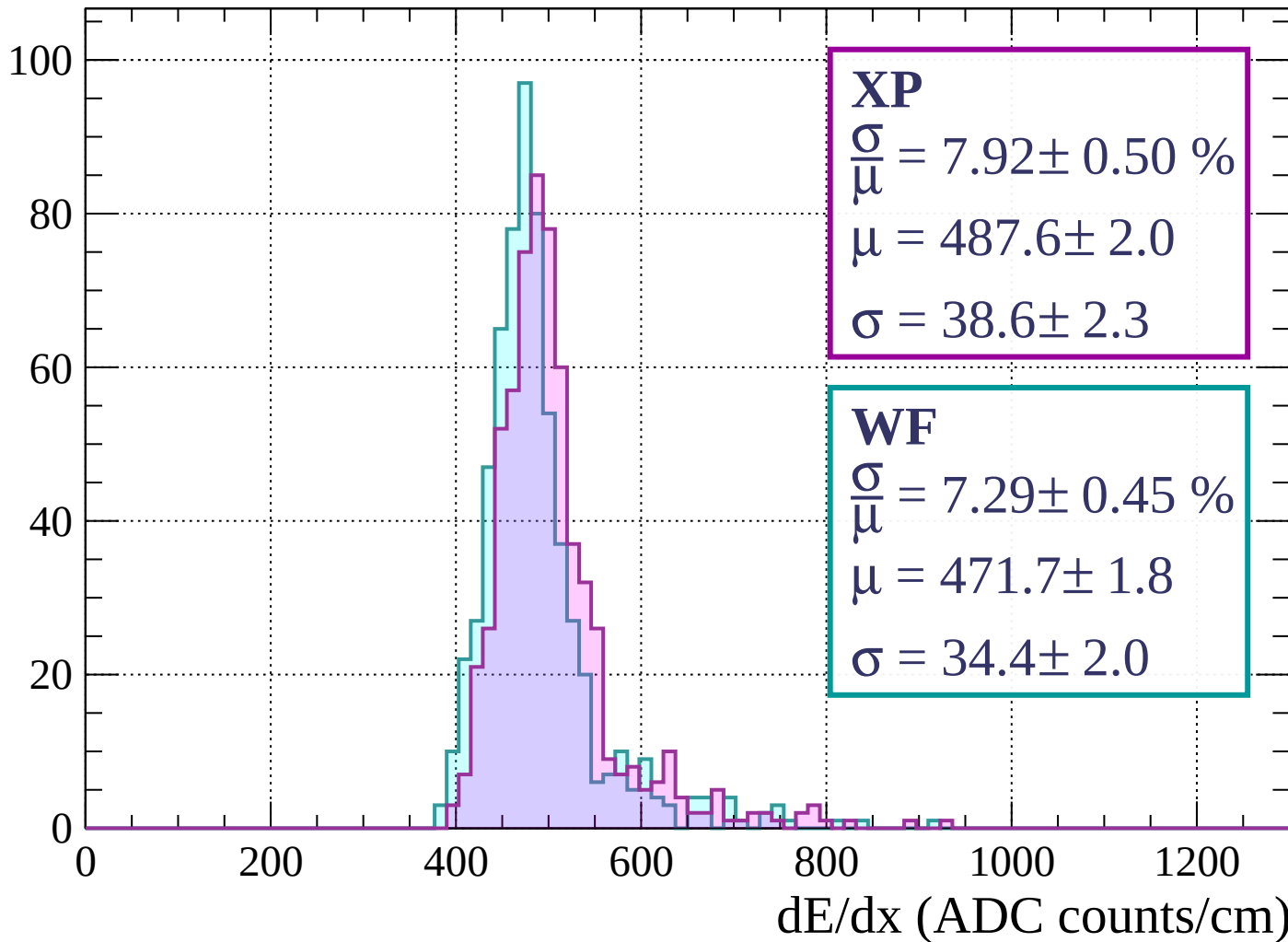
$$\mu = 466.5 \pm 1.8$$

$$\sigma = 36.1 \pm 1.7$$

dE/dx (ADC counts/cm)

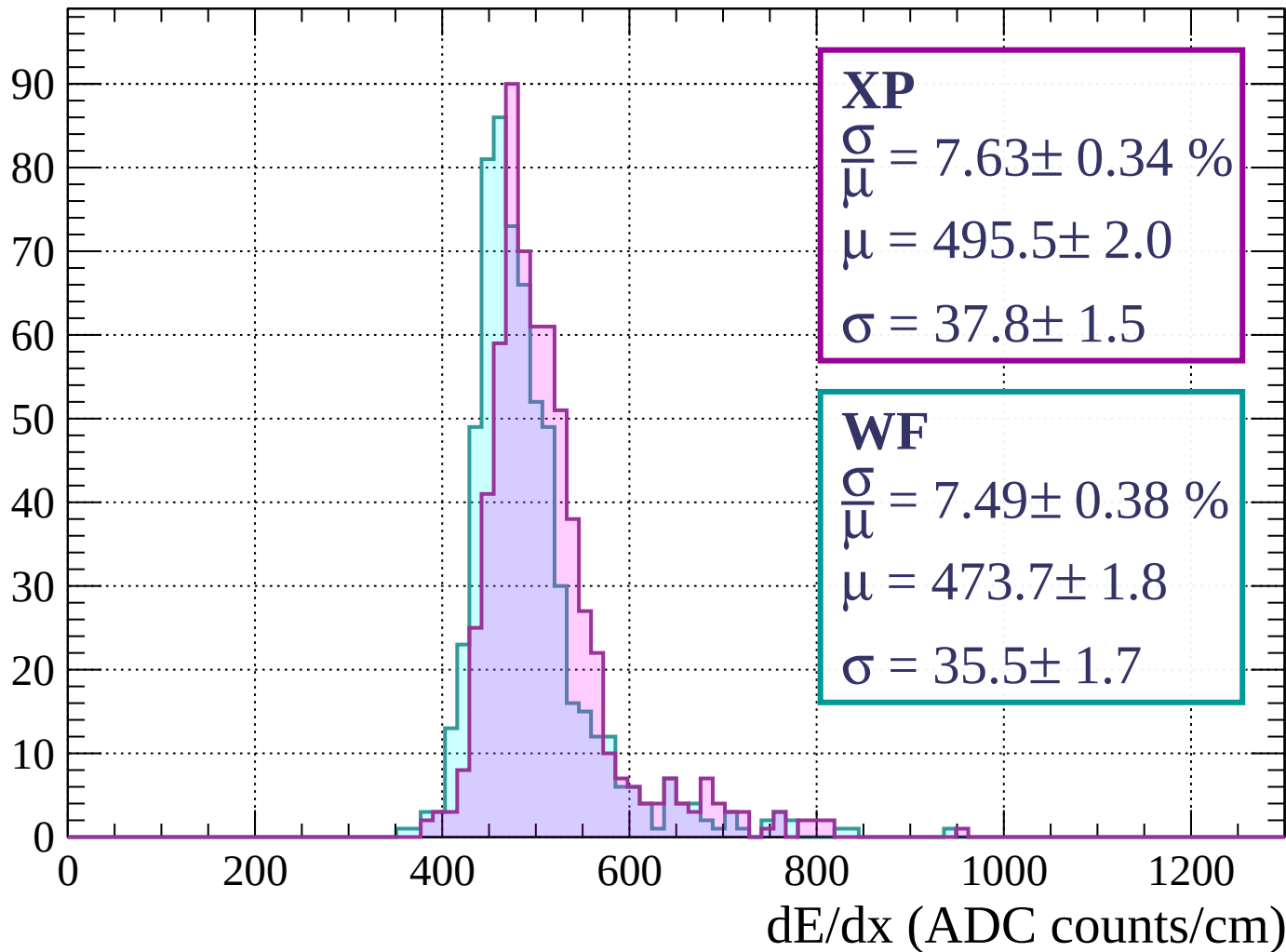
Energy loss | $1520 < p < 1600$

Count



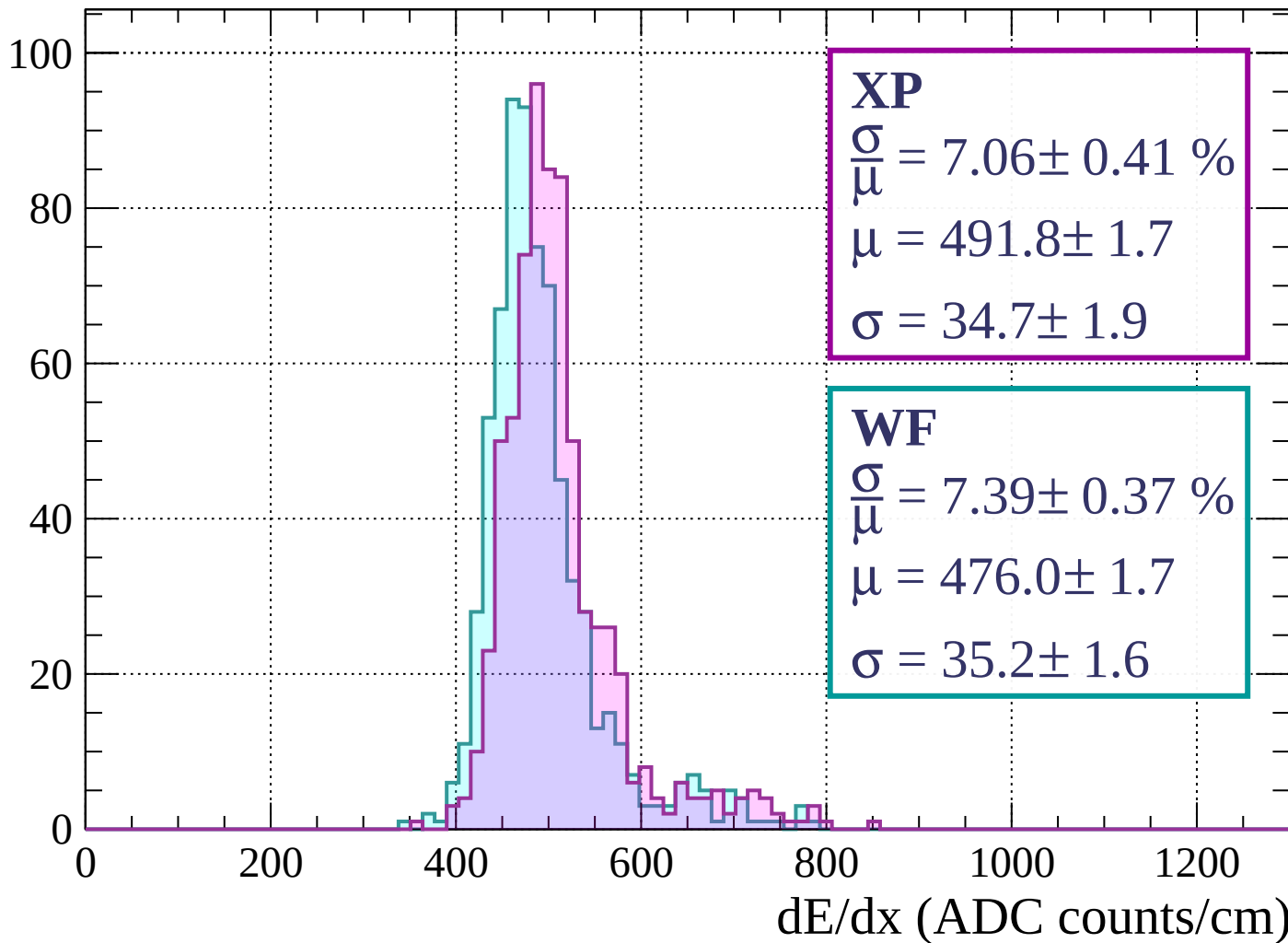
Energy loss | $1600 < p < 1680$

Count



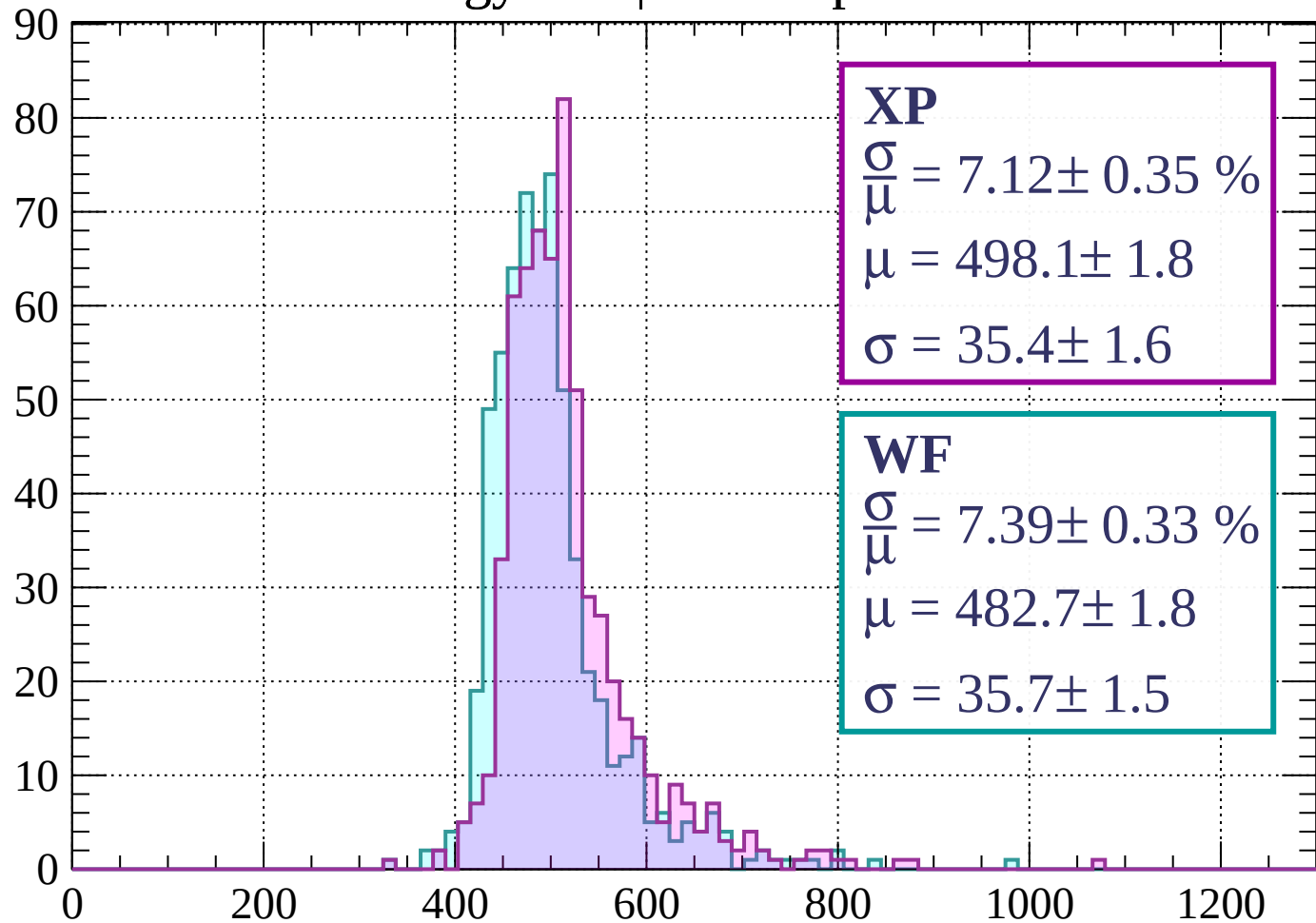
Energy loss | $1680 < p < 1760$

Count



Energy loss | $1760 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 7.12 \pm 0.35 \%$$

$$\mu = 498.1 \pm 1.8$$

$$\sigma = 35.4 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 7.39 \pm 0.33 \%$$

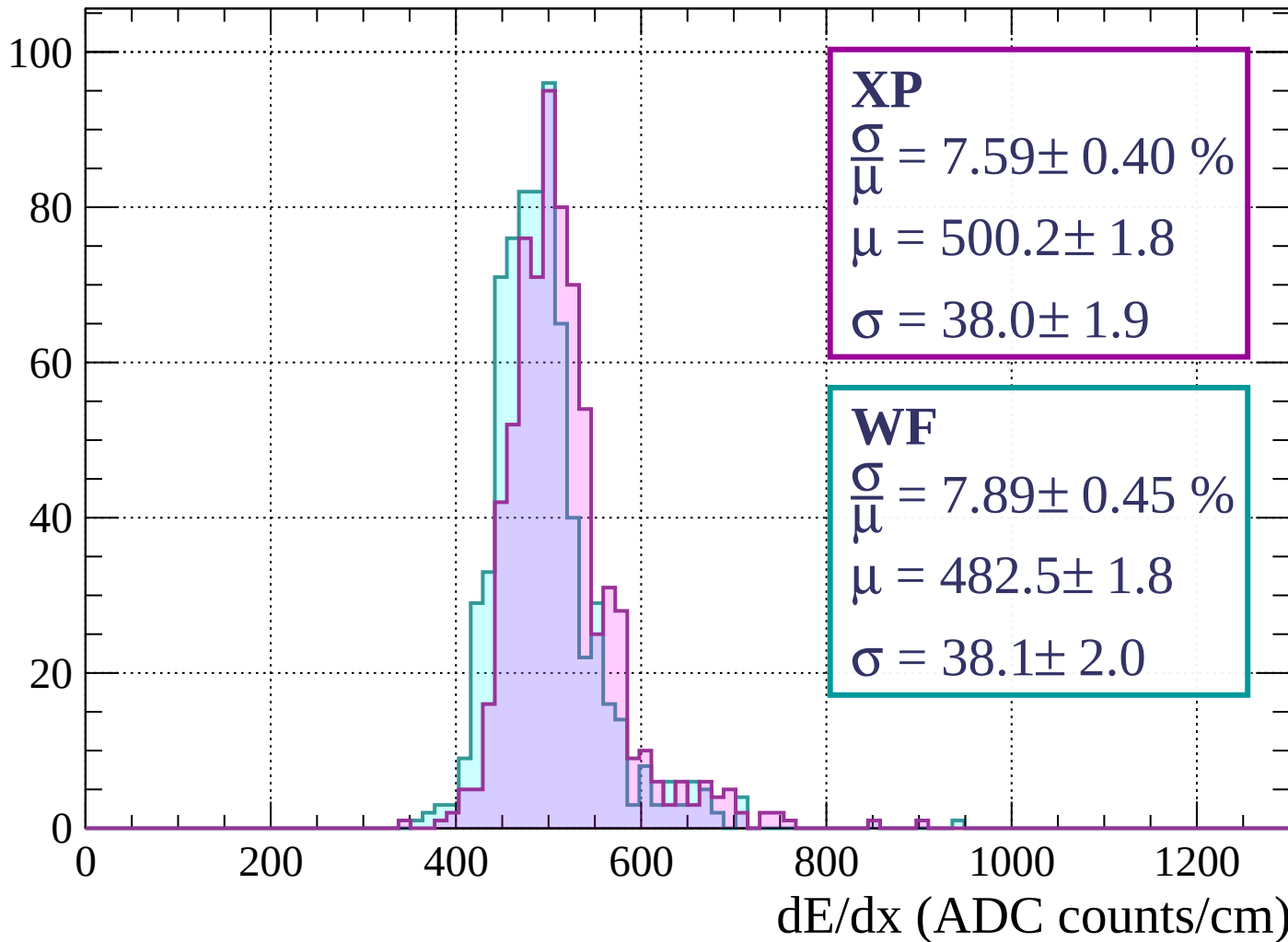
$$\mu = 482.7 \pm 1.8$$

$$\sigma = 35.7 \pm 1.5$$

dE/dx (ADC counts/cm)

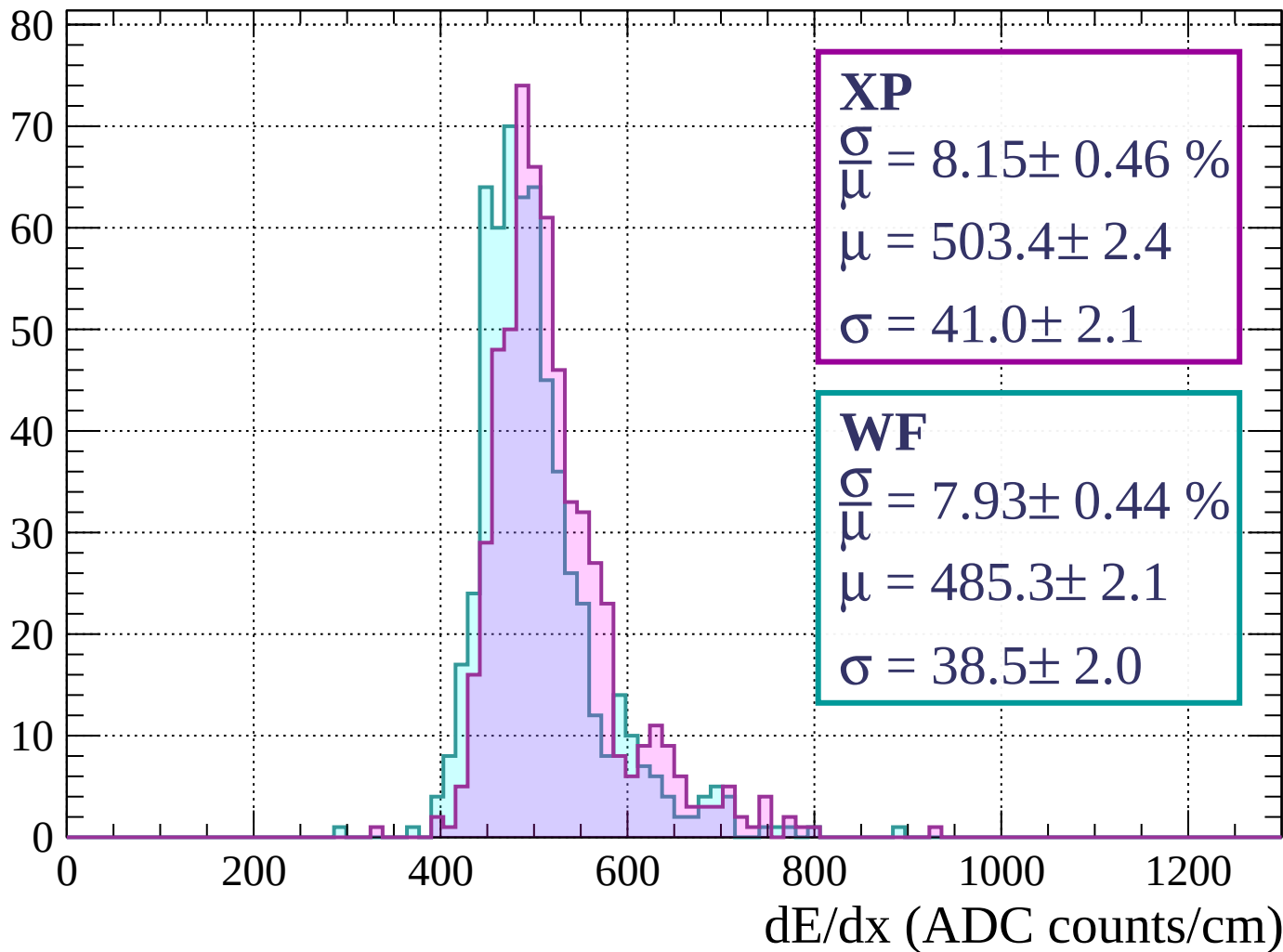
Energy loss | $1840 < p < 1920$

Count



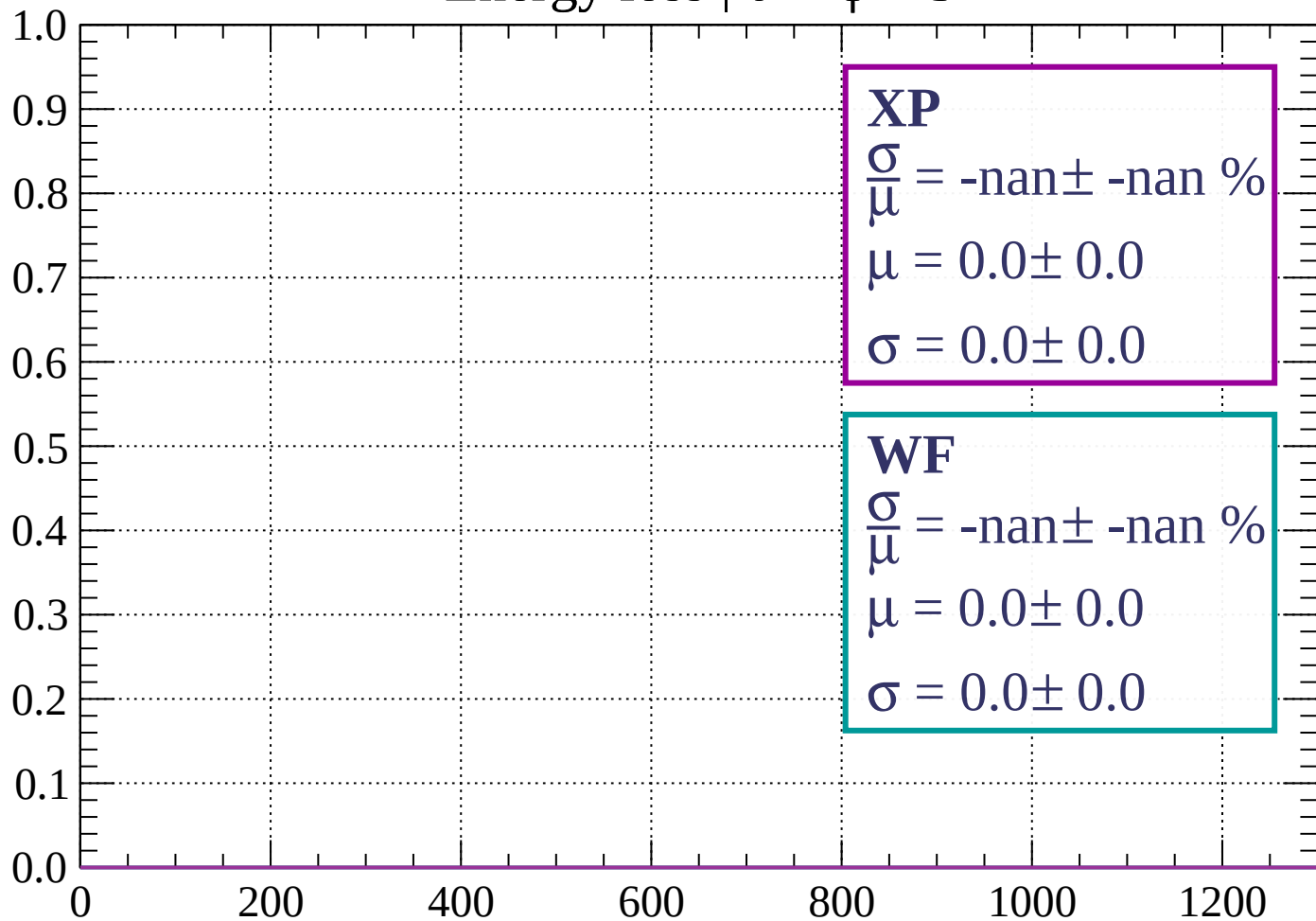
Energy loss | $1920 < p < 2000$

Count



Energy loss | $0 < \phi < 3$

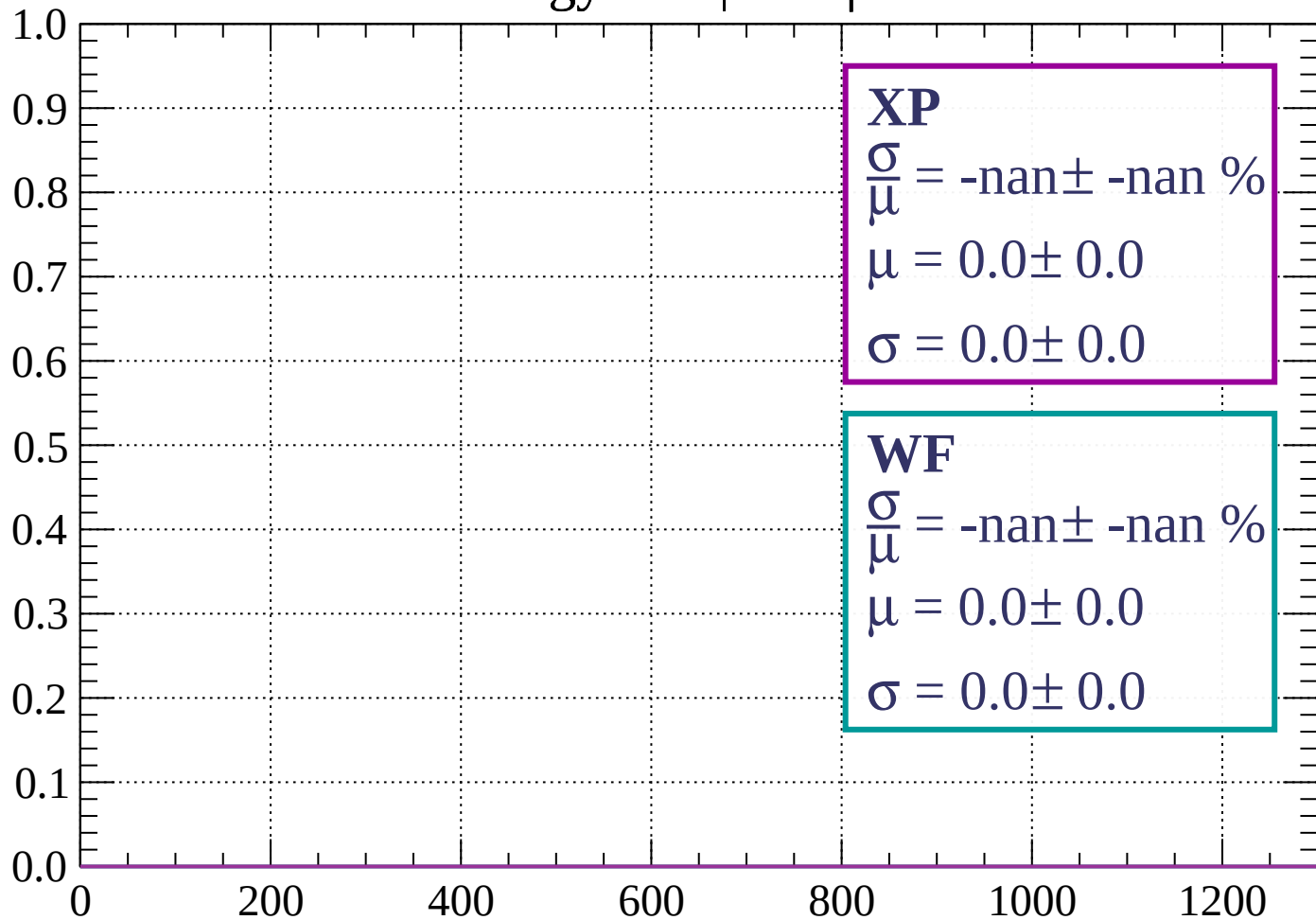
Count



dE/dx (ADC counts/cm)

Energy loss | $3 < \phi < 6$

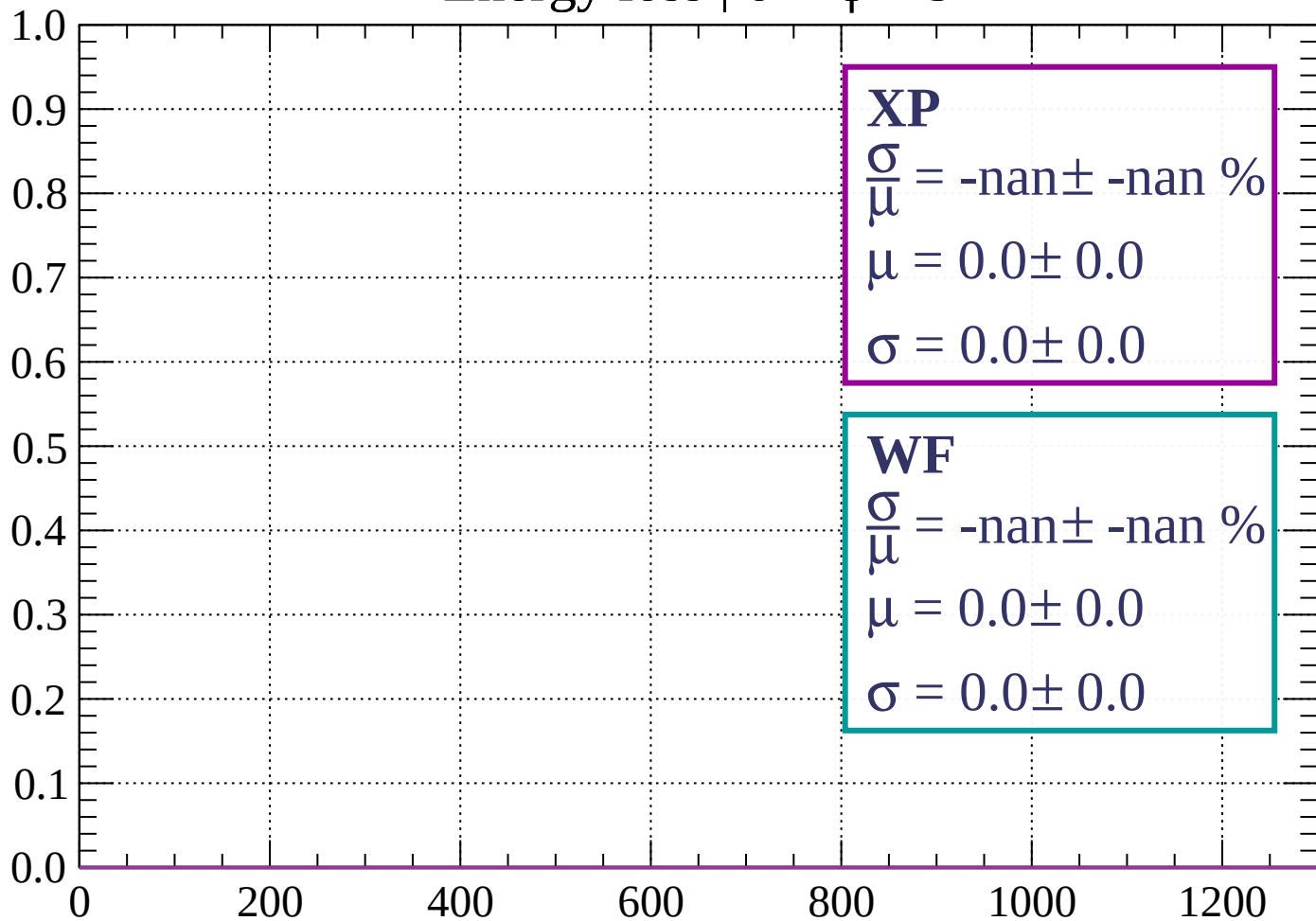
Count



dE/dx (ADC counts/cm)

Energy loss | $6 < \phi < 9$

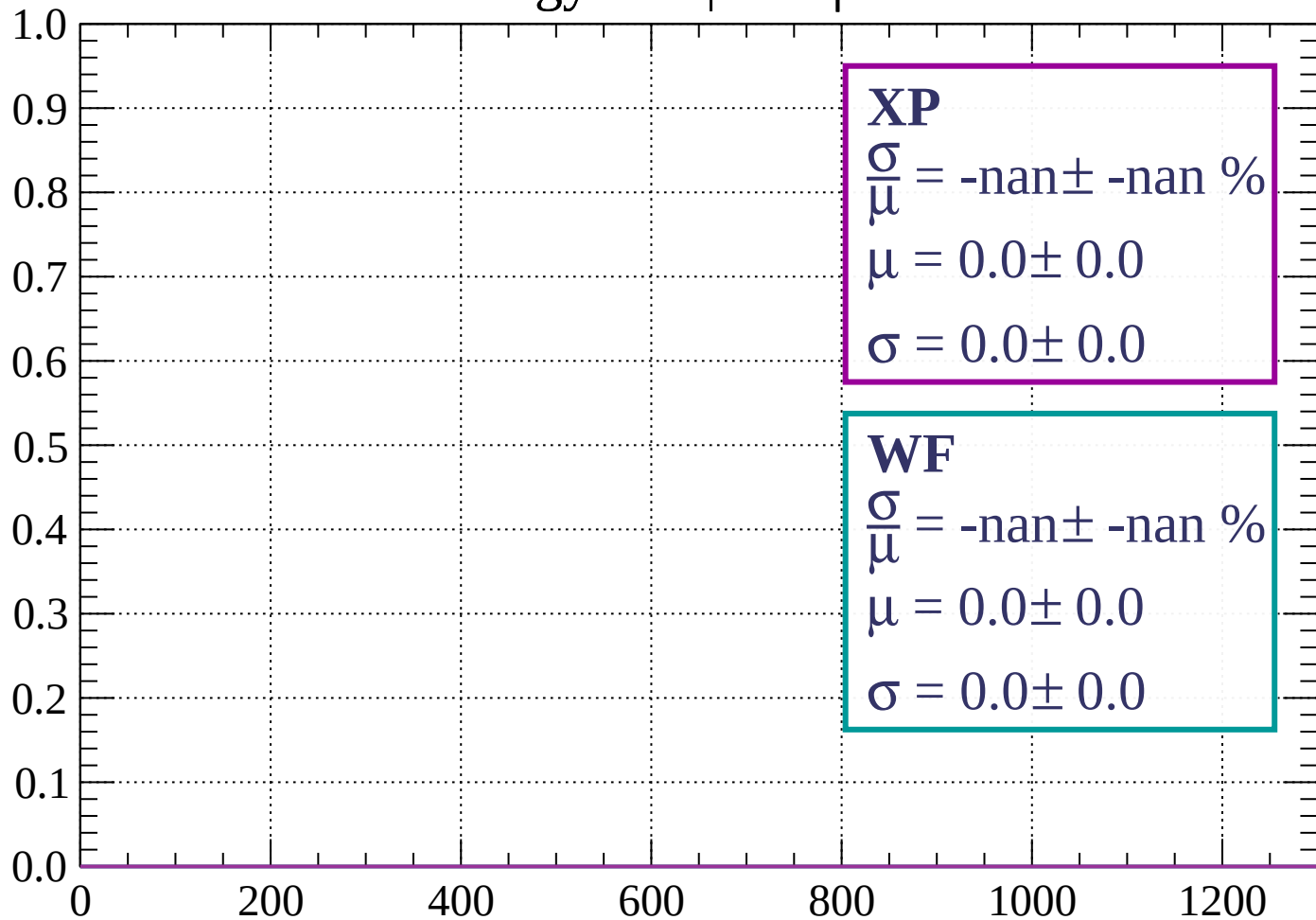
Count



dE/dx (ADC counts/cm)

Energy loss | $9 < \phi < 12$

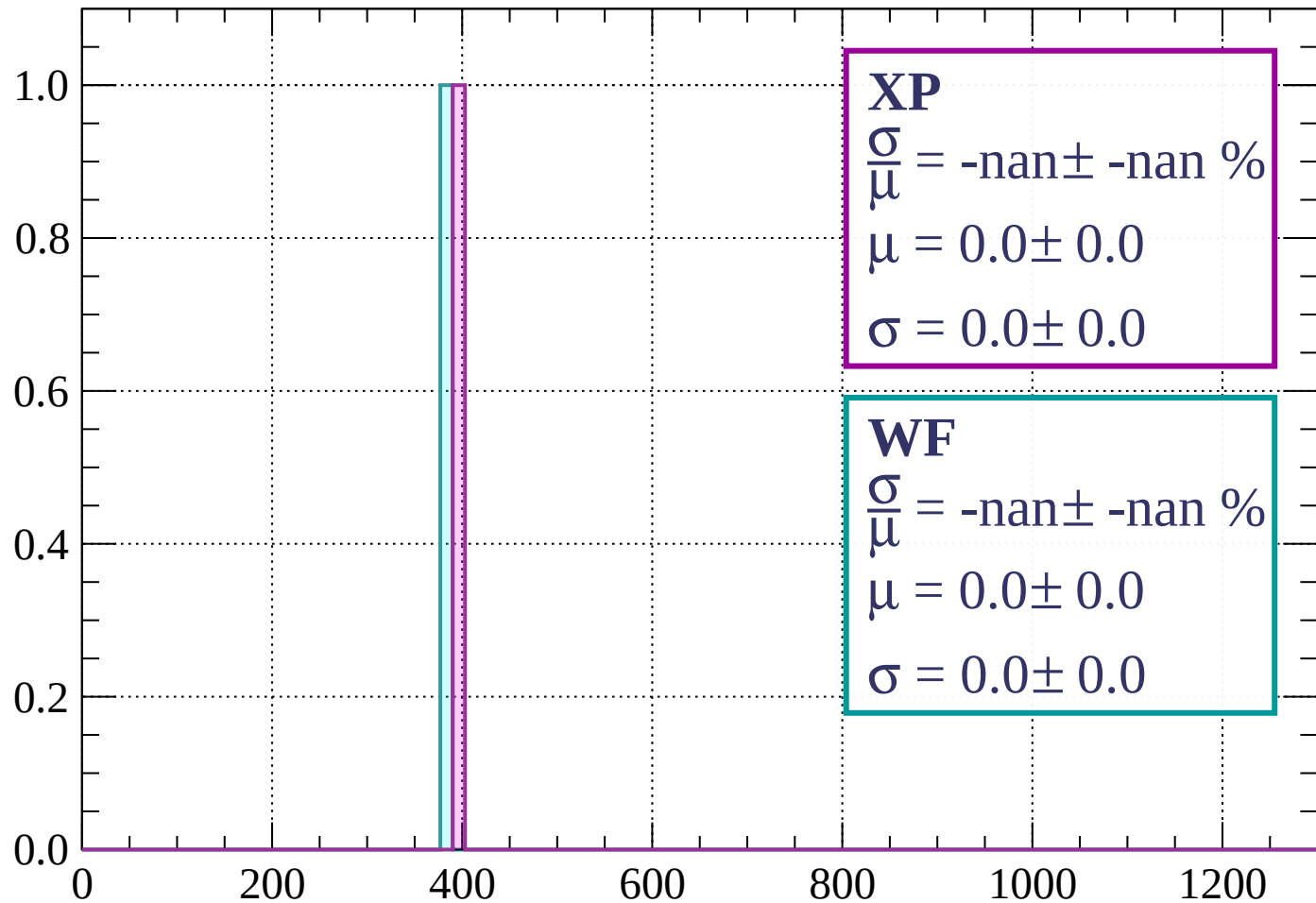
Count



dE/dx (ADC counts/cm)

Energy loss | $12 < \phi < 15$

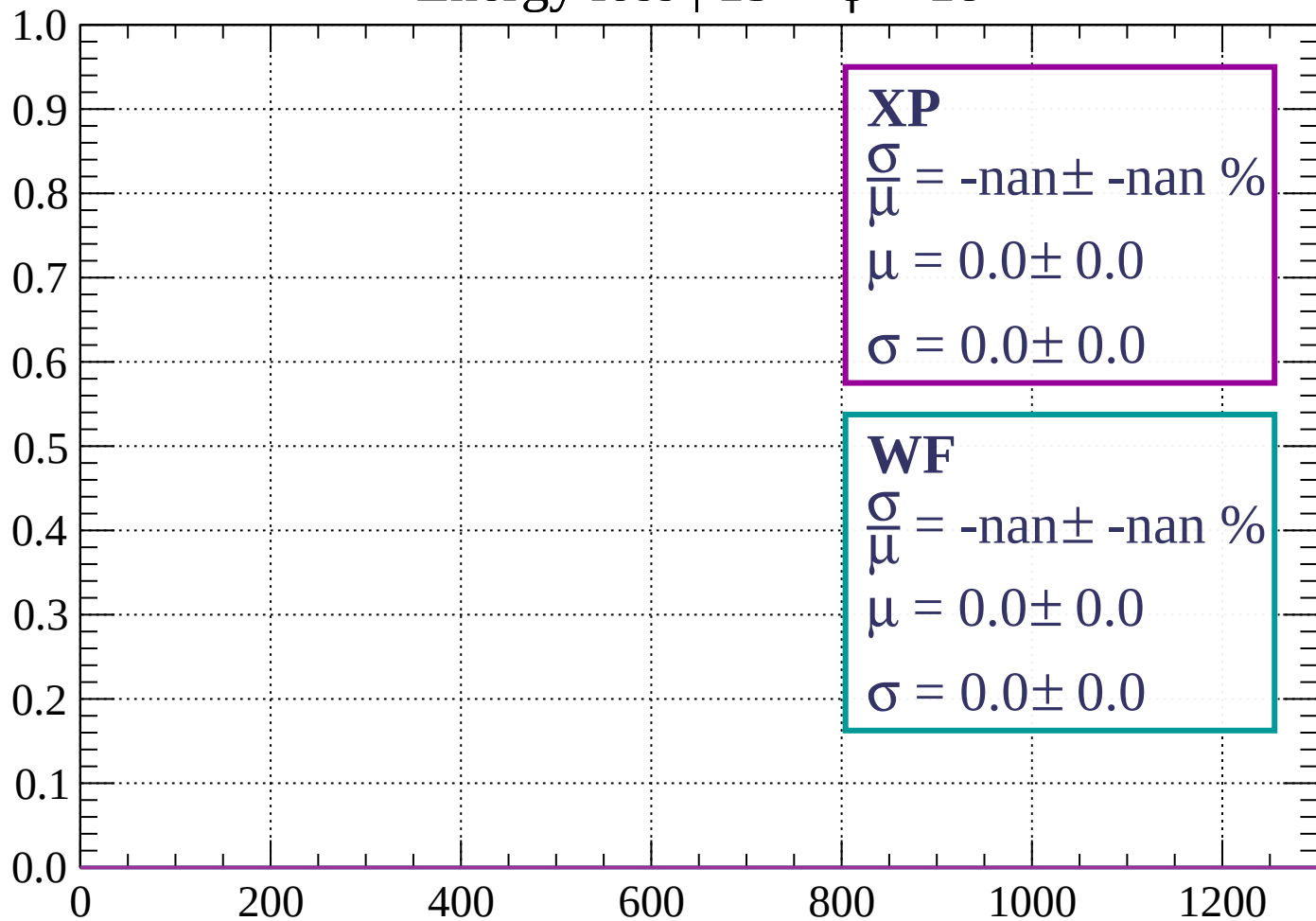
Count



dE/dx (ADC counts/cm)

Energy loss | $15 < \phi < 18$

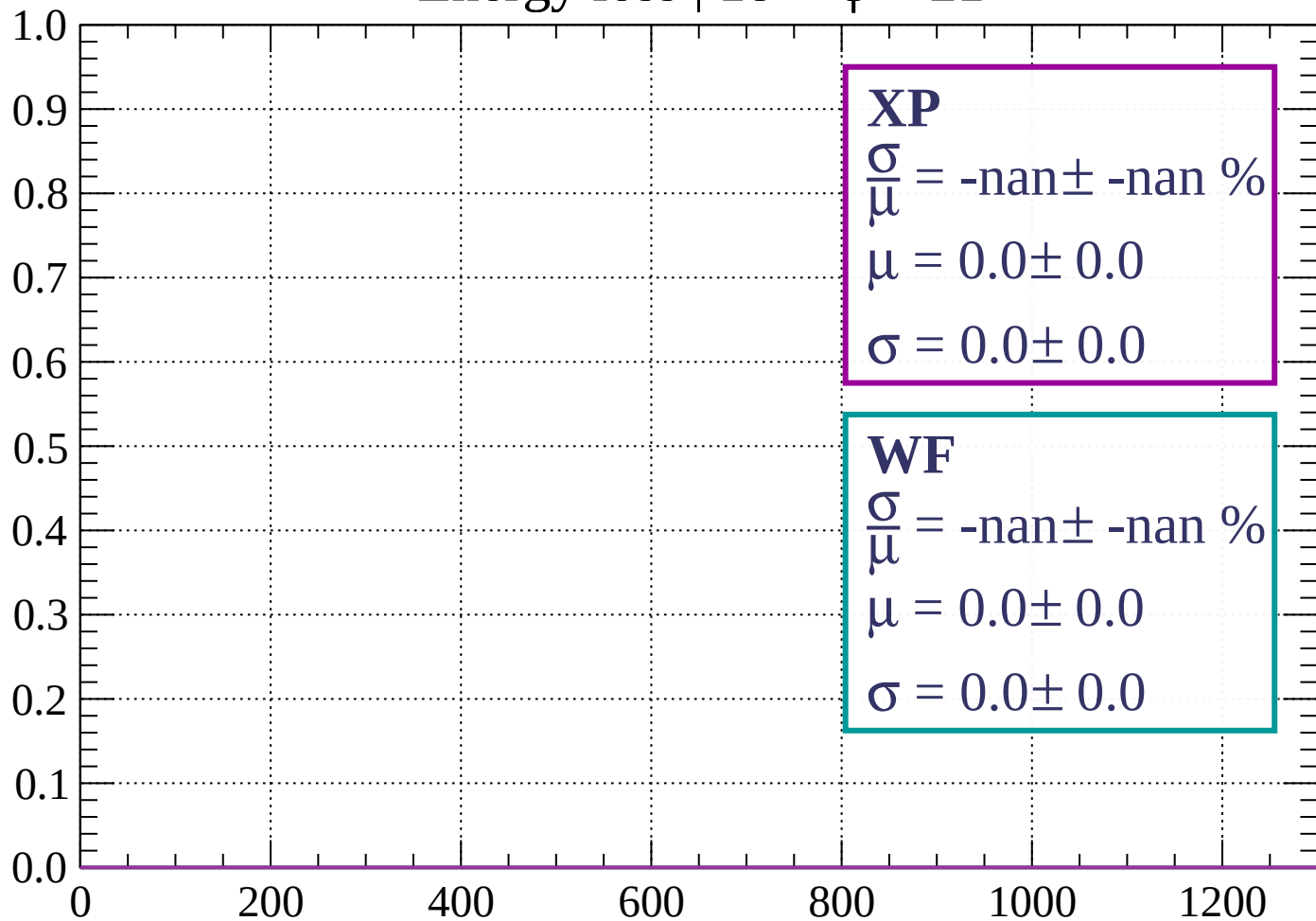
Count



dE/dx (ADC counts/cm)

Energy loss | $18 < \phi < 21$

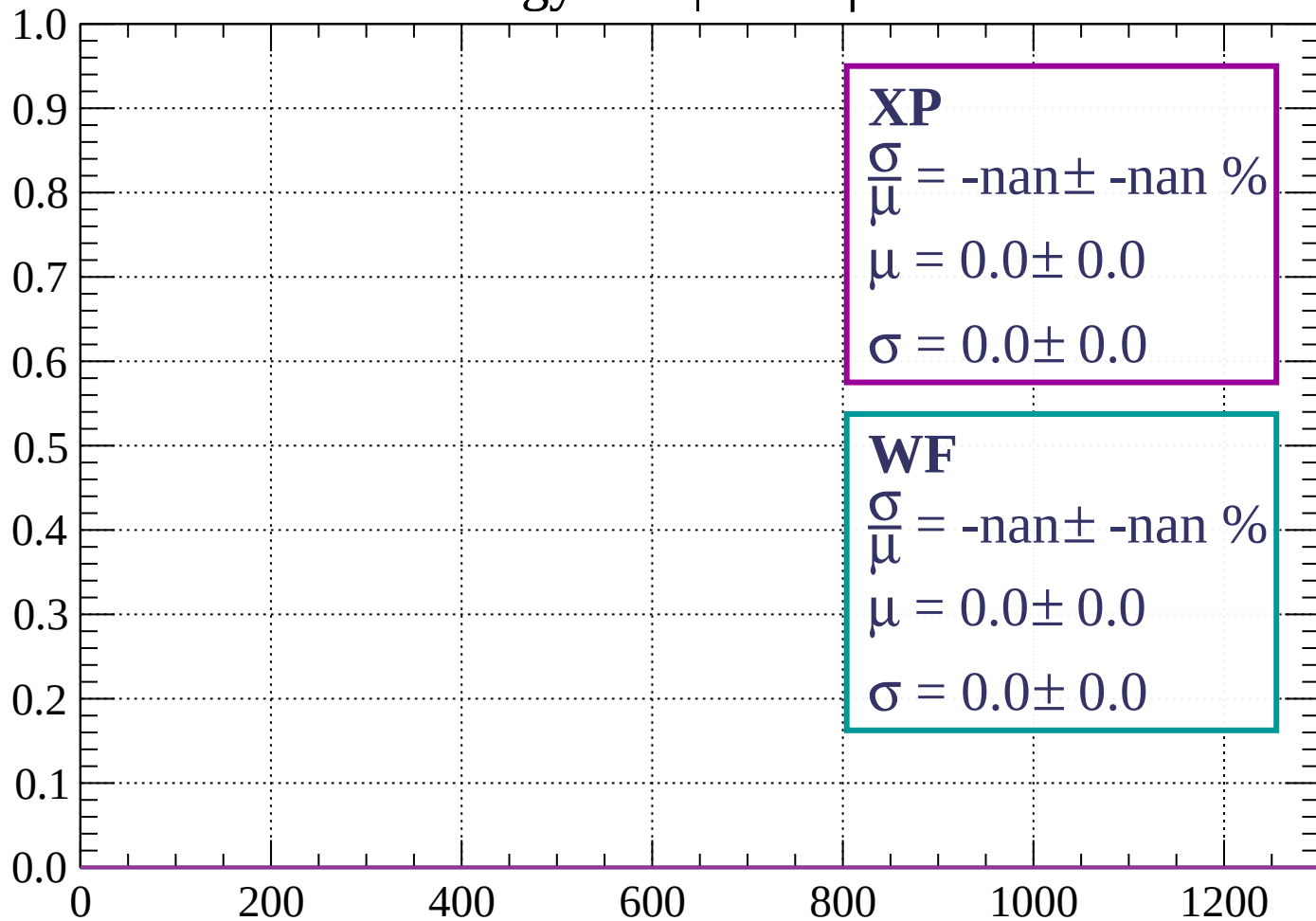
Count



dE/dx (ADC counts/cm)

Energy loss | $21 < \phi < 24$

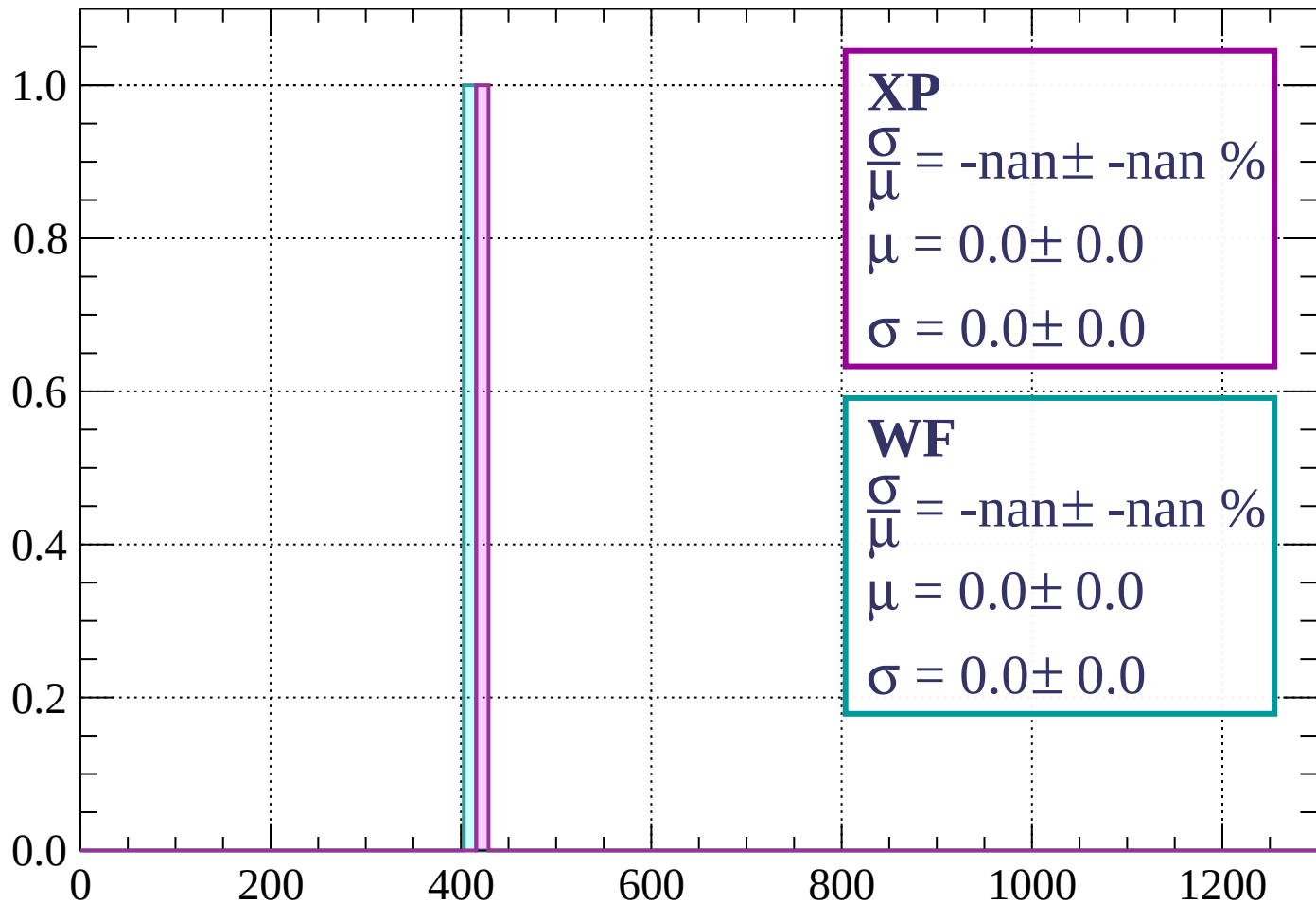
Count



dE/dx (ADC counts/cm)

Energy loss | $24 < \phi < 27$

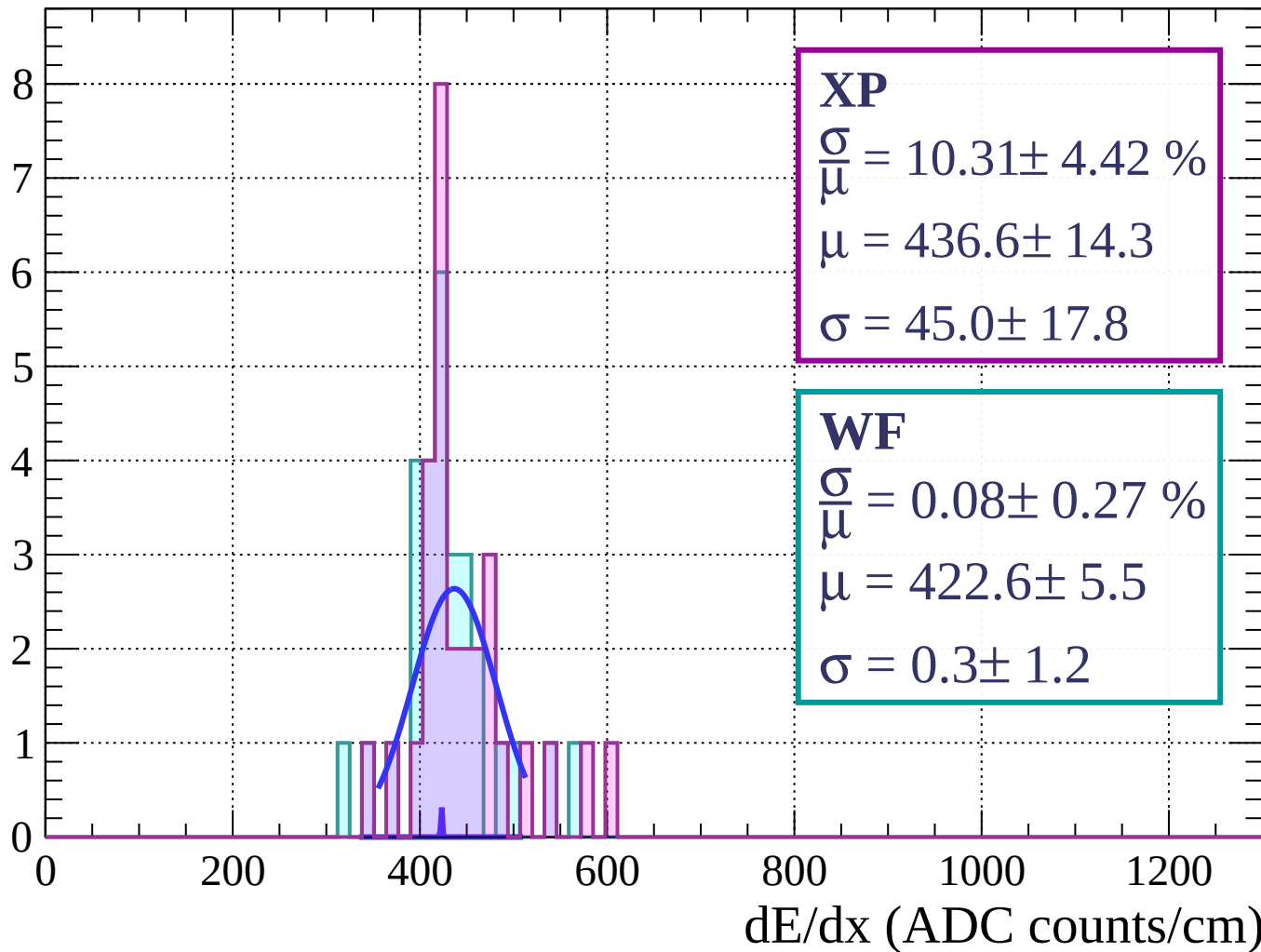
Count



dE/dx (ADC counts/cm)

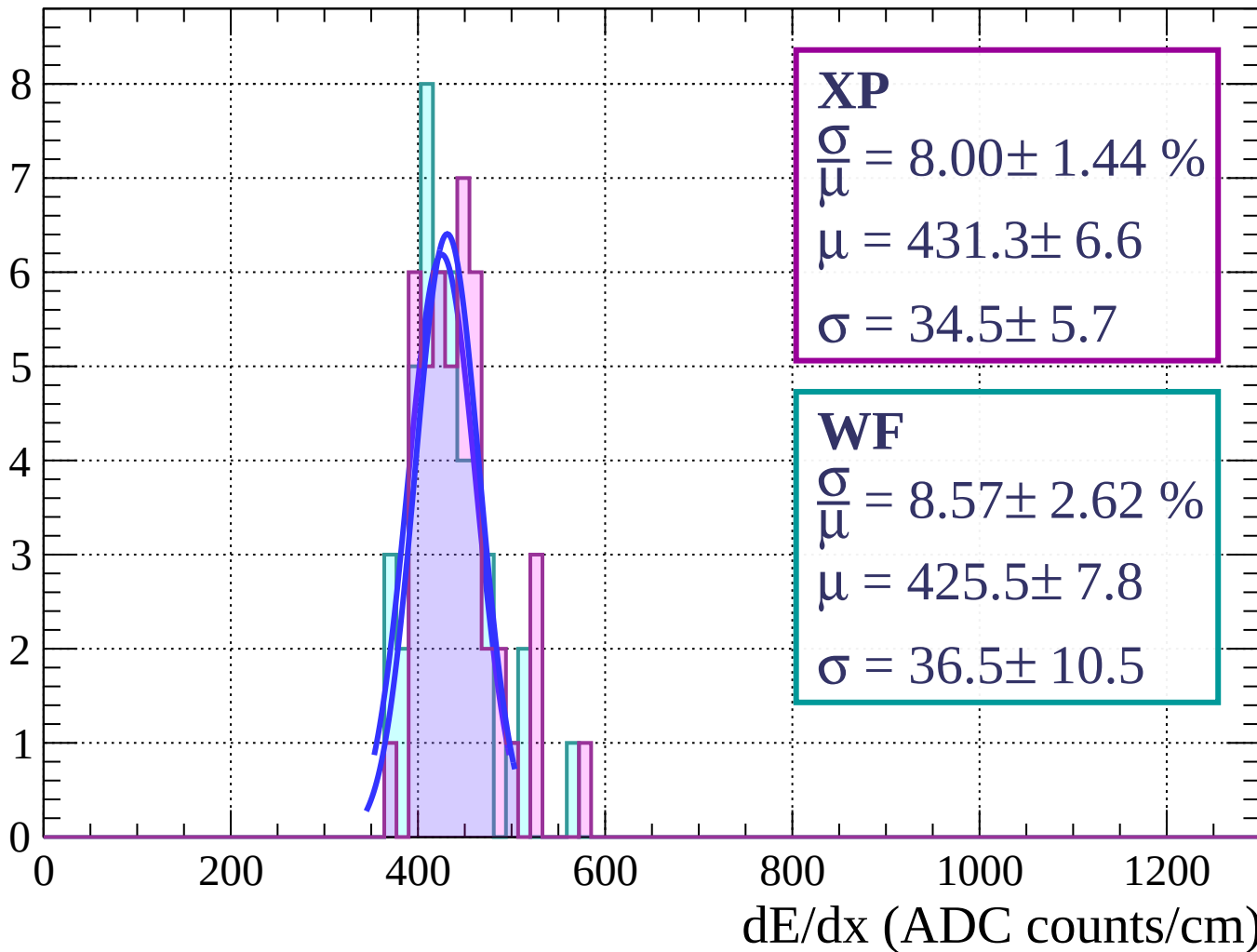
Energy loss | $27 < \phi < 30$

Count



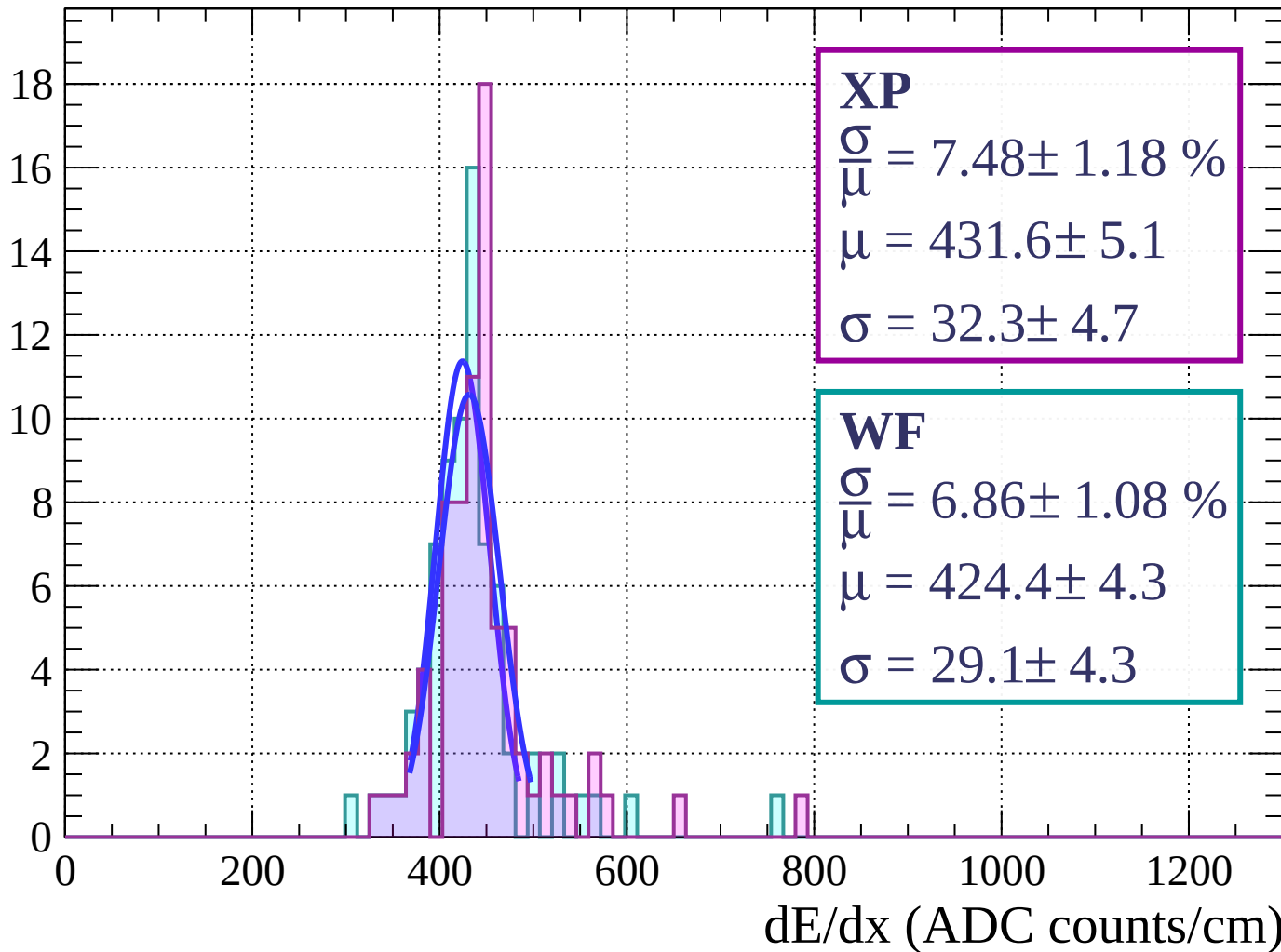
Energy loss | $30 < \phi < 33$

Count



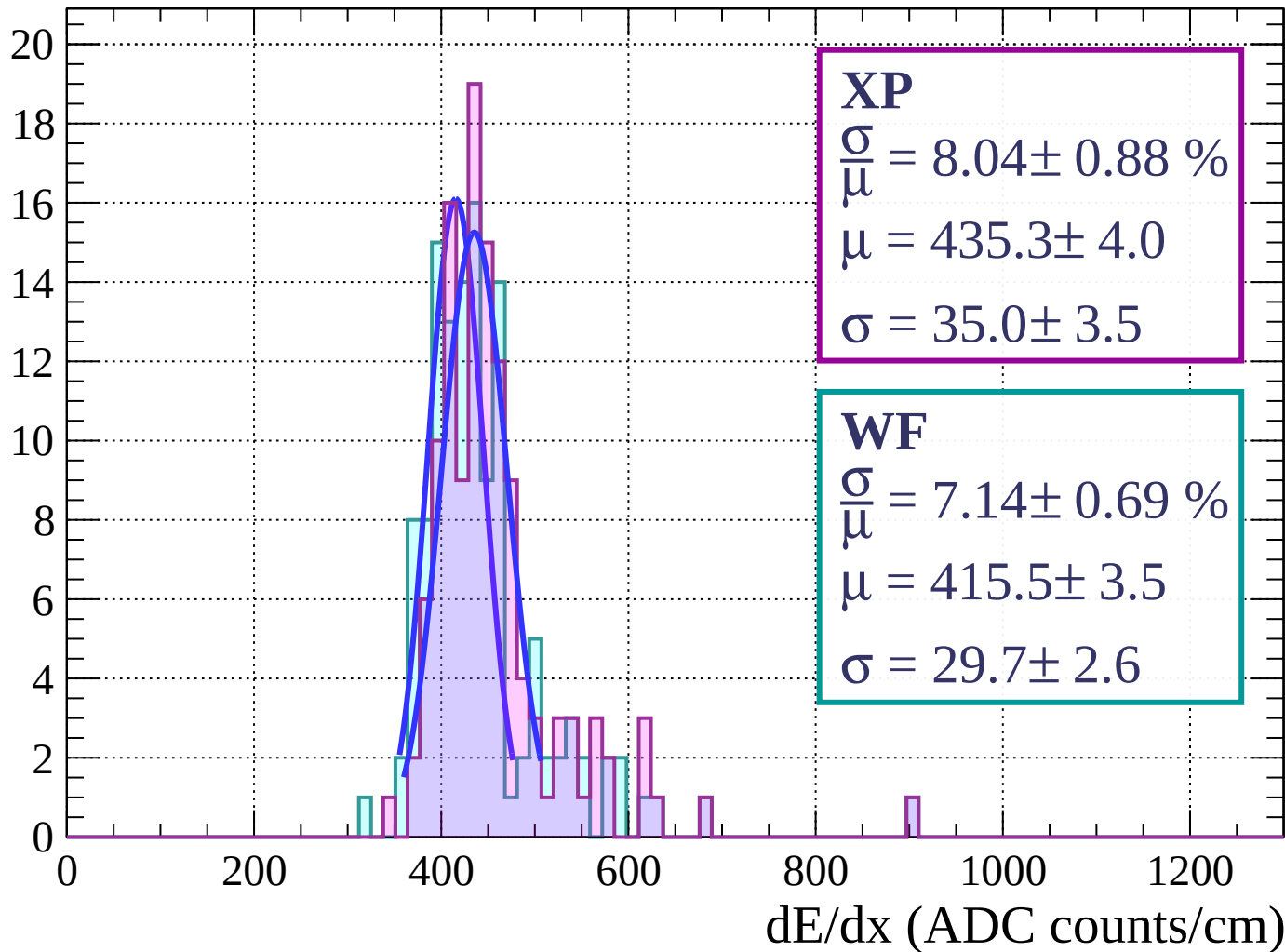
Energy loss | $33 < \phi < 36$

Count



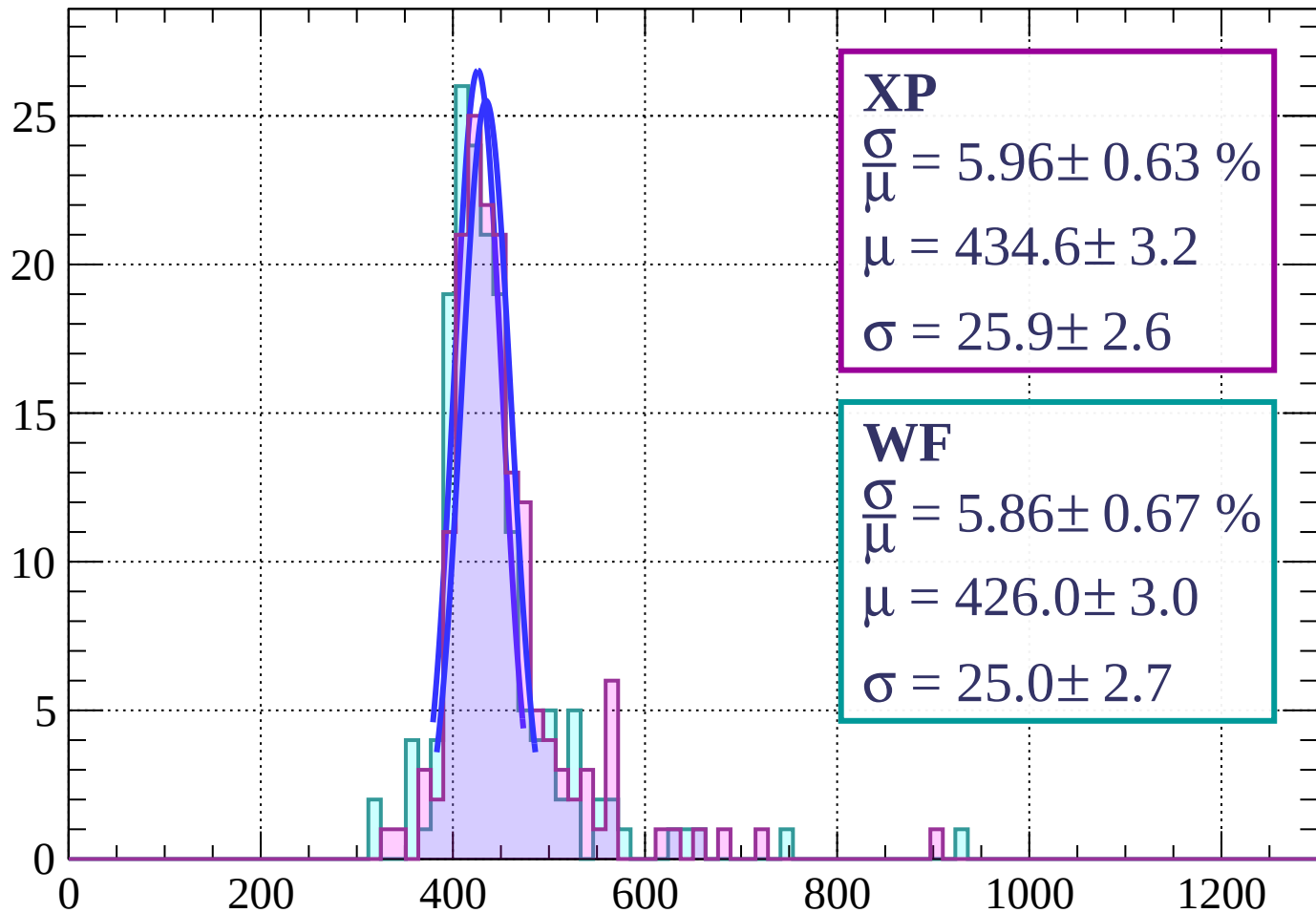
Energy loss | $36 < \phi < 39$

Count



Energy loss | $39 < \phi < 42$

Count



XP

$$\frac{\sigma}{\mu} = 5.96 \pm 0.63 \%$$

$$\mu = 434.6 \pm 3.2$$

$$\sigma = 25.9 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 5.86 \pm 0.67 \%$$

$$\mu = 426.0 \pm 3.0$$

$$\sigma = 25.0 \pm 2.7$$

dE/dx (ADC counts/cm)

Energy loss | $42 < \phi < 45$

Count

XP

$$\frac{\sigma}{\mu} = 6.72 \pm 0.68 \%$$

$$\mu = 436.9 \pm 2.9$$

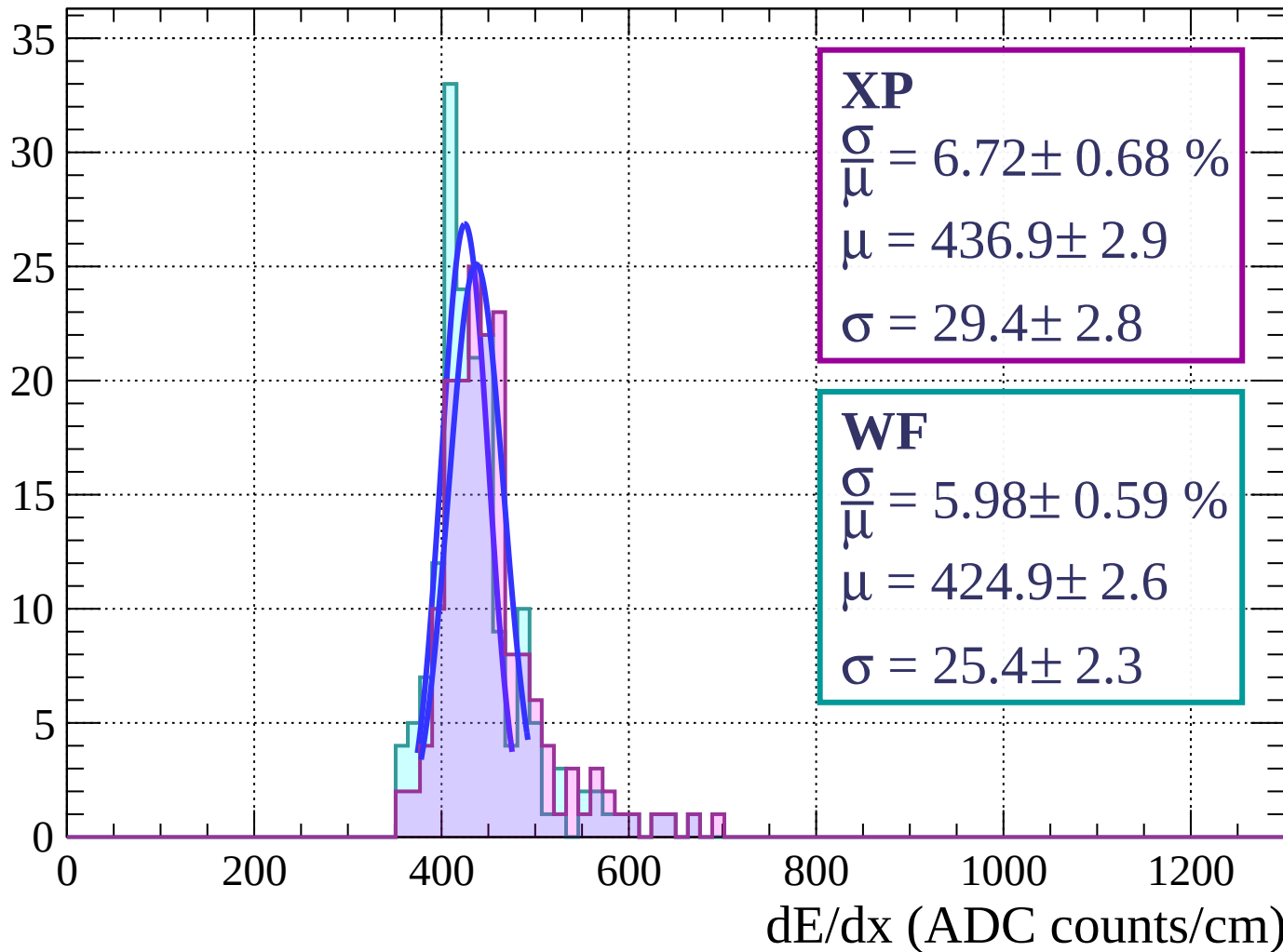
$$\sigma = 29.4 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 5.98 \pm 0.59 \%$$

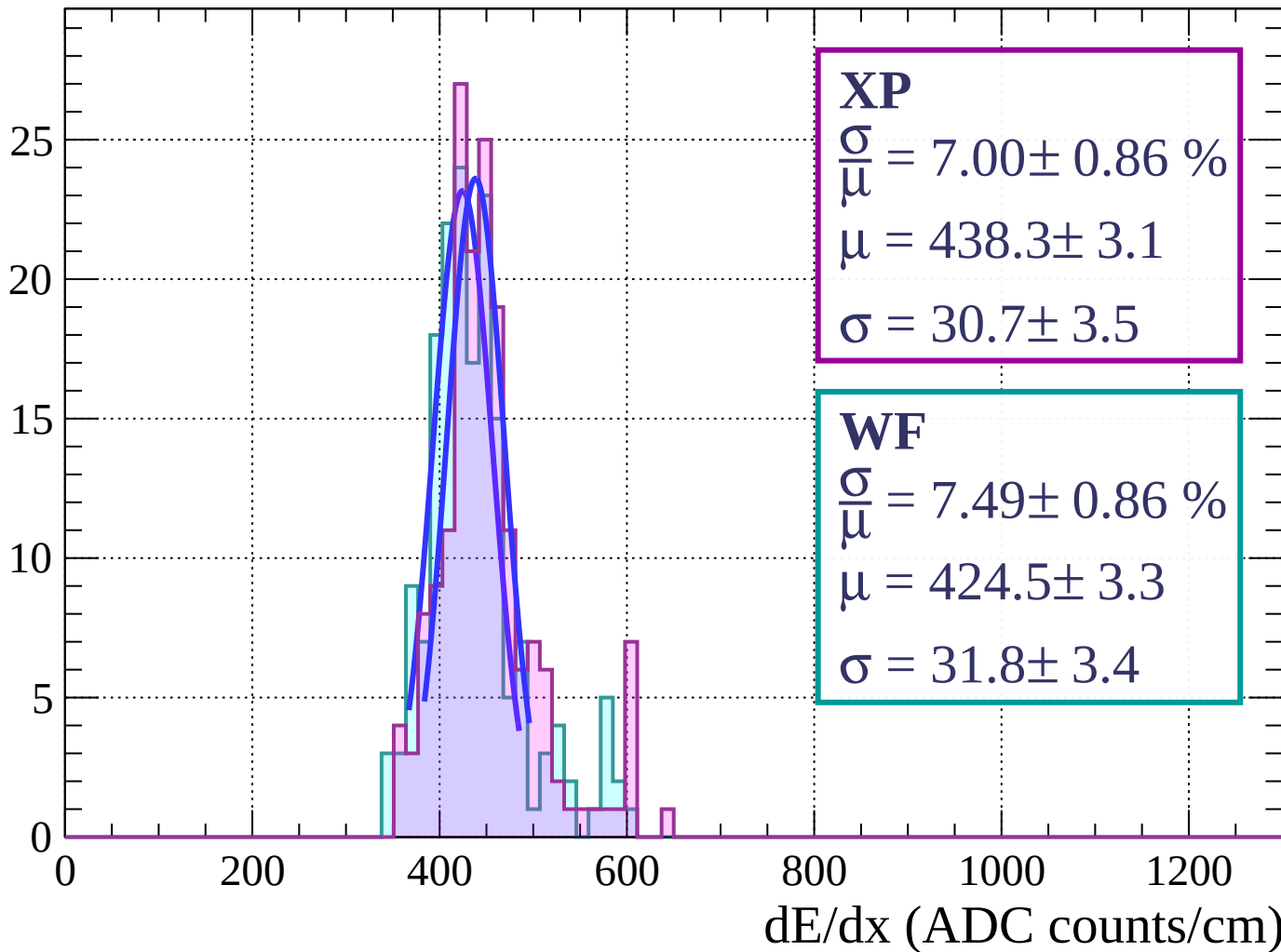
$$\mu = 424.9 \pm 2.6$$

$$\sigma = 25.4 \pm 2.3$$



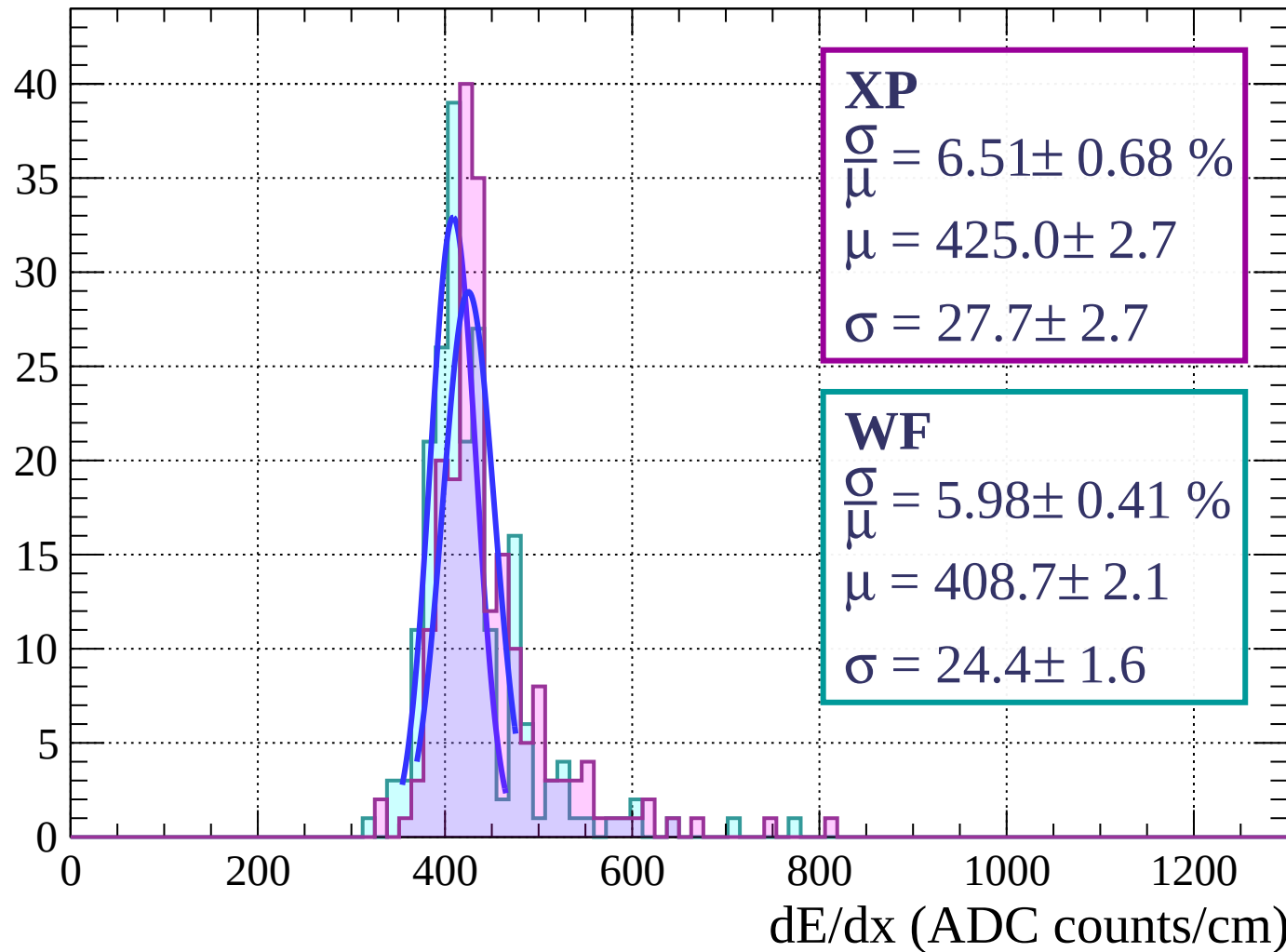
Energy loss | $45 < \phi < 48$

Count



Energy loss | $48 < \phi < 51$

Count



Energy loss | $51 < \phi < 54$

Count

XP

$$\frac{\sigma}{\mu} = 7.60 \pm 0.63 \%$$

$$\mu = 441.1 \pm 3.5$$

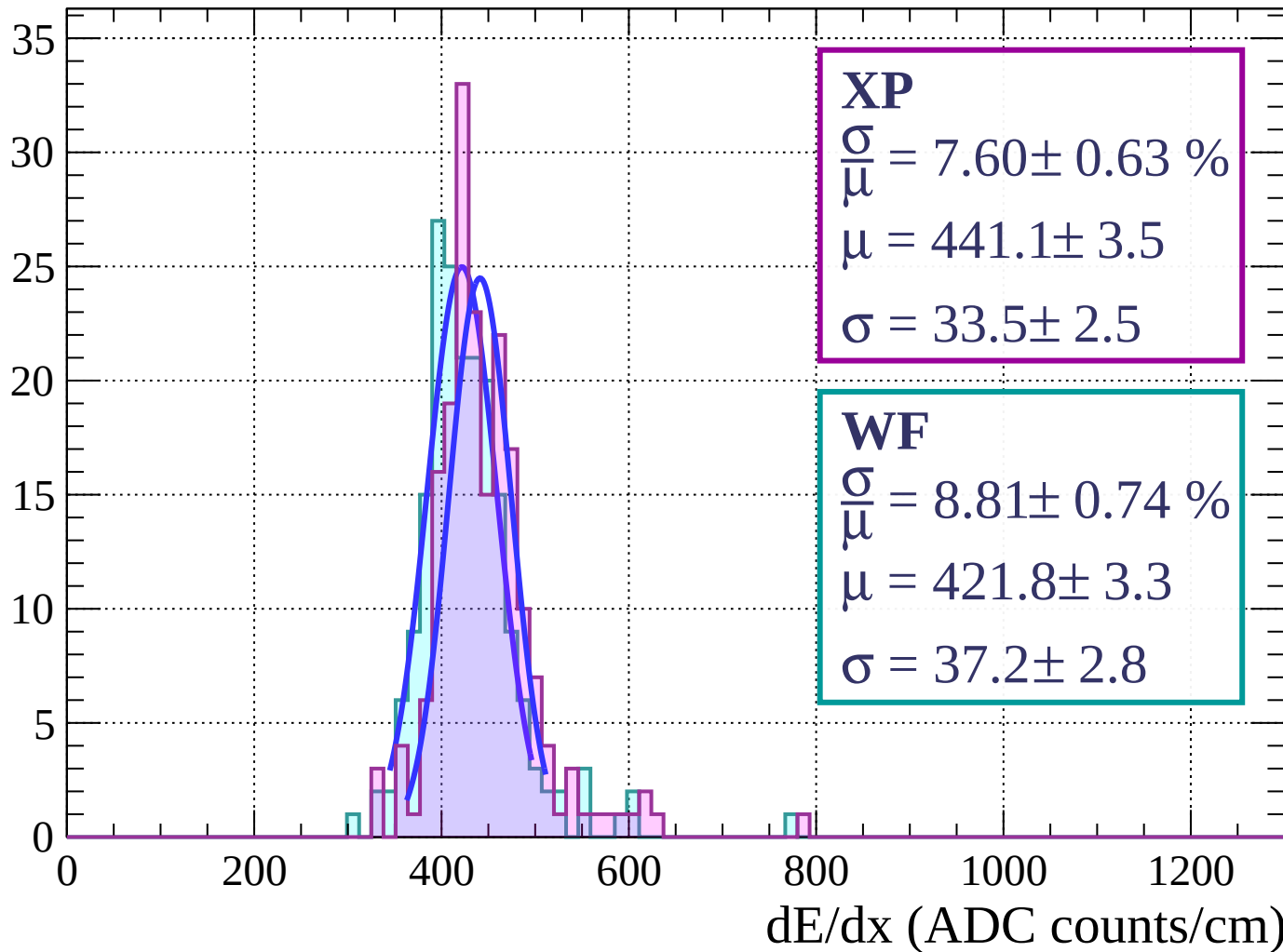
$$\sigma = 33.5 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 8.81 \pm 0.74 \%$$

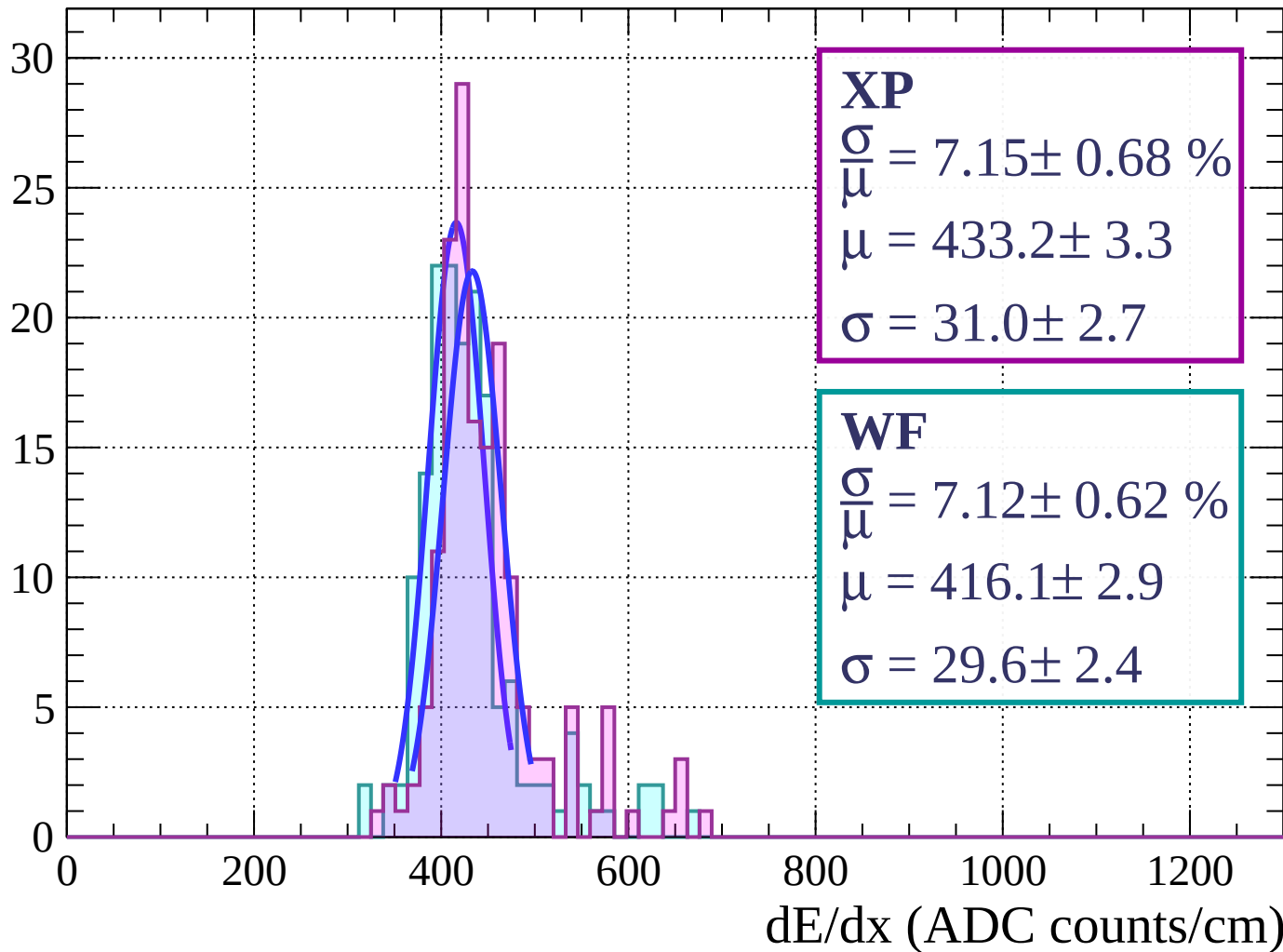
$$\mu = 421.8 \pm 3.3$$

$$\sigma = 37.2 \pm 2.8$$



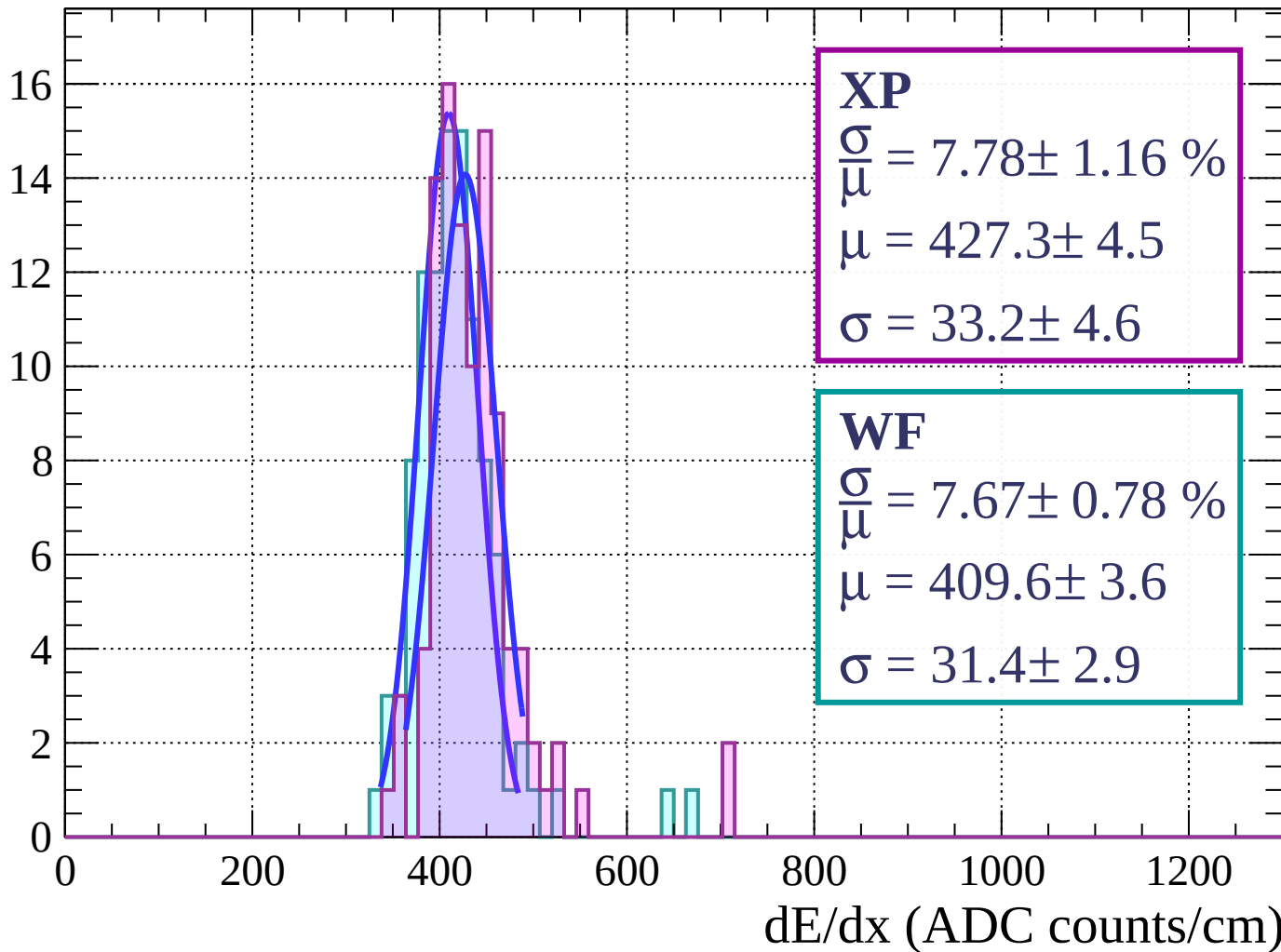
Energy loss | $54 < \phi < 57$

Count



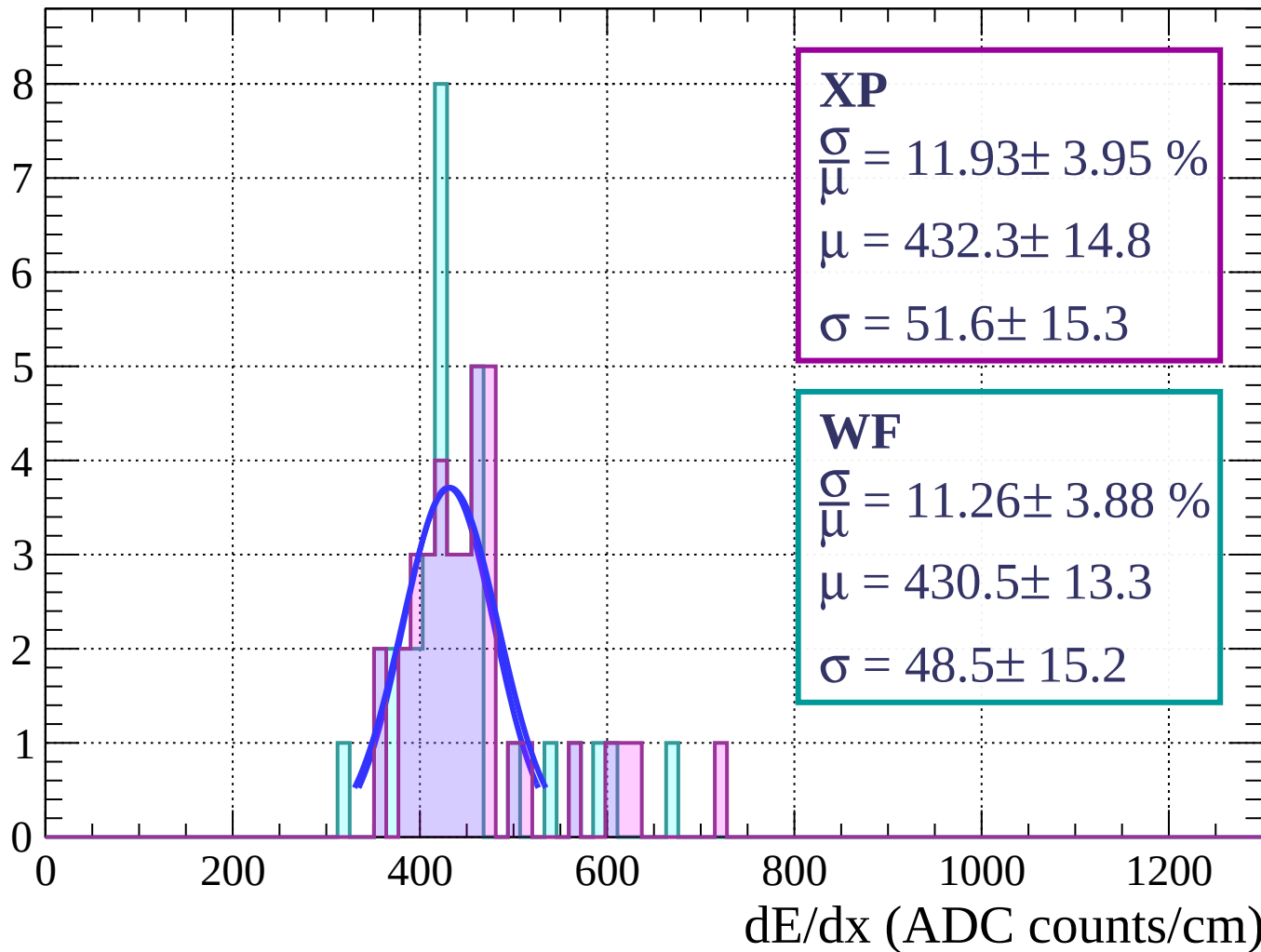
Energy loss | $57 < \phi < 60$

Count



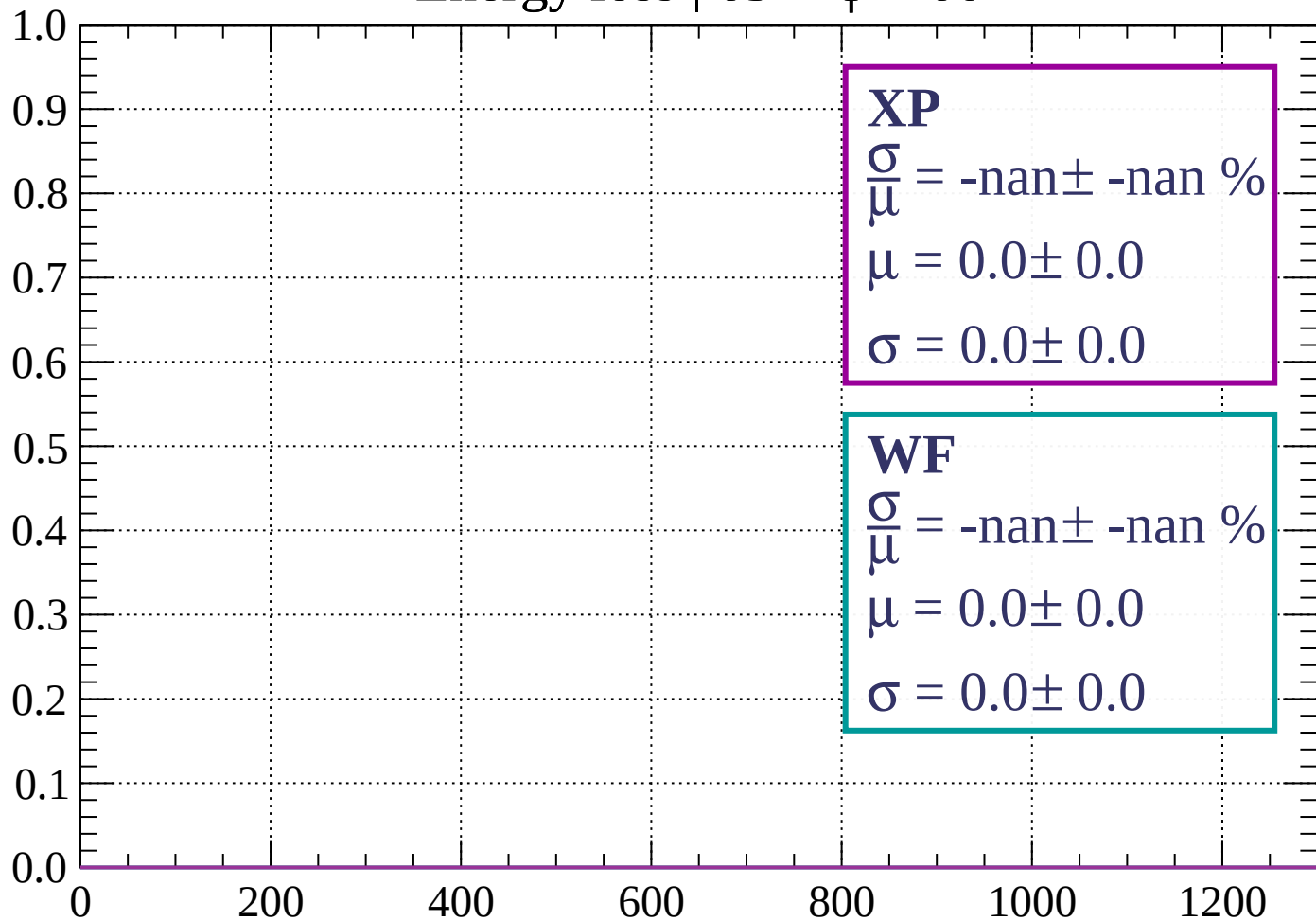
Energy loss | $60 < \phi < 63$

Count



Energy loss | $63 < \phi < 66$

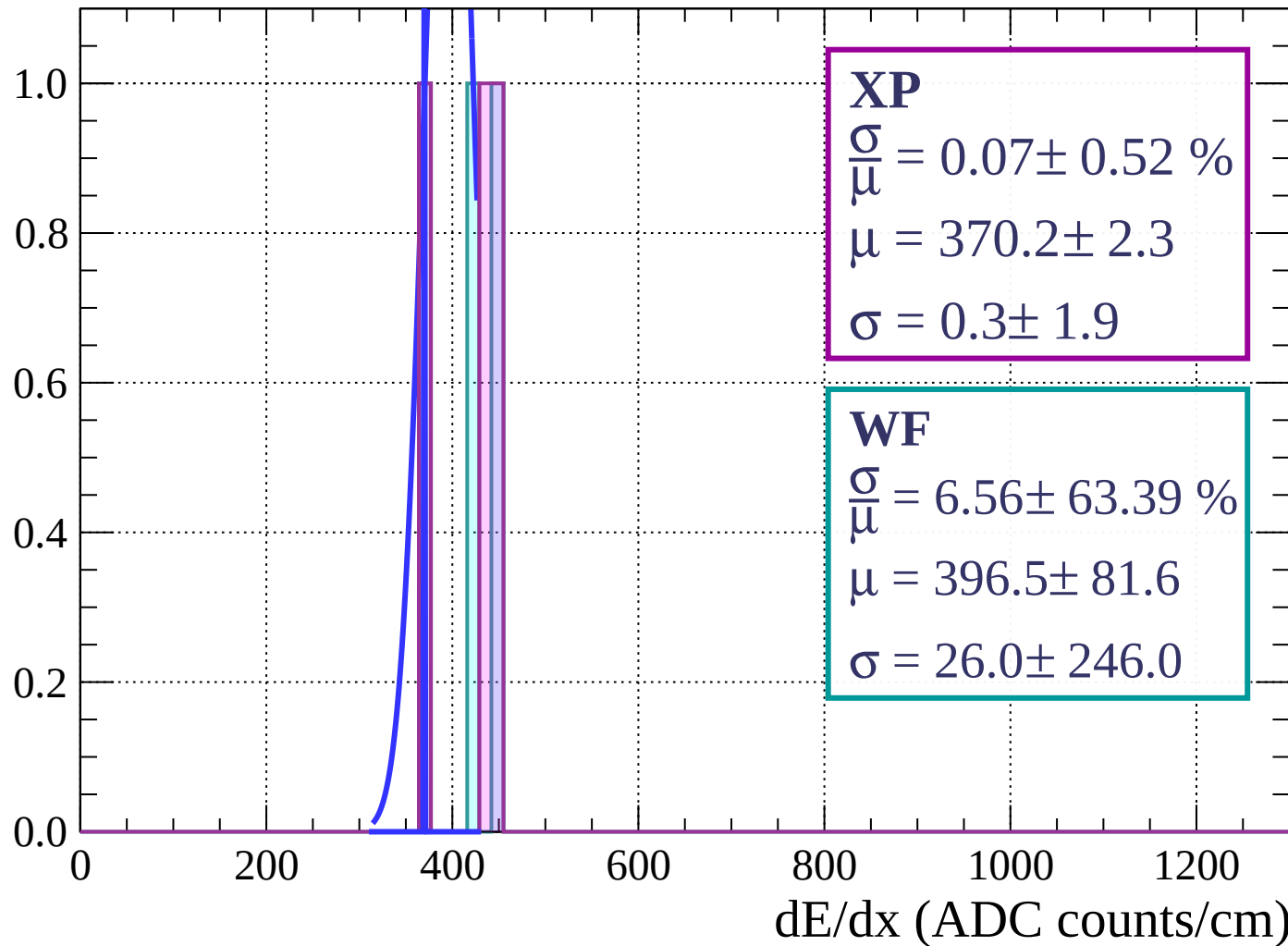
Count



dE/dx (ADC counts/cm)

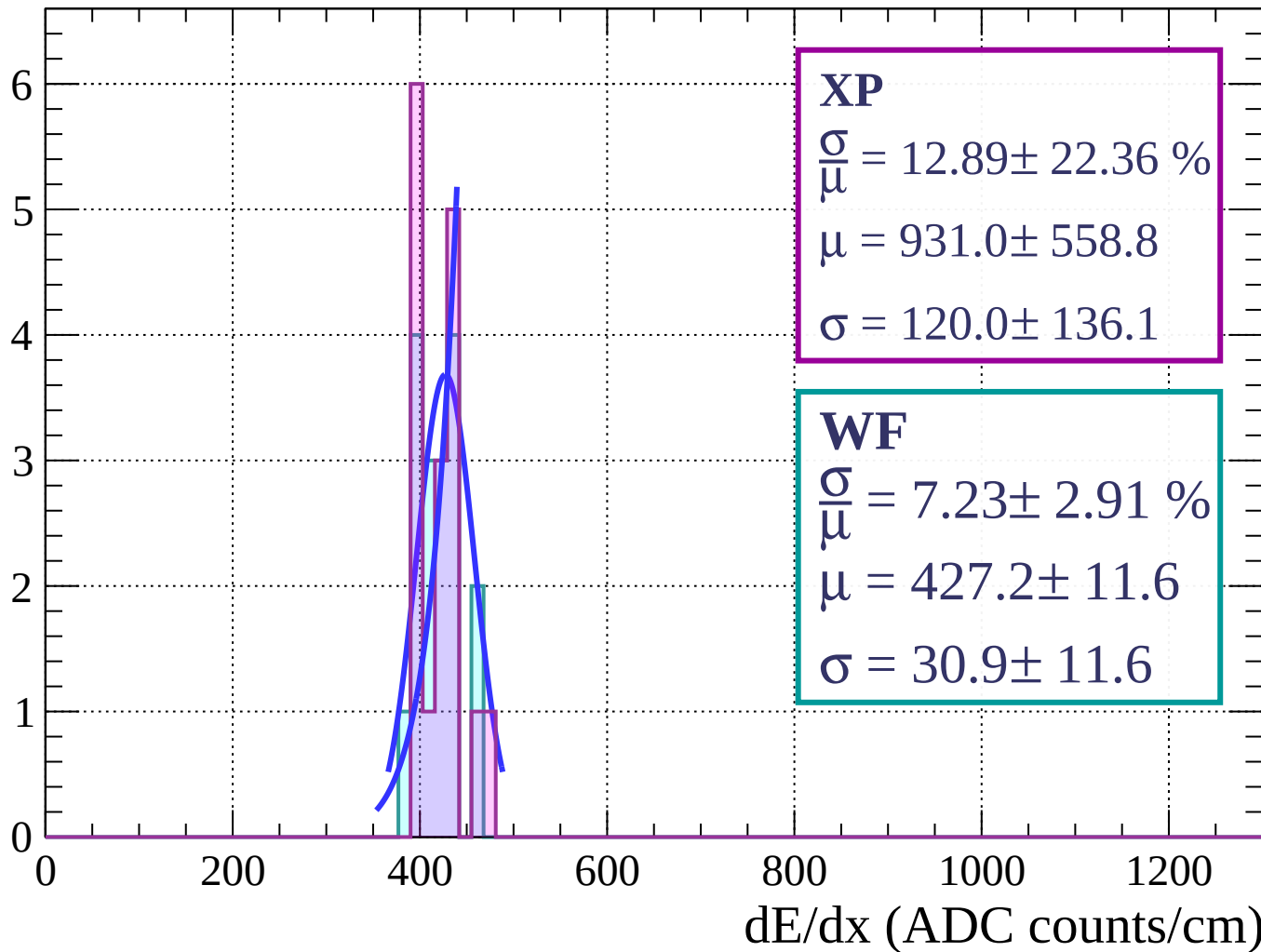
Energy loss | $66 < \phi < 69$

Count



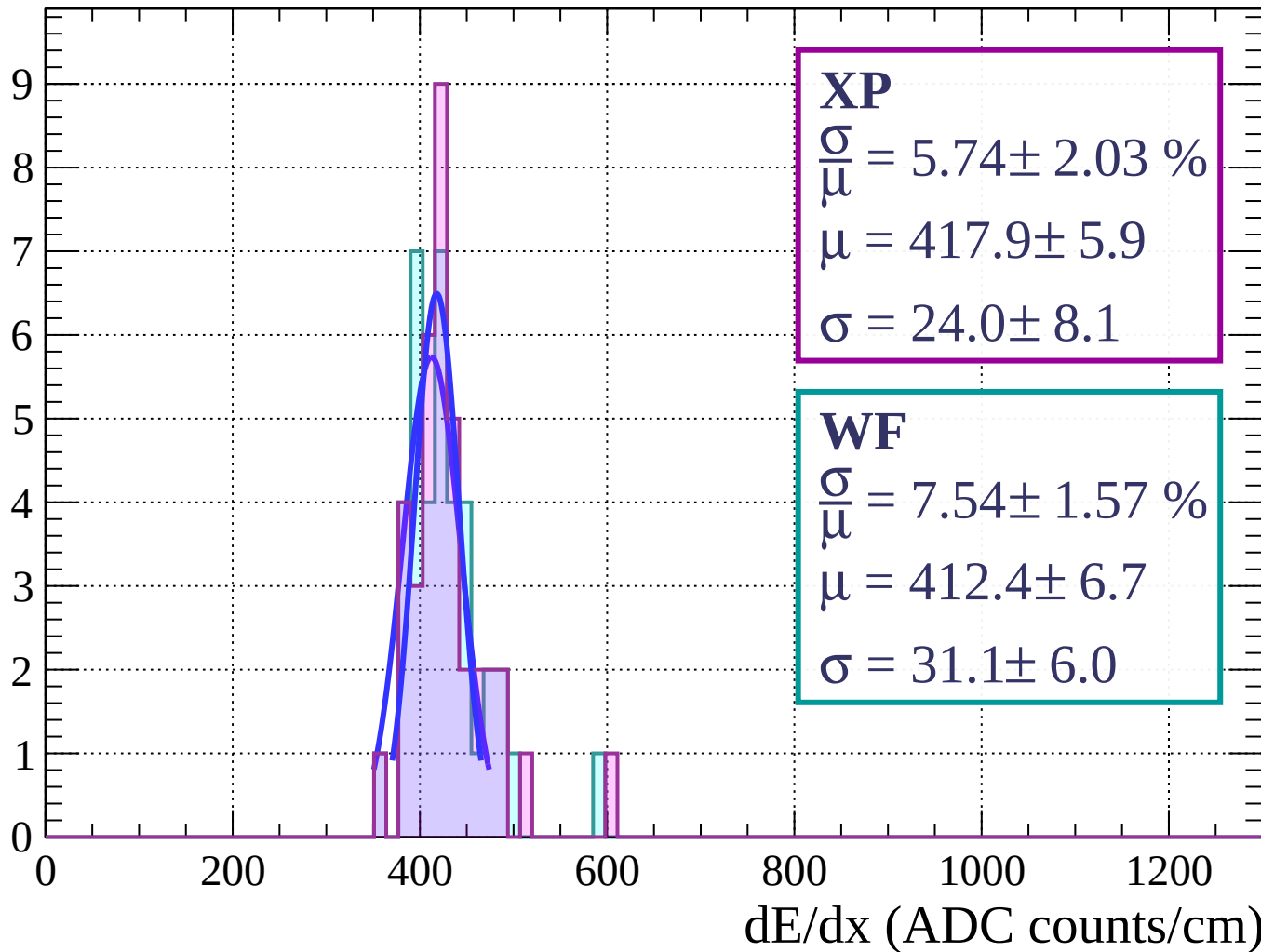
Energy loss | $69 < \phi < 72$

Count



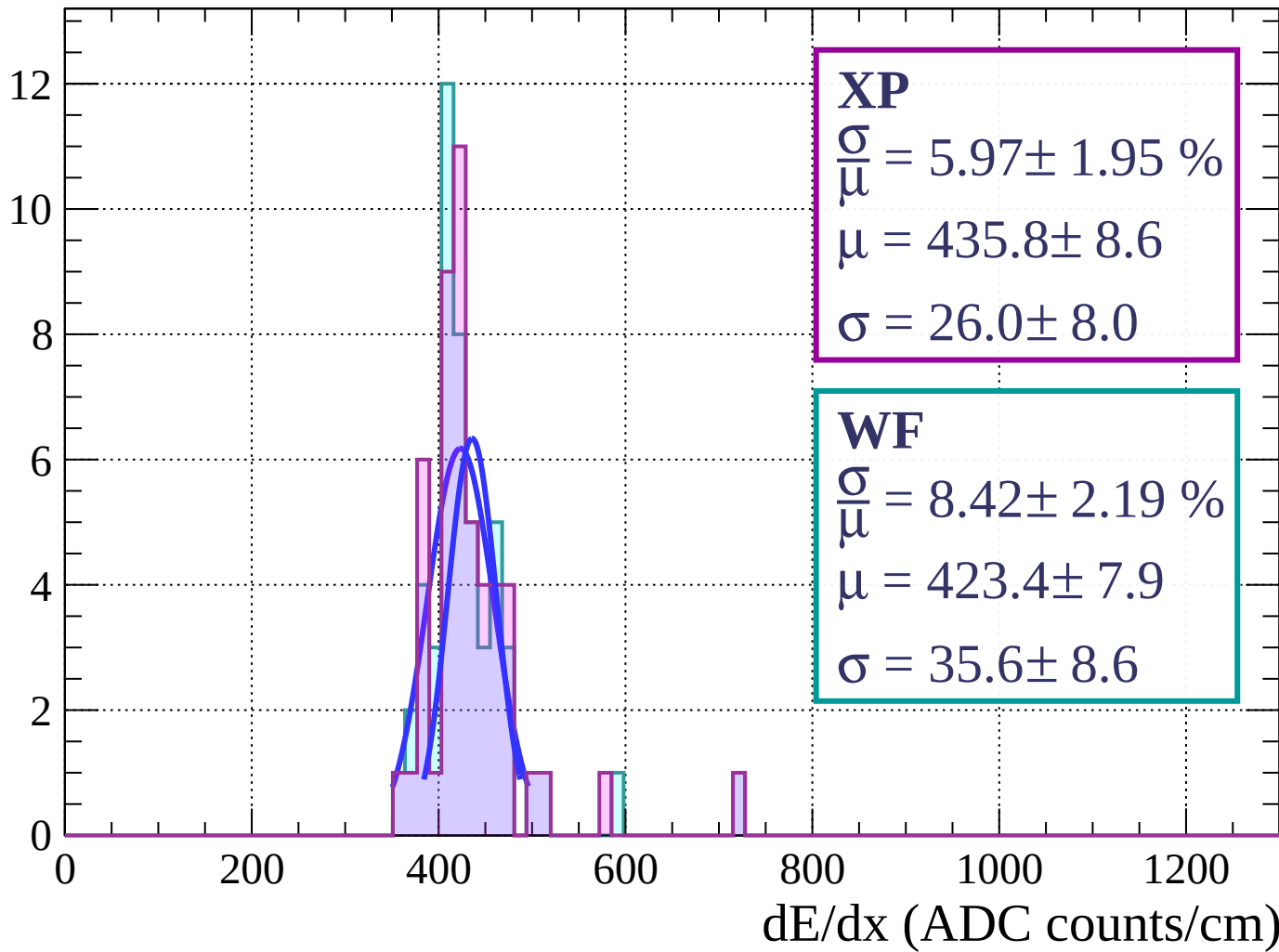
Energy loss | $72 < \phi < 75$

Count



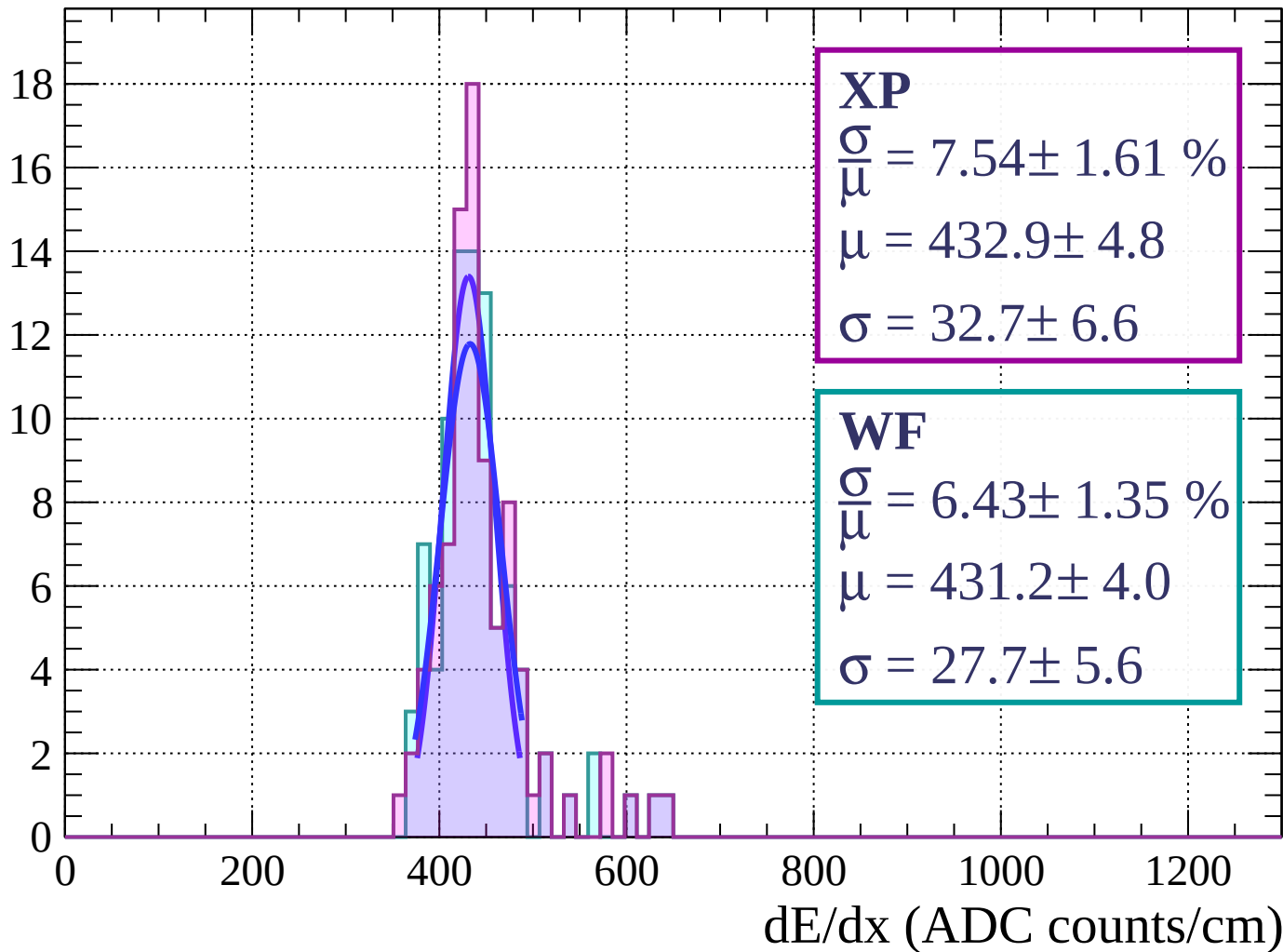
Energy loss | $75 < \phi < 78$

Count



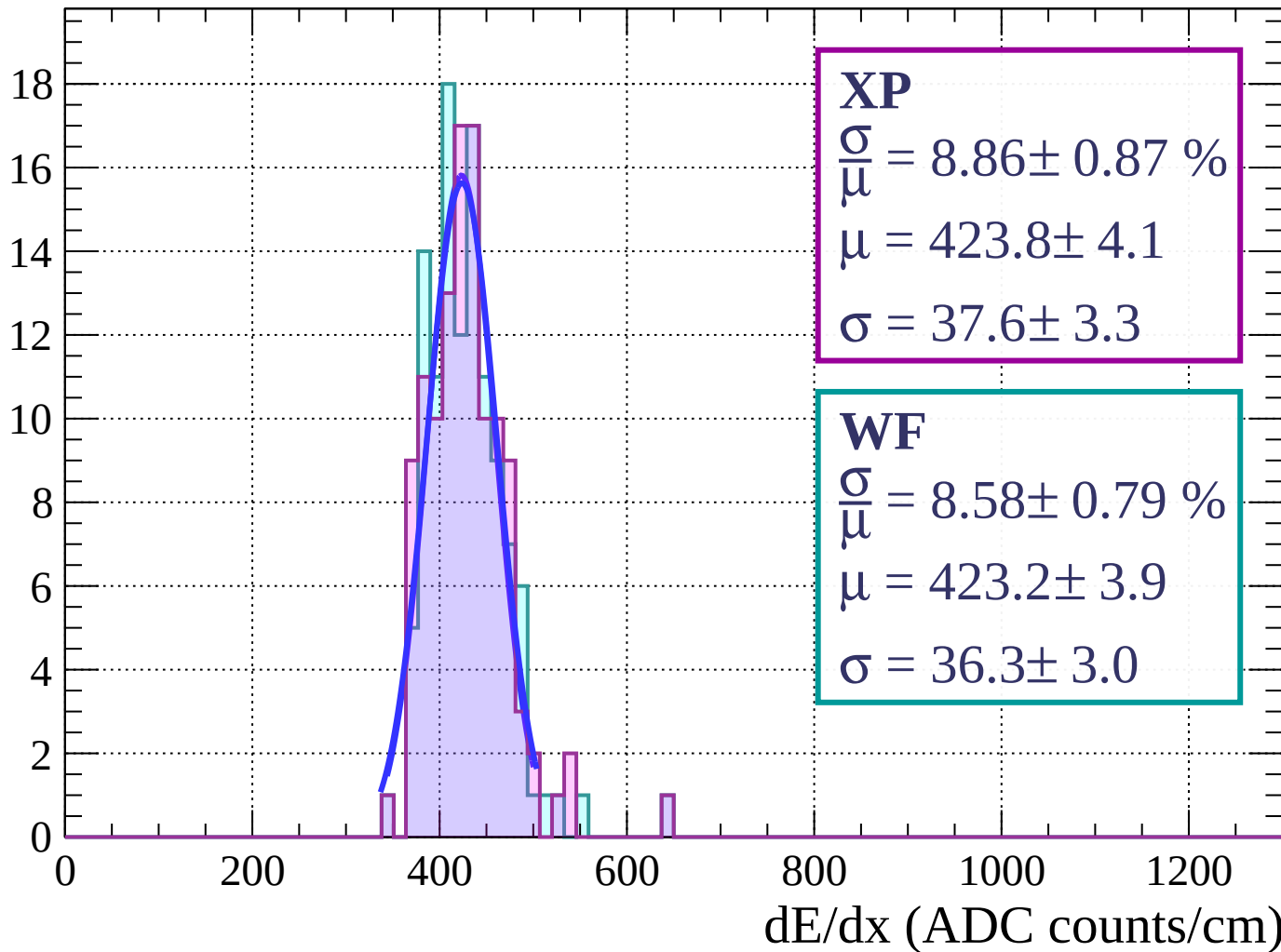
Energy loss | $78 < \phi < 81$

Count



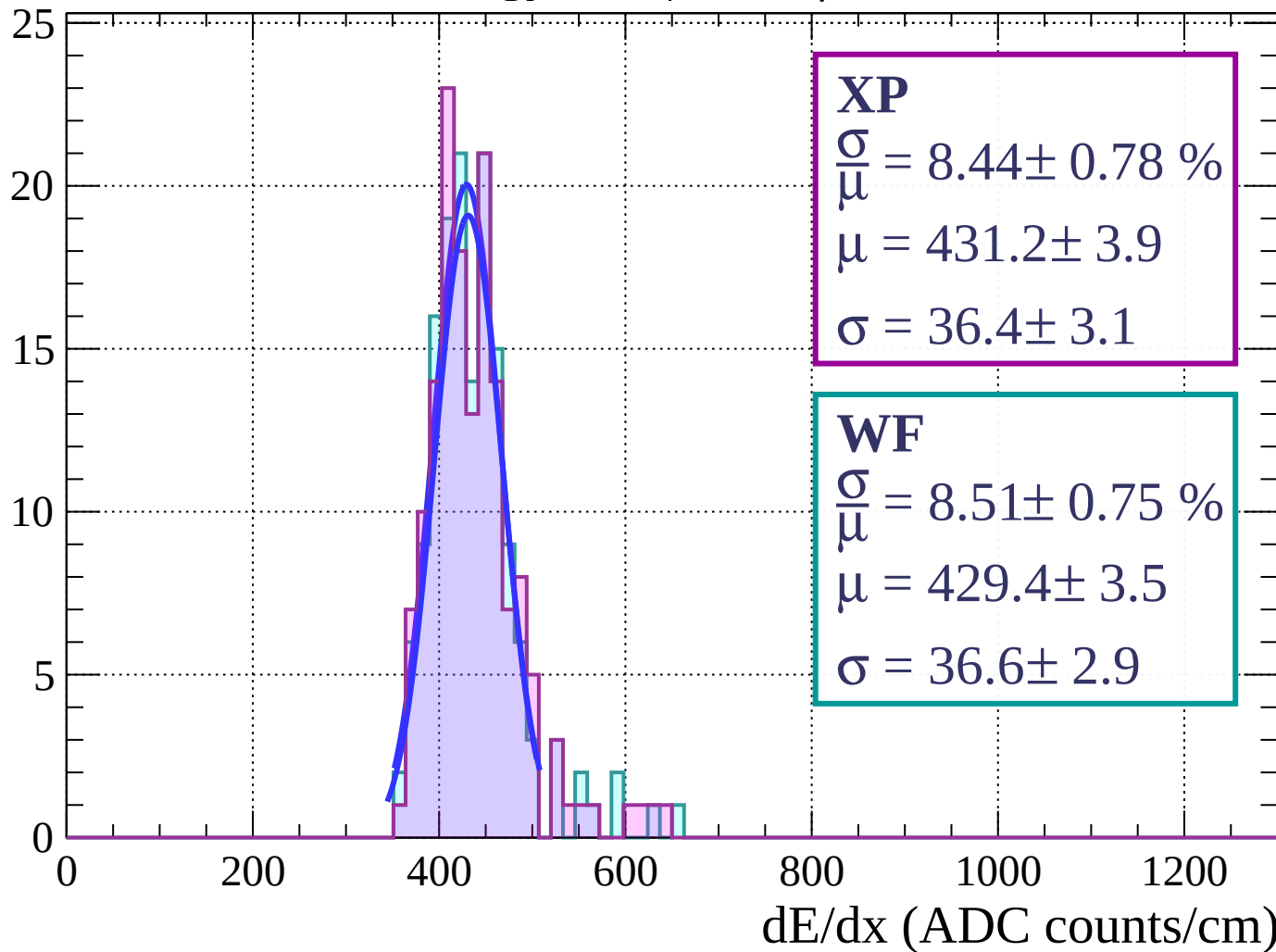
Energy loss | $81 < \phi < 84$

Count



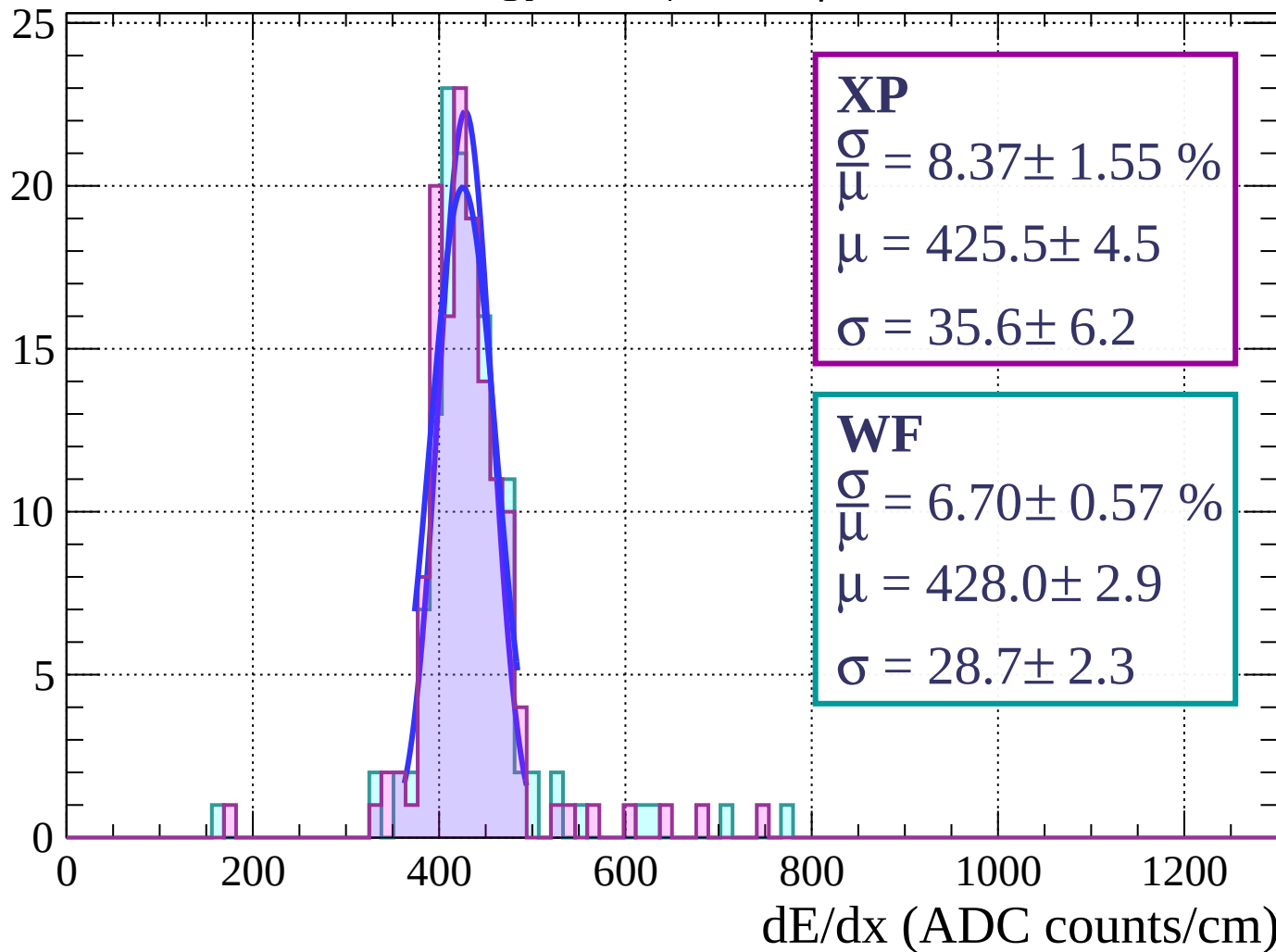
Energy loss | $84 < \phi < 87$

Count



Energy loss | $87 < \phi < 90$

Count



Energy loss | $0 < \theta < 3$

Count

XP

$$\frac{\sigma}{\mu} = 8.05 \pm 0.71 \%$$

$$\mu = 425.6 \pm 3.1$$

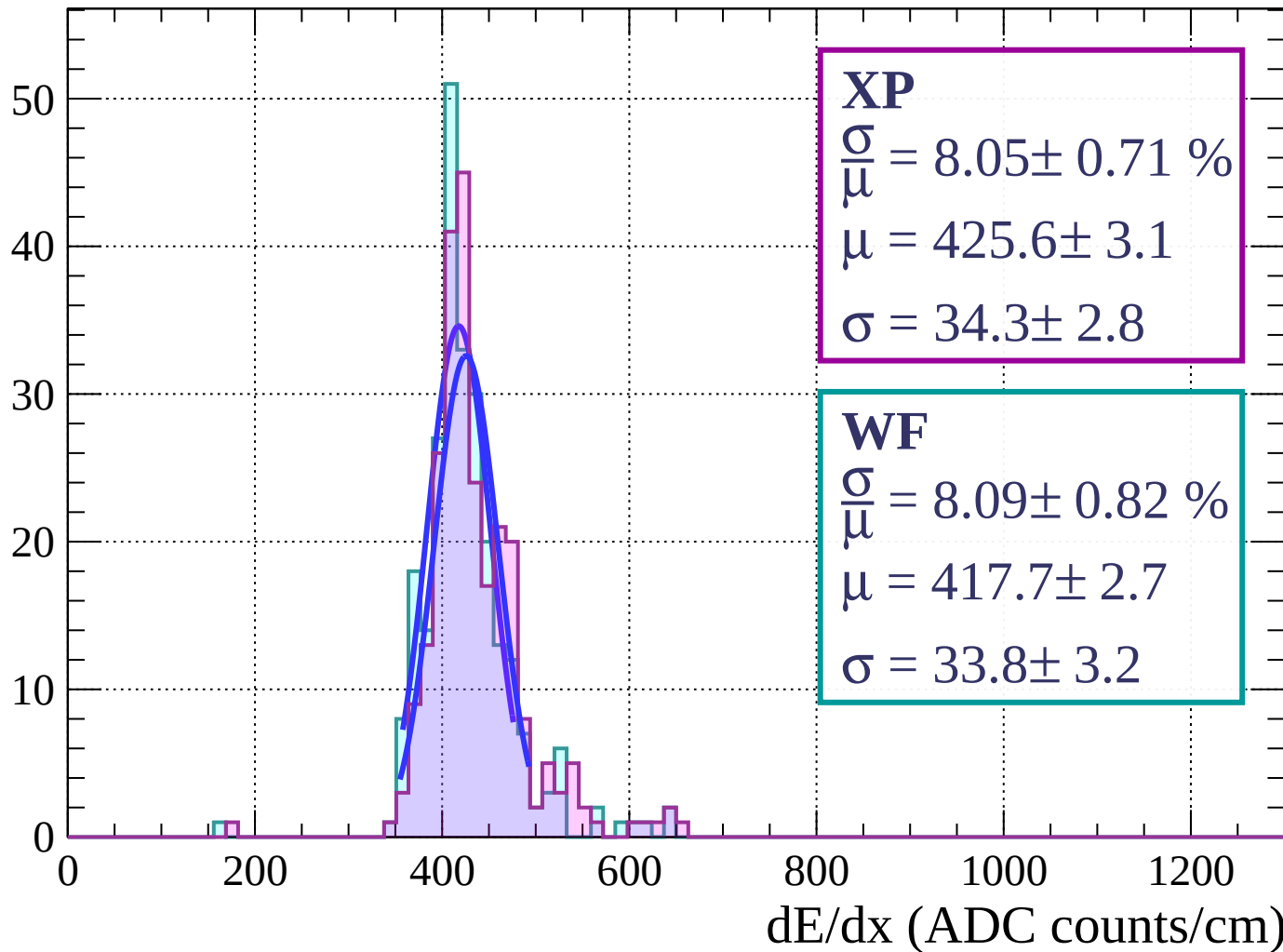
$$\sigma = 34.3 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 8.09 \pm 0.82 \%$$

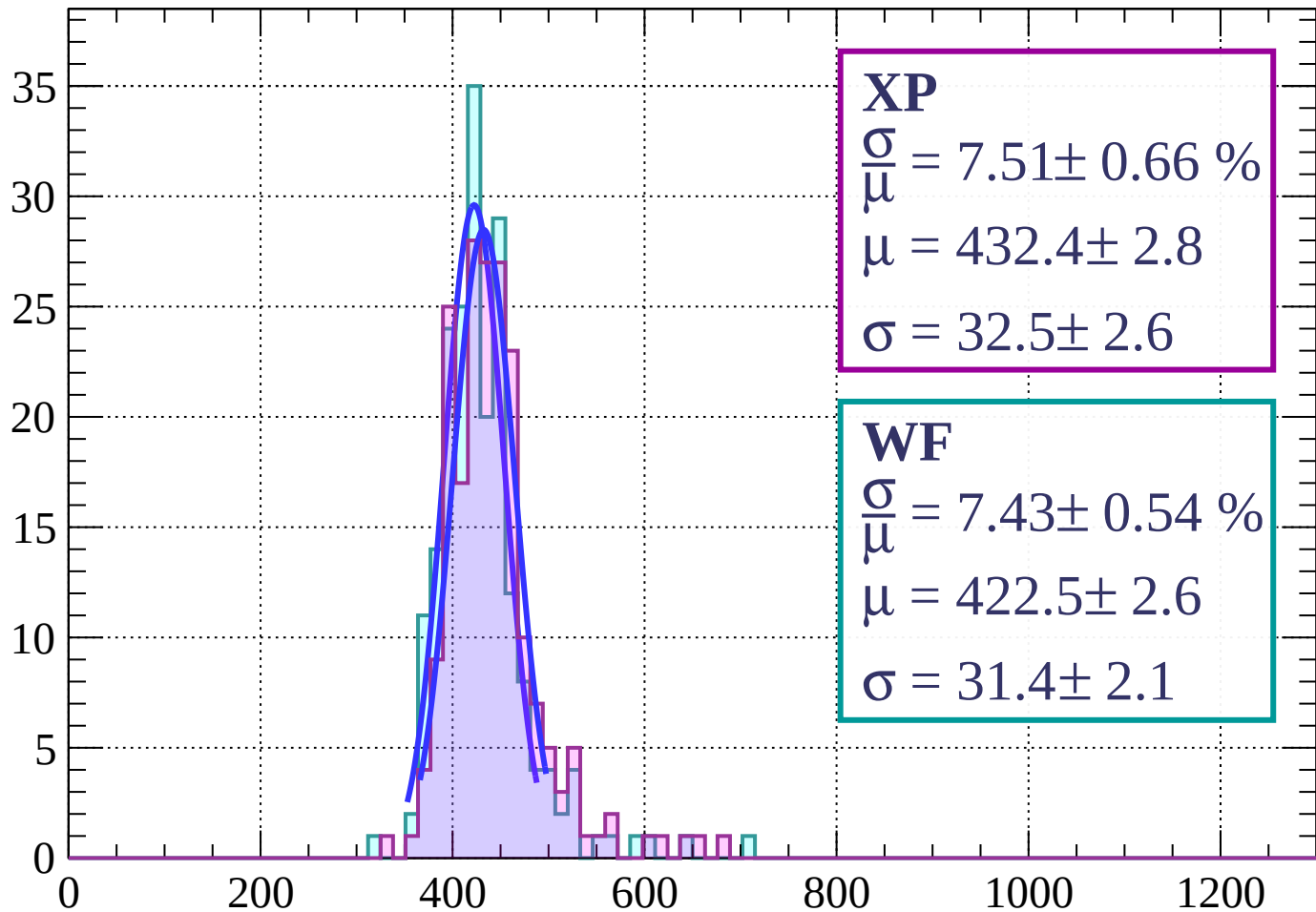
$$\mu = 417.7 \pm 2.7$$

$$\sigma = 33.8 \pm 3.2$$



Energy loss | $3 < \theta < 6$

Count



XP

$$\frac{\sigma}{\mu} = 7.51 \pm 0.66 \%$$

$$\mu = 432.4 \pm 2.8$$

$$\sigma = 32.5 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 7.43 \pm 0.54 \%$$

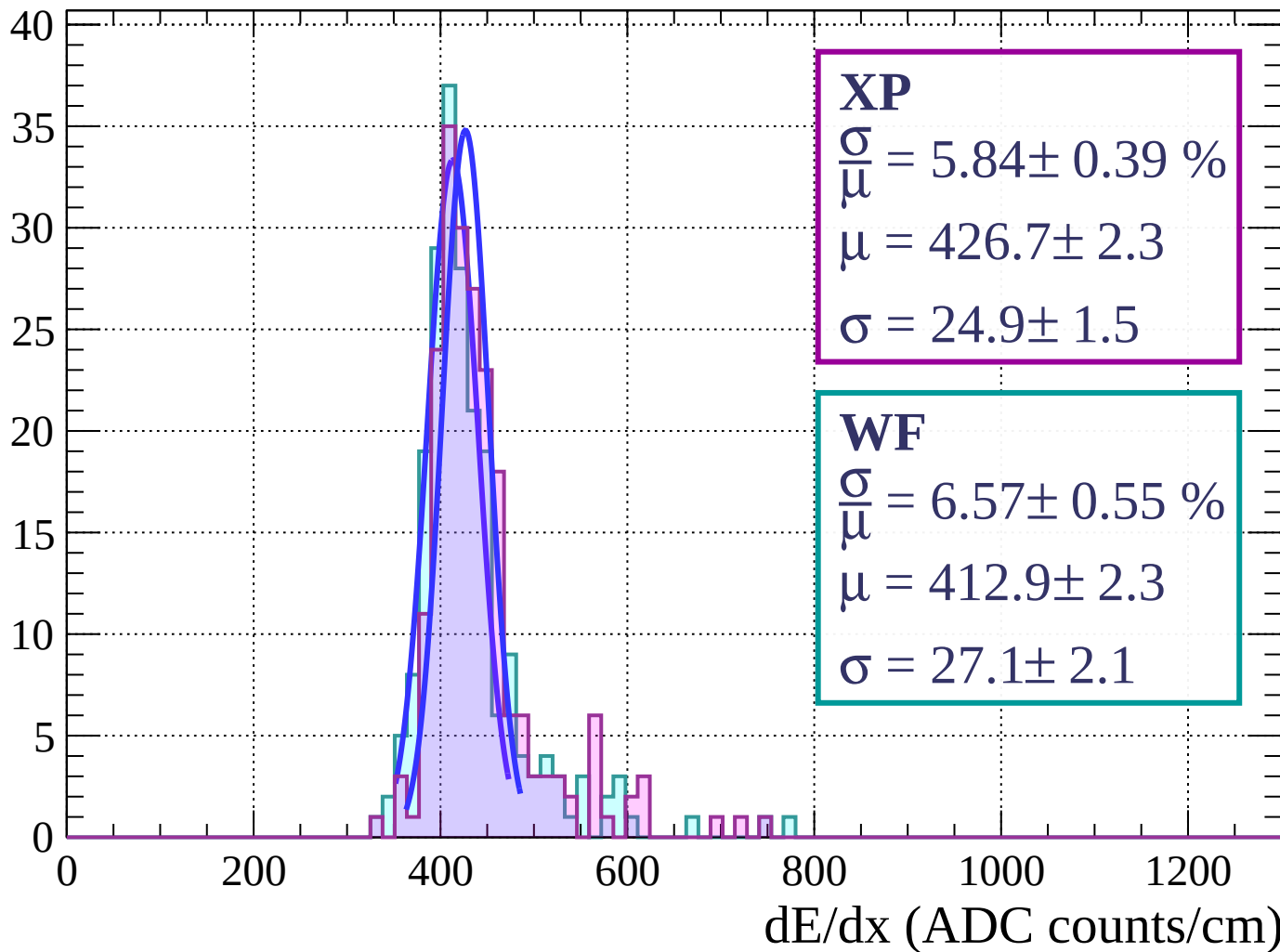
$$\mu = 422.5 \pm 2.6$$

$$\sigma = 31.4 \pm 2.1$$

dE/dx (ADC counts/cm)

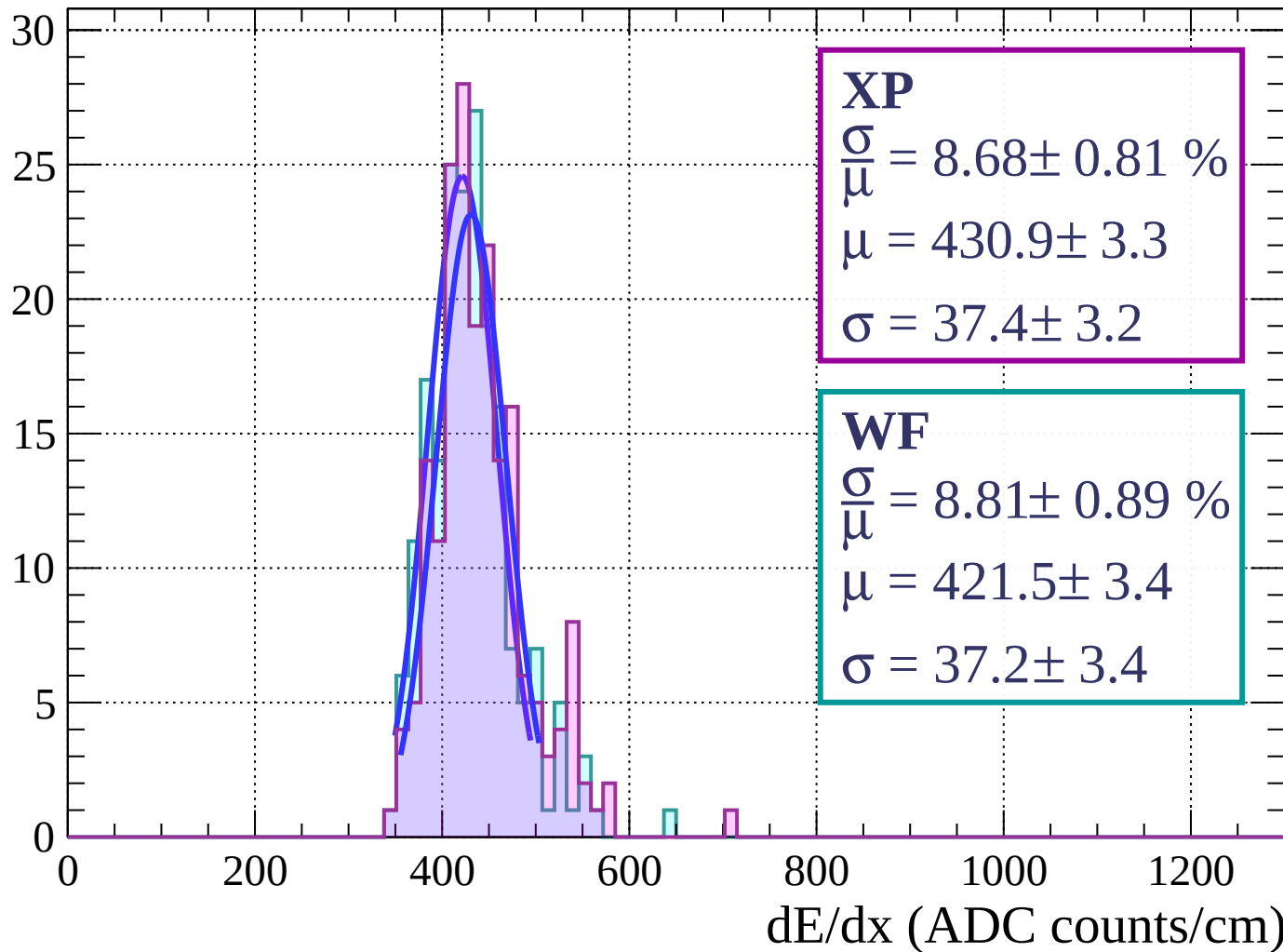
Energy loss | $6 < \theta < 9$

Count



Energy loss | $9 < \theta < 12$

Count



Energy loss | $12 < \theta < 15$

Count

XP

$$\frac{\sigma}{\mu} = 6.64 \pm 0.60 \%$$

$$\mu = 433.1 \pm 2.6$$

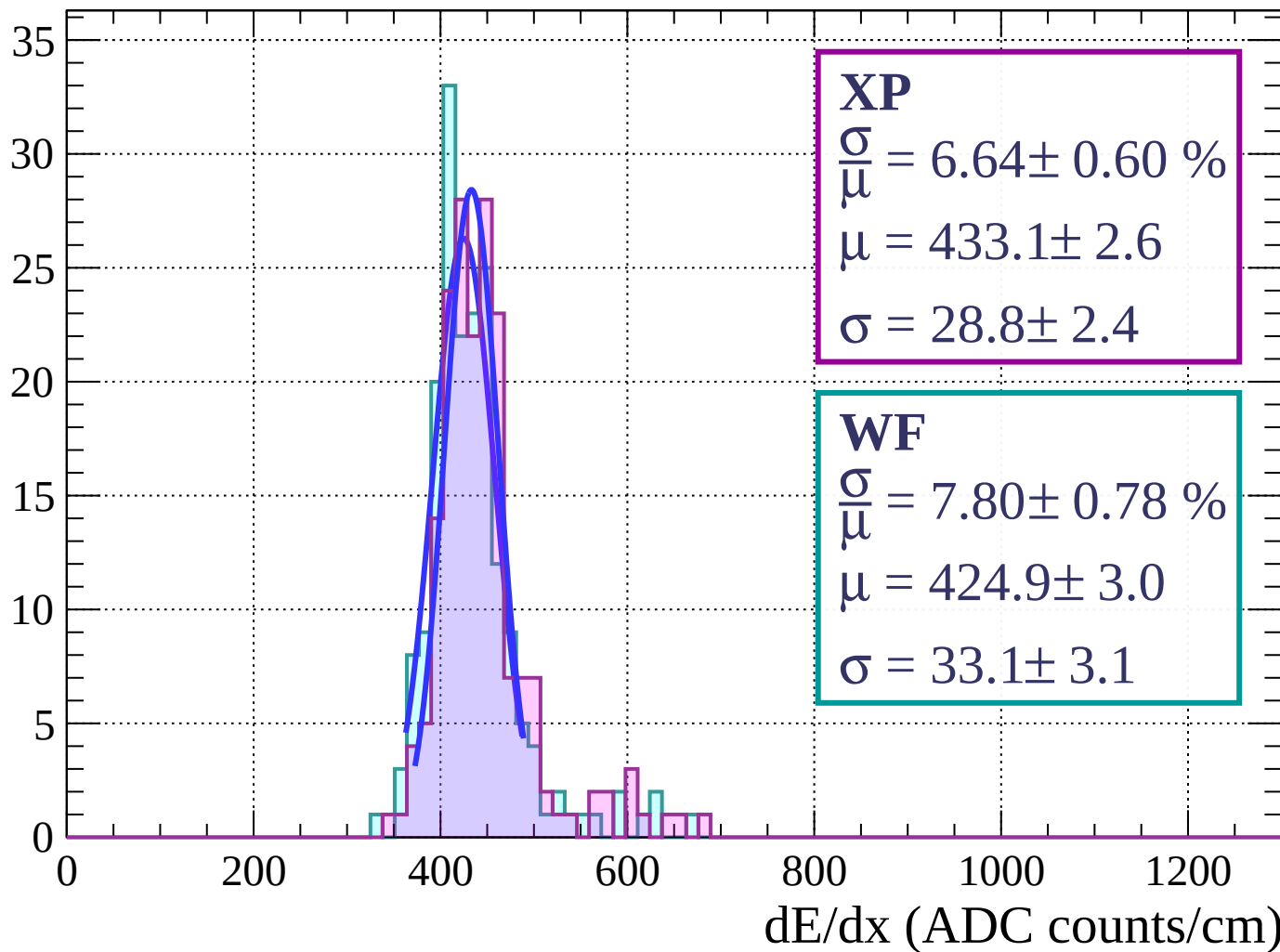
$$\sigma = 28.8 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 7.80 \pm 0.78 \%$$

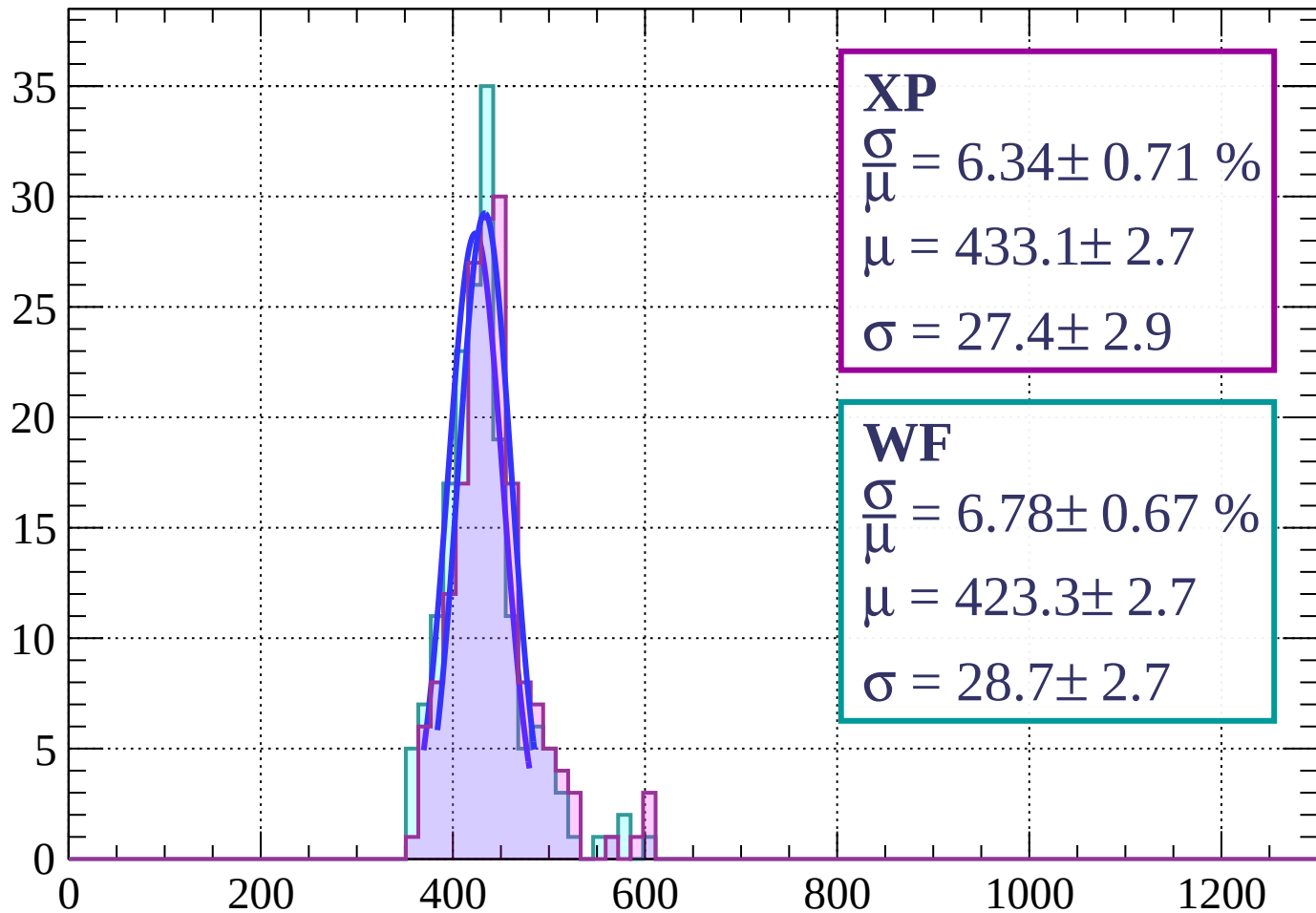
$$\mu = 424.9 \pm 3.0$$

$$\sigma = 33.1 \pm 3.1$$



Energy loss | $15 < \theta < 18$

Count



XP

$$\frac{\sigma}{\mu} = 6.34 \pm 0.71 \%$$

$$\mu = 433.1 \pm 2.7$$

$$\sigma = 27.4 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 6.78 \pm 0.67 \%$$

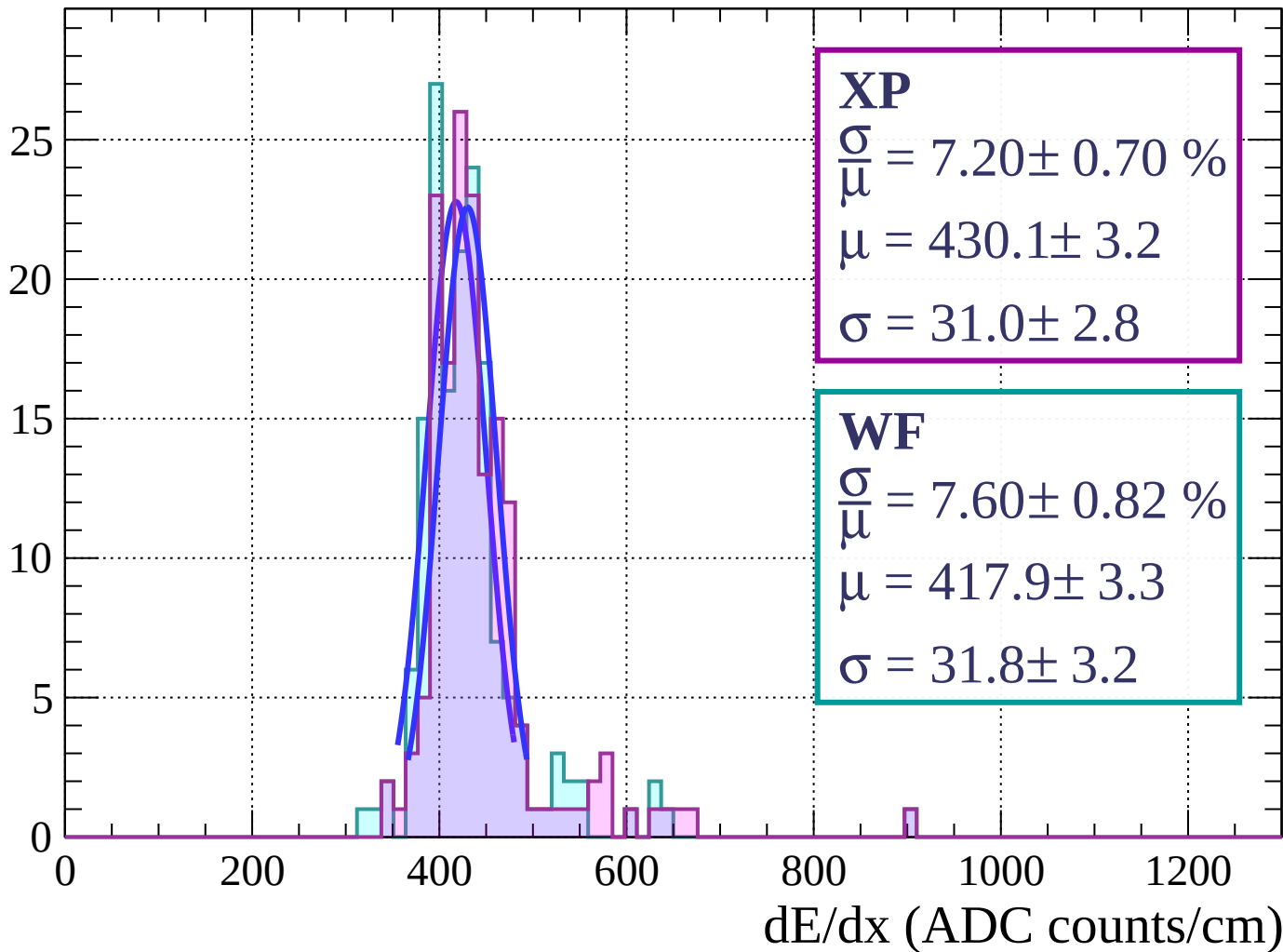
$$\mu = 423.3 \pm 2.7$$

$$\sigma = 28.7 \pm 2.7$$

dE/dx (ADC counts/cm)

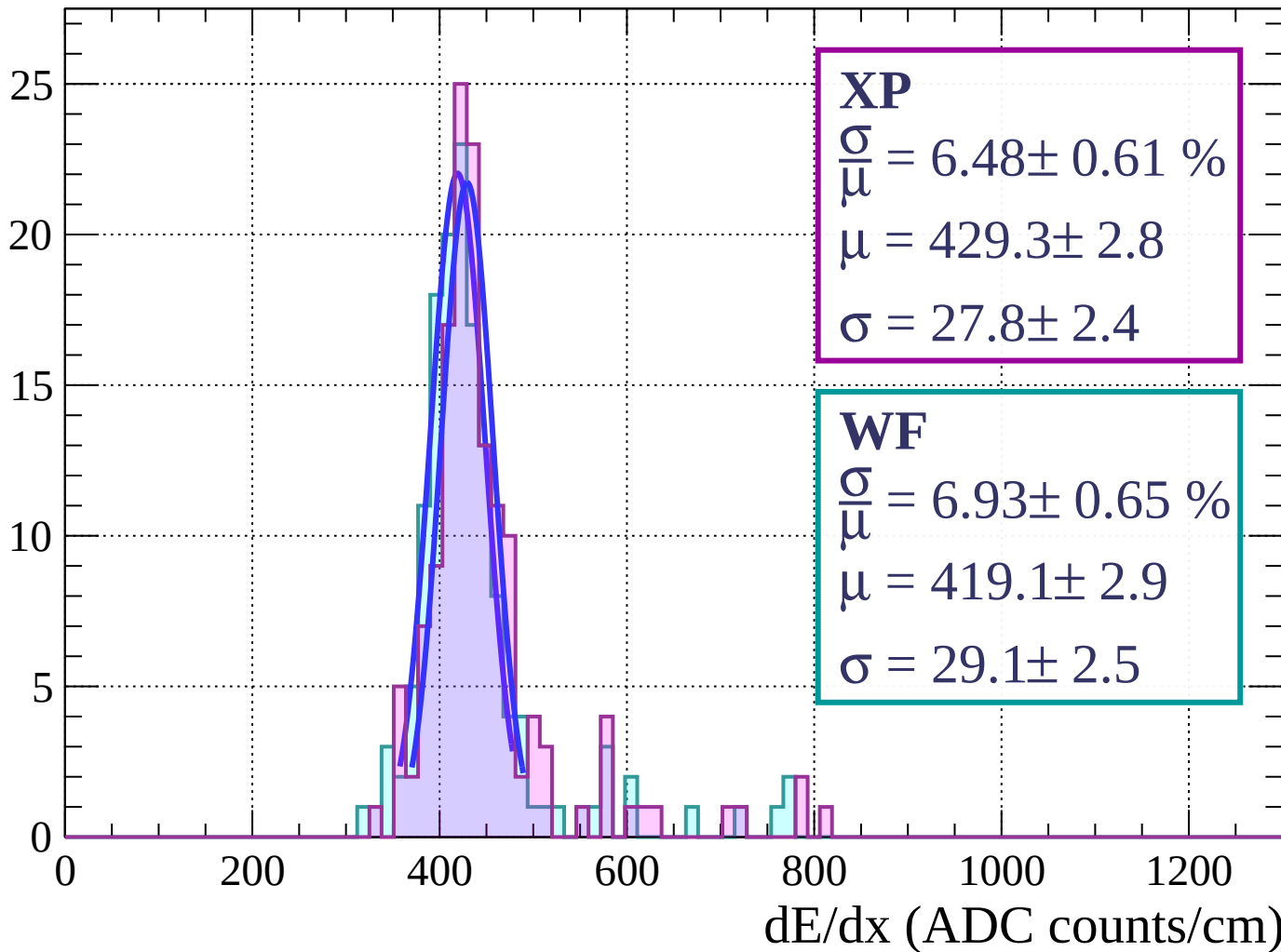
Energy loss | $18 < \theta < 21$

Count



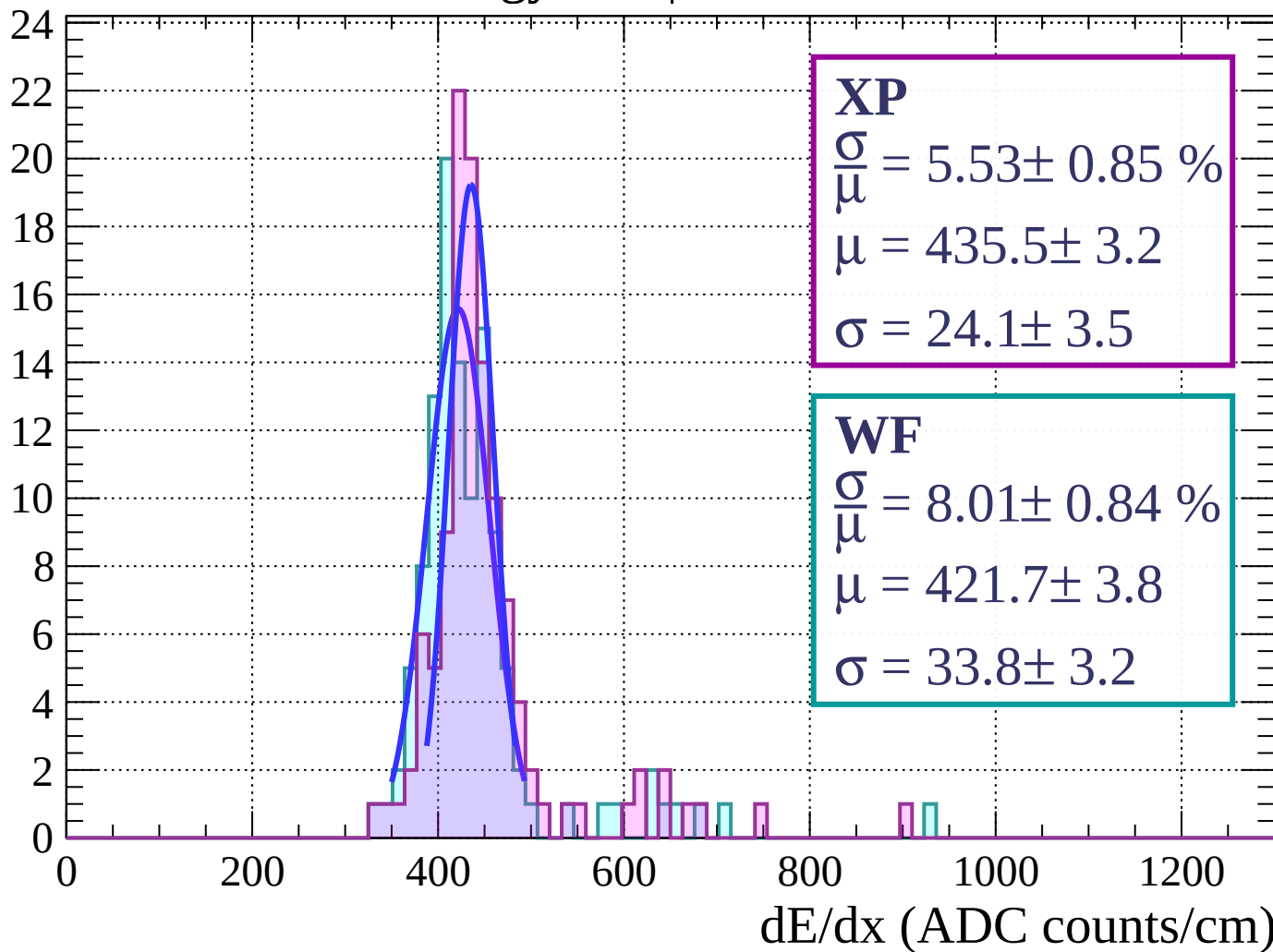
Energy loss | $21 < \theta < 24$

Count



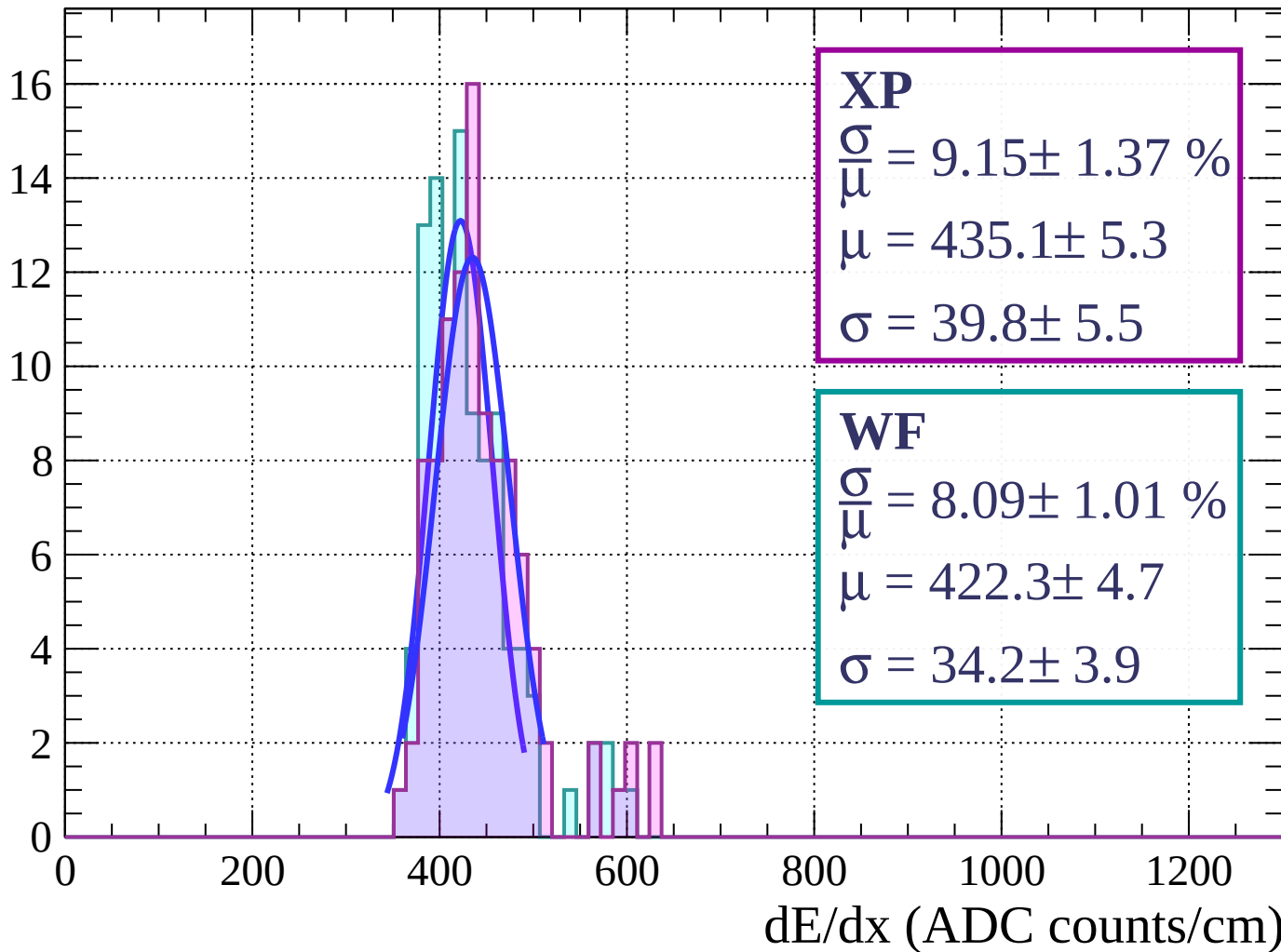
Energy loss | $24 < \theta < 27$

Count



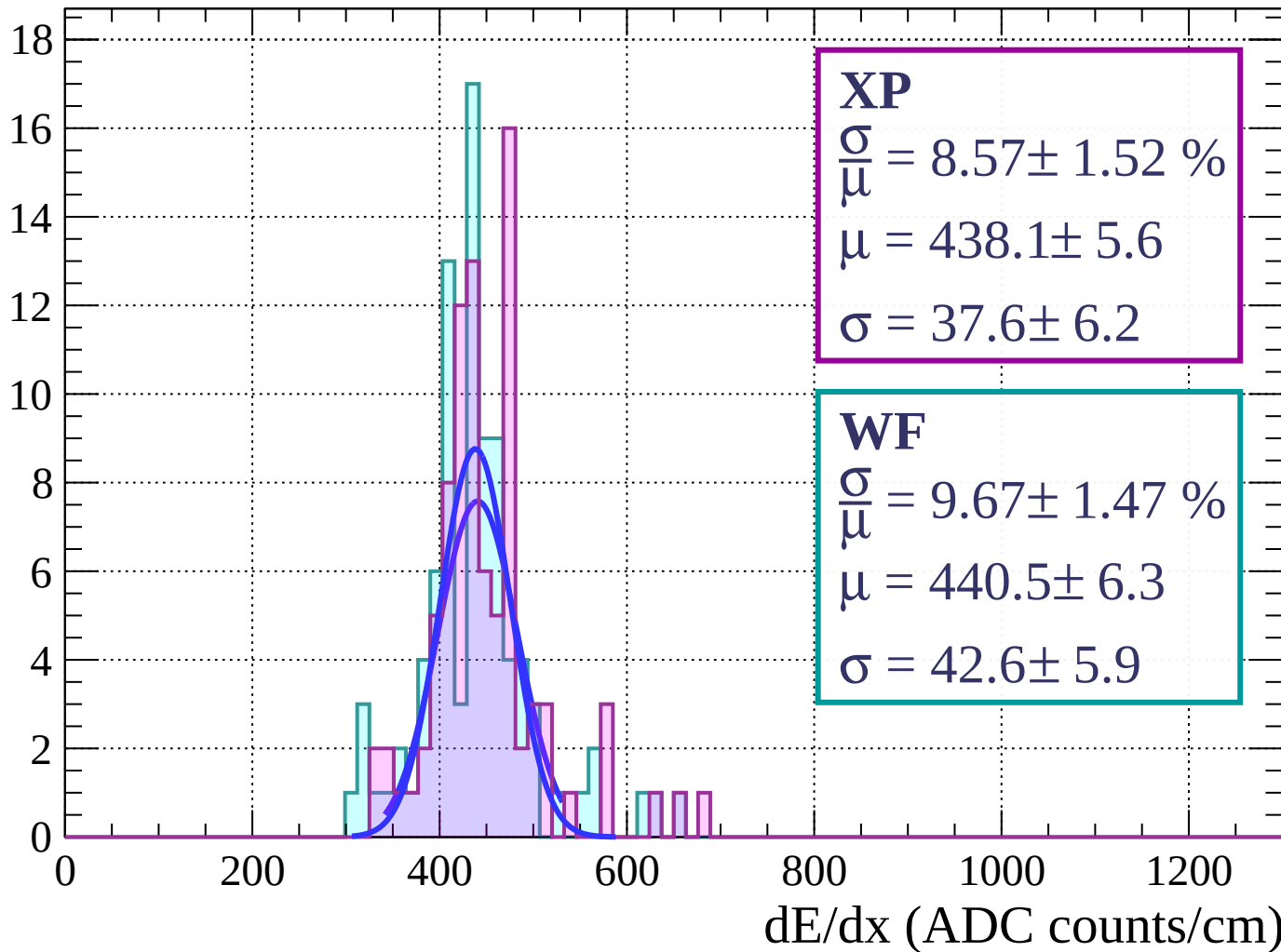
Energy loss | $27 < \theta < 30$

Count



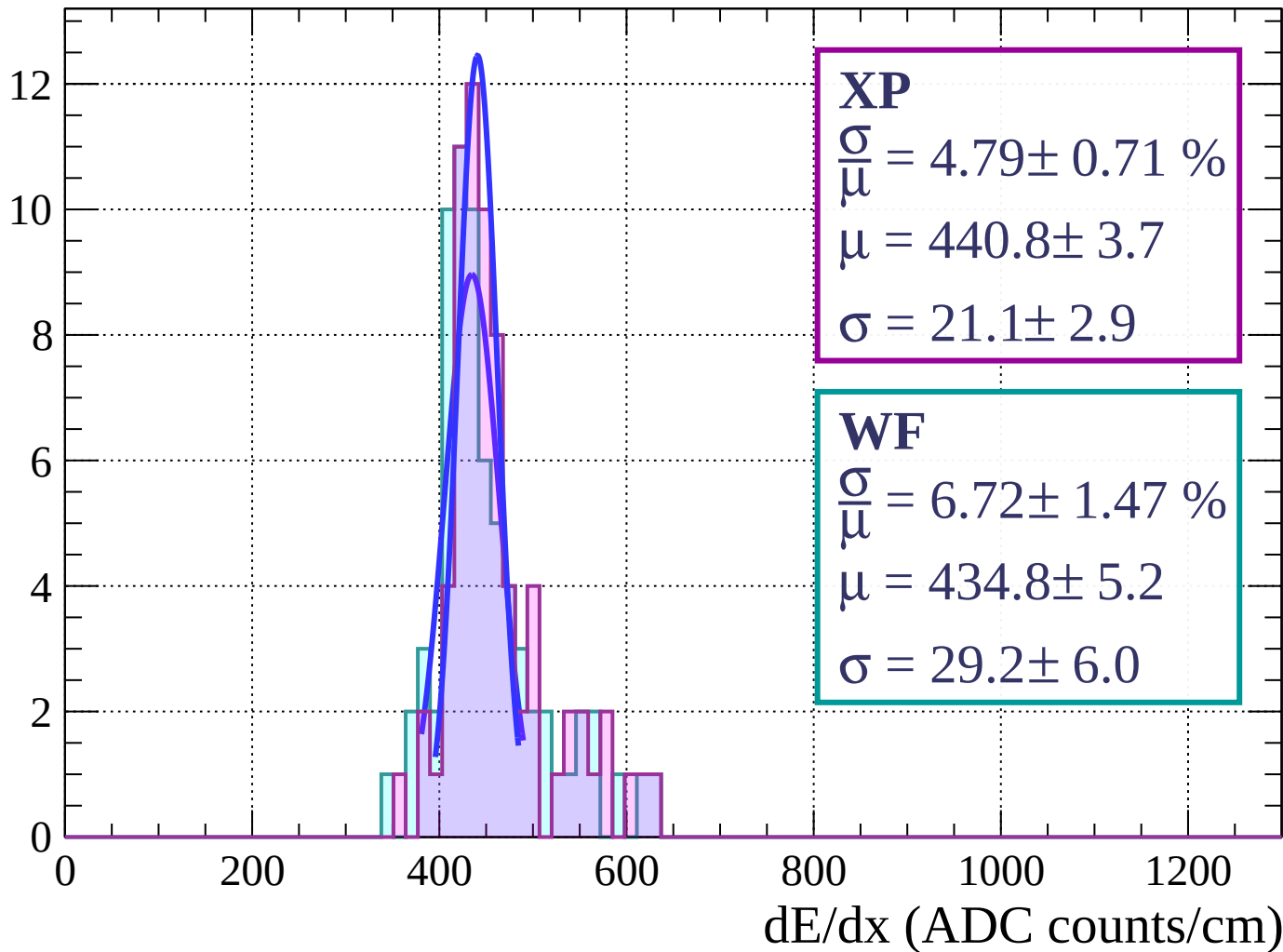
Energy loss | $30 < \theta < 33$

Count



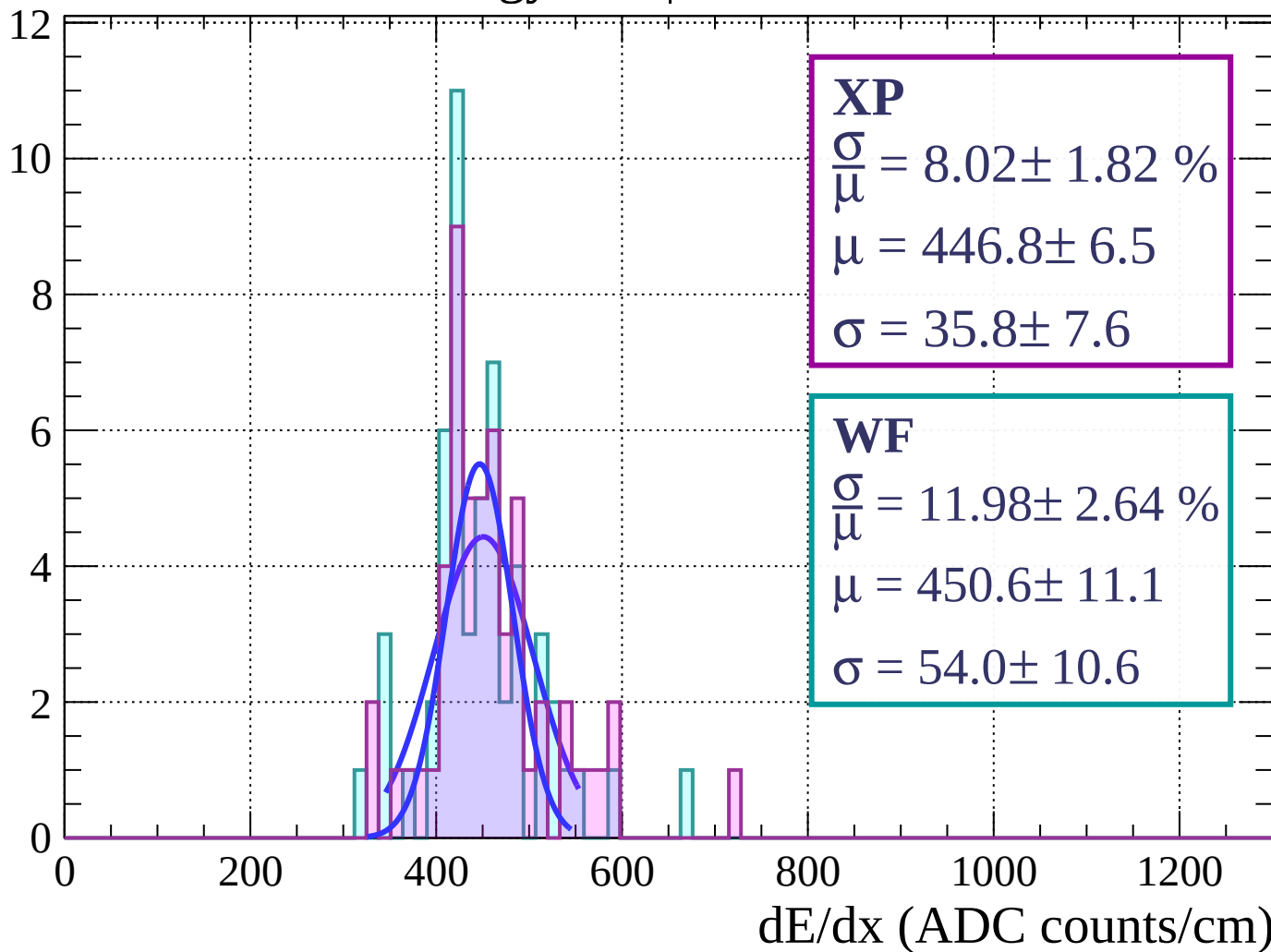
Energy loss | $33 < \theta < 36$

Count



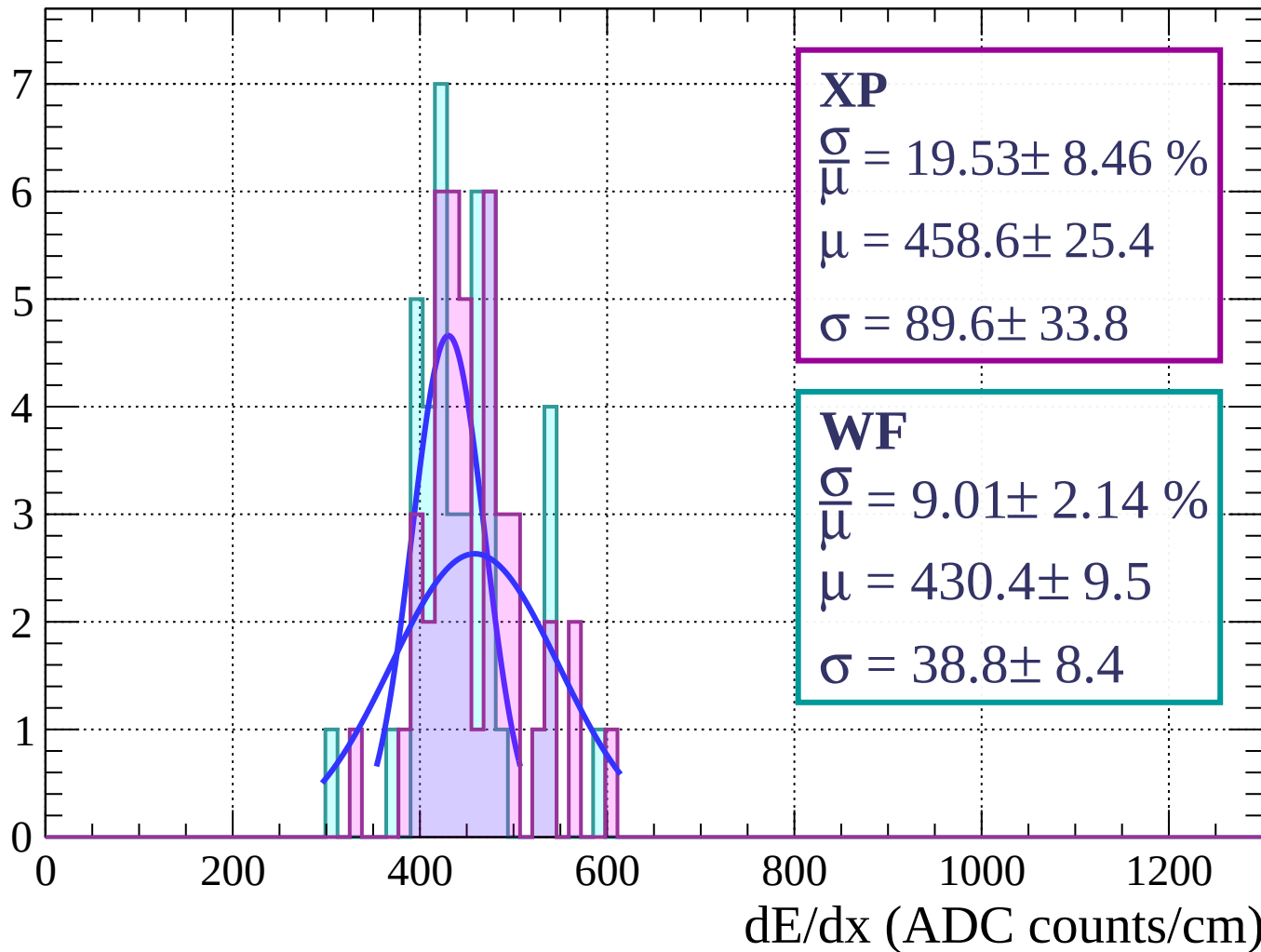
Energy loss | $36 < \theta < 39$

Count



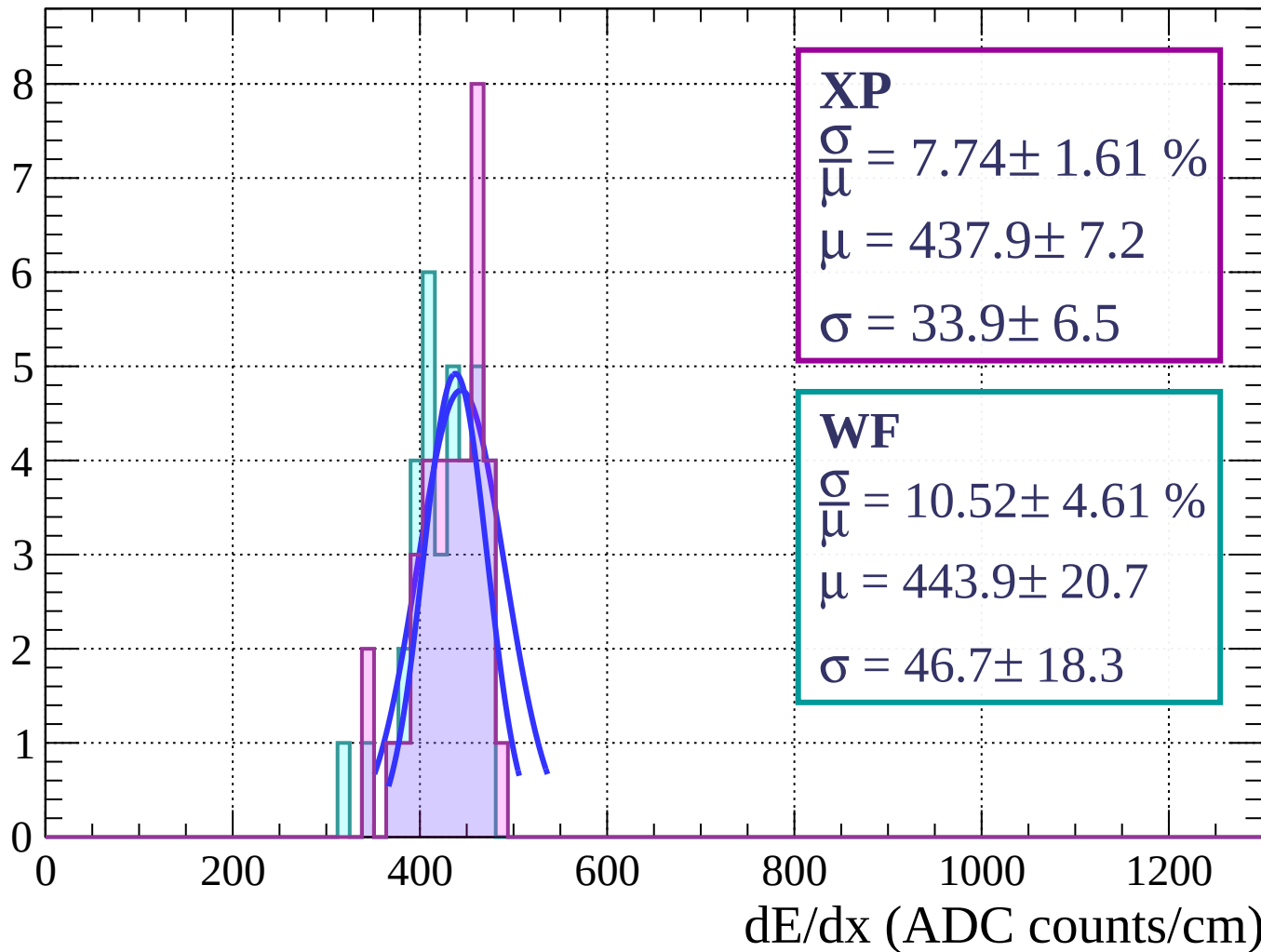
Energy loss | $39 < \theta < 42$

Count



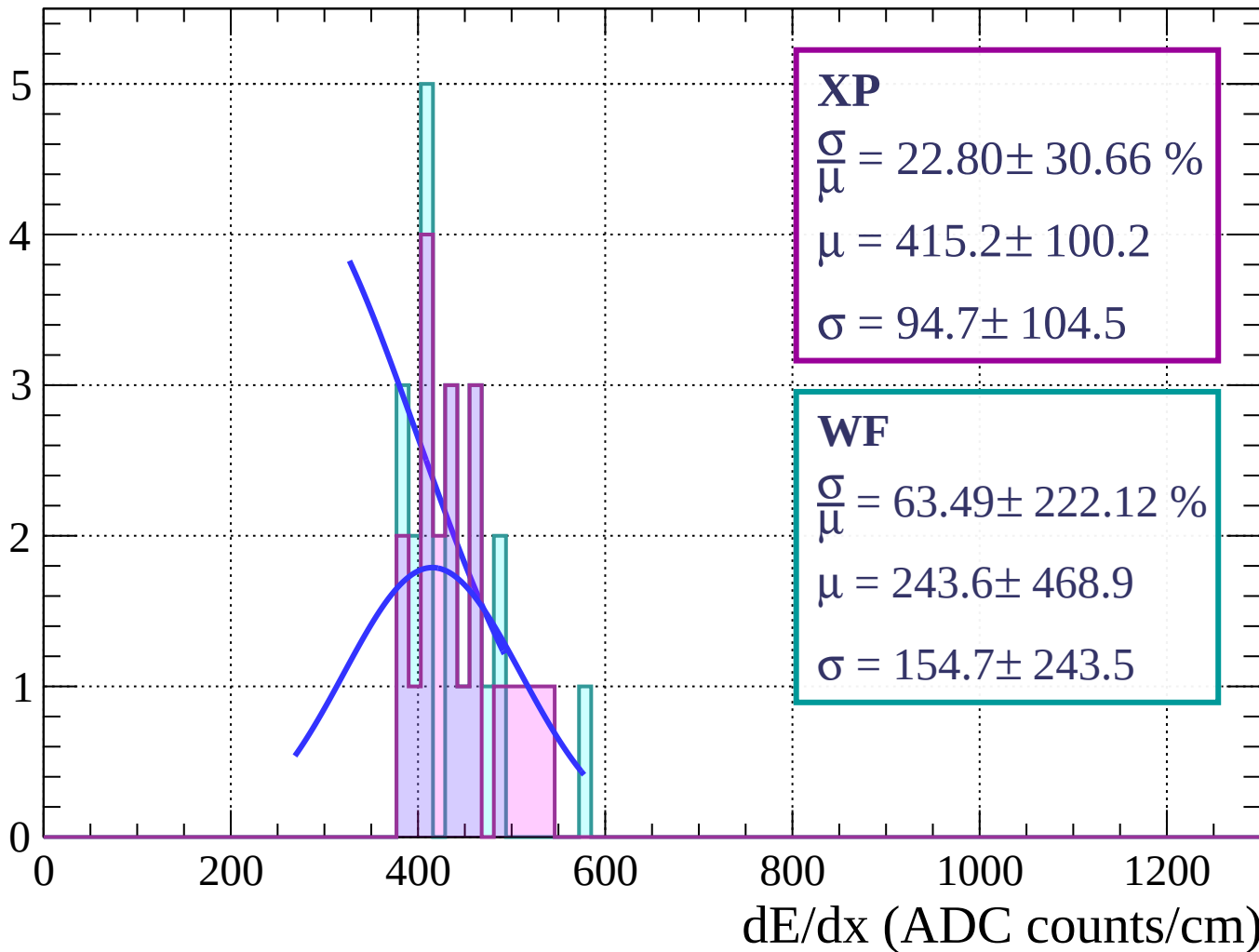
Energy loss | $42 < \theta < 45$

Count



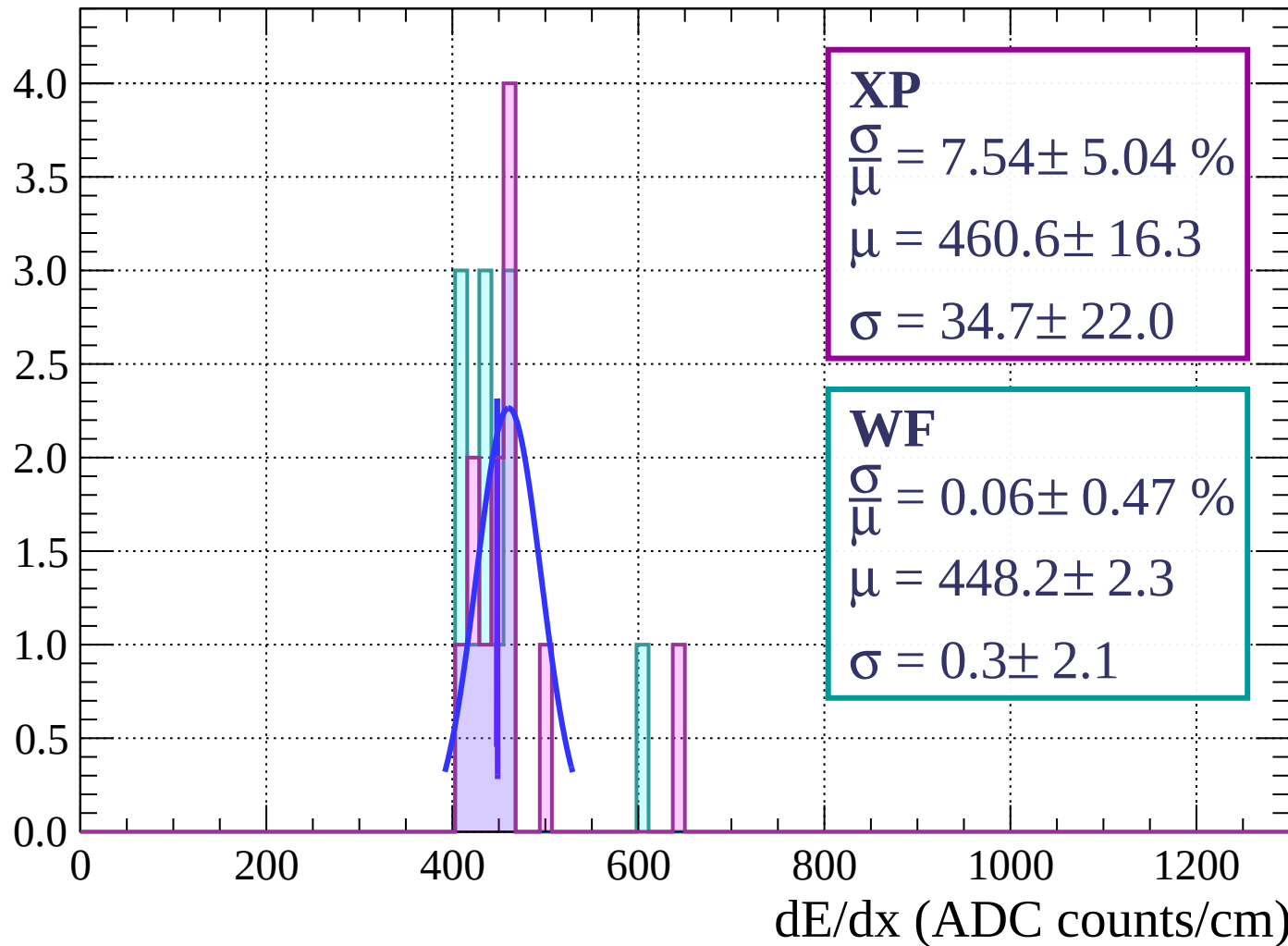
Energy loss | $45 < \theta < 48$

Count



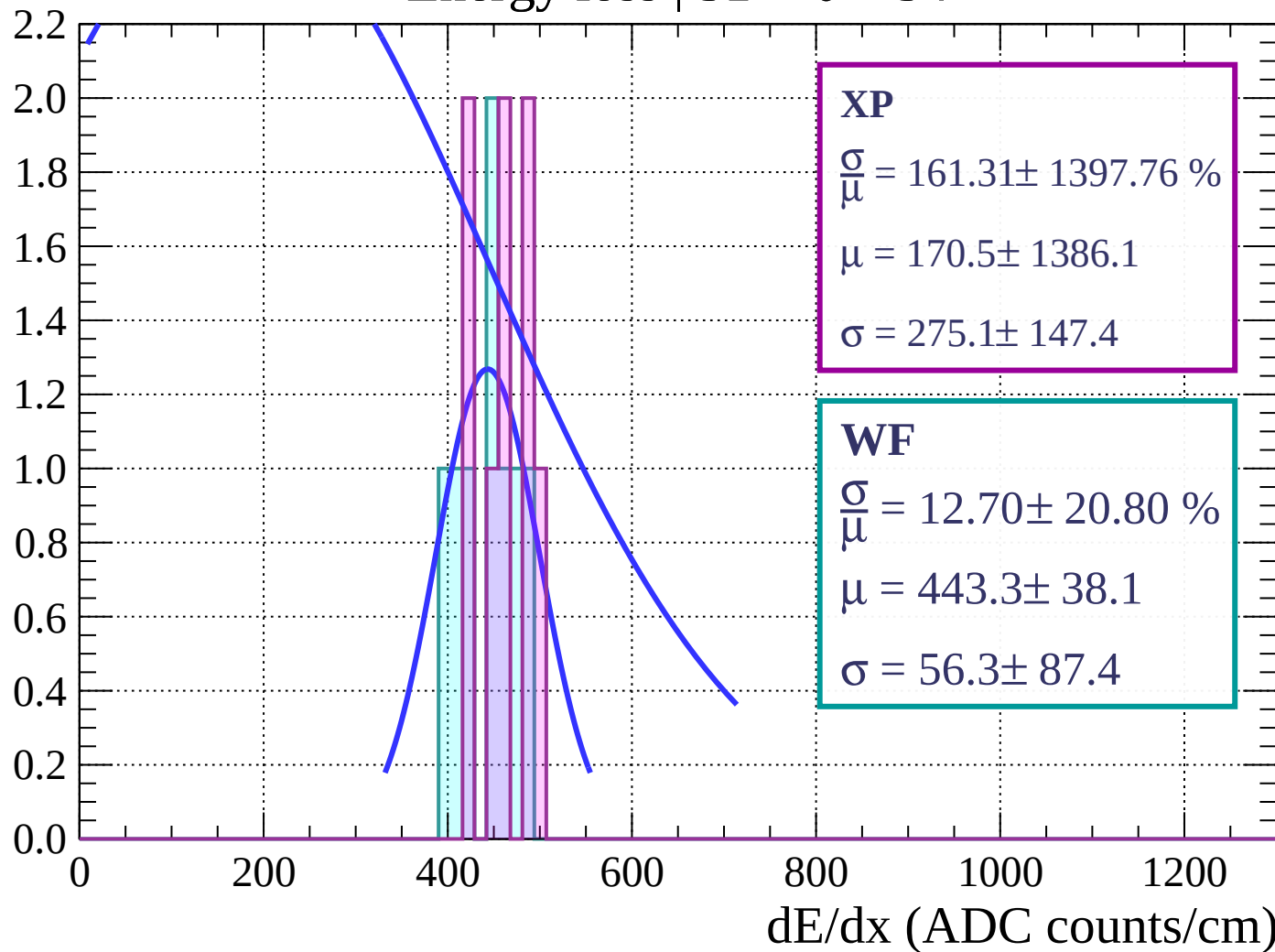
Energy loss | $48 < \theta < 51$

Count



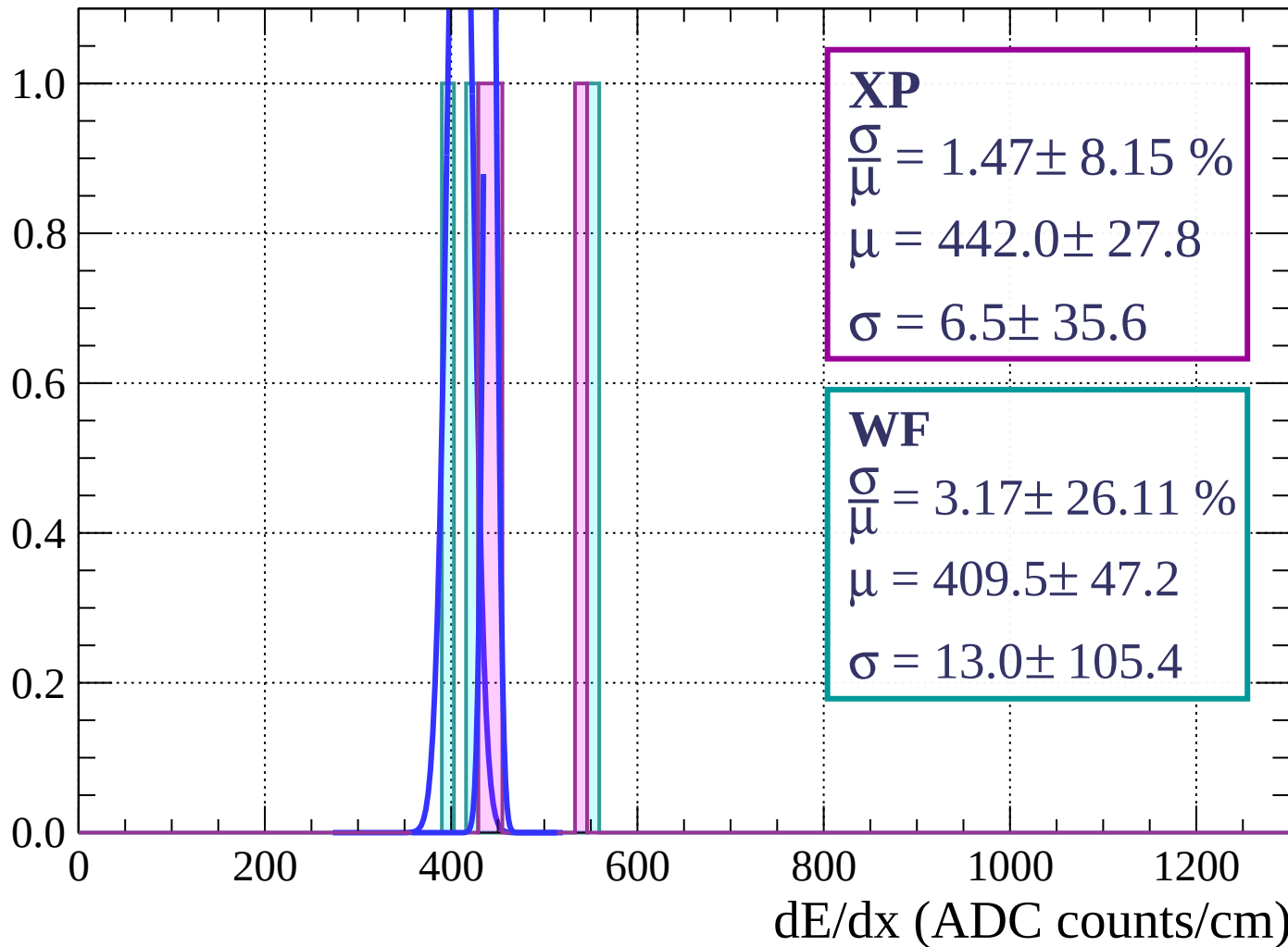
Energy loss | $51 < \theta < 54$

Count



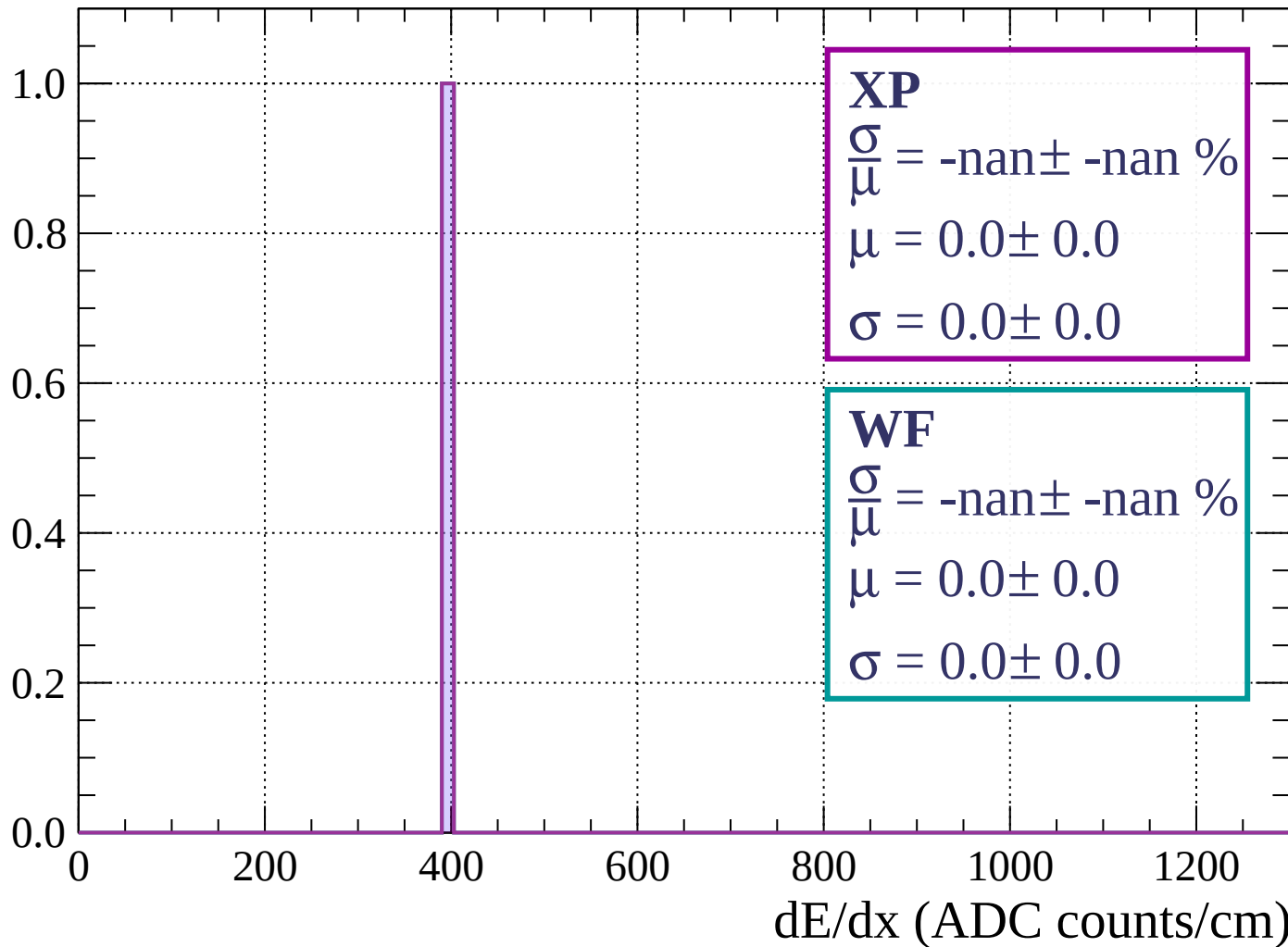
Energy loss | $54 < \theta < 57$

Count



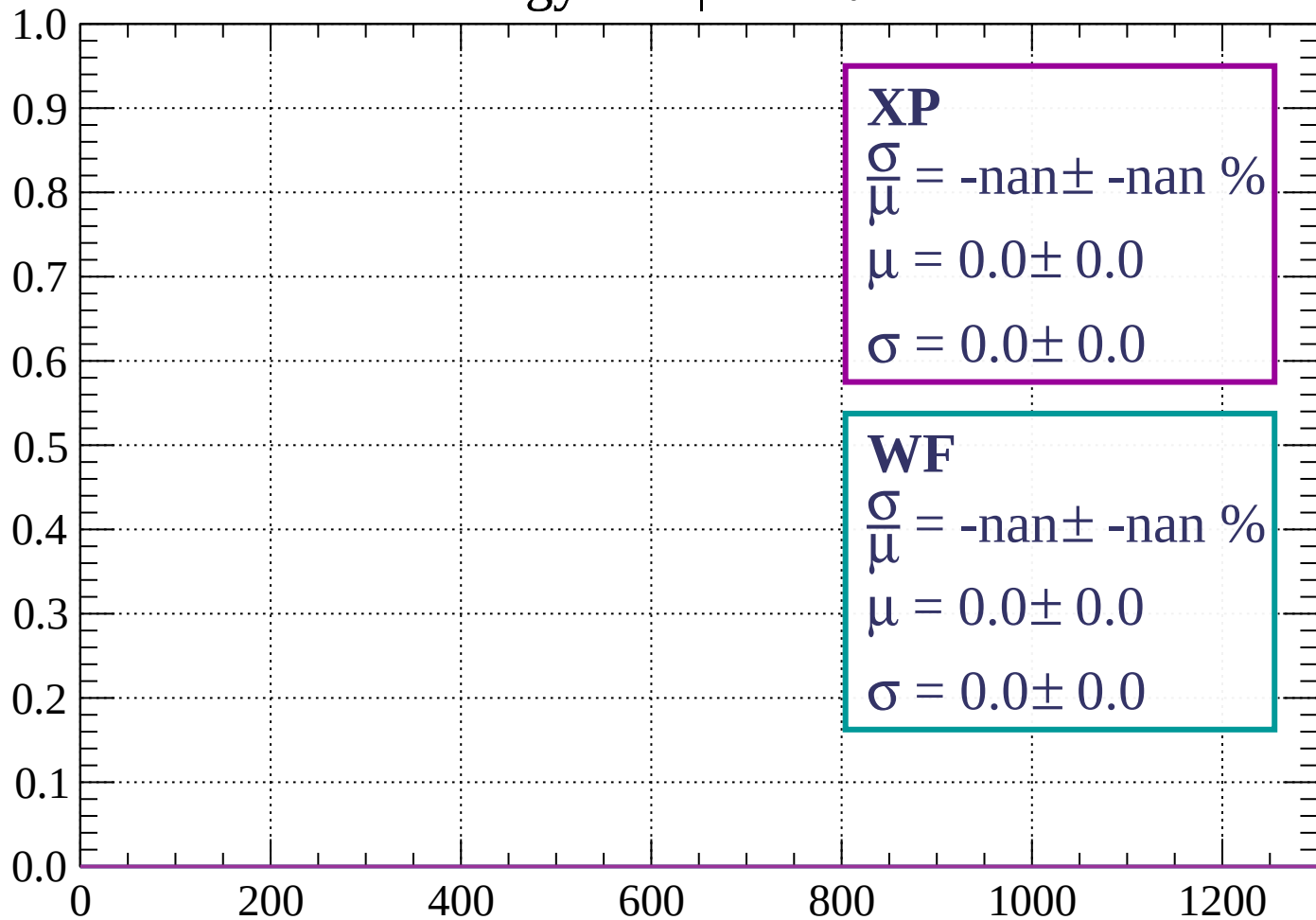
Energy loss | $57 < \theta < 60$

Count



Energy loss | $60 < \theta < 63$

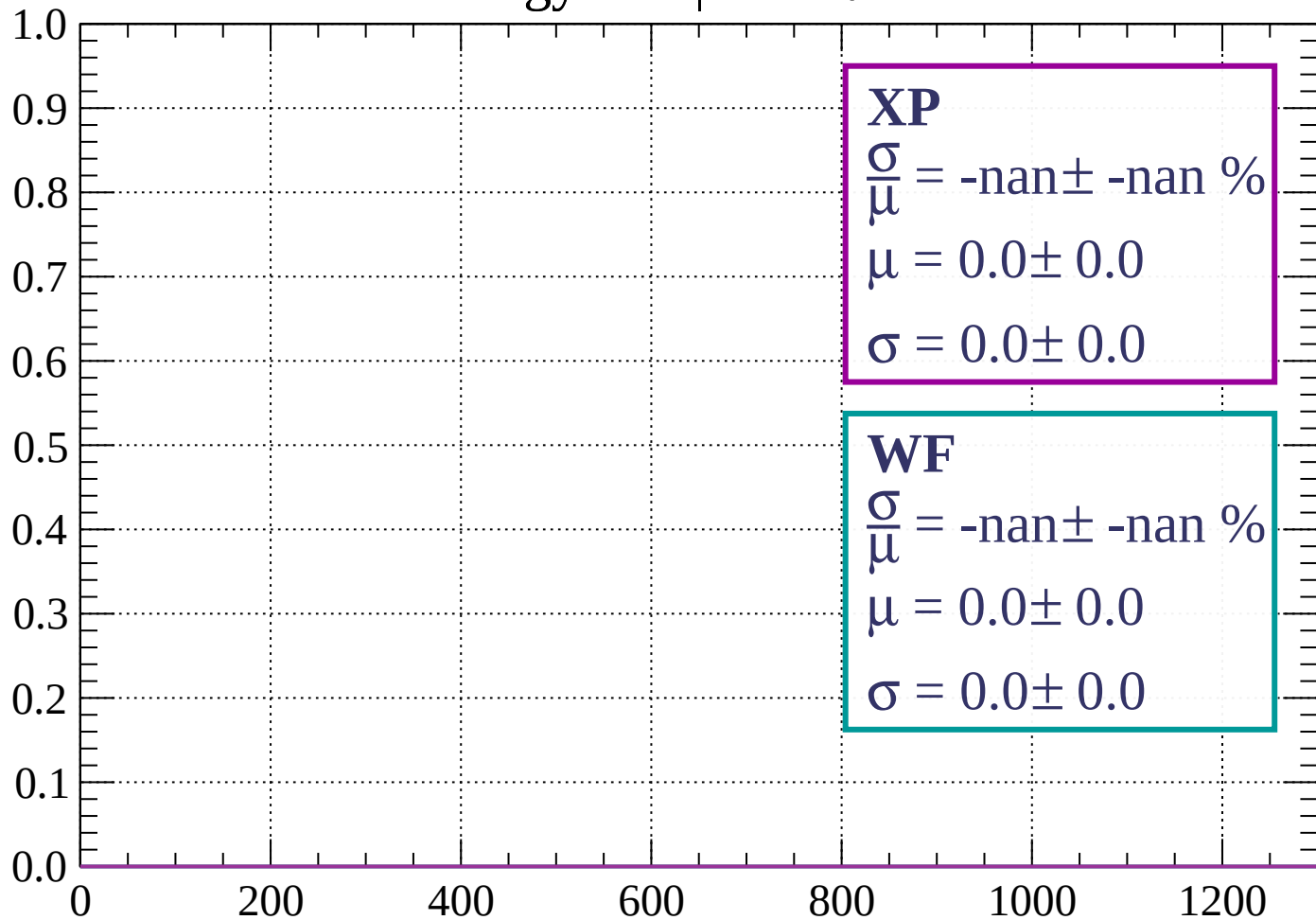
Count



dE/dx (ADC counts/cm)

Energy loss | $63 < \theta < 66$

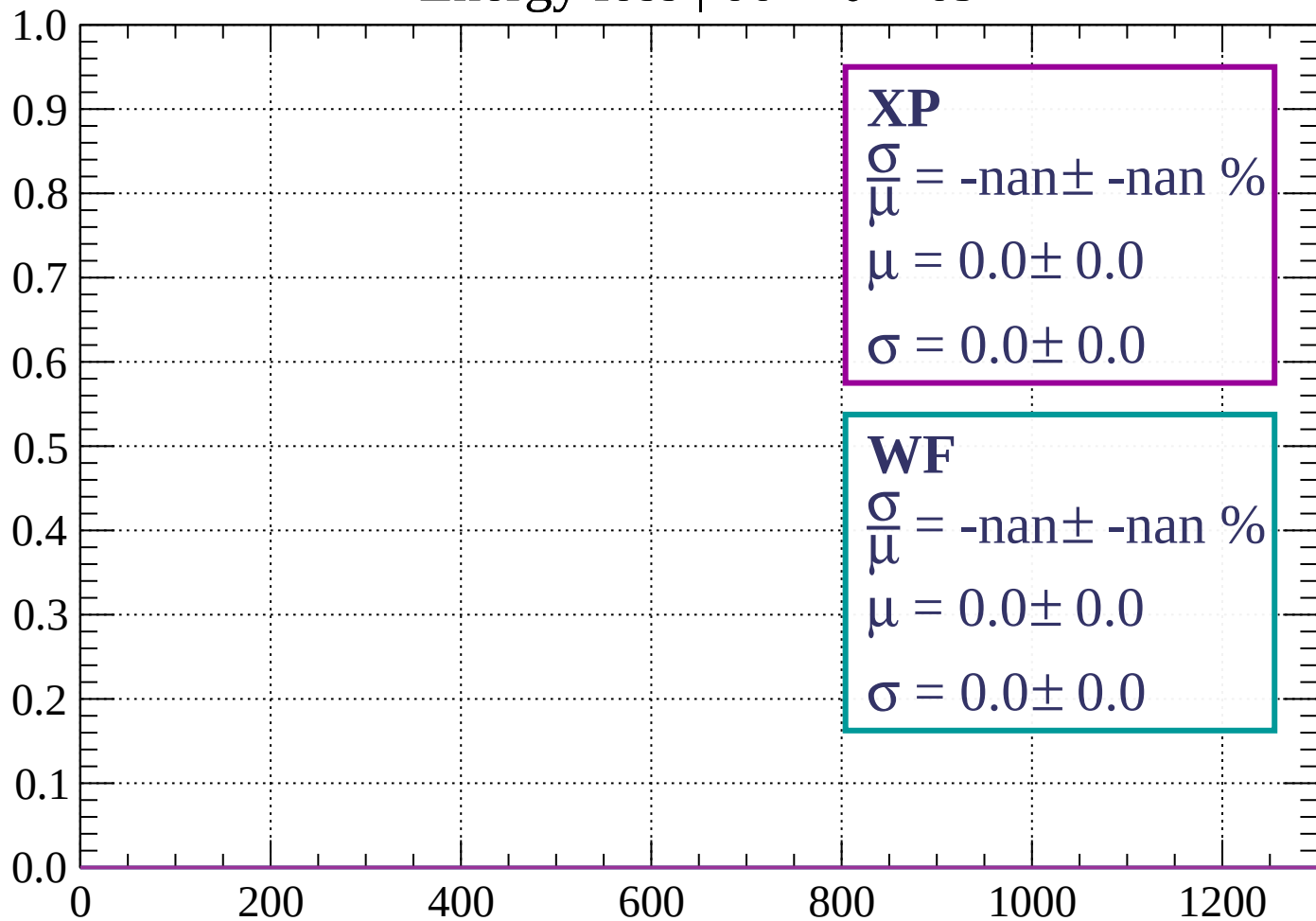
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 69$

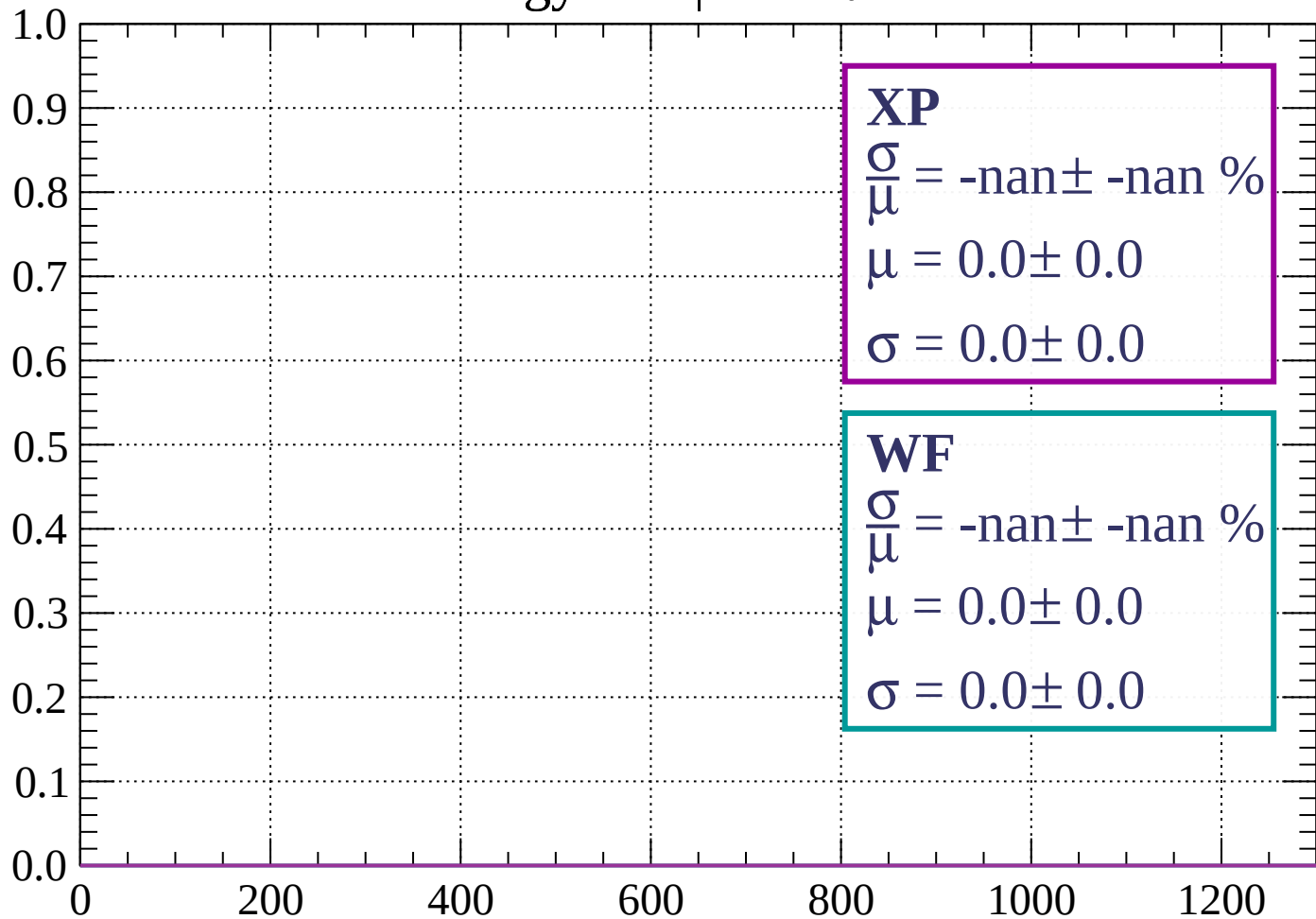
Count



dE/dx (ADC counts/cm)

Energy loss | $69 < \theta < 72$

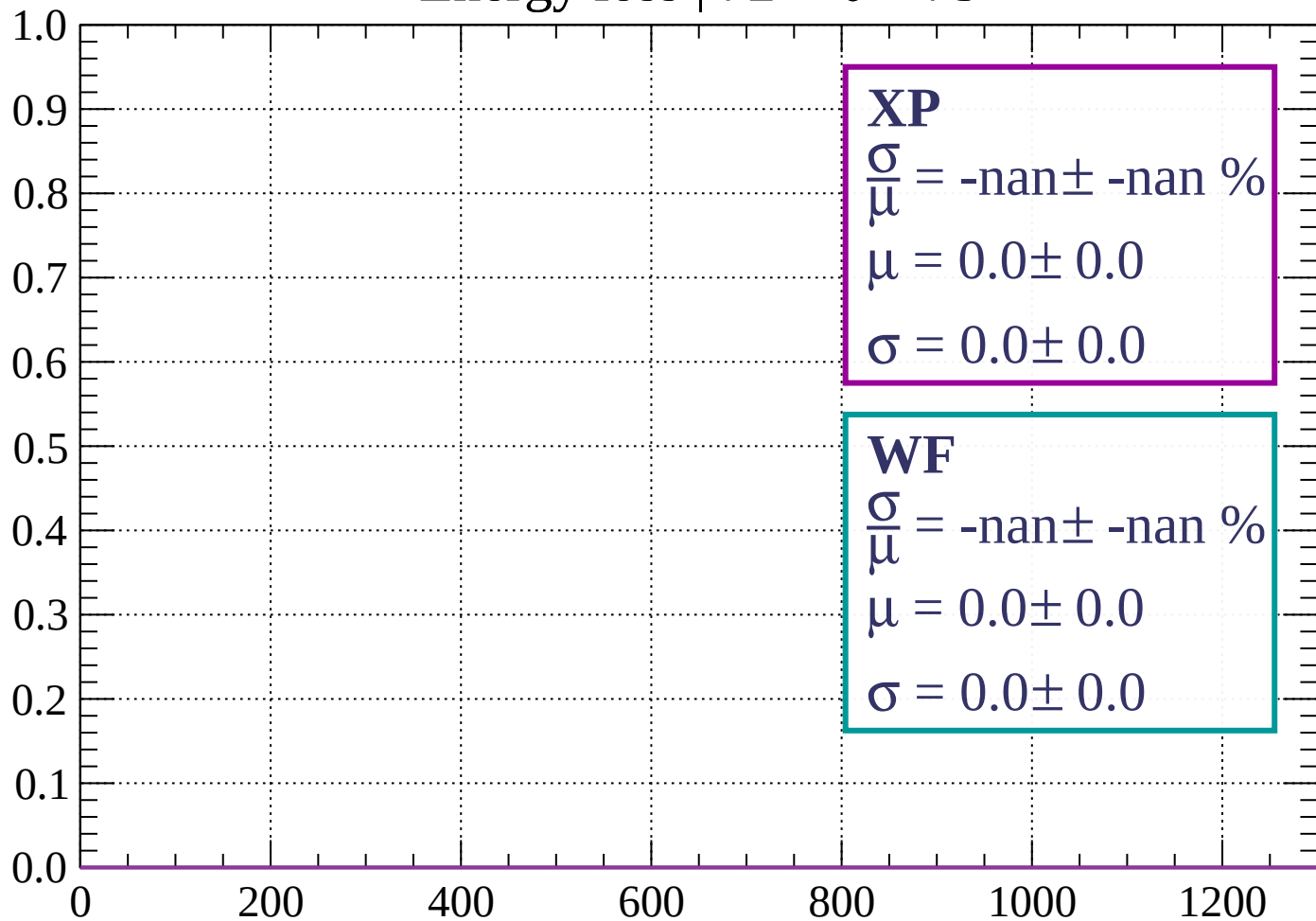
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 75$

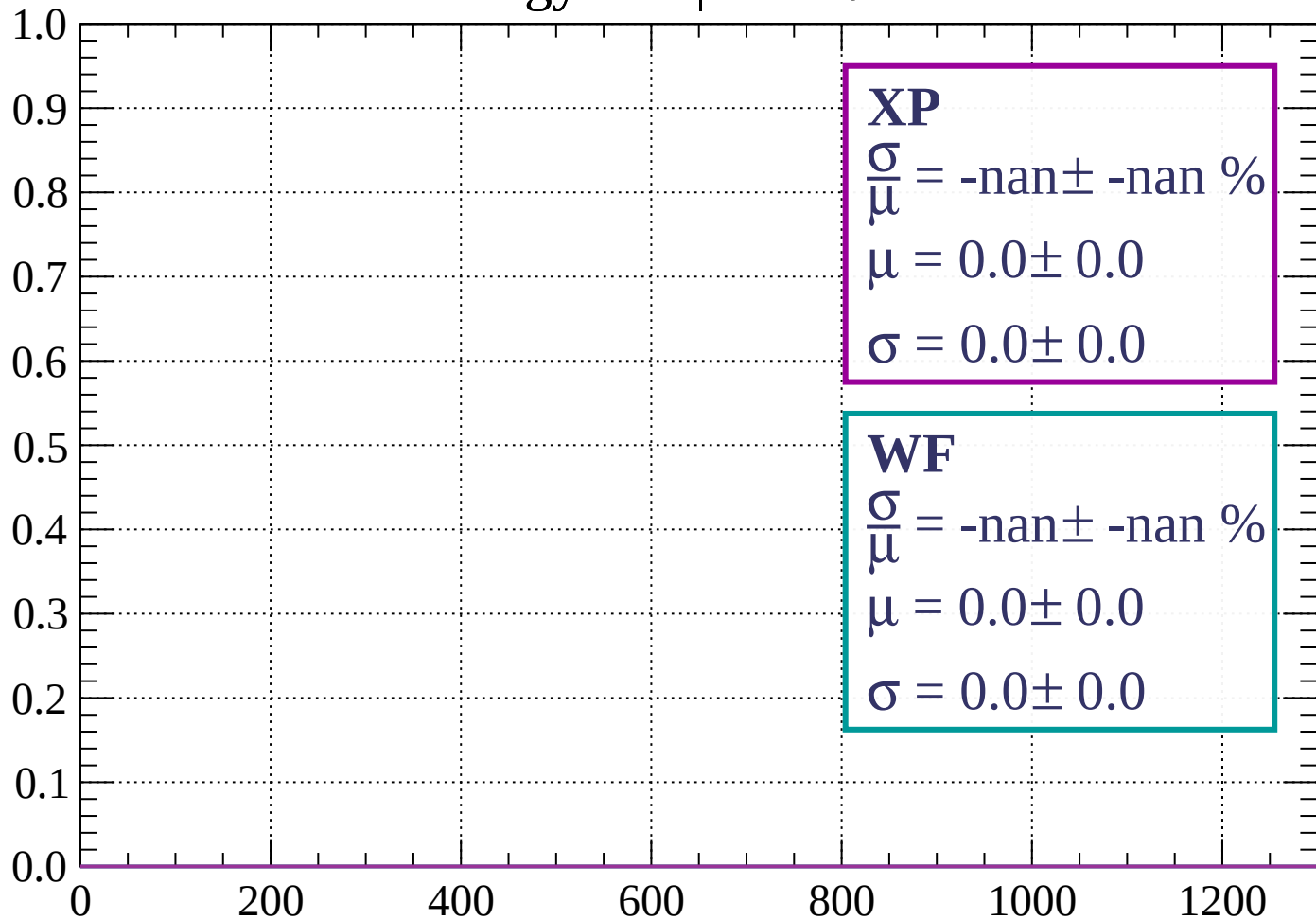
Count



dE/dx (ADC counts/cm)

Energy loss | $75 < \theta < 78$

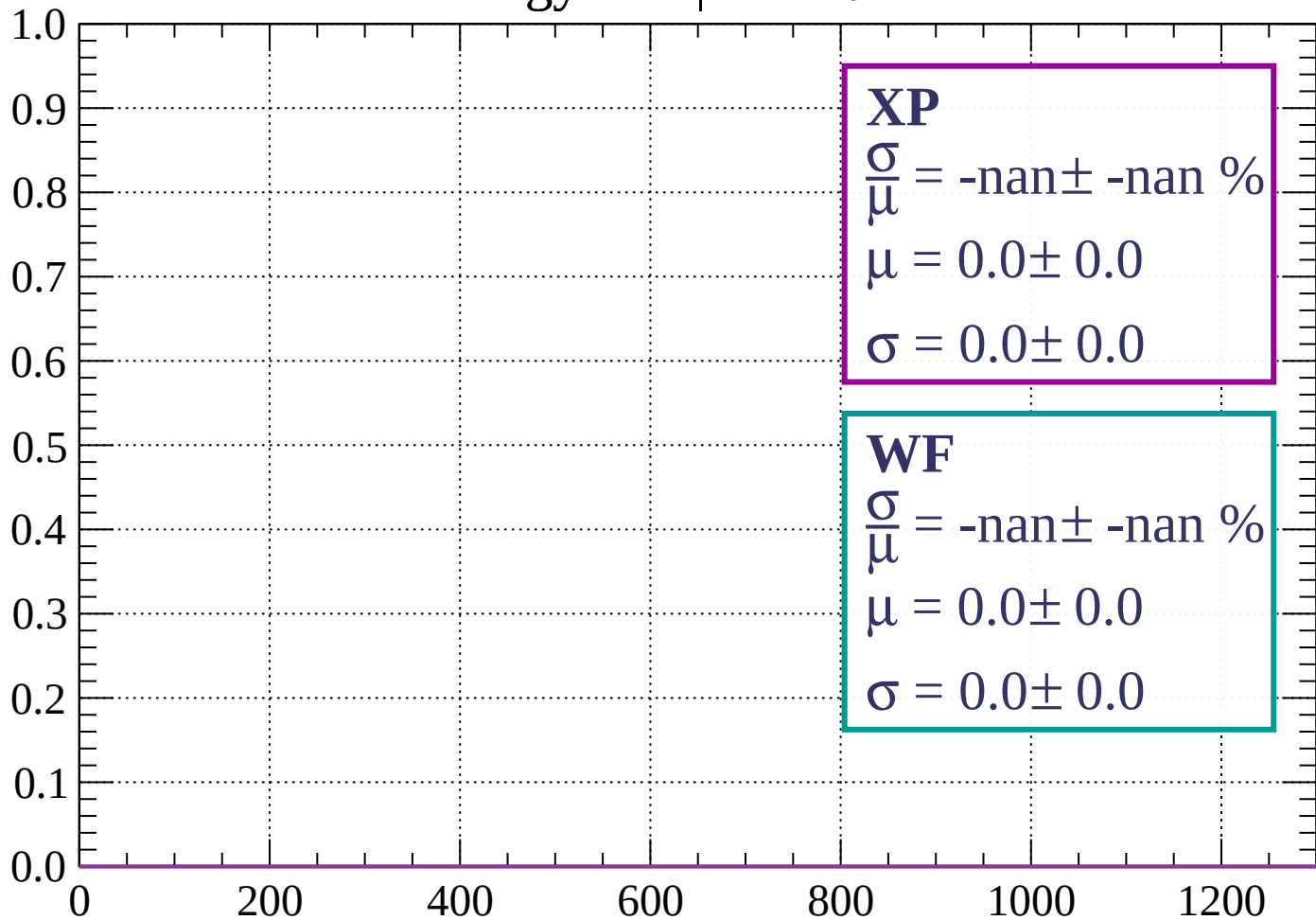
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$

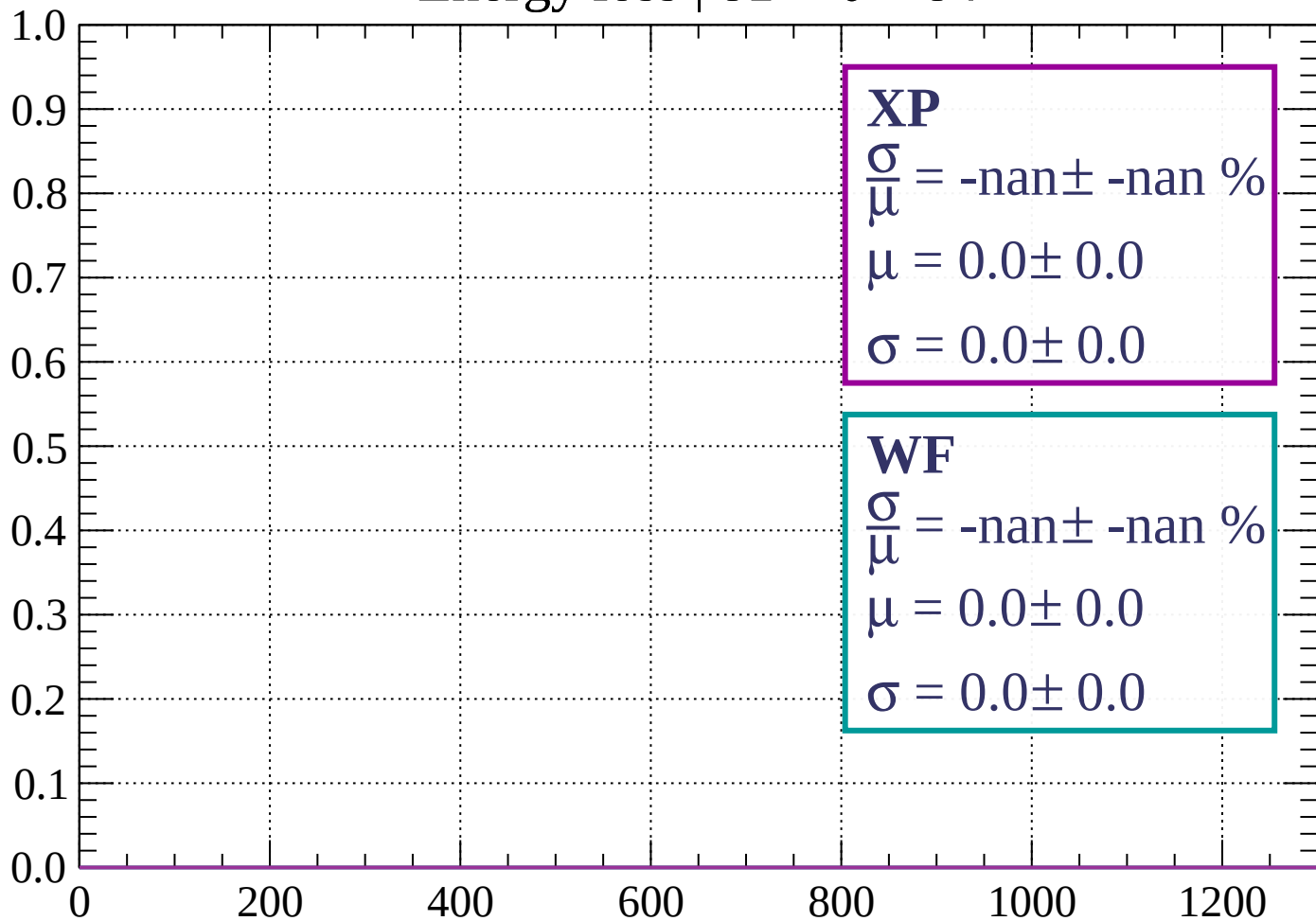
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \theta < 84$

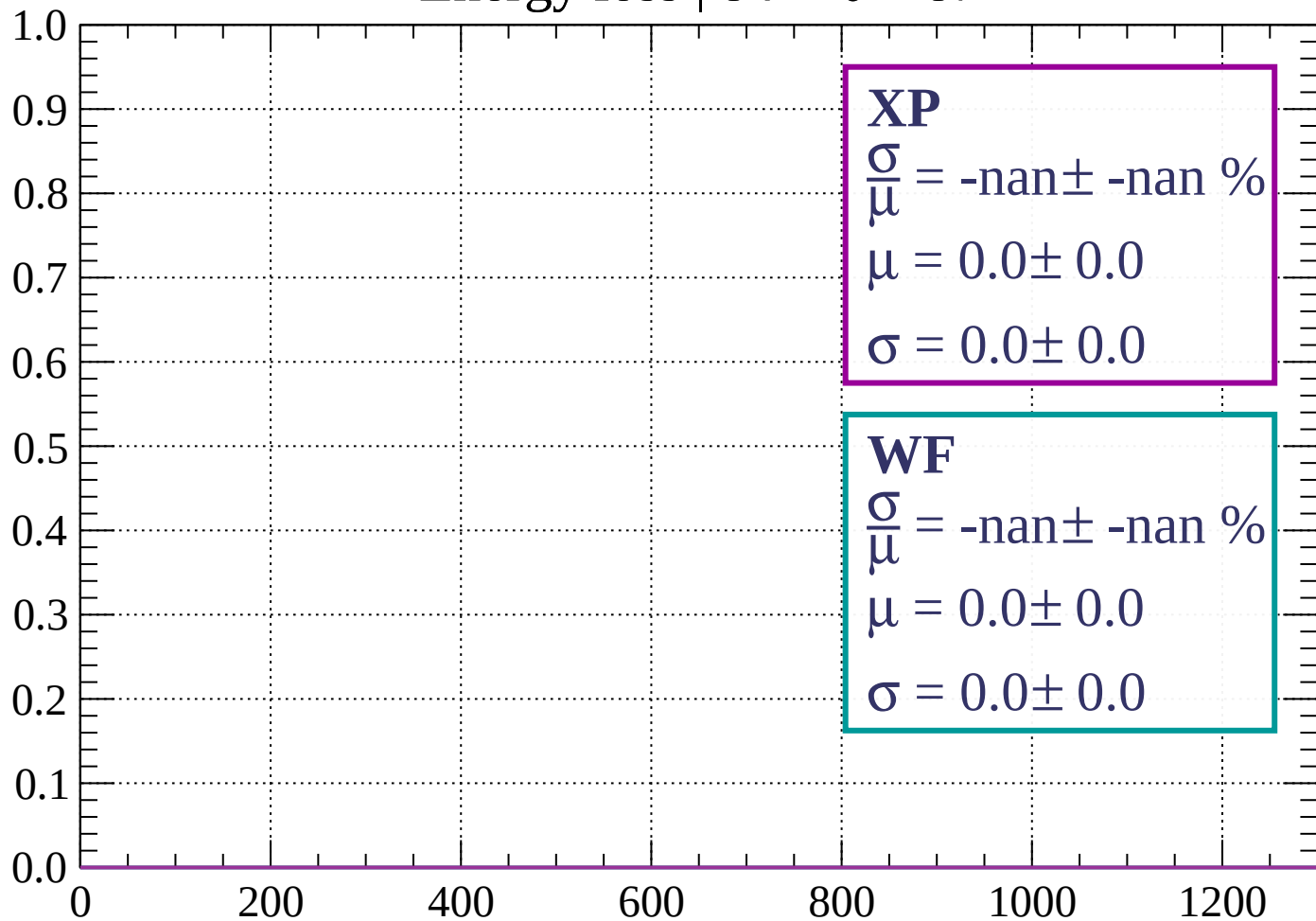
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

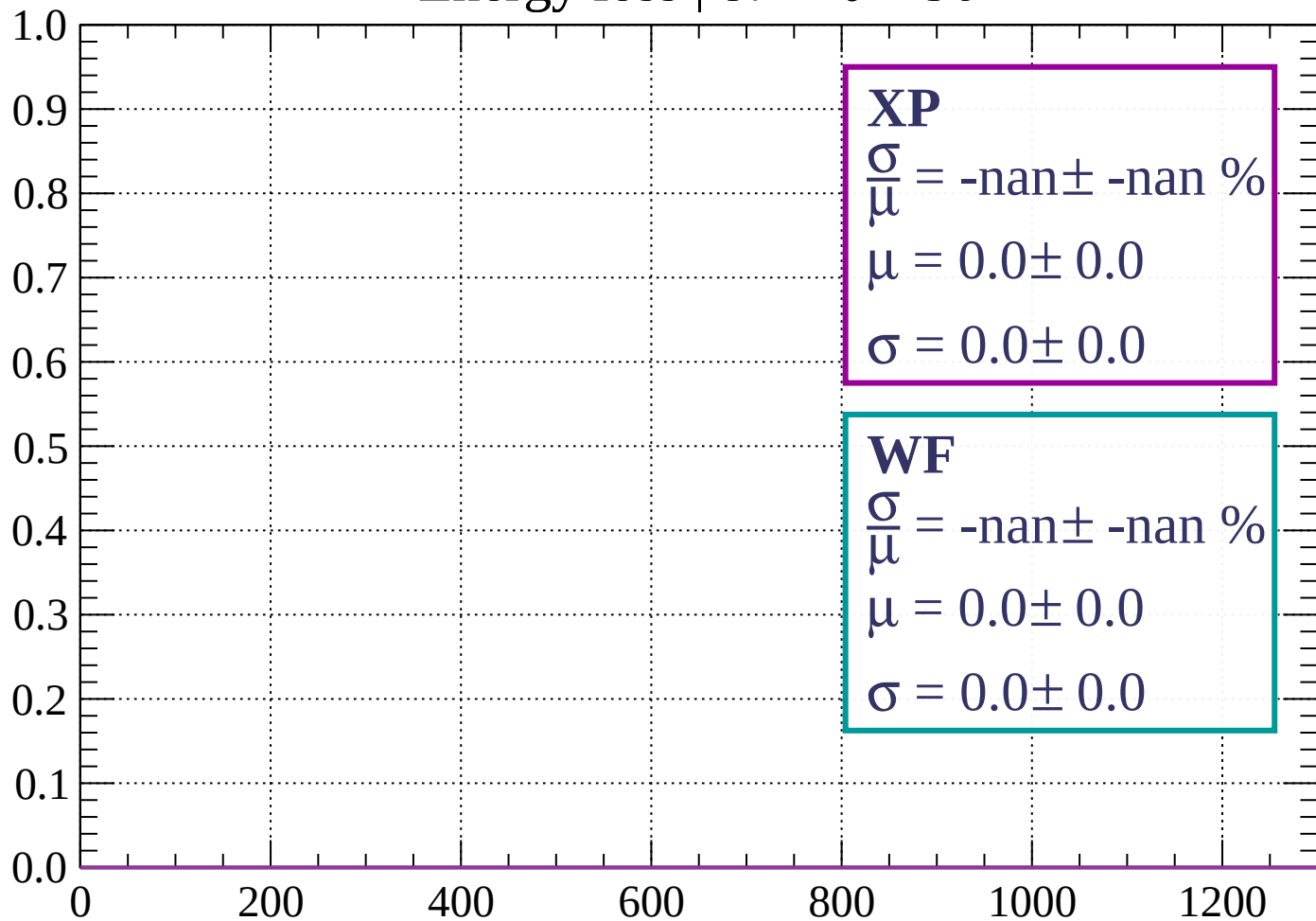
Count



dE/dx (ADC counts/cm)

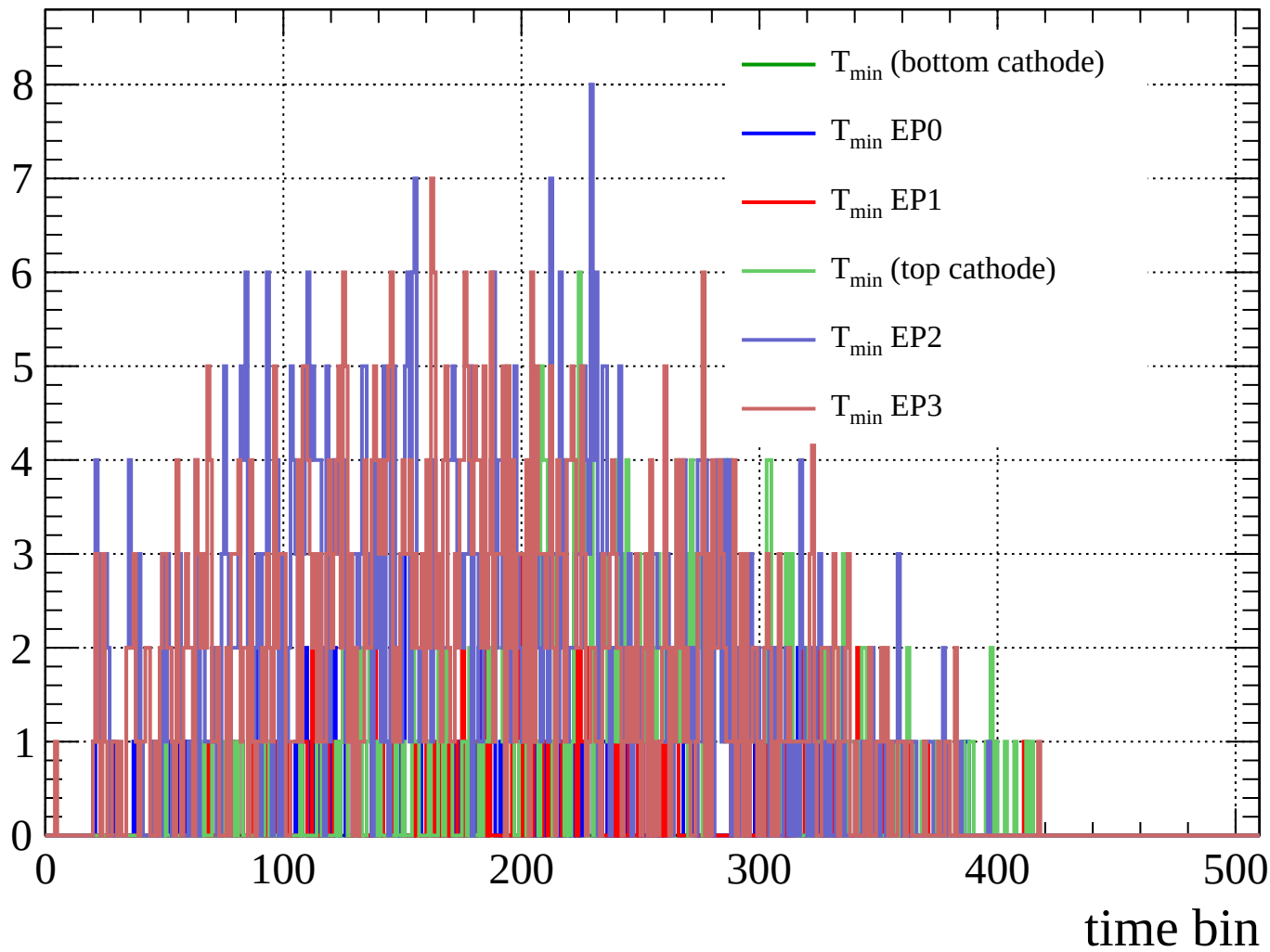
Energy loss | $87 < \theta < 90$

Count



dE/dx (ADC counts/cm)

Count



Count

